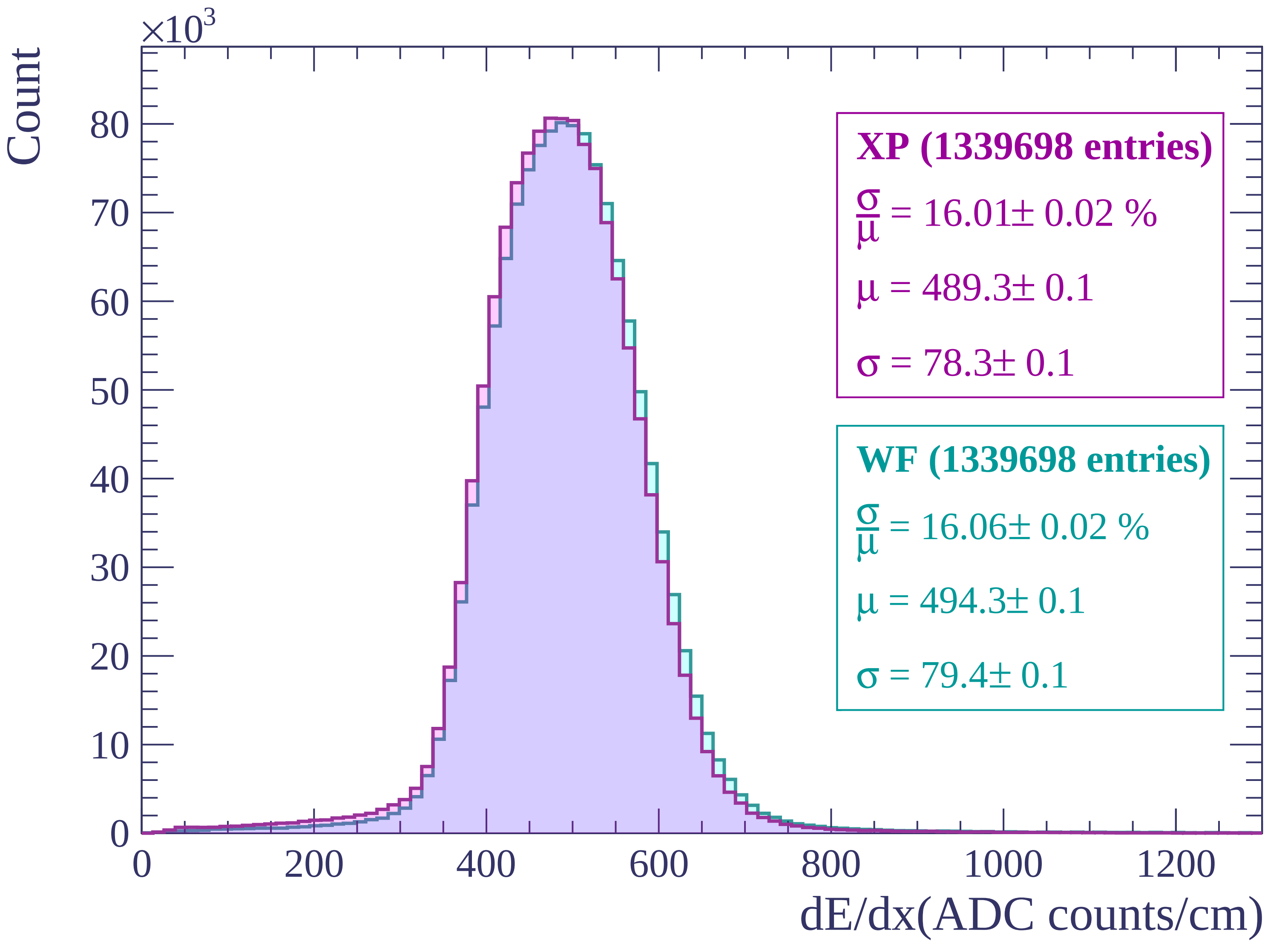
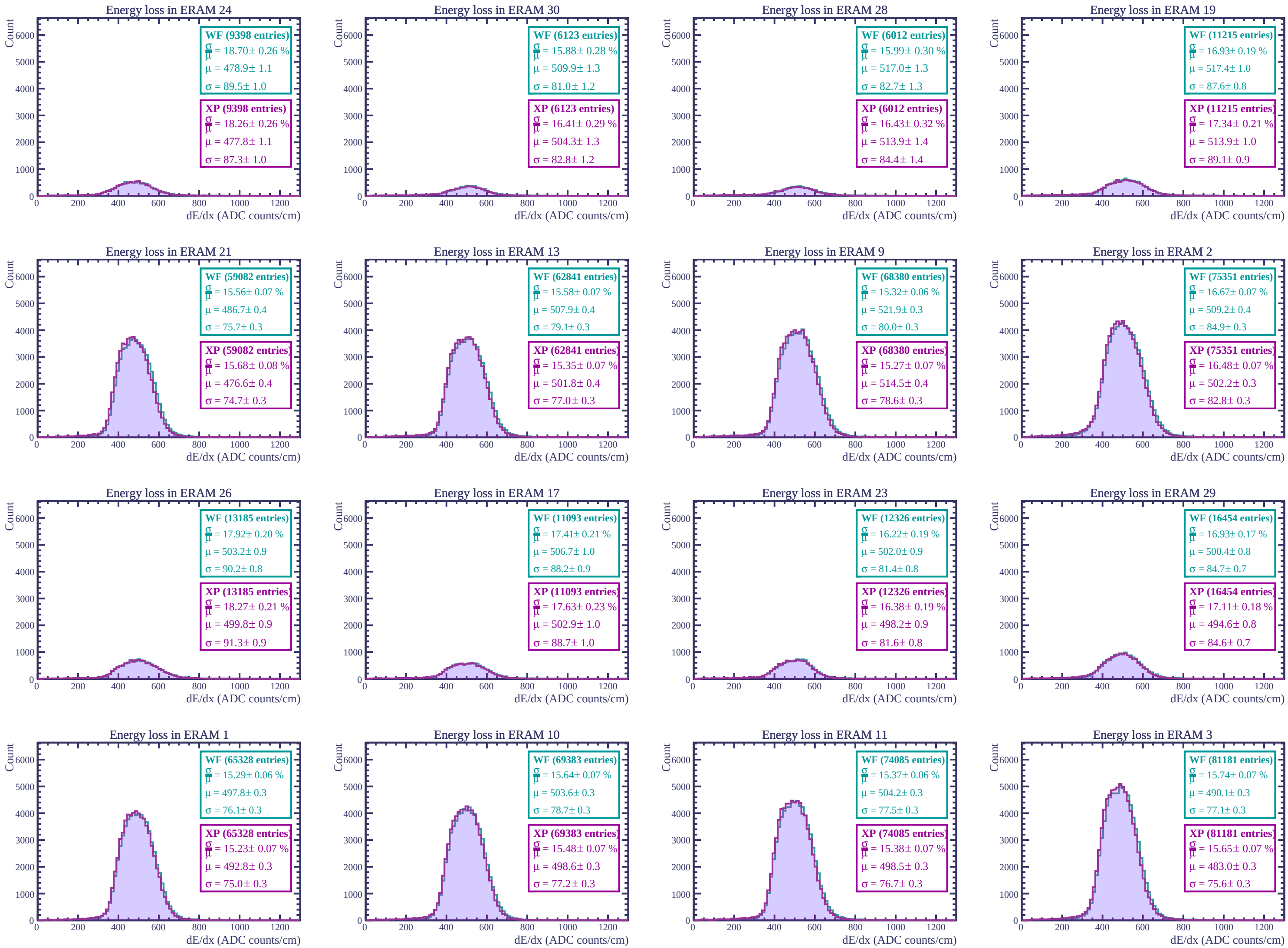
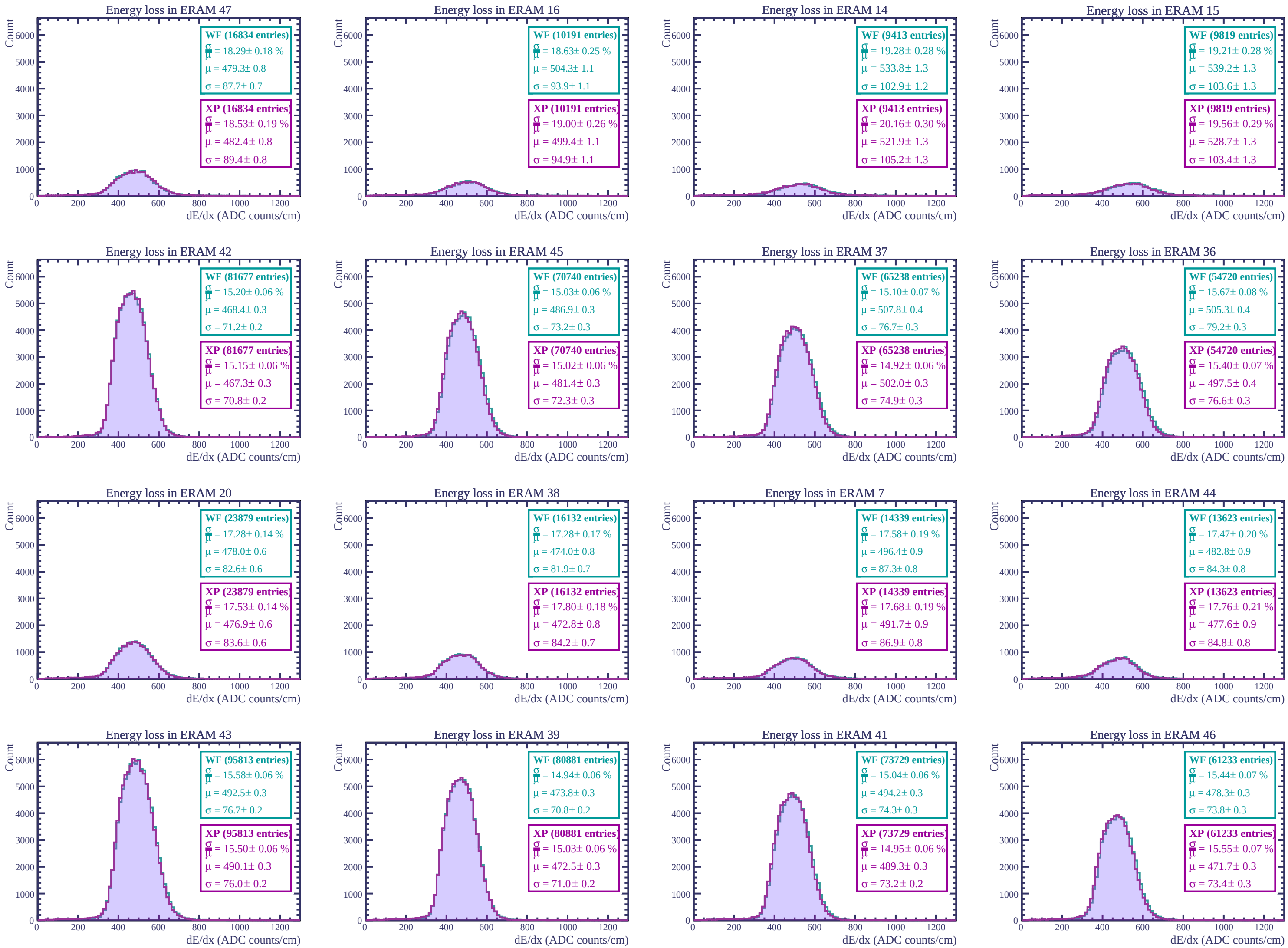
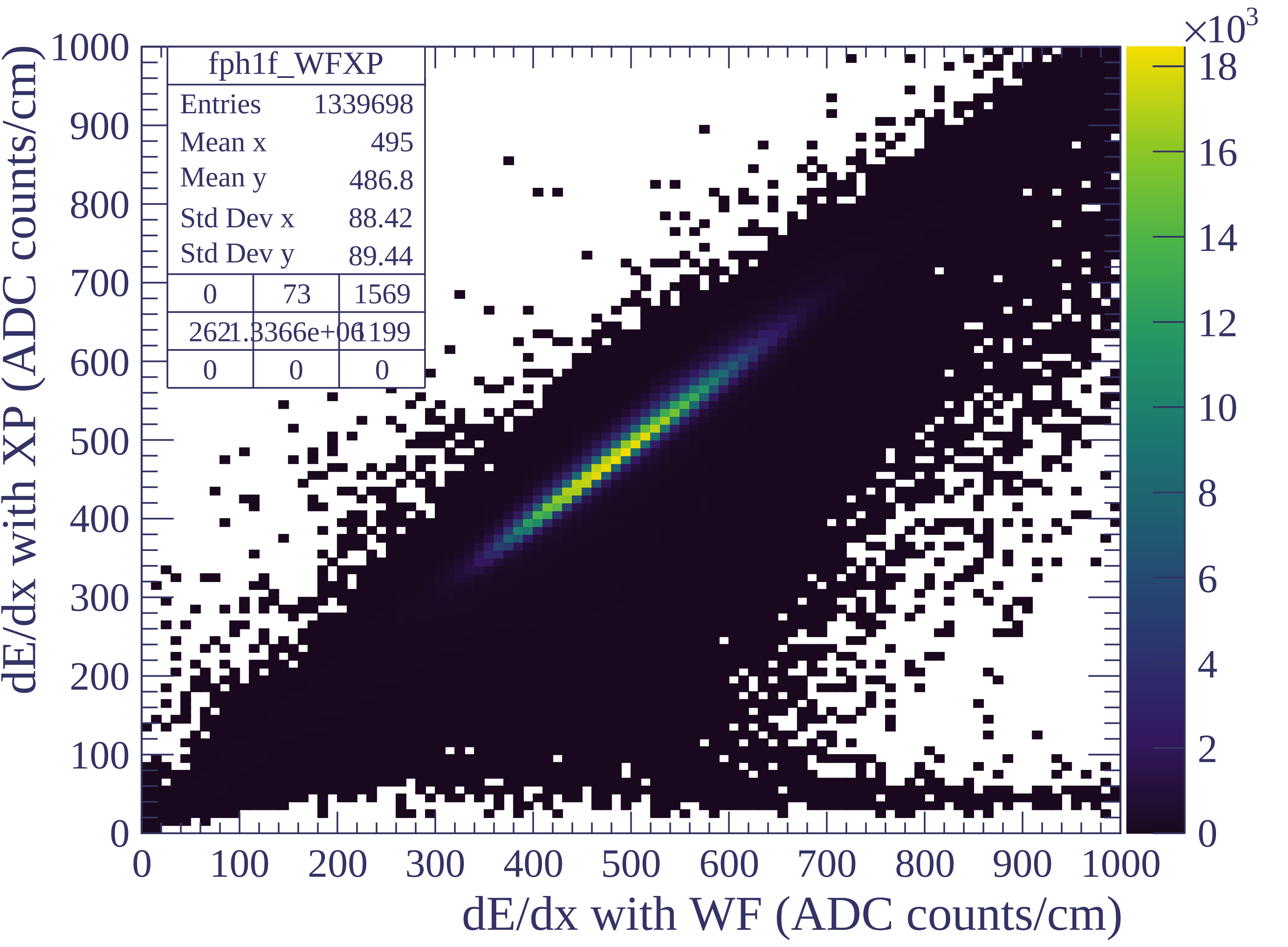


×10^3
Count
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (1339698 entries)
σ/μ = 16.01± 0.02 %
μ = 489.3± 0.1
σ = 78.3± 0.1
WF (1339698 entries)
σ/μ = 16.06± 0.02 %
μ = 494.3± 0.1
σ = 79.4± 0.1

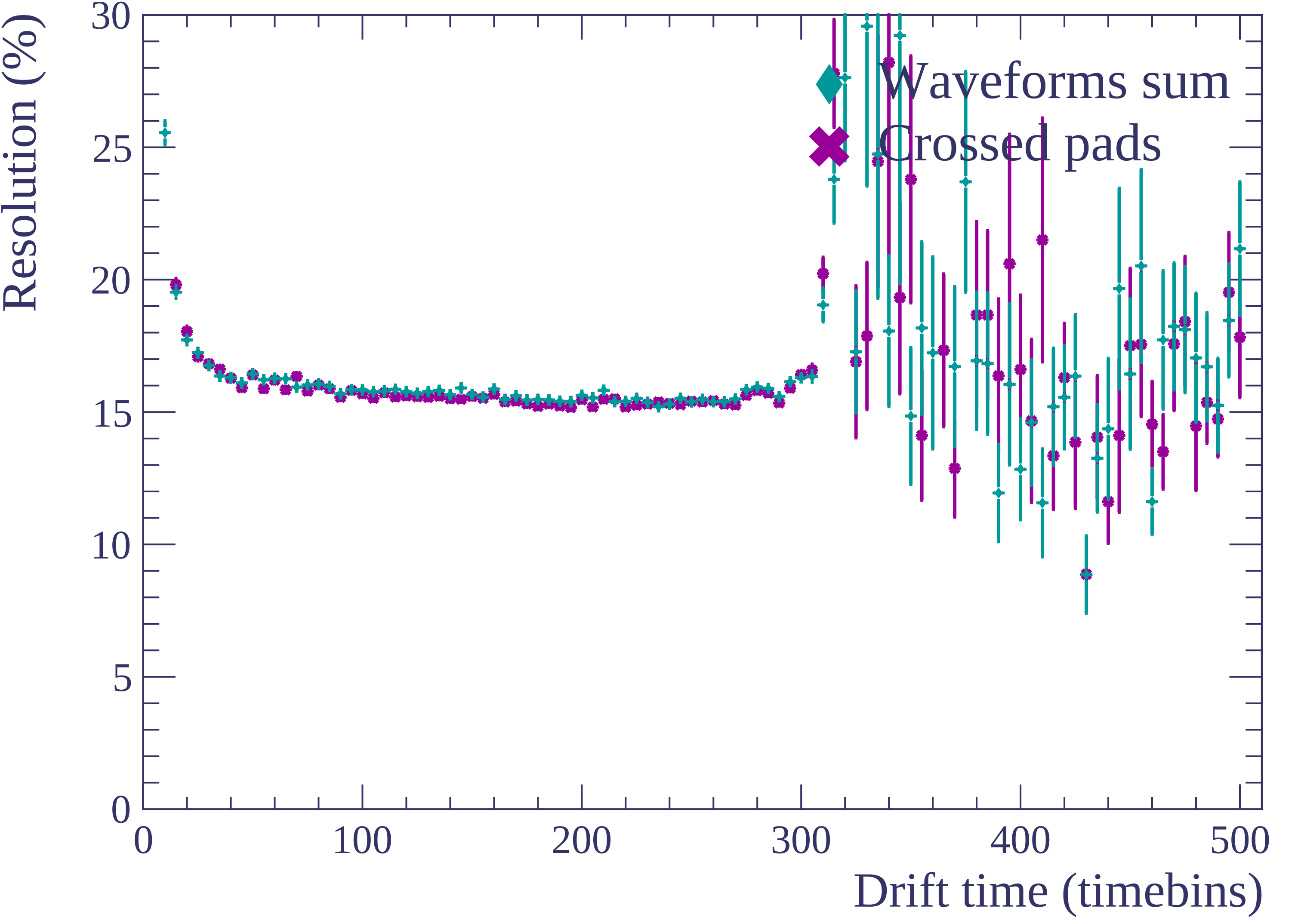


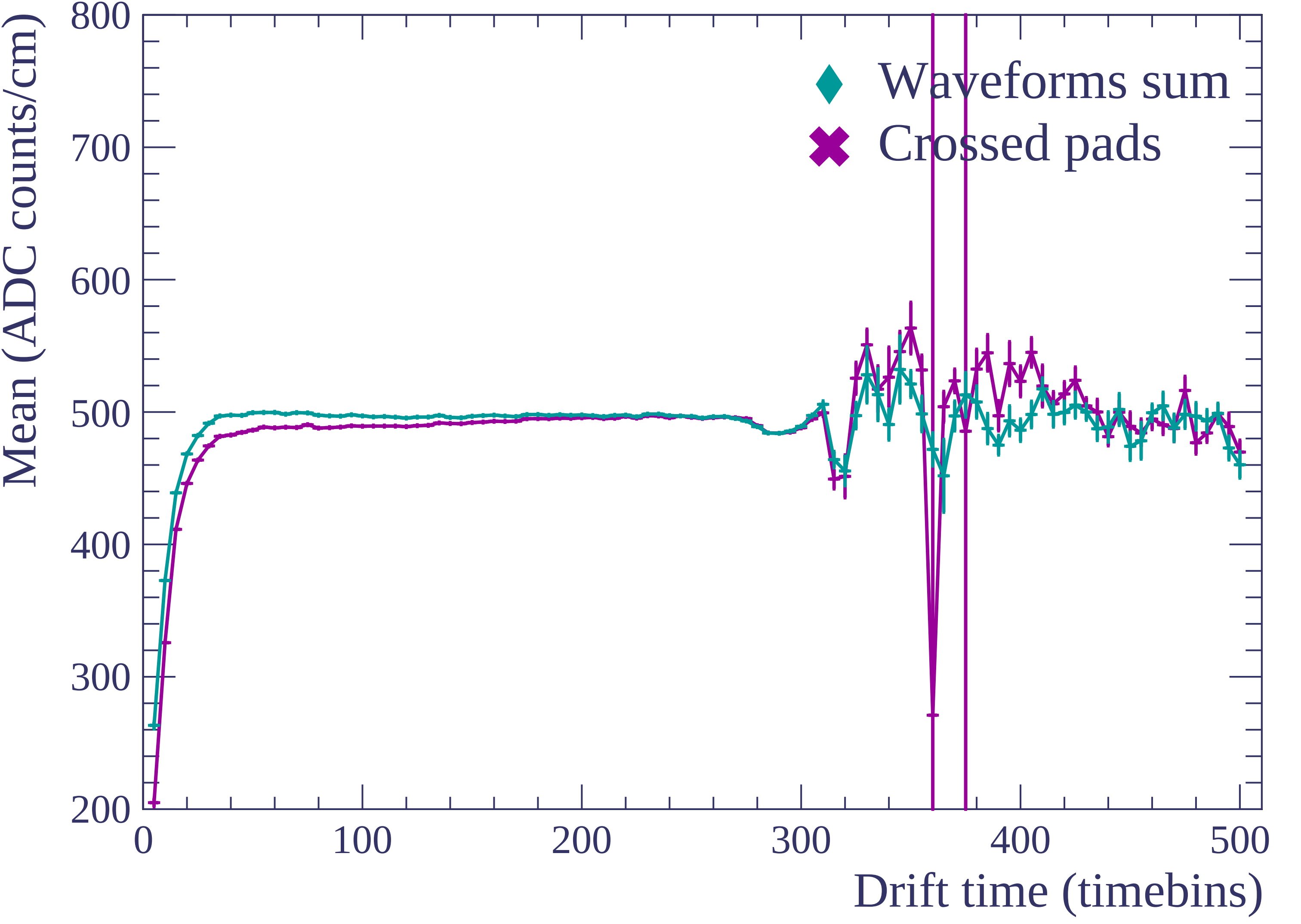
Energy loss in ERAM 24
WF (9398 entries)
σ/μ = 18.70± 0.26 %
μ = 478.9± 1.1
σ = 89.5± 1.0
XP (9398 entries)
σ/μ = 18.26± 0.26 %
μ = 477.8± 1.1
σ = 87.3± 1.0
Energy loss in ERAM 30
WF (6123 entries)
σ/μ = 15.88± 0.28 %
μ = 509.9± 1.3
σ = 81.0± 1.2
XP (6123 entries)
σ/μ = 16.41± 0.29 %
μ = 504.3± 1.3
σ = 82.8± 1.2
Energy loss in ERAM 28
WF (6012 entries)
σ/μ = 15.99± 0.30 %
μ = 517.0± 1.3
σ = 82.7± 1.3
XP (6012 entries)
σ/μ = 16.43± 0.32 %
μ = 513.9± 1.4
σ = 84.4± 1.4
Energy loss in ERAM 19
WF (11215 entries)
σ/μ = 16.93± 0.19 %
μ = 517.4± 1.0
σ = 87.6± 0.8
XP (11215 entries)
σ/μ = 17.34± 0.21 %
μ = 513.9± 1.0
σ = 89.1± 0.9
Energy loss in ERAM 21
WF (59082 entries)
σ/μ = 15.56± 0.07 %
μ = 486.7± 0.4
σ = 75.7± 0.3
XP (59082 entries)
σ/μ = 15.68± 0.08 %
μ = 476.6± 0.4
σ = 74.7± 0.3
Energy loss in ERAM 13
WF (62841 entries)
σ/μ = 15.58± 0.07 %
μ = 507.9± 0.4
σ = 79.1± 0.3
XP (62841 entries)
σ/μ = 15.35± 0.07 %
μ = 501.8± 0.4
σ = 77.0± 0.3
Energy loss in ERAM 9
WF (68380 entries)
σ/μ = 15.32± 0.06 %
μ = 521.9± 0.3
σ = 80.0± 0.3
XP (68380 entries)
σ/μ = 15.27± 0.07 %
μ = 514.5± 0.4
σ = 78.6± 0.3
Energy loss in ERAM 2
WF (75351 entries)
σ/μ = 16.67± 0.07 %
μ = 509.2± 0.4
σ = 84.9± 0.3
XP (75351 entries)
σ/μ = 16.48± 0.07 %
μ = 502.2± 0.3
σ = 82.8± 0.3
Energy loss in ERAM 26
WF (13185 entries)
σ/μ = 17.92± 0.20 %
μ = 503.2± 0.9
σ = 90.2± 0.8
XP (13185 entries)
σ/μ = 18.27± 0.21 %
μ = 499.8± 0.9
σ = 91.3± 0.9
Energy loss in ERAM 17
WF (11093 entries)
σ/μ = 17.41± 0.21 %
μ = 506.7± 1.0
σ = 88.2± 0.9
XP (11093 entries)
σ/μ = 17.63± 0.23 %
μ = 502.9± 1.0
σ = 88.7± 1.0
Energy loss in ERAM 23
WF (12326 entries)
σ/μ = 16.22± 0.19 %
μ = 502.0± 0.9
σ = 81.4± 0.8
XP (12326 entries)
σ/μ = 16.38± 0.19 %
μ = 498.2± 0.9
σ = 81.6± 0.8
Energy loss in ERAM 29
WF (16454 entries)
σ/μ = 16.93± 0.17 %
μ = 500.4± 0.8
σ = 84.7± 0.7
XP (16454 entries)
σ/μ = 17.11± 0.18 %
μ = 494.6± 0.8
σ = 84.6± 0.7
Energy loss in ERAM 1
WF (65328 entries)
σ/μ = 15.29± 0.06 %
μ = 497.8± 0.3
σ = 76.1± 0.3
XP (65328 entries)
σ/μ = 15.23± 0.07 %
μ = 492.8± 0.3
σ = 75.0± 0.3
Energy loss in ERAM 10
WF (69383 entries)
σ/μ = 15.64± 0.07 %
μ = 503.6± 0.3
σ = 78.7± 0.3
XP (69383 entries)
σ/μ = 15.48± 0.07 %
μ = 498.6± 0.3
σ = 77.2± 0.3
Energy loss in ERAM 11
WF (74085 entries)
σ/μ = 15.37± 0.06 %
μ = 504.2± 0.3
σ = 77.5± 0.3
XP (74085 entries)
σ/μ = 15.38± 0.07 %
μ = 498.5± 0.3
σ = 76.7± 0.3
Energy loss in ERAM 3
WF (81181 entries)
σ/μ = 15.74± 0.07 %
μ = 490.1± 0.3
σ = 77.1± 0.3
XP (81181 entries)
σ/μ = 15.65± 0.07 %
μ = 483.0± 0.3
σ = 75.6± 0.3
Count
dE/dx (ADC counts/cm)


Energy loss in ERAM 47
WF (16834 entries)
σ/μ = 18.29± 0.18 %
μ = 479.3± 0.8
σ = 87.7± 0.7
XP (16834 entries)
σ/μ = 18.53± 0.19 %
μ = 482.4± 0.8
σ = 89.4± 0.8
Energy loss in ERAM 16
WF (10191 entries)
σ/μ = 18.63± 0.25 %
μ = 504.3± 1.1
σ = 93.9± 1.1
XP (10191 entries)
σ/μ = 19.00± 0.26 %
μ = 499.4± 1.1
σ = 94.9± 1.1
Energy loss in ERAM 14
WF (9413 entries)
σ/μ = 19.28± 0.28 %
μ = 533.8± 1.3
σ = 102.9± 1.2
XP (9413 entries)
σ/μ = 20.16± 0.30 %
μ = 521.9± 1.3
σ = 105.2± 1.3
Energy loss in ERAM 15
WF (9819 entries)
σ/μ = 19.21± 0.28 %
μ = 539.2± 1.3
σ = 103.6± 1.3
XP (9819 entries)
σ/μ = 19.56± 0.29 %
μ = 528.7± 1.3
σ = 103.4± 1.3
Energy loss in ERAM 42
WF (81677 entries)
σ/μ = 15.20± 0.06 %
μ = 468.4± 0.3
σ = 71.2± 0.2
XP (81677 entries)
σ/μ = 15.15± 0.06 %
μ = 467.3± 0.3
σ = 70.8± 0.2
Energy loss in ERAM 45
WF (70740 entries)
σ/μ = 15.03± 0.06 %
μ = 486.9± 0.3
σ = 73.2± 0.3
XP (70740 entries)
σ/μ = 15.02± 0.06 %
μ = 481.4± 0.3
σ = 72.3± 0.3
Energy loss in ERAM 37
WF (65238 entries)
σ/μ = 15.10± 0.07 %
μ = 507.8± 0.4
σ = 76.7± 0.3
XP (65238 entries)
σ/μ = 14.92± 0.06 %
μ = 502.0± 0.3
σ = 74.9± 0.3
Energy loss in ERAM 36
WF (54720 entries)
σ/μ = 15.67± 0.08 %
μ = 505.3± 0.4
σ = 79.2± 0.3
XP (54720 entries)
σ/μ = 15.40± 0.07 %
μ = 497.5± 0.4
σ = 76.6± 0.3
Energy loss in ERAM 20
WF (23879 entries)
σ/μ = 17.28± 0.14 %
μ = 478.0± 0.6
σ = 82.6± 0.6
XP (23879 entries)
σ/μ = 17.53± 0.14 %
μ = 476.9± 0.6
σ = 83.6± 0.6
Energy loss in ERAM 38
WF (16132 entries)
σ/μ = 17.28± 0.17 %
μ = 474.0± 0.8
σ = 81.9± 0.7
XP (16132 entries)
σ/μ = 17.80± 0.18 %
μ = 472.8± 0.8
σ = 84.2± 0.7
Energy loss in ERAM 7
WF (14339 entries)
σ/μ = 17.58± 0.19 %
μ = 496.4± 0.9
σ = 87.3± 0.8
XP (14339 entries)
σ/μ = 17.68± 0.19 %
μ = 491.7± 0.9
σ = 86.9± 0.8
Energy loss in ERAM 44
WF (13623 entries)
σ/μ = 17.47± 0.20 %
μ = 482.8± 0.9
σ = 84.3± 0.8
XP (13623 entries)
σ/μ = 17.76± 0.21 %
μ = 477.6± 0.9
σ = 84.8± 0.8
Energy loss in ERAM 43
WF (95813 entries)
σ/μ = 15.58± 0.06 %
μ = 492.5± 0.3
σ = 76.7± 0.2
XP (95813 entries)
σ/μ = 15.50± 0.06 %
μ = 490.1± 0.3
σ = 76.0± 0.2
Energy loss in ERAM 39
WF (80881 entries)
σ/μ = 14.94± 0.06 %
μ = 473.8± 0.3
σ = 70.8± 0.2
XP (80881 entries)
σ/μ = 15.03± 0.06 %
μ = 472.5± 0.3
σ = 71.0± 0.2
Energy loss in ERAM 41
WF (73729 entries)
σ/μ = 15.04± 0.06 %
μ = 494.2± 0.3
σ = 74.3± 0.3
XP (73729 entries)
σ/μ = 14.95± 0.06 %
μ = 489.3± 0.3
σ = 73.2± 0.2
Energy loss in ERAM 46
WF (61233 entries)
σ/μ = 15.44± 0.07 %
μ = 478.3± 0.3
σ = 73.8± 0.3
XP (61233 entries)
σ/μ = 15.55± 0.07 %
μ = 471.7± 0.3
σ = 73.4± 0.3
Count
dE/dx (ADC counts/cm)

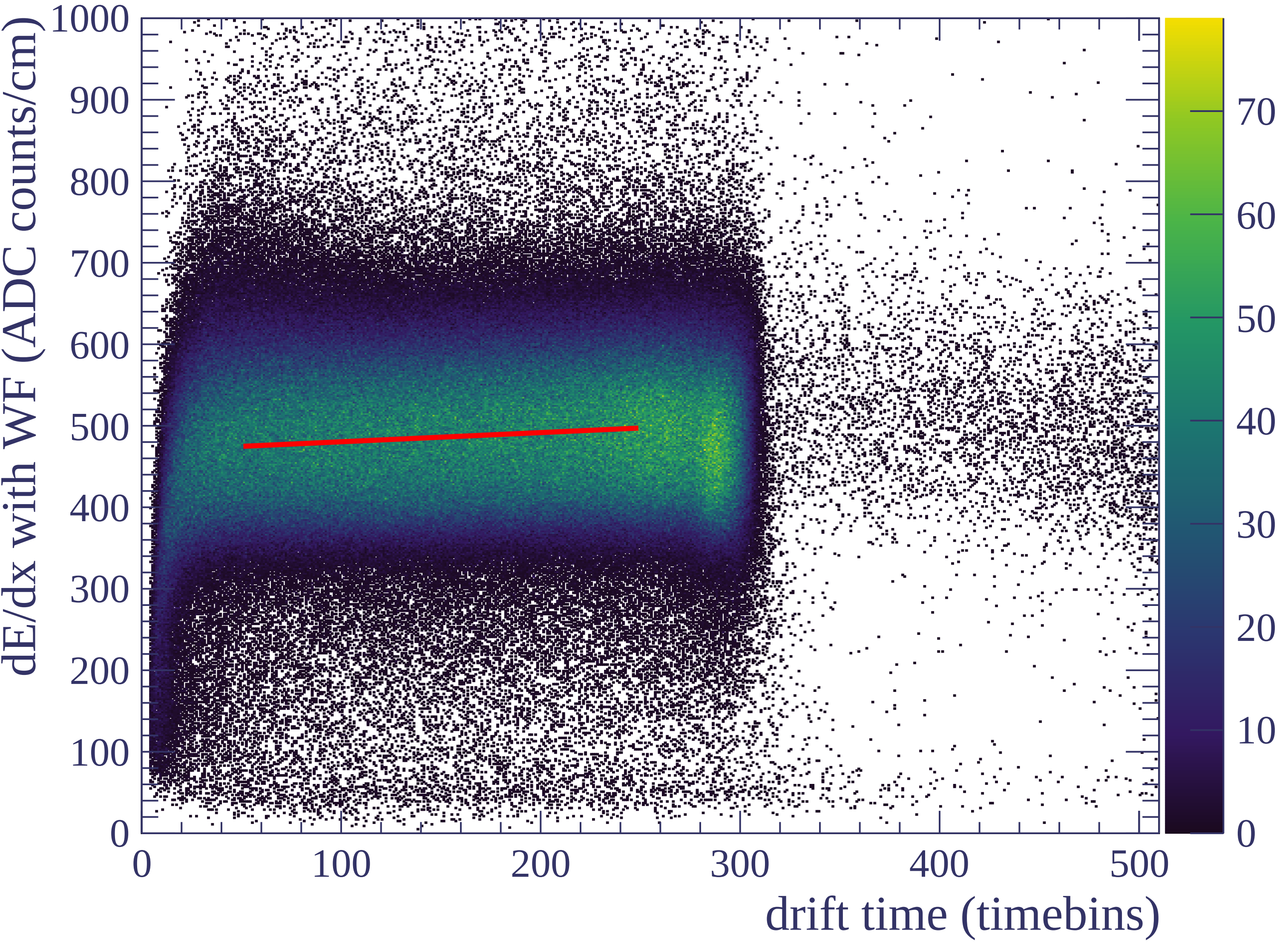


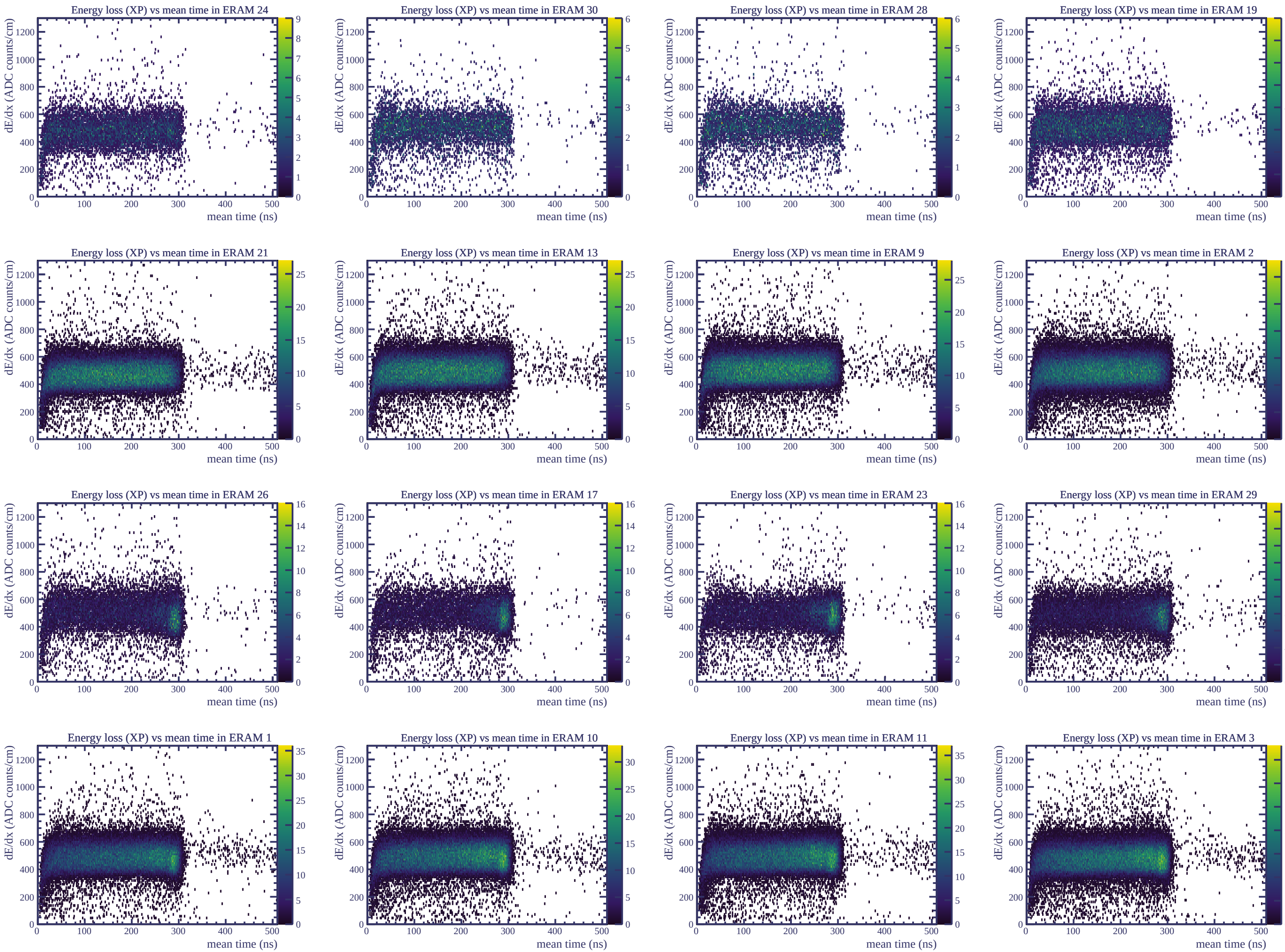
fph1f_WFXP
Entries 1339698
Mean x 495
Mean y 486.8
Std Dev x 88.42
Std Dev y 89.44
0 73 1569
2621.3366e+061199
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0 100 200 300 400 500 600 700 800 900 1000
0 2 4 6 8 10 12 14 16 18

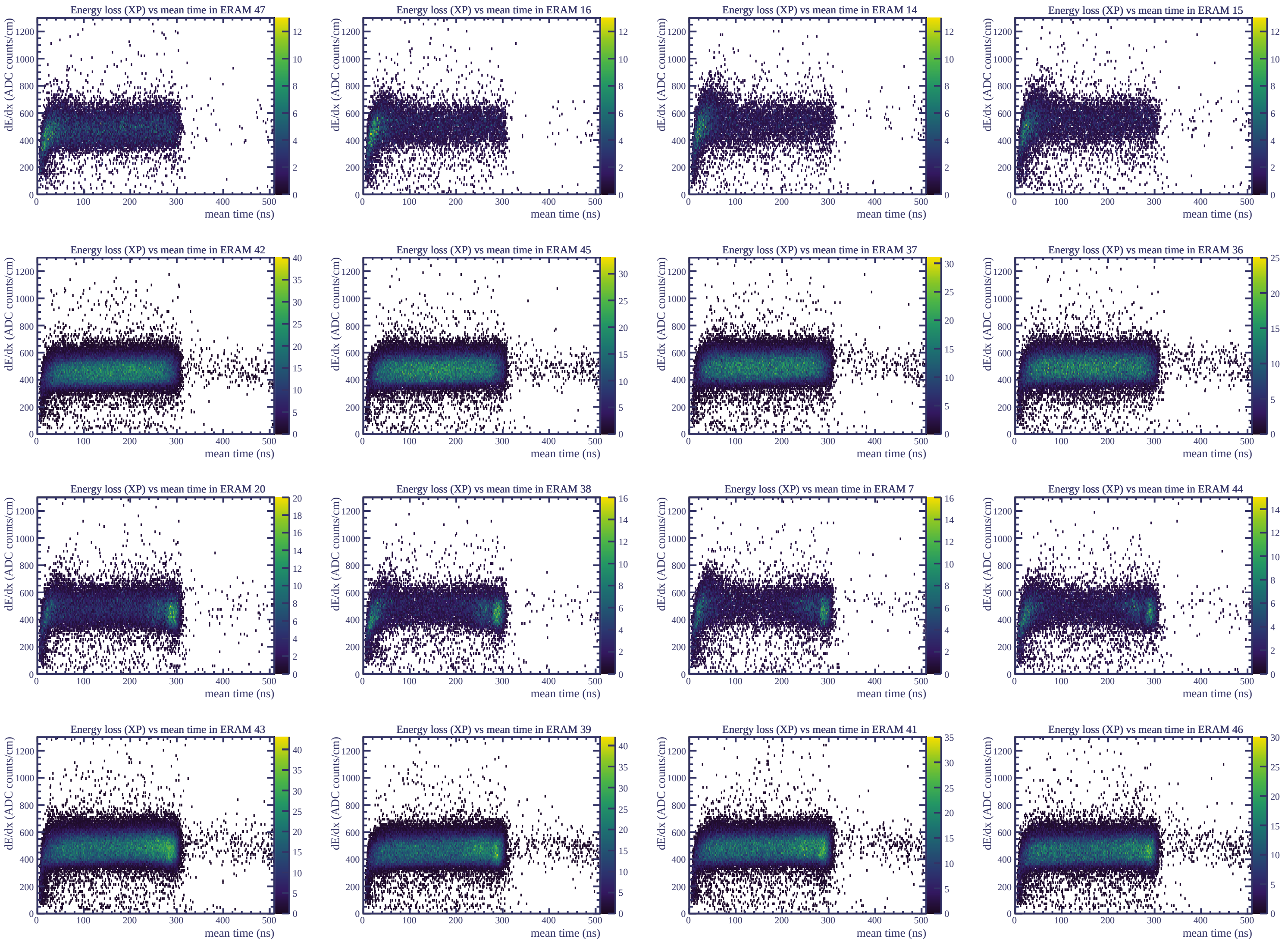

Resolution (%)
Drift time (timebins)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
0
100
200
300
400
500

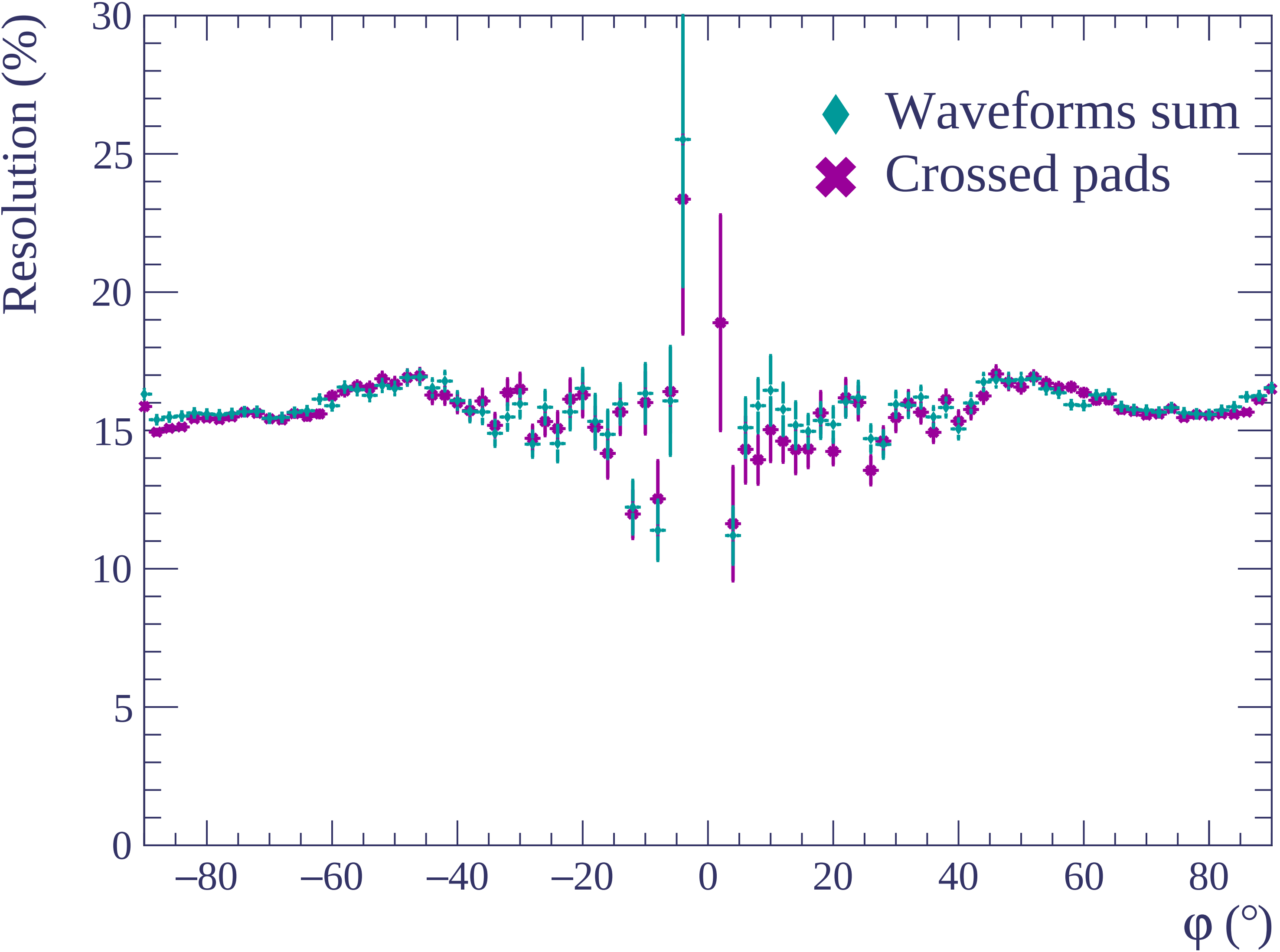


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

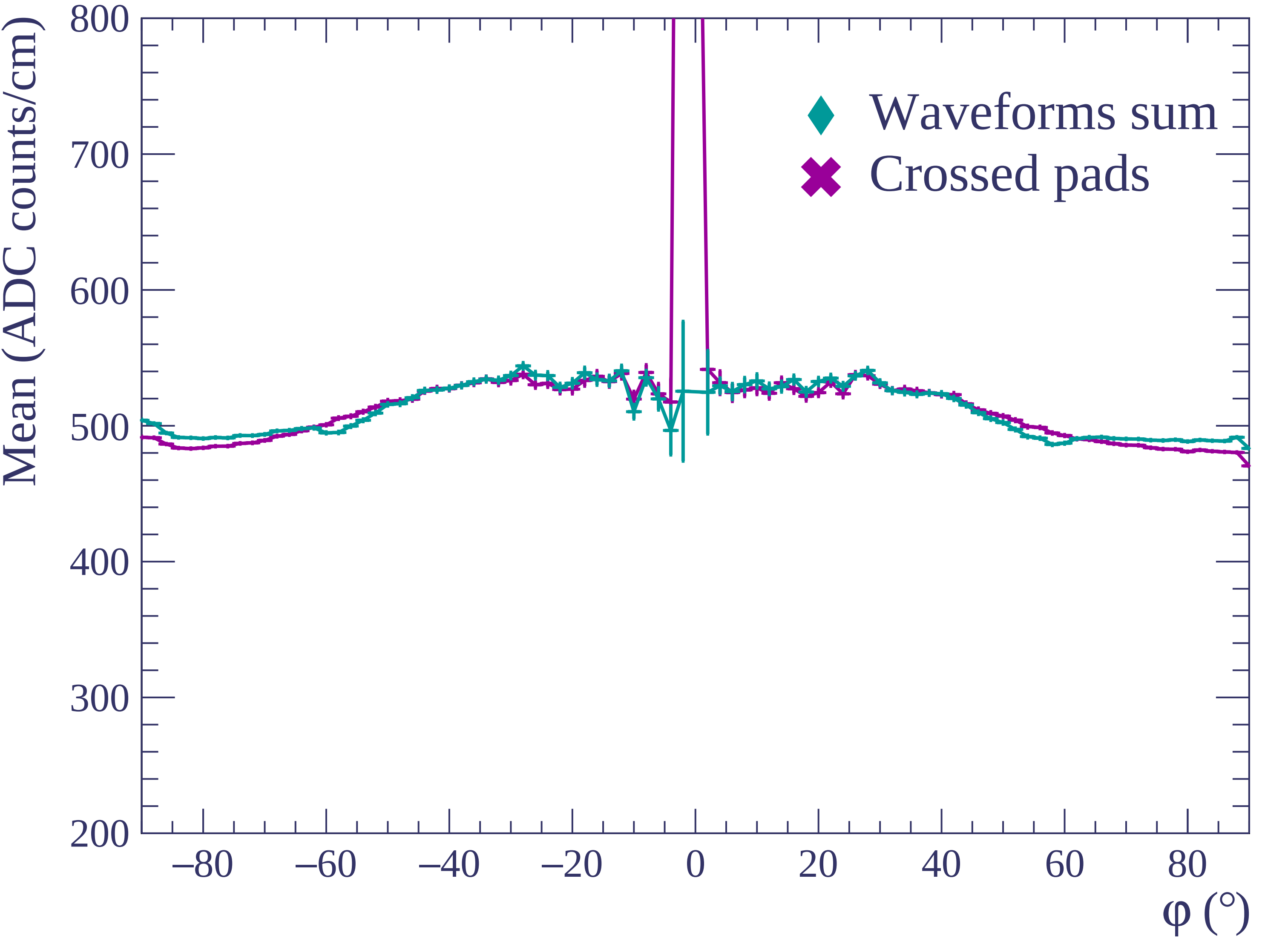

dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50
60
70

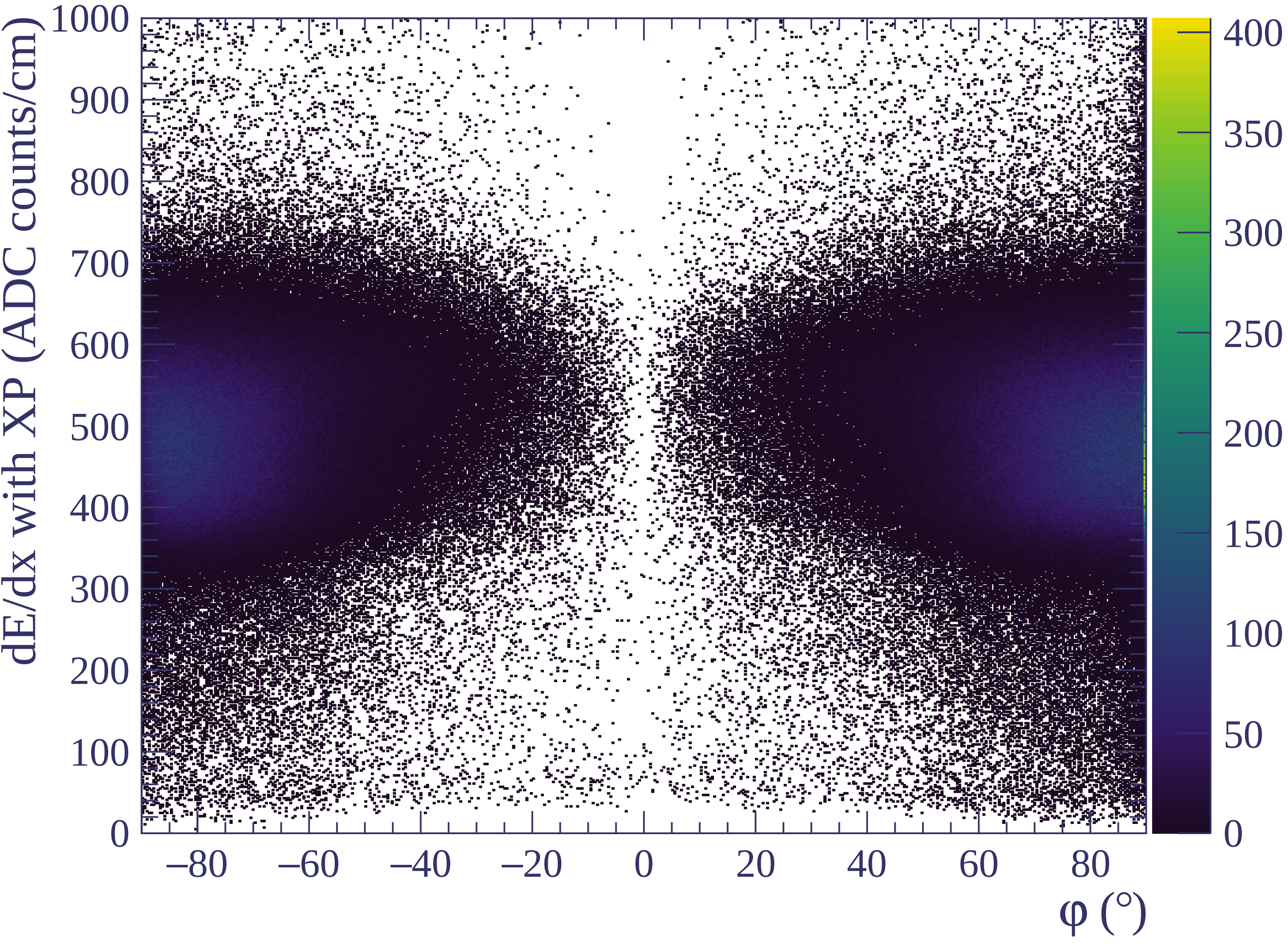

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

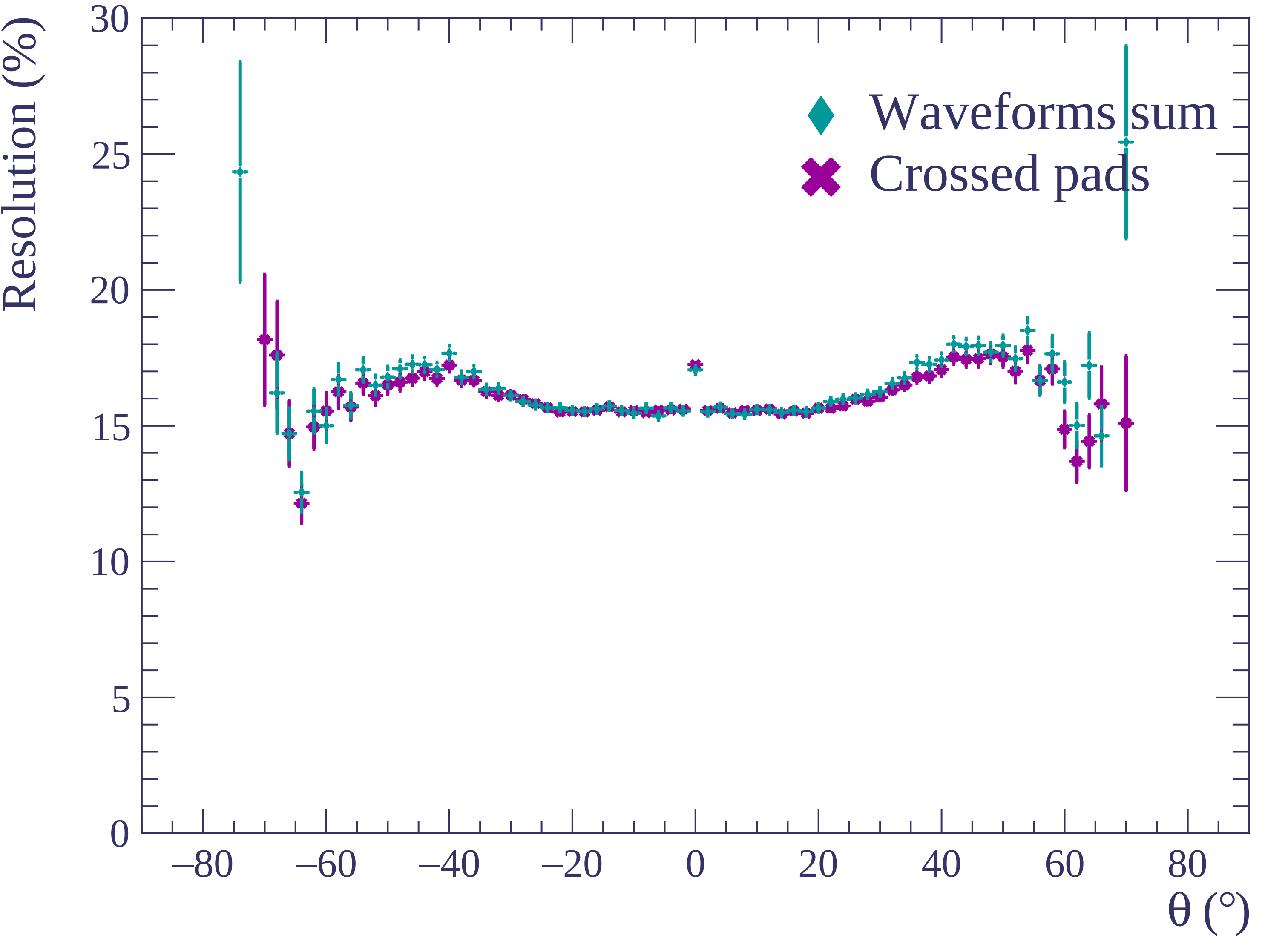


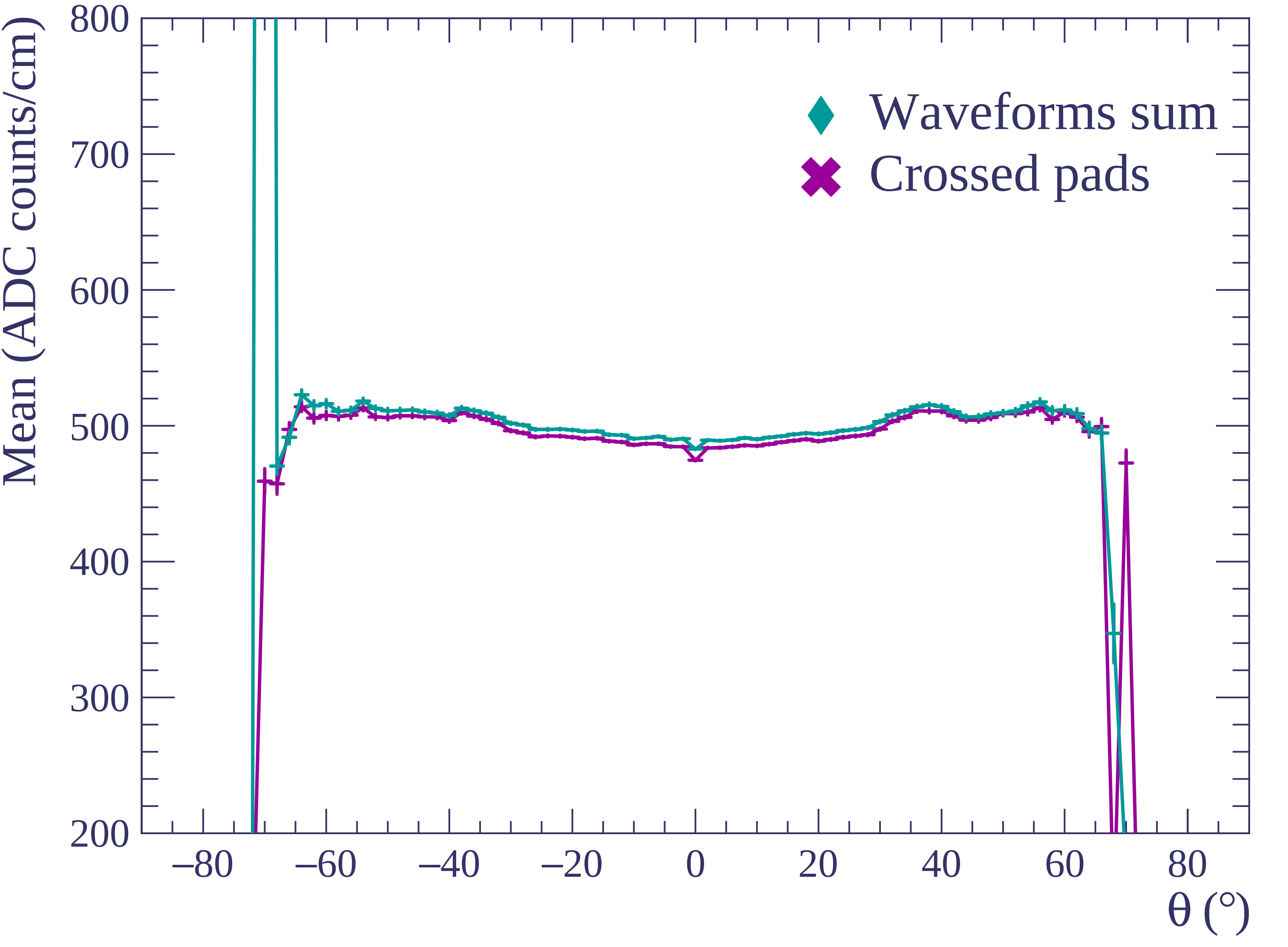
Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

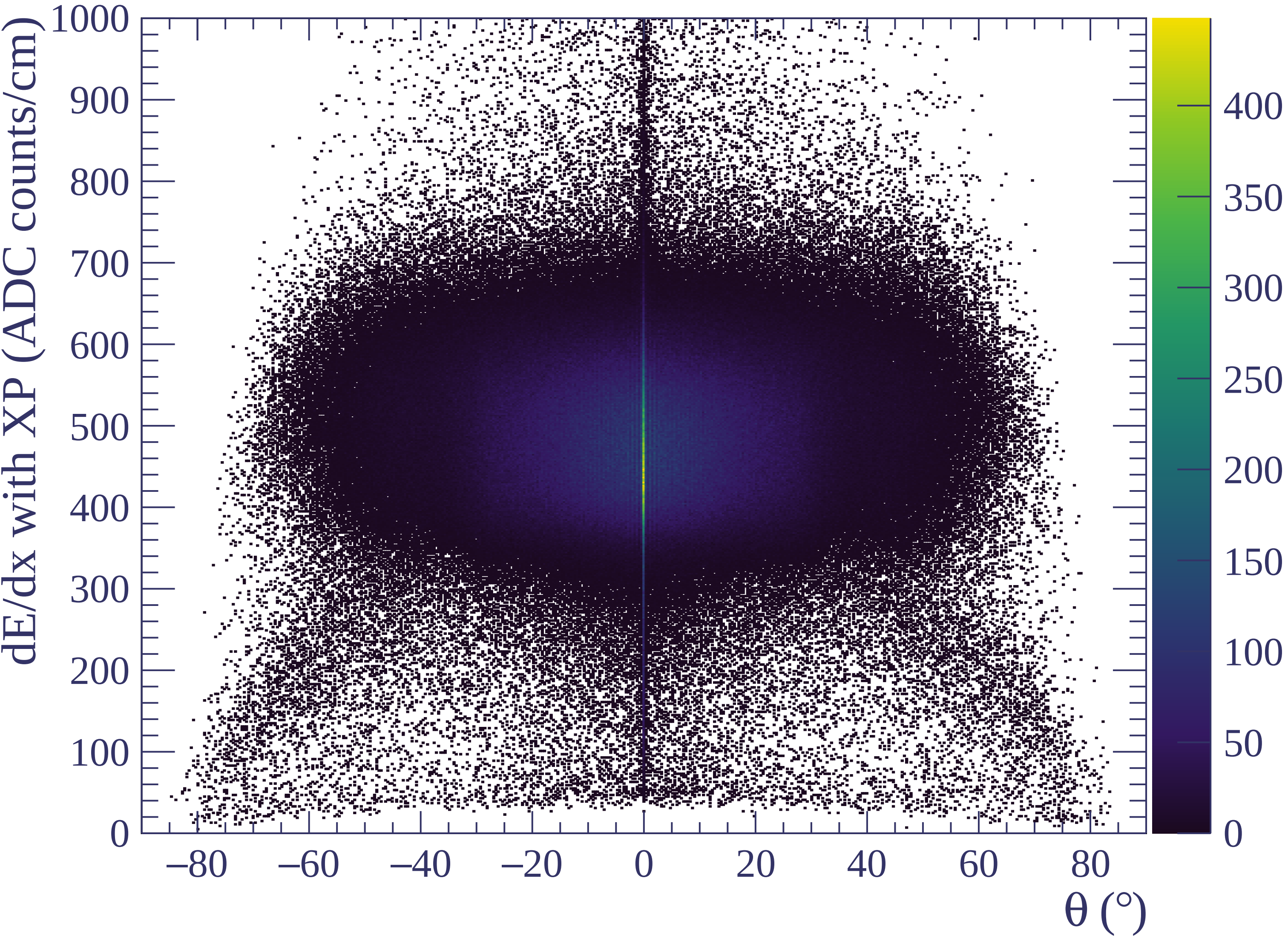

Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

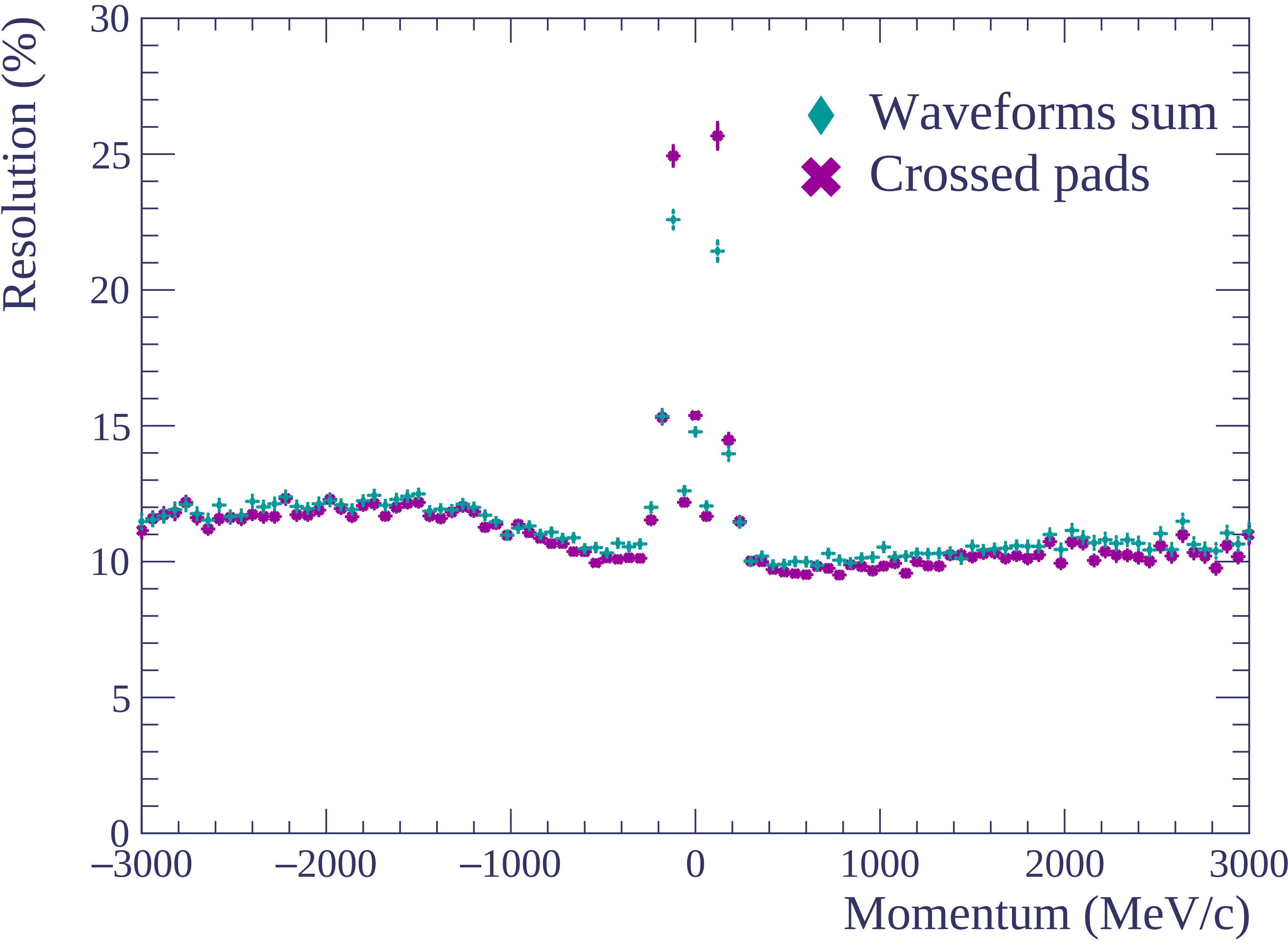


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
400
350
300
250
200
150
100
50
0

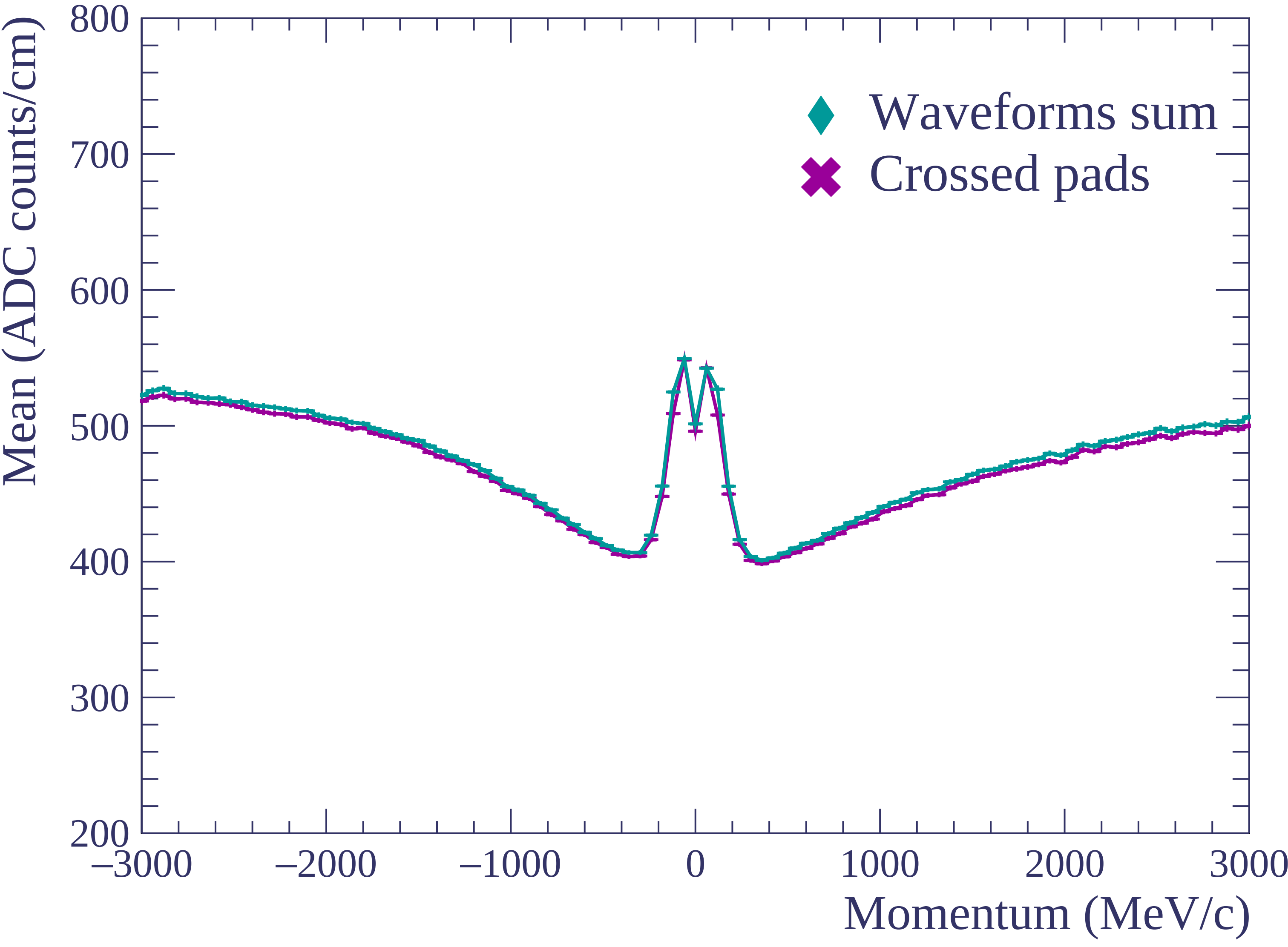

Resolution (%)
θ (°)
Waveforms sum
Crossed pads
0
5
10
15
20
25
30
−80
−60
−40
−20
0
20
40
60
80


Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads

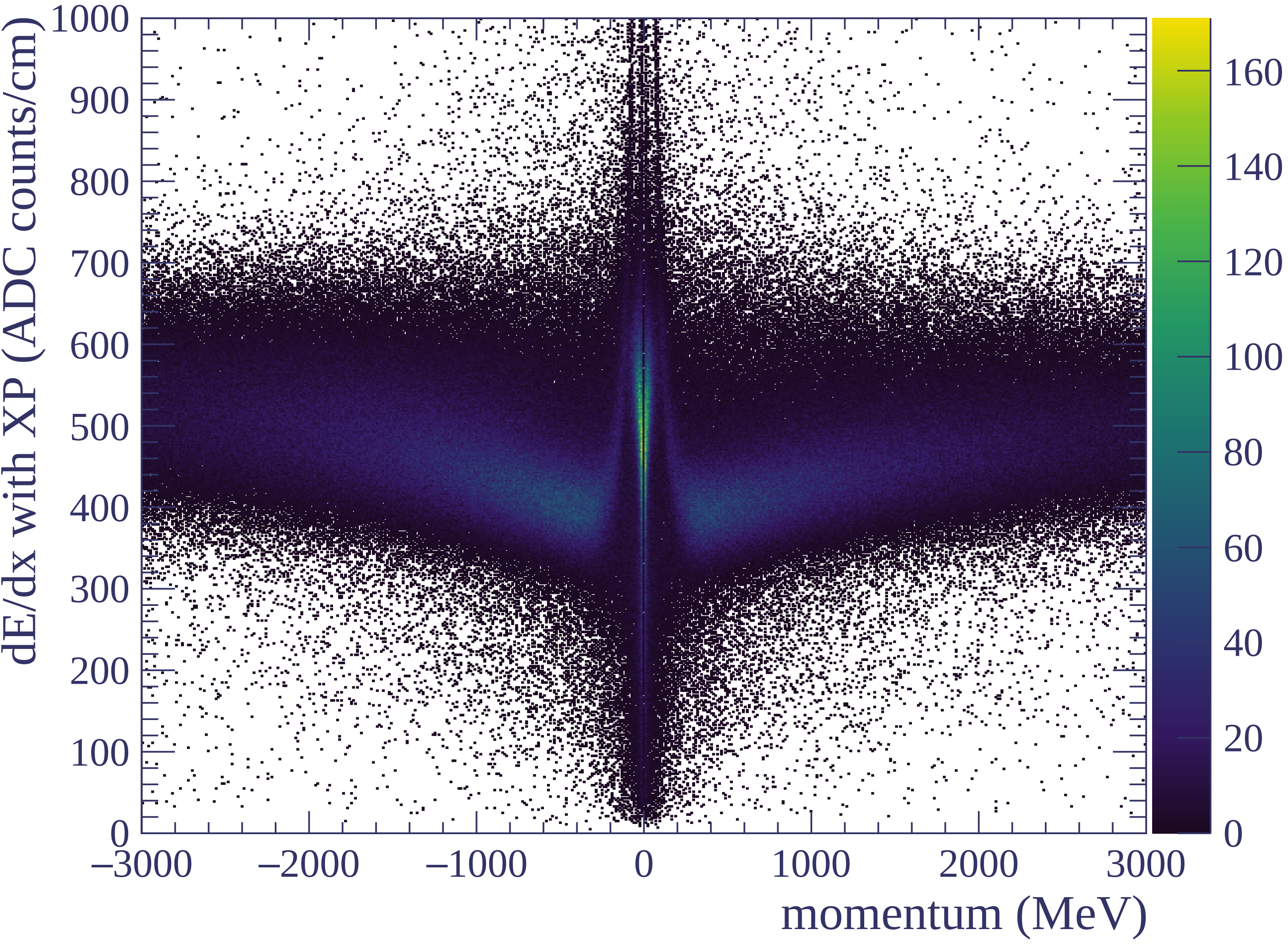

dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300
350
400



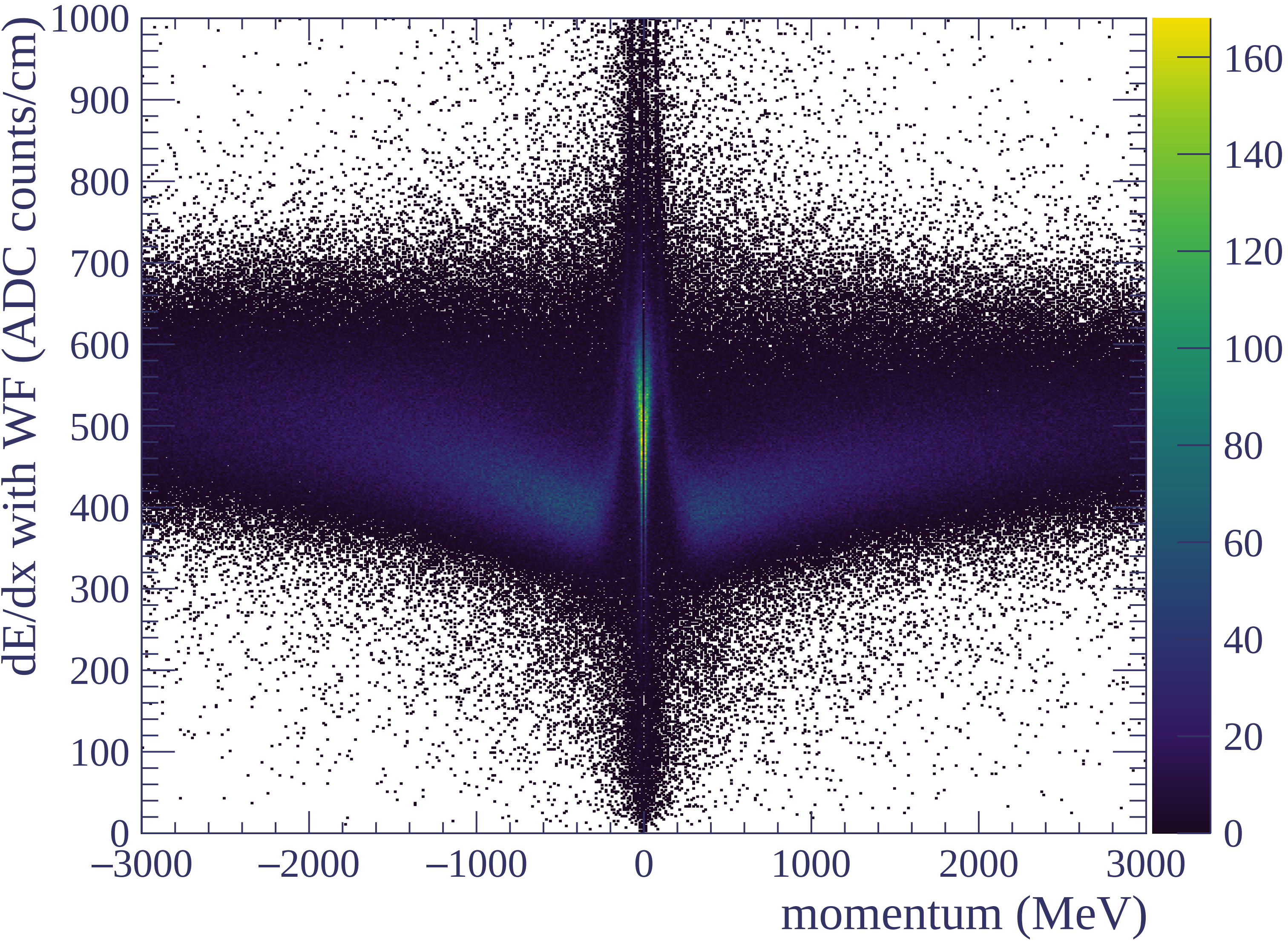
Resolution (%)
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



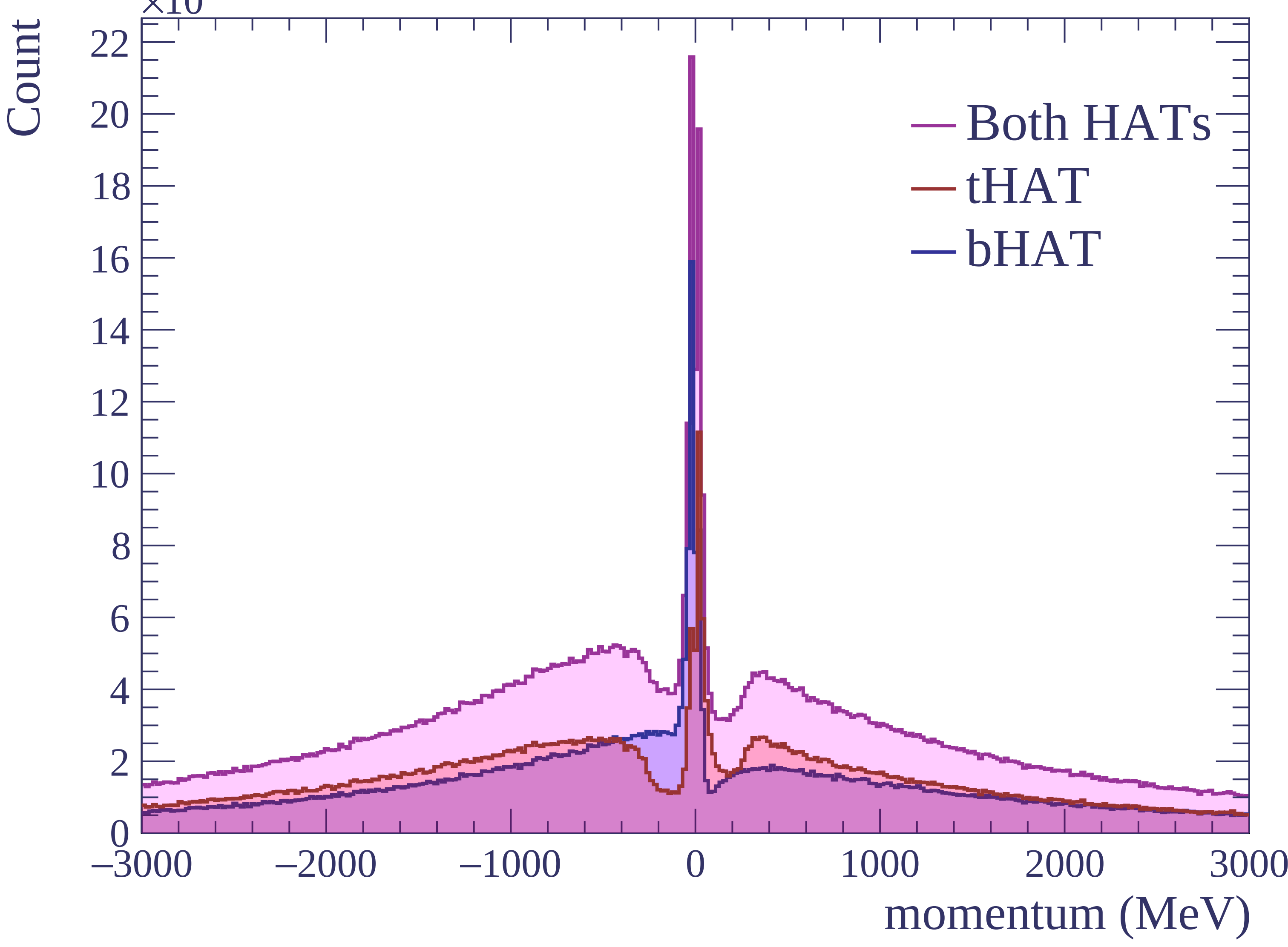
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



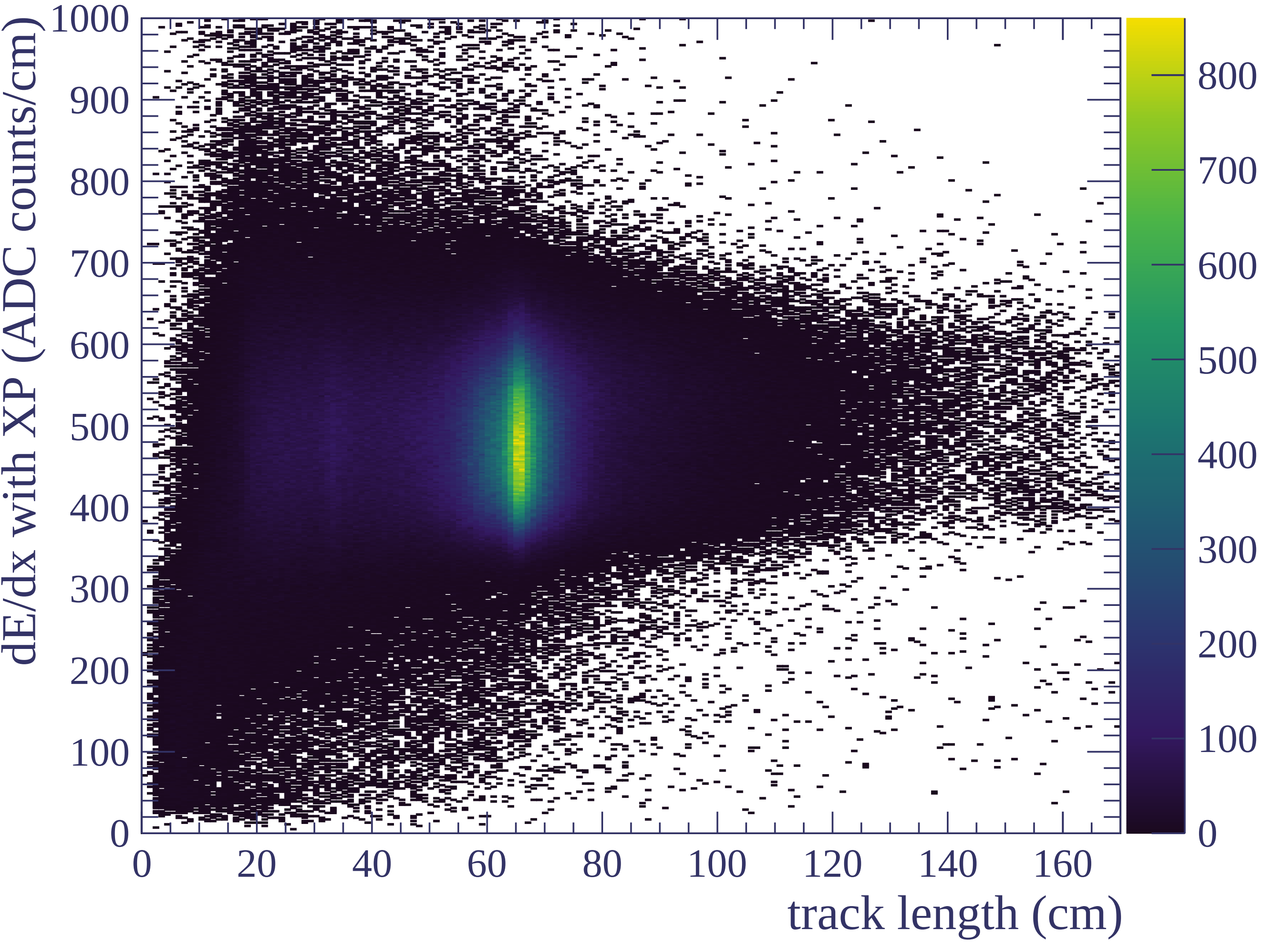
dE/dx with XP (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
0
20
40
60
80
100
120
140
160



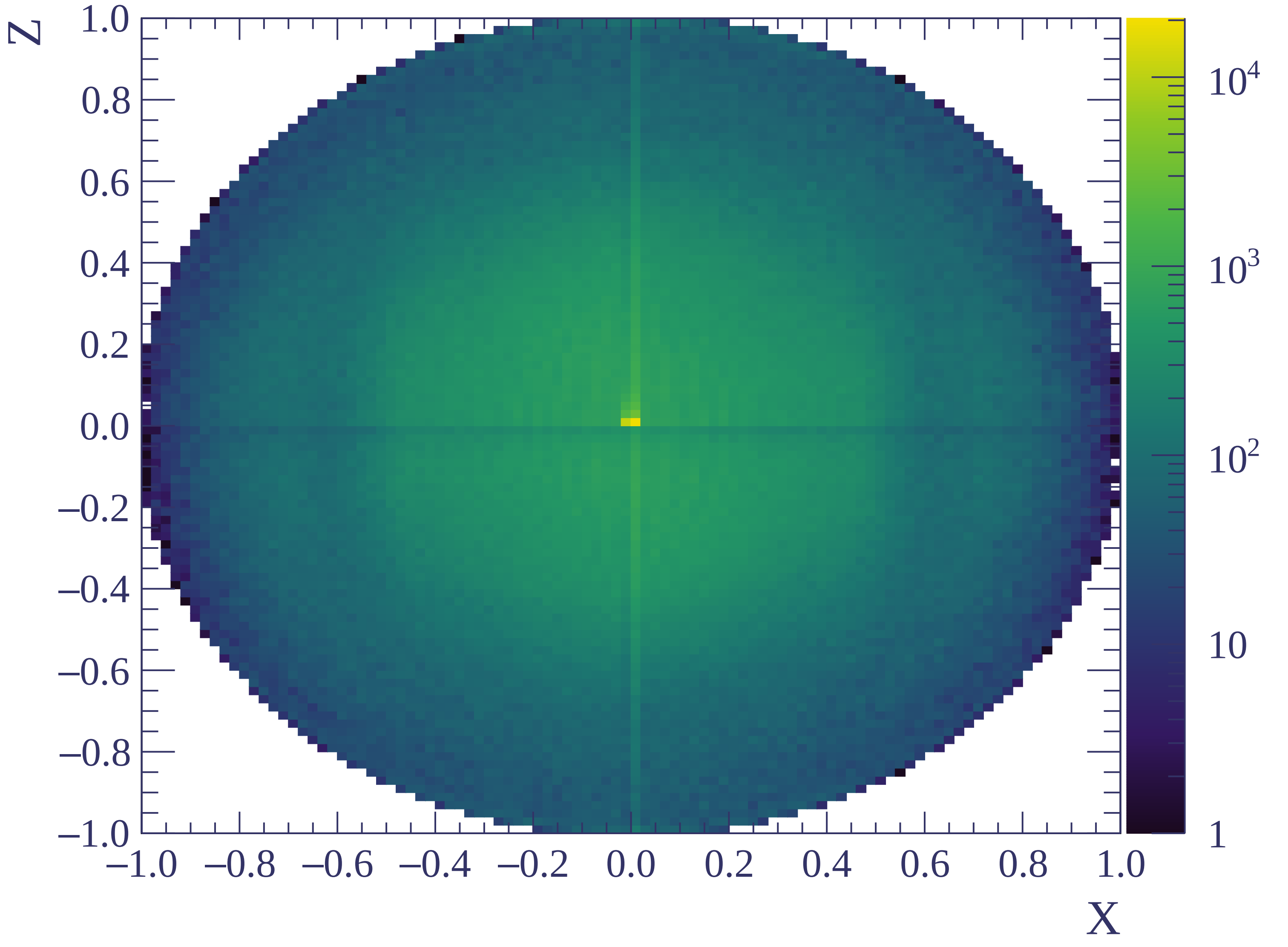
dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
20
40
60
80
100
120
140
160

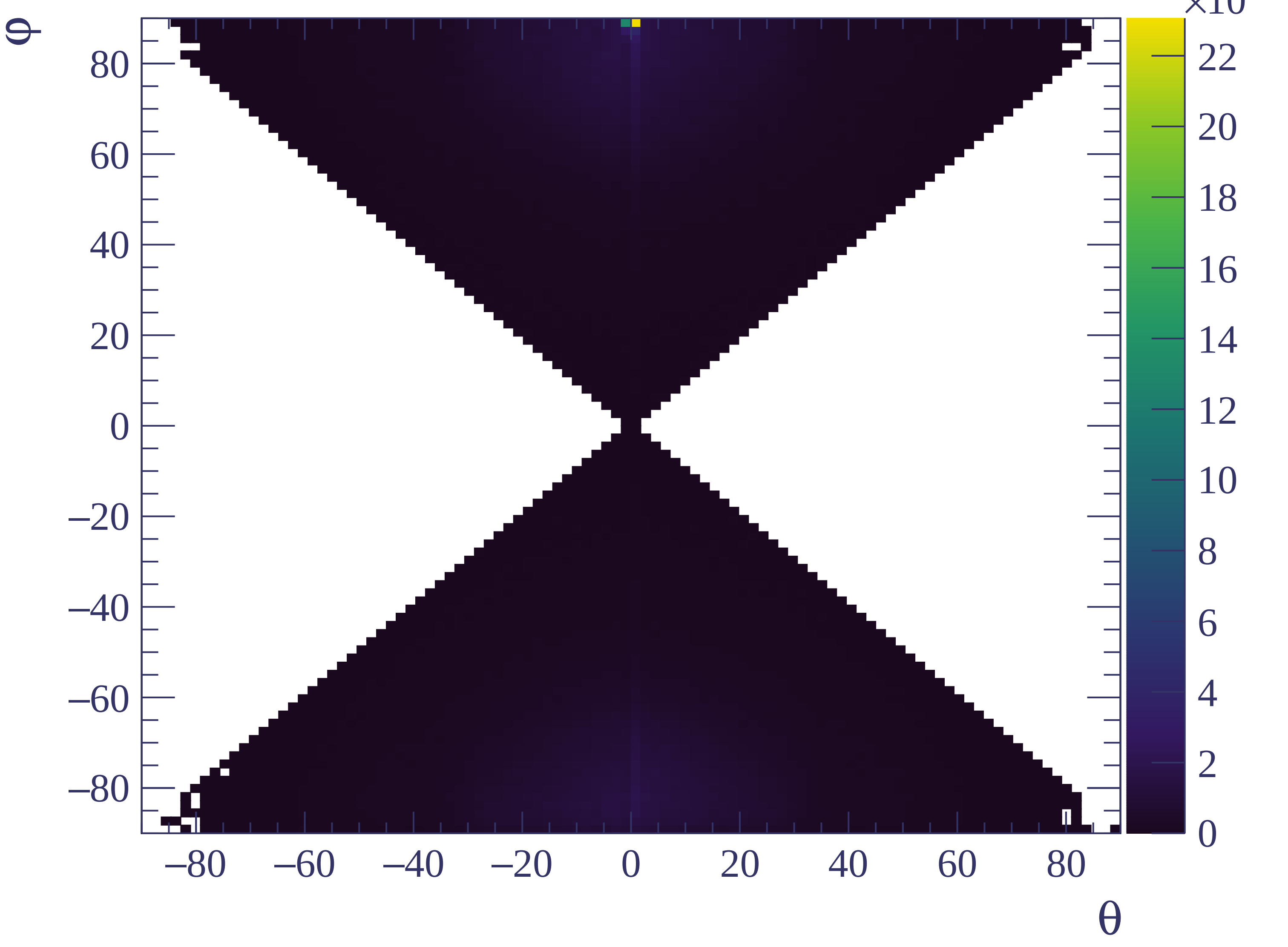


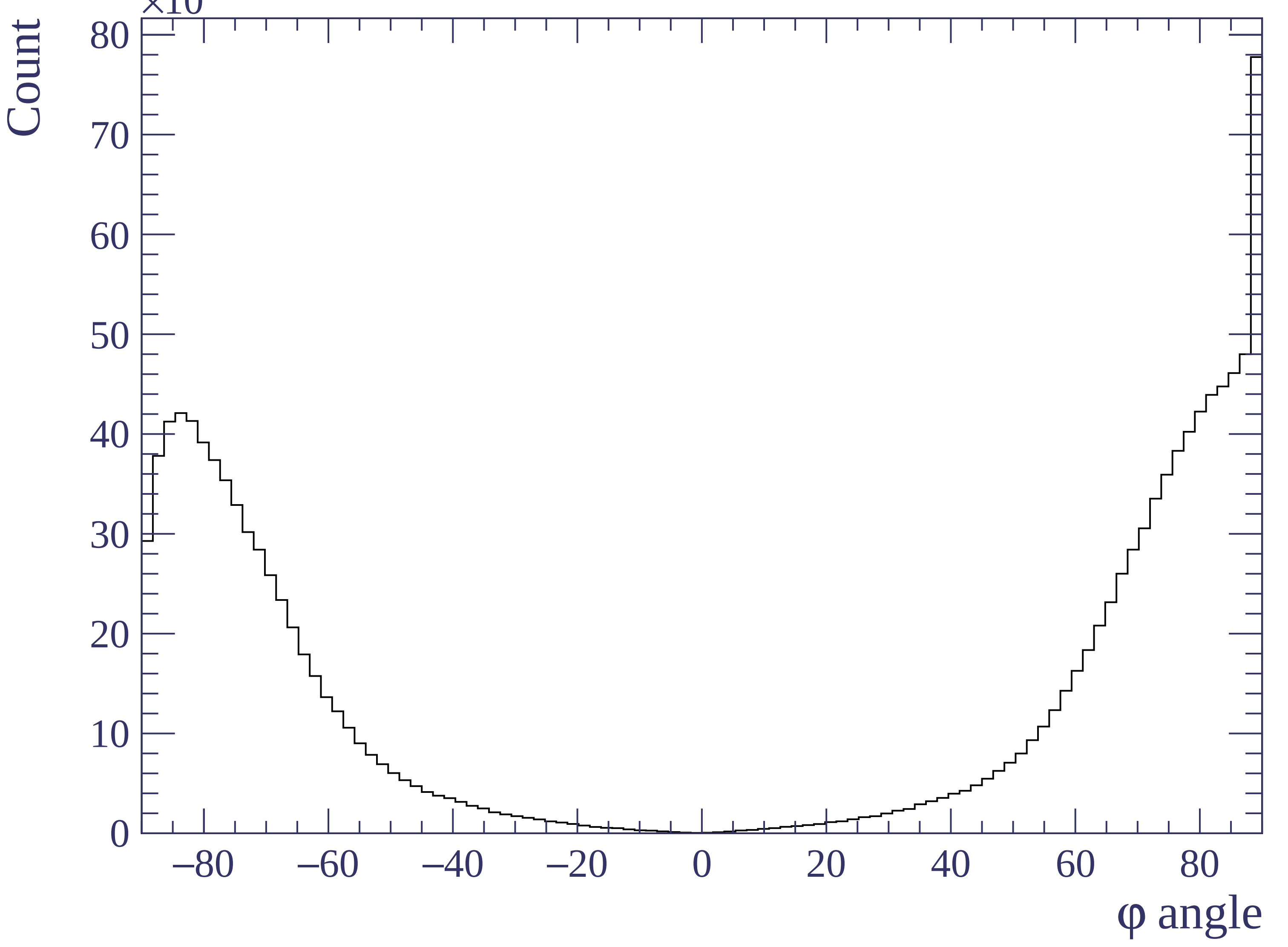
Count
×10
0
2
4
6
8
10
12
14
16
18
20
22
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



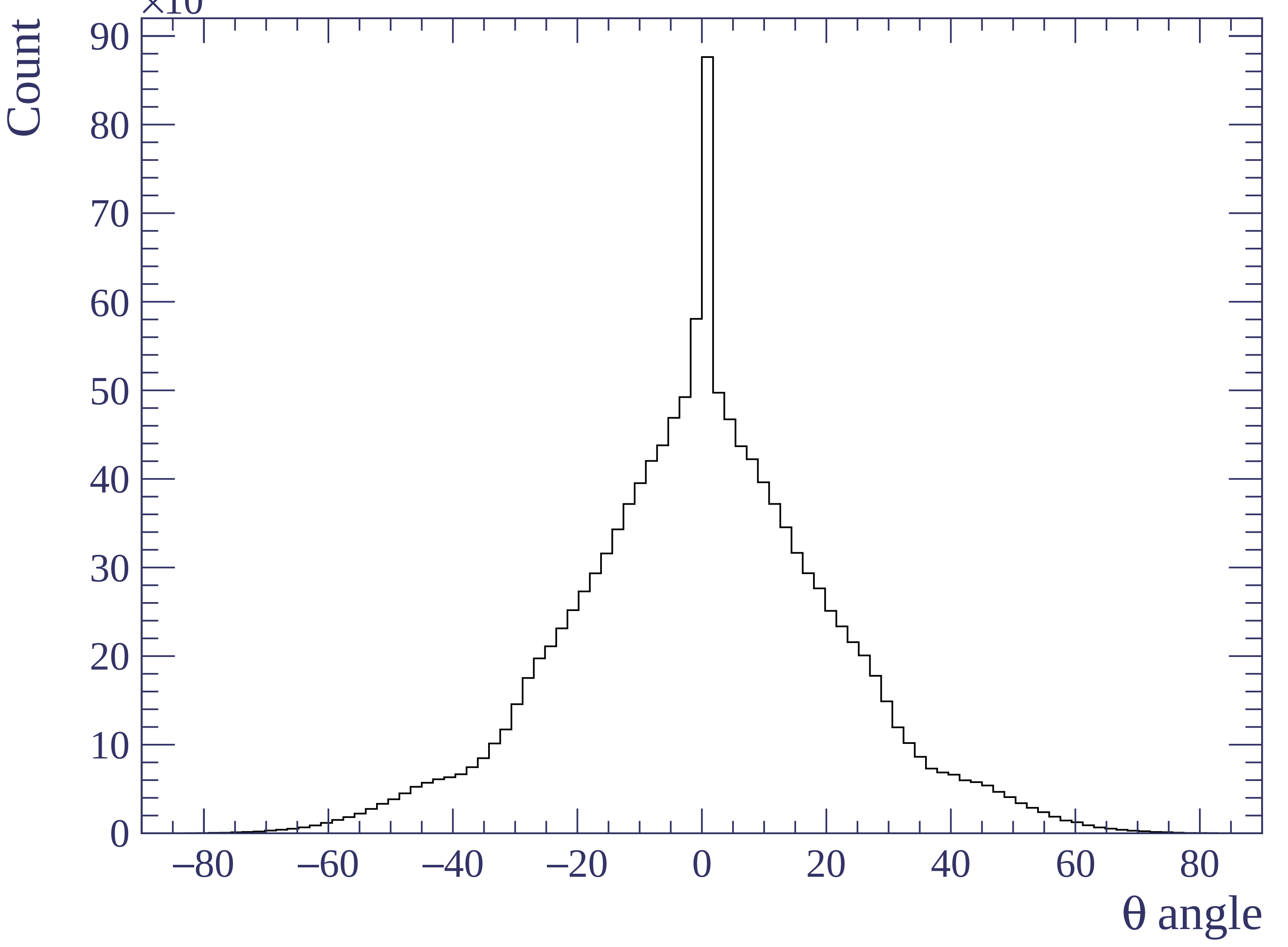
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
$10^4$
$10^3$
$10^2$
10
1

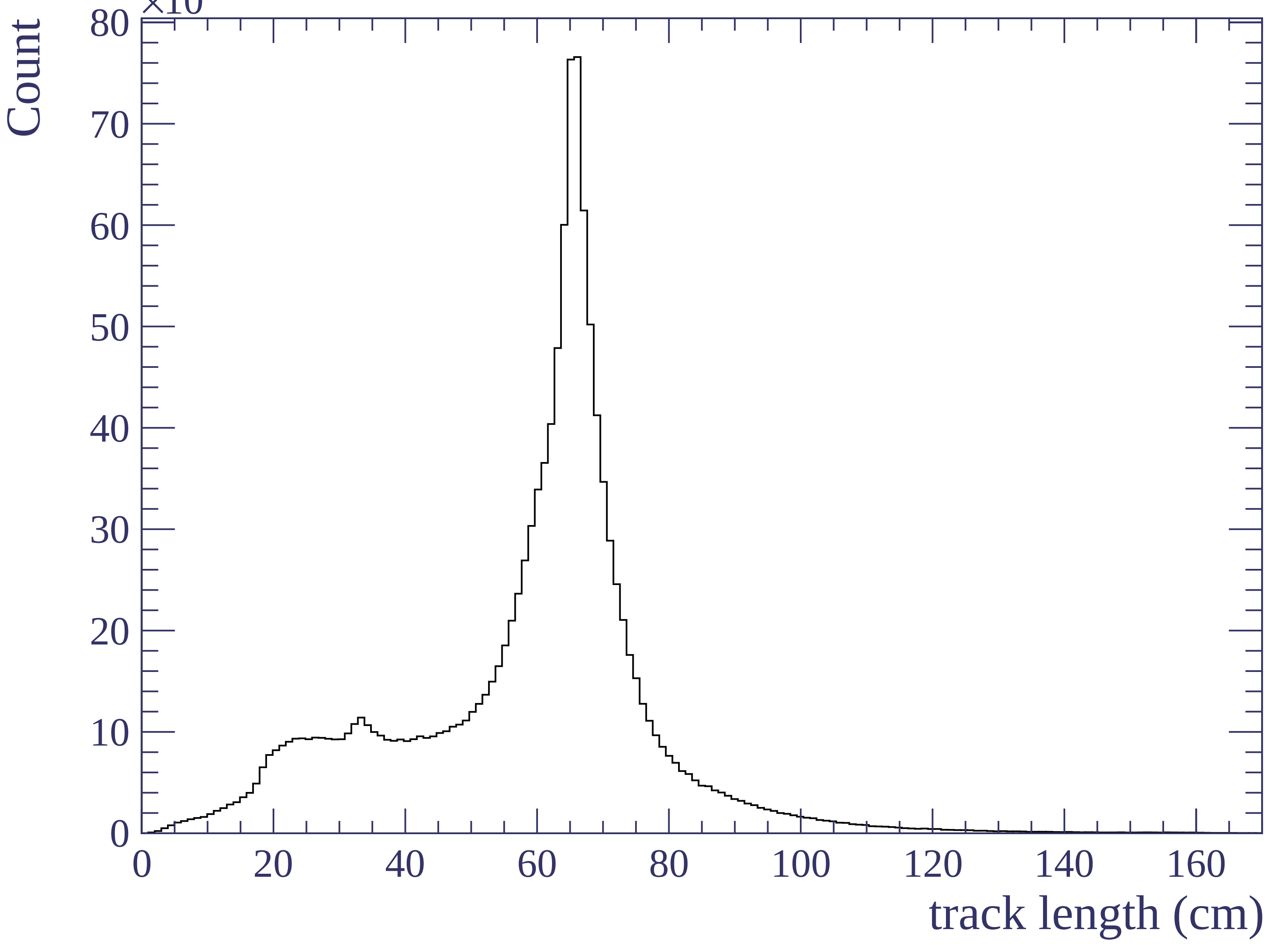

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
22
20
18
16
14
12
10
8
6
4
2
0



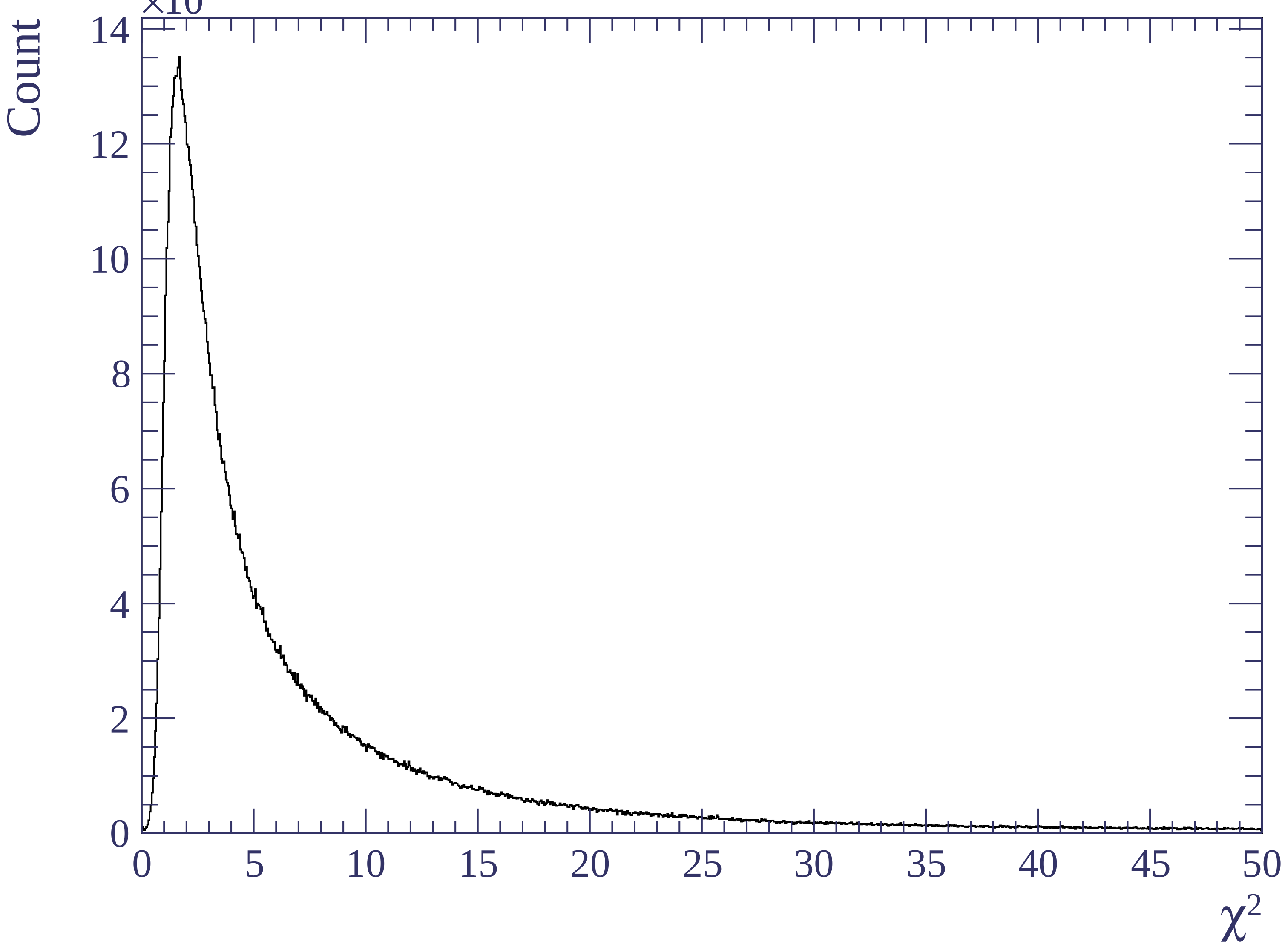
×10
Count
80
70
60
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



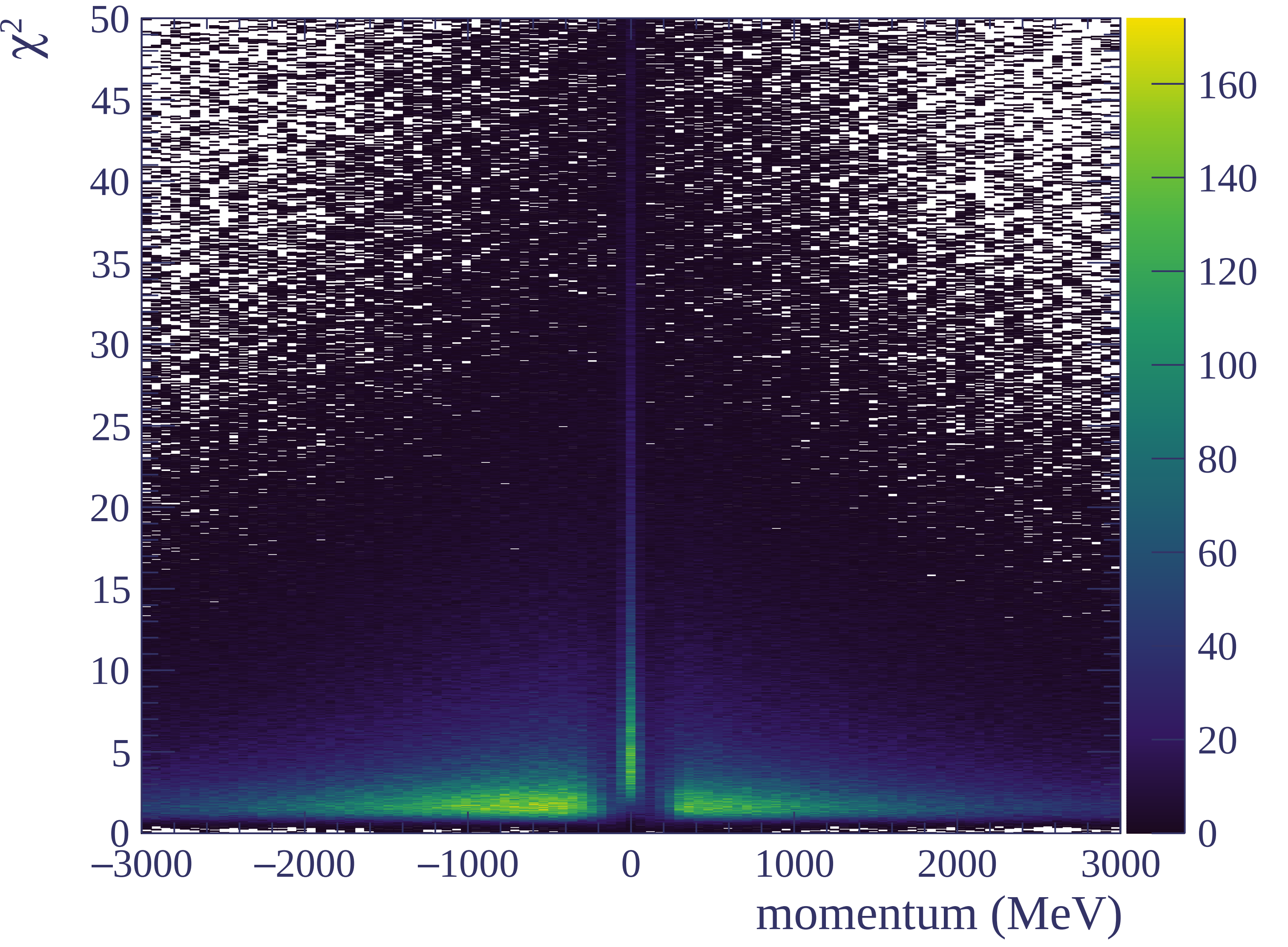
Count
×10
90
80
70
60
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



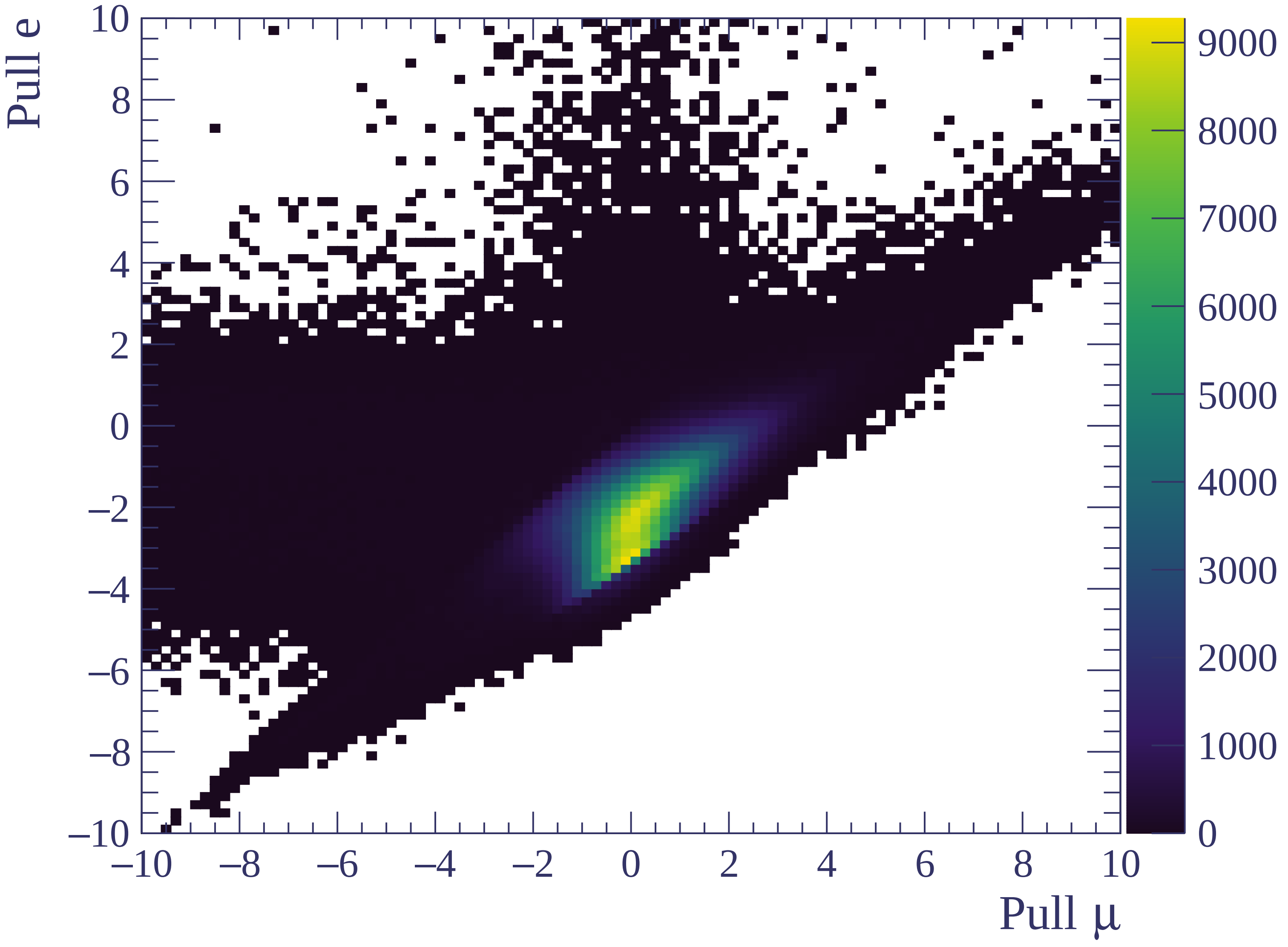
×10
Count
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
120
140
160
track length (cm)



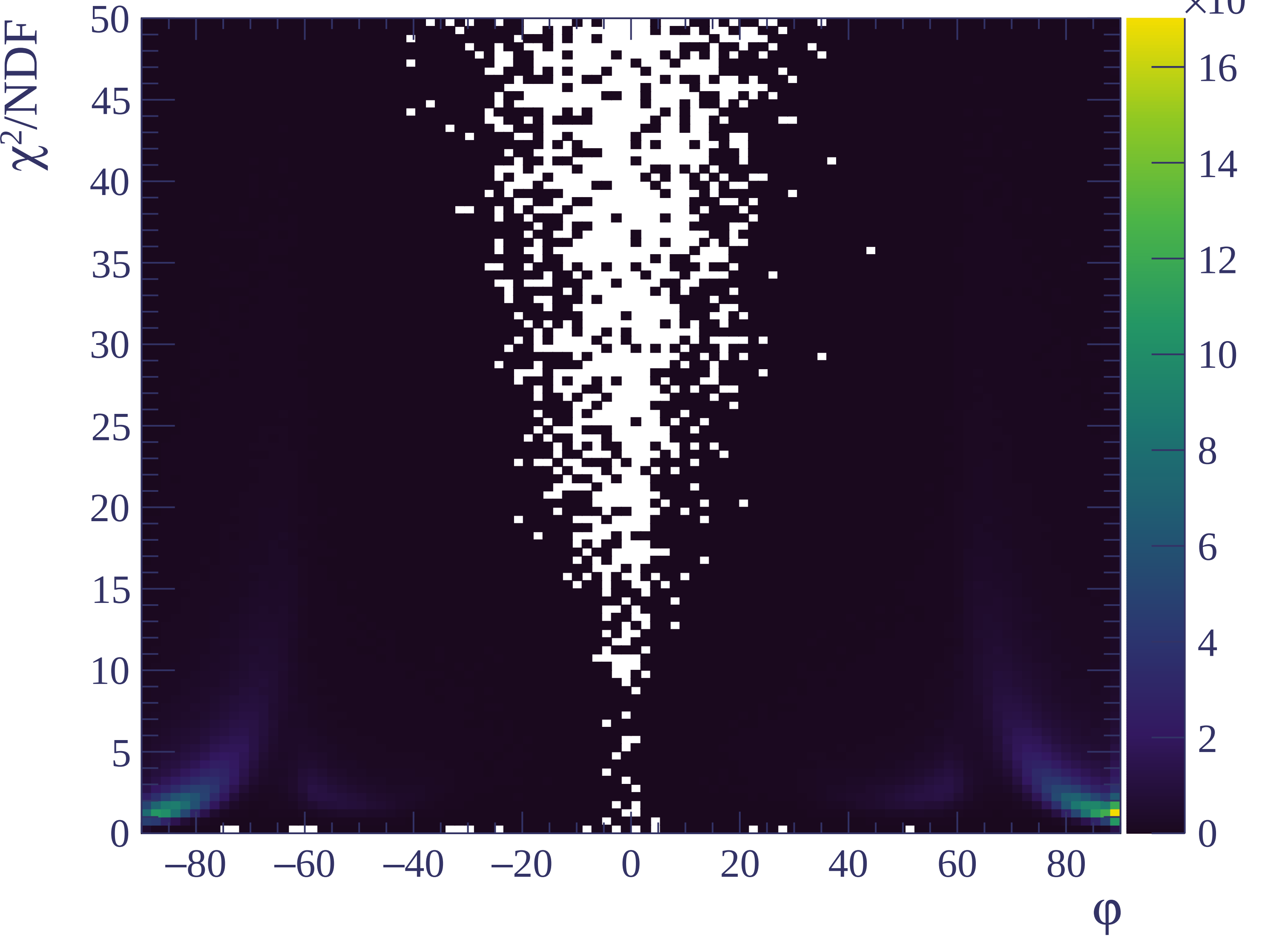
Count
×10
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
$\chi^2$



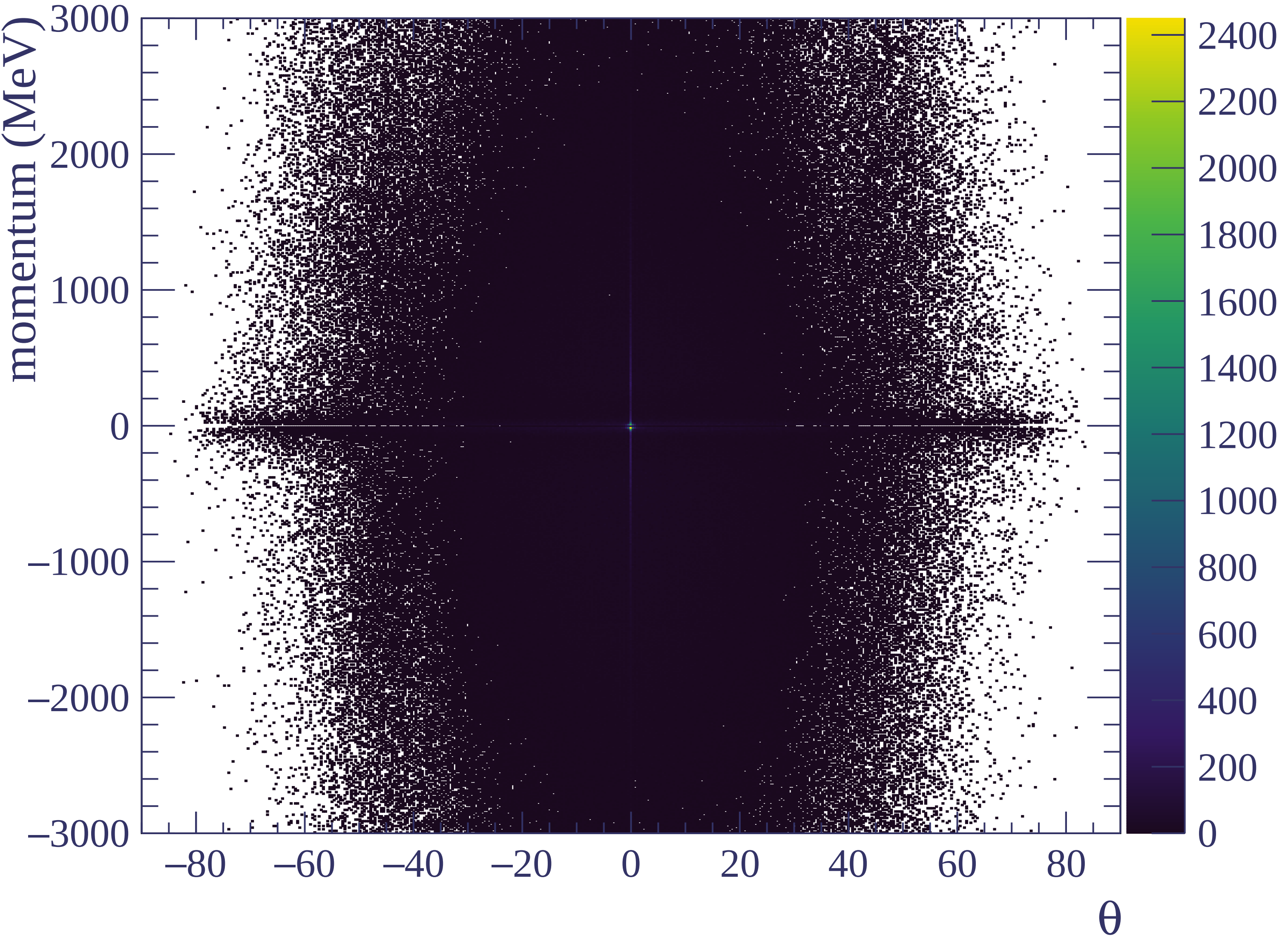
$\chi^2$
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
160
140
120
100
80
60
40
20
0



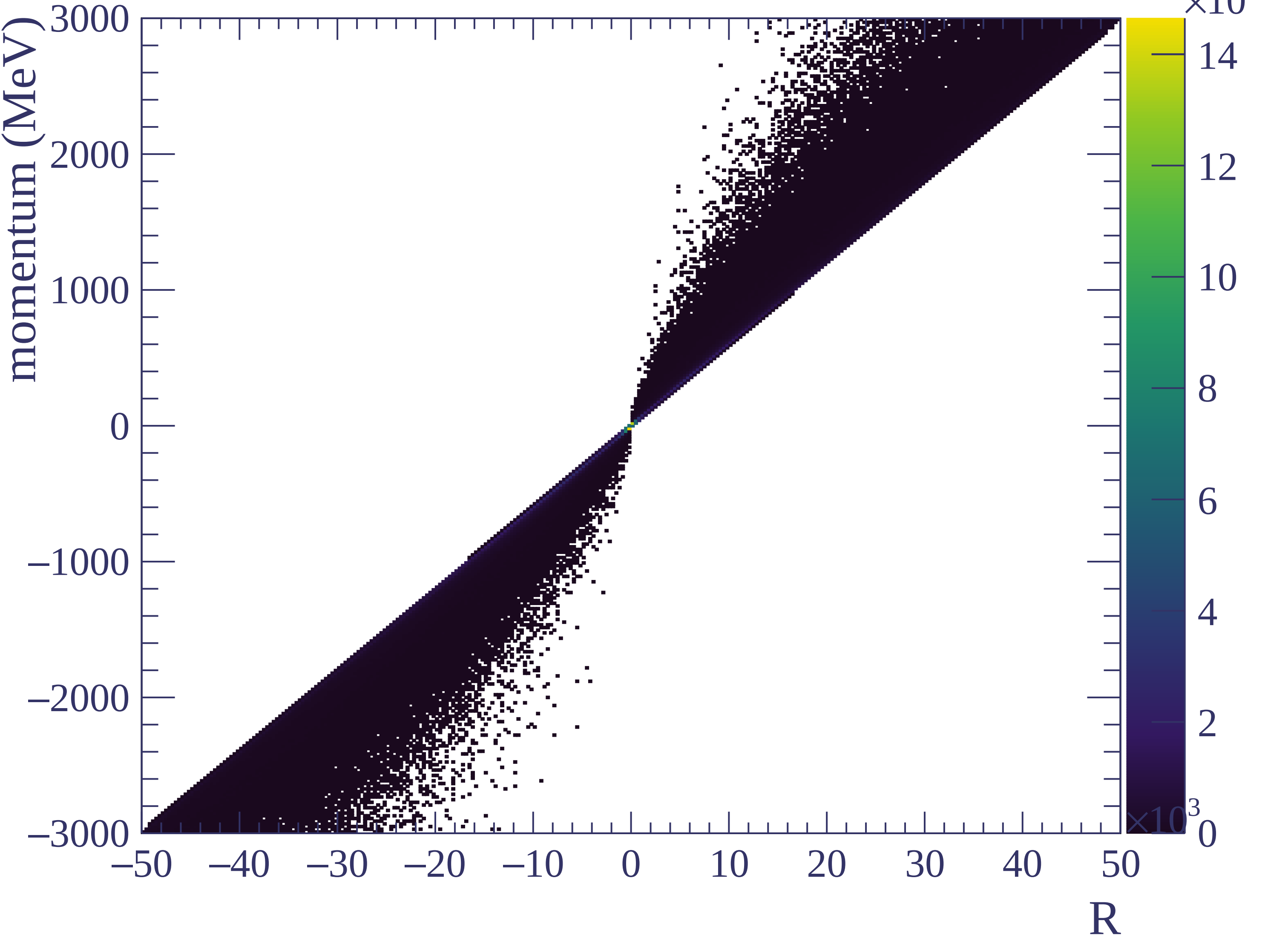
Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
9000
8000
7000
6000
5000
4000
3000
2000
1000

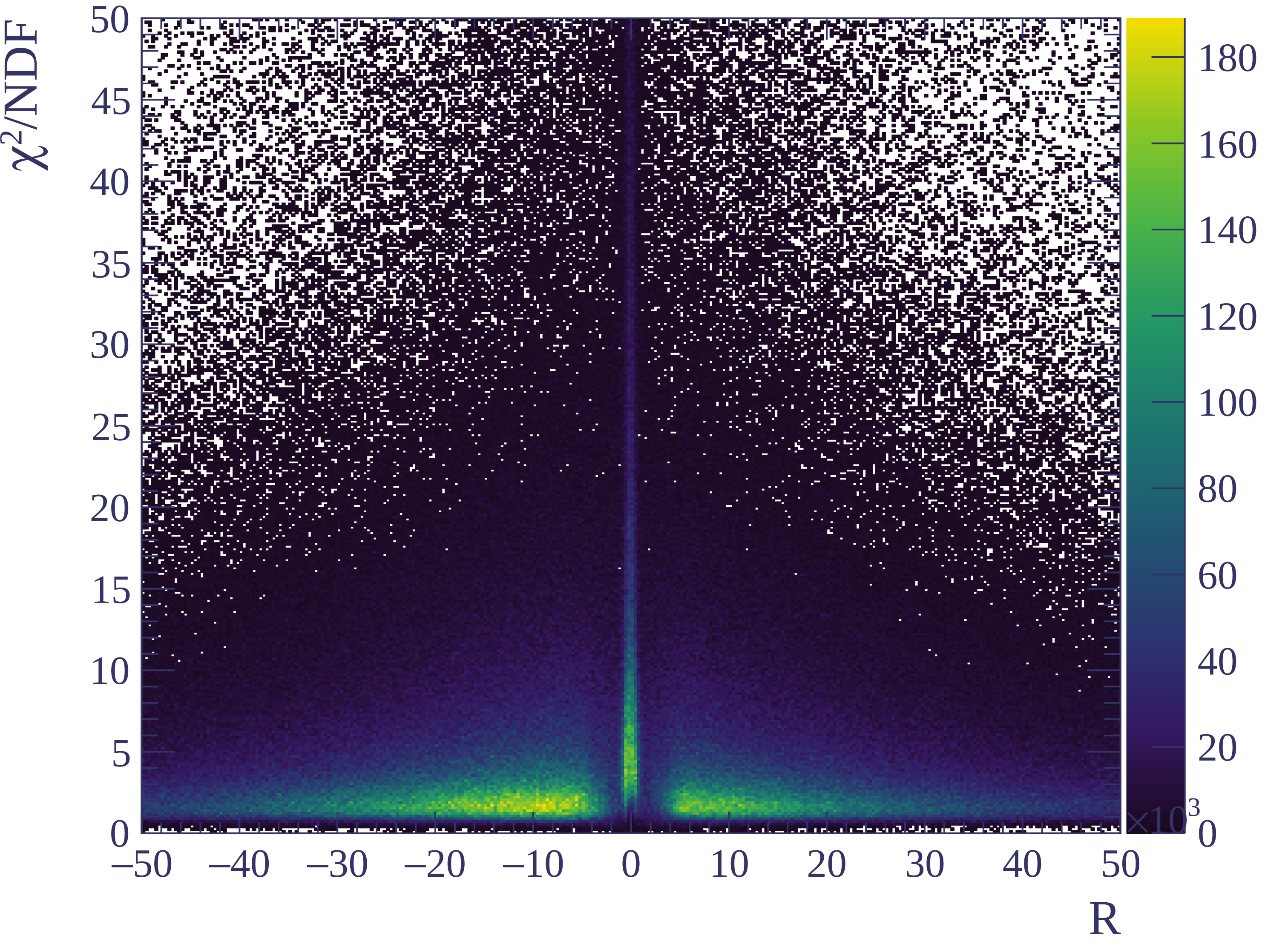


χ²/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
16
14
12
10
8
6
4
2
0

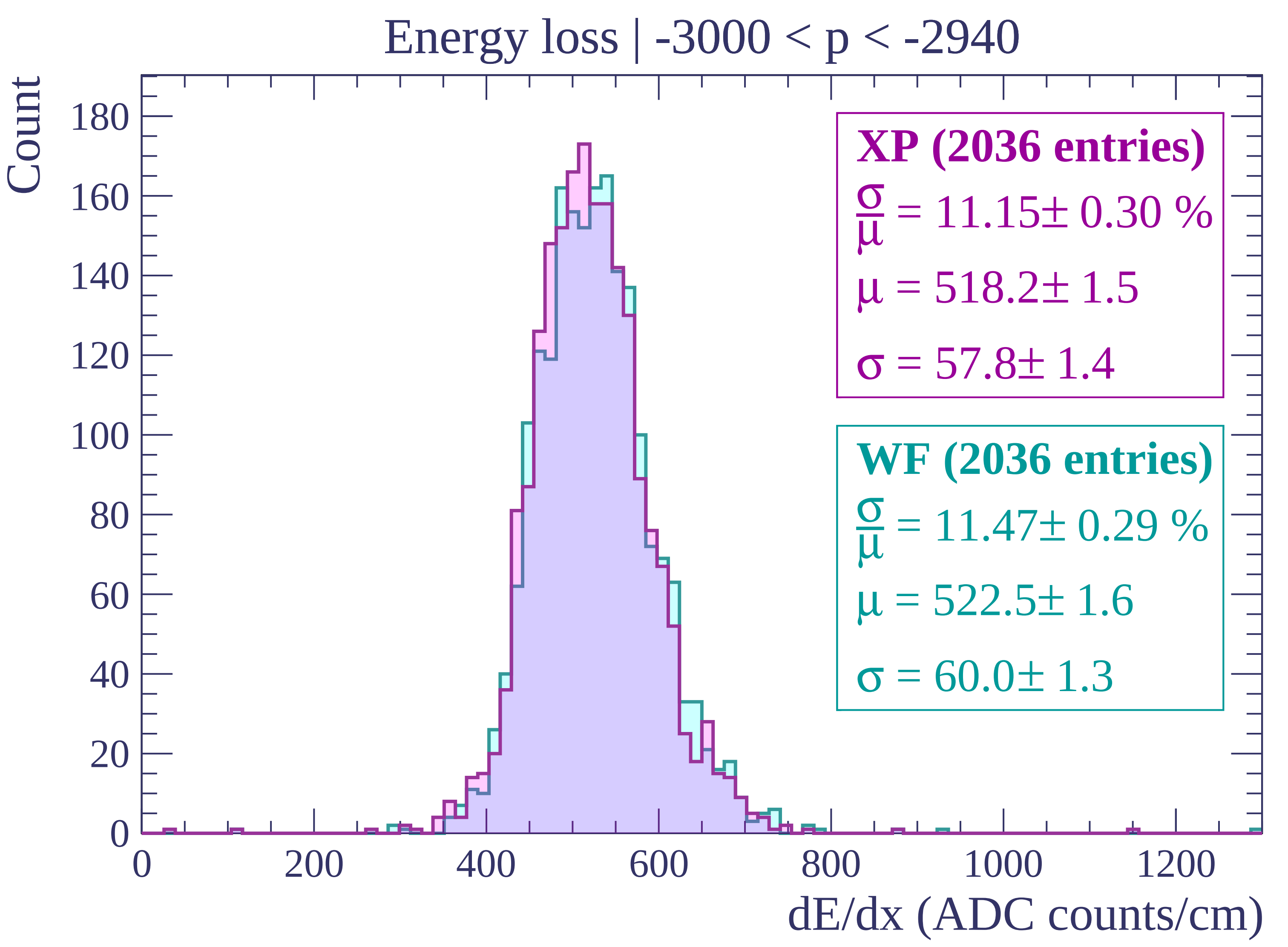


momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

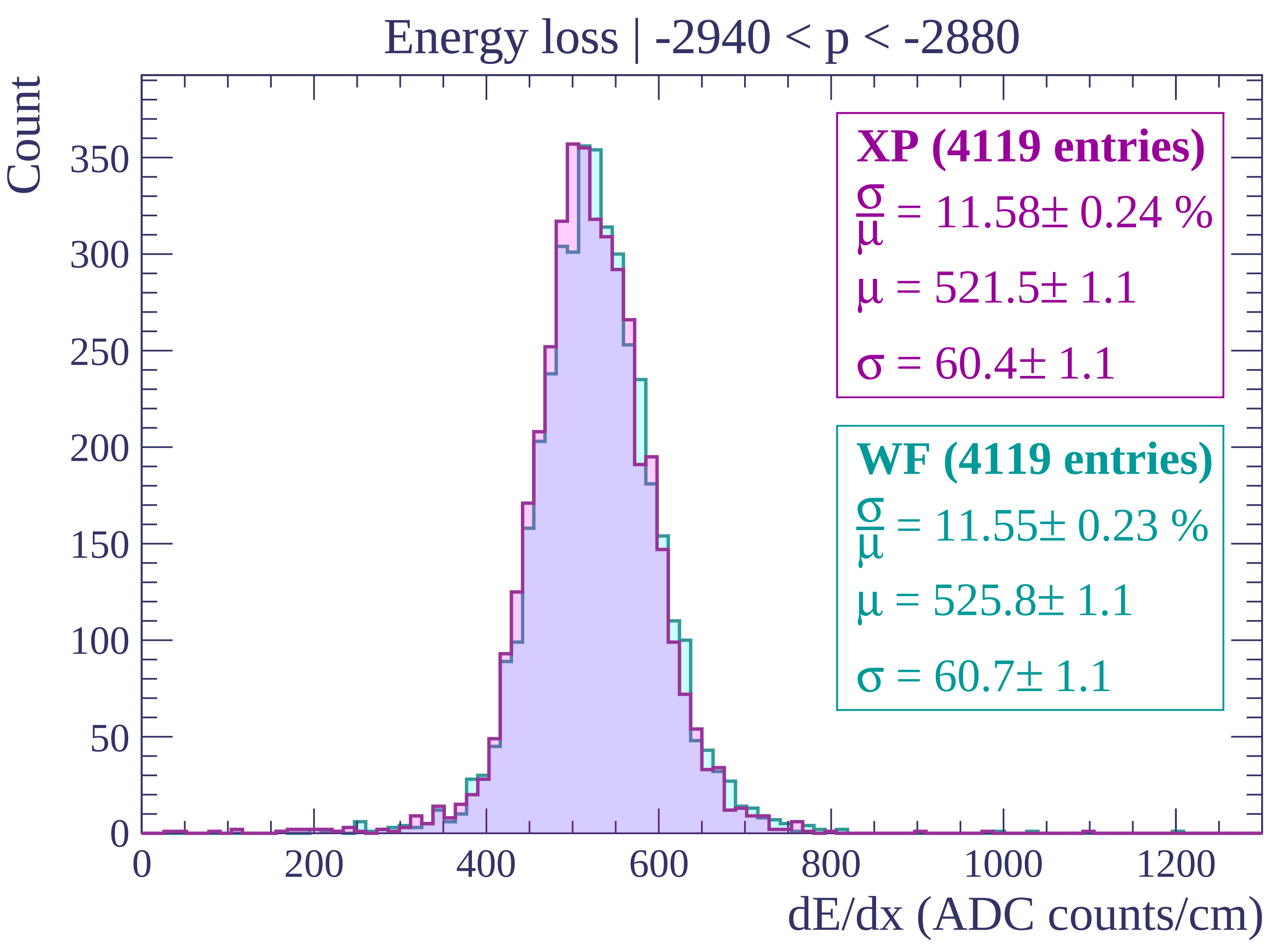

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
×10³
14
12
10
8
6
4
2
0



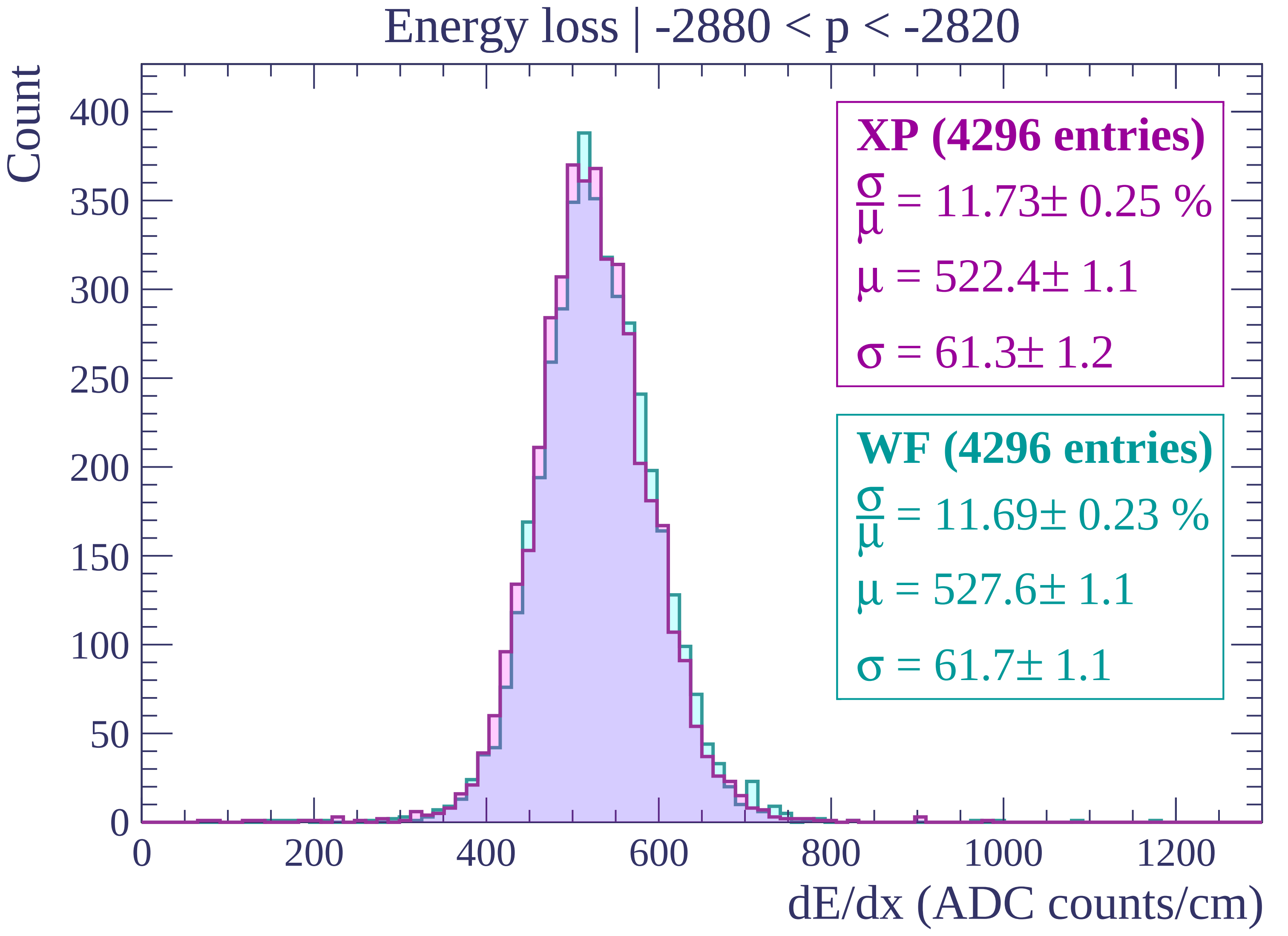
$\chi^2$/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
180
160
140
120
100
80
60
40
20
0
$\times10^3$



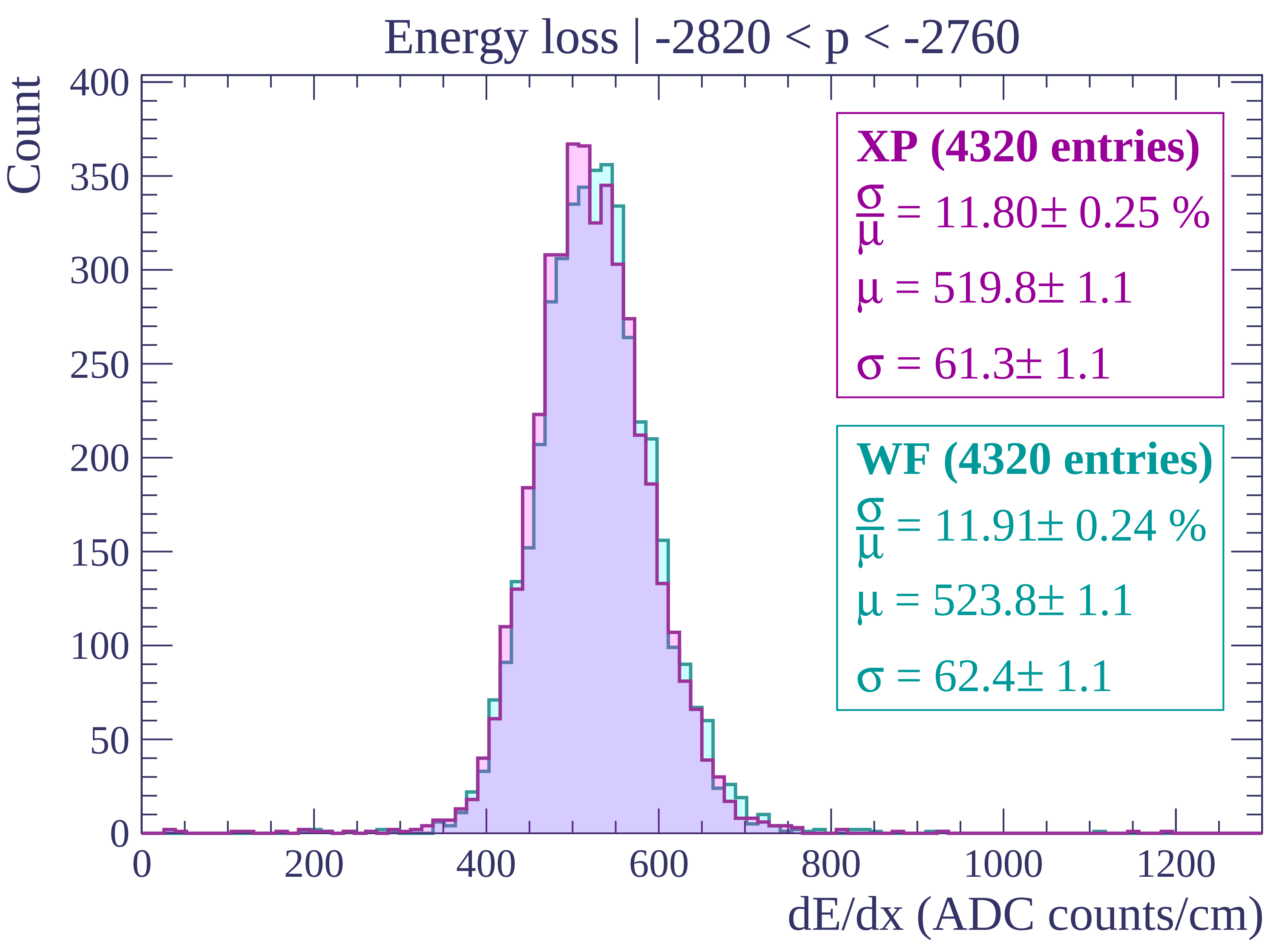
Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2036 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.30 %
μ = 518.2± 1.5
σ = 57.8± 1.4
WF (2036 entries)
$\frac{\sigma}{\mu}$ = 11.47± 0.29 %
μ = 522.5± 1.6
σ = 60.0± 1.3



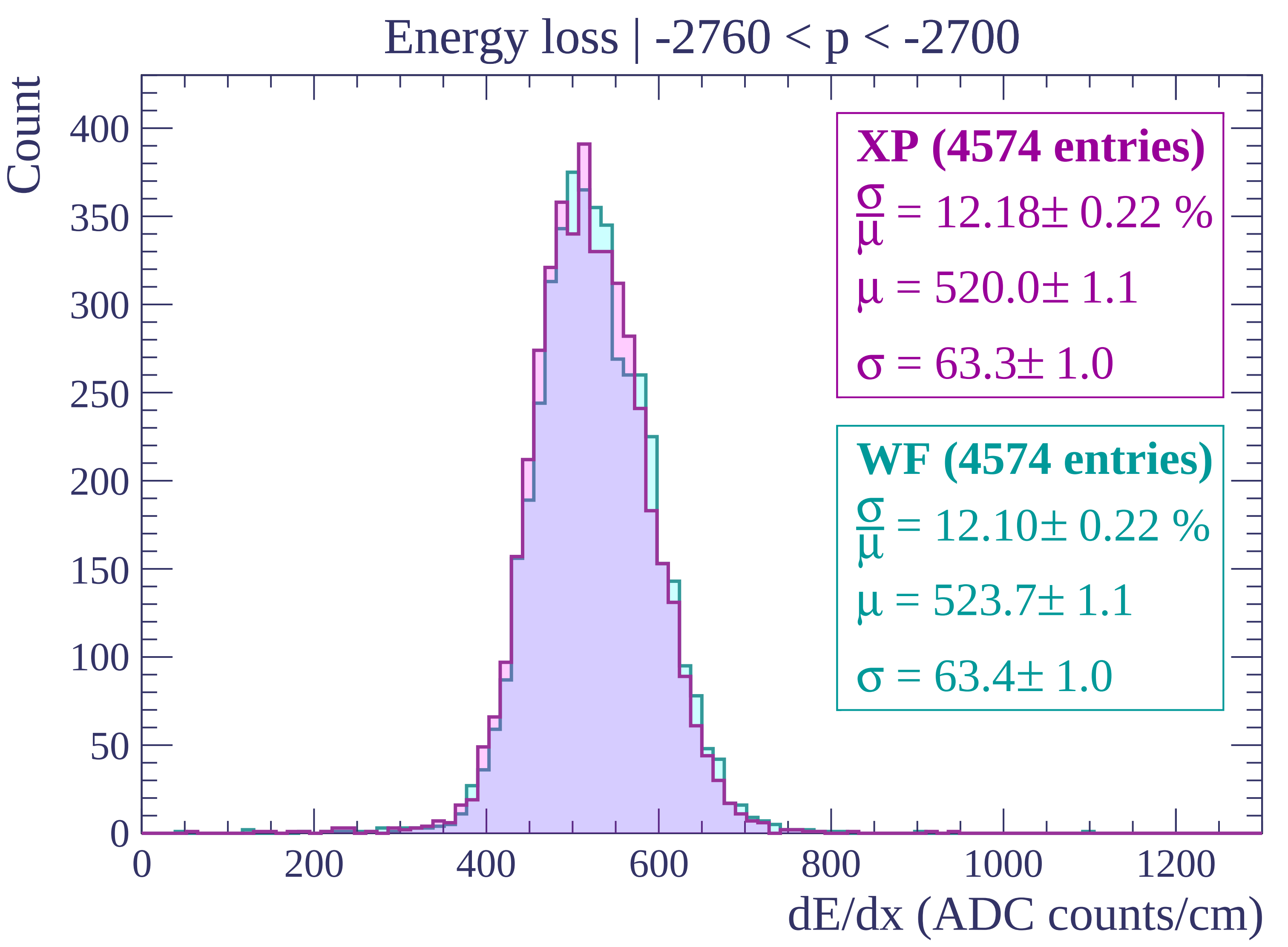
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (4119 entries)
σ/μ = 11.58± 0.24 %
μ = 521.5± 1.1
σ = 60.4± 1.1
WF (4119 entries)
σ/μ = 11.55± 0.23 %
μ = 525.8± 1.1
σ = 60.7± 1.1

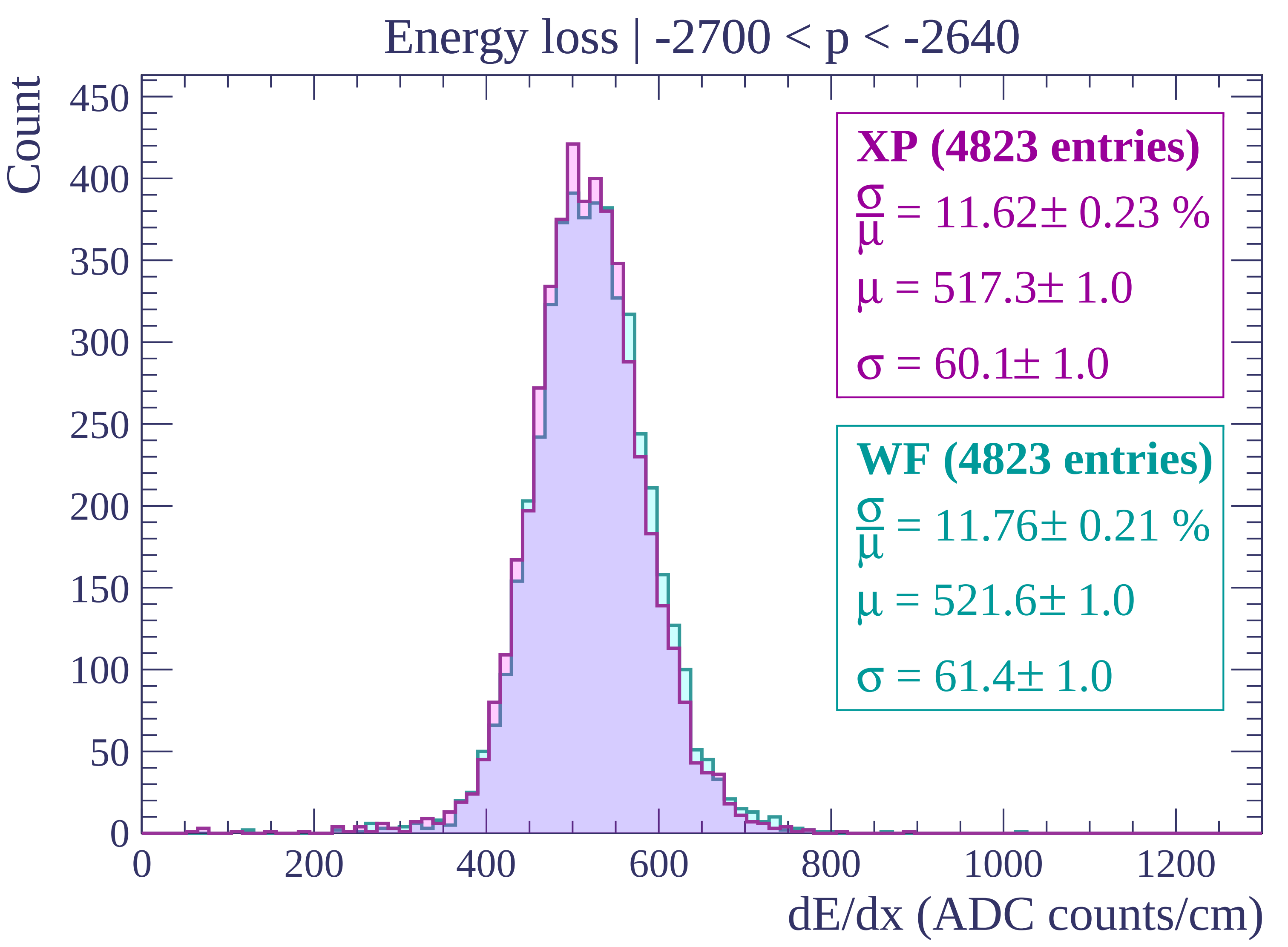


Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4296 entries)
σ/μ = 11.73± 0.25 %
μ = 522.4± 1.1
σ = 61.3± 1.2
WF (4296 entries)
σ/μ = 11.69± 0.23 %
μ = 527.6± 1.1
σ = 61.7± 1.1

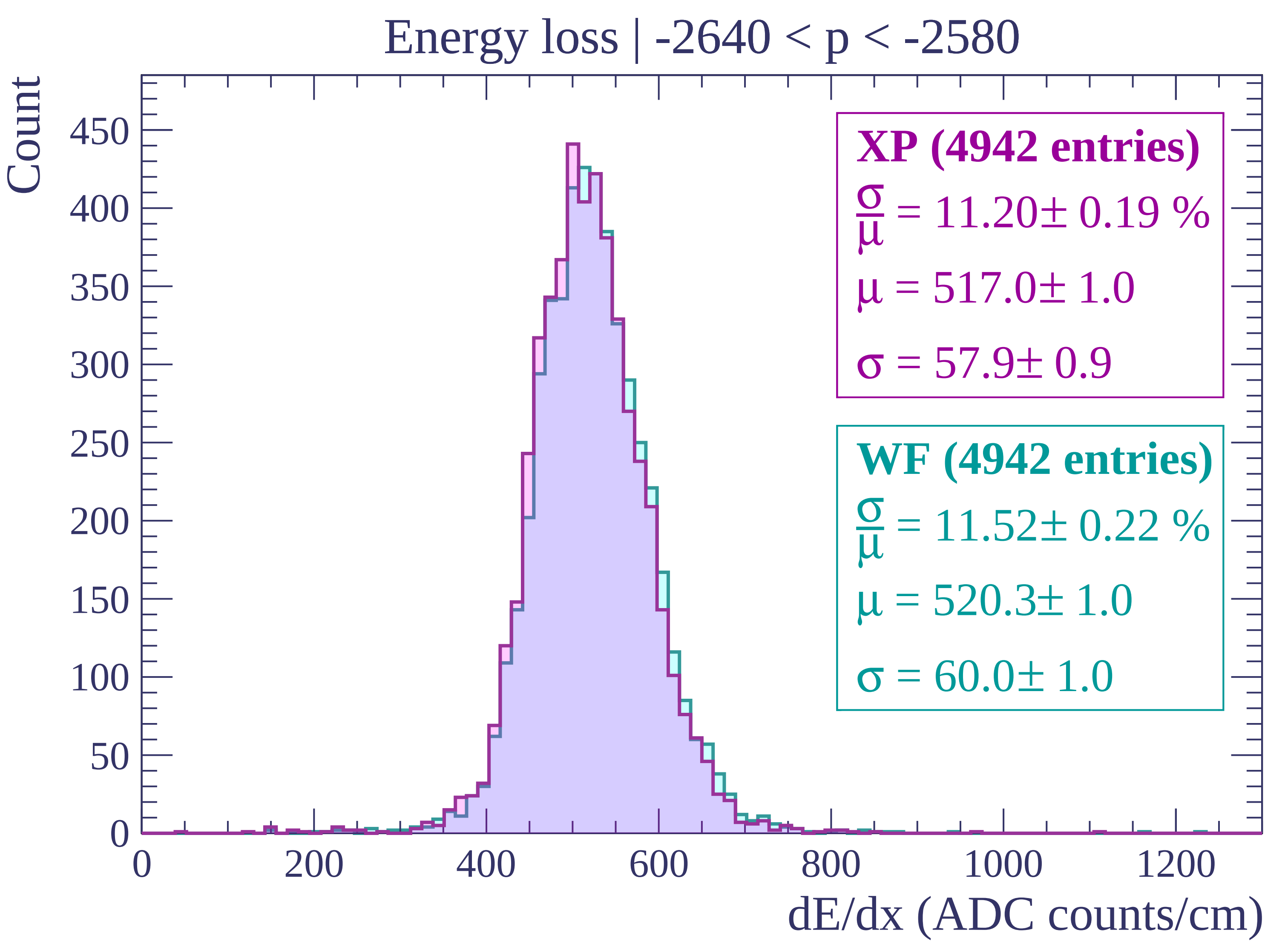


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4320 entries)
σ/μ = 11.80± 0.25 %
μ = 519.8± 1.1
σ = 61.3± 1.1
WF (4320 entries)
σ/μ = 11.91± 0.24 %
μ = 523.8± 1.1
σ = 62.4± 1.1

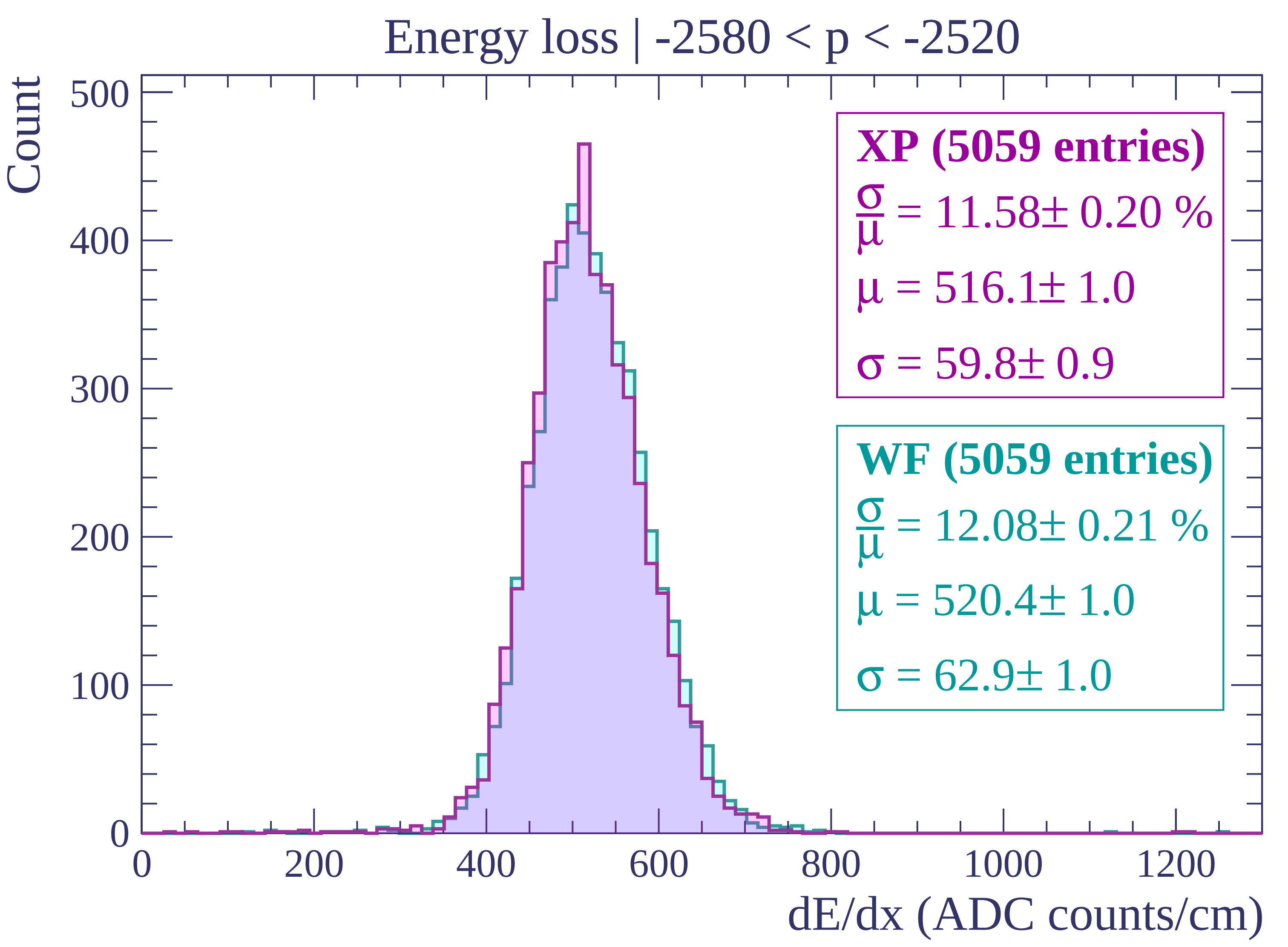

Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (4574 entries)
σ/μ = 12.18± 0.22 %
μ = 520.0± 1.1
σ = 63.3± 1.0
WF (4574 entries)
σ/μ = 12.10± 0.22 %
μ = 523.7± 1.1
σ = 63.4± 1.0

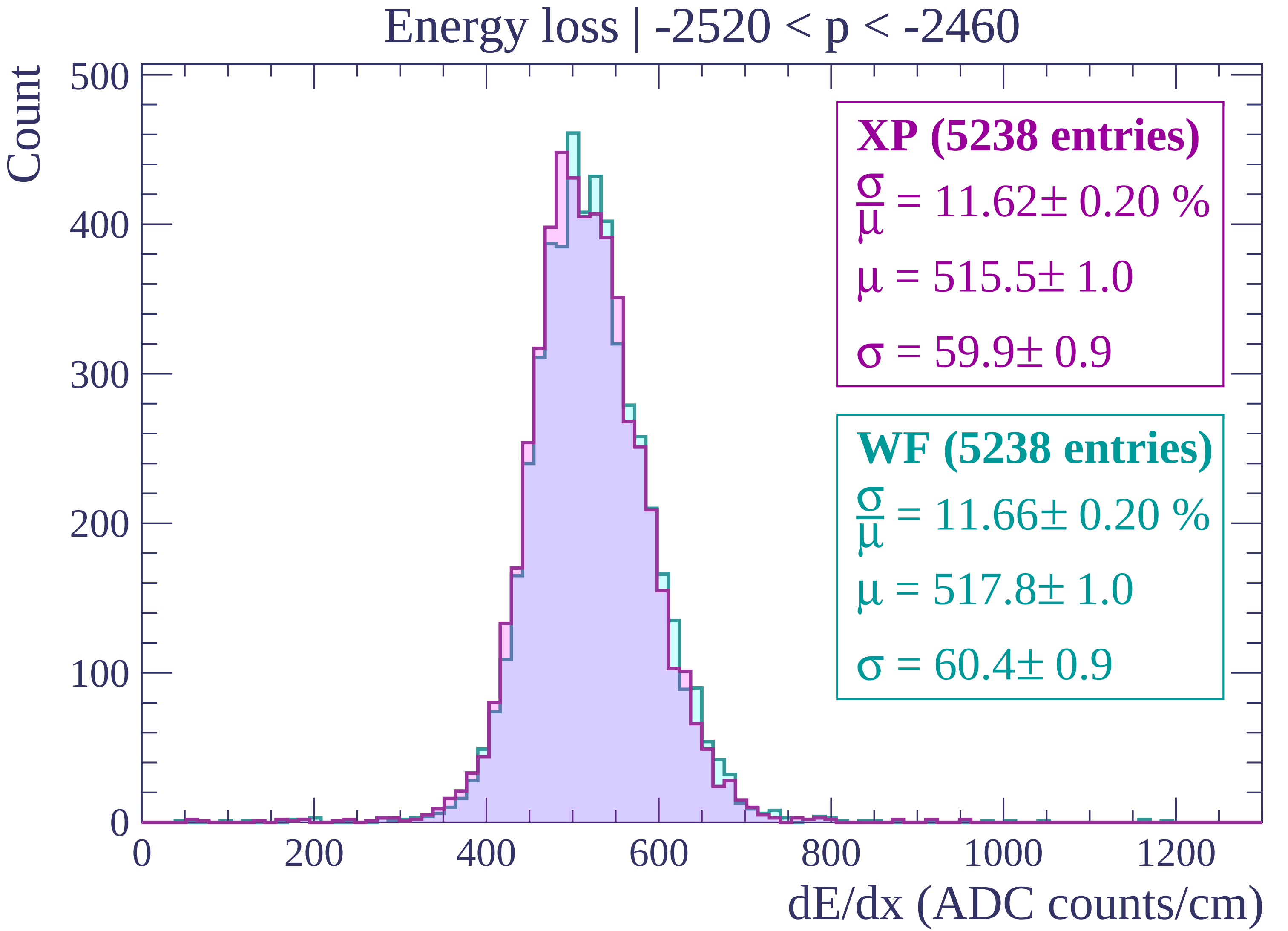


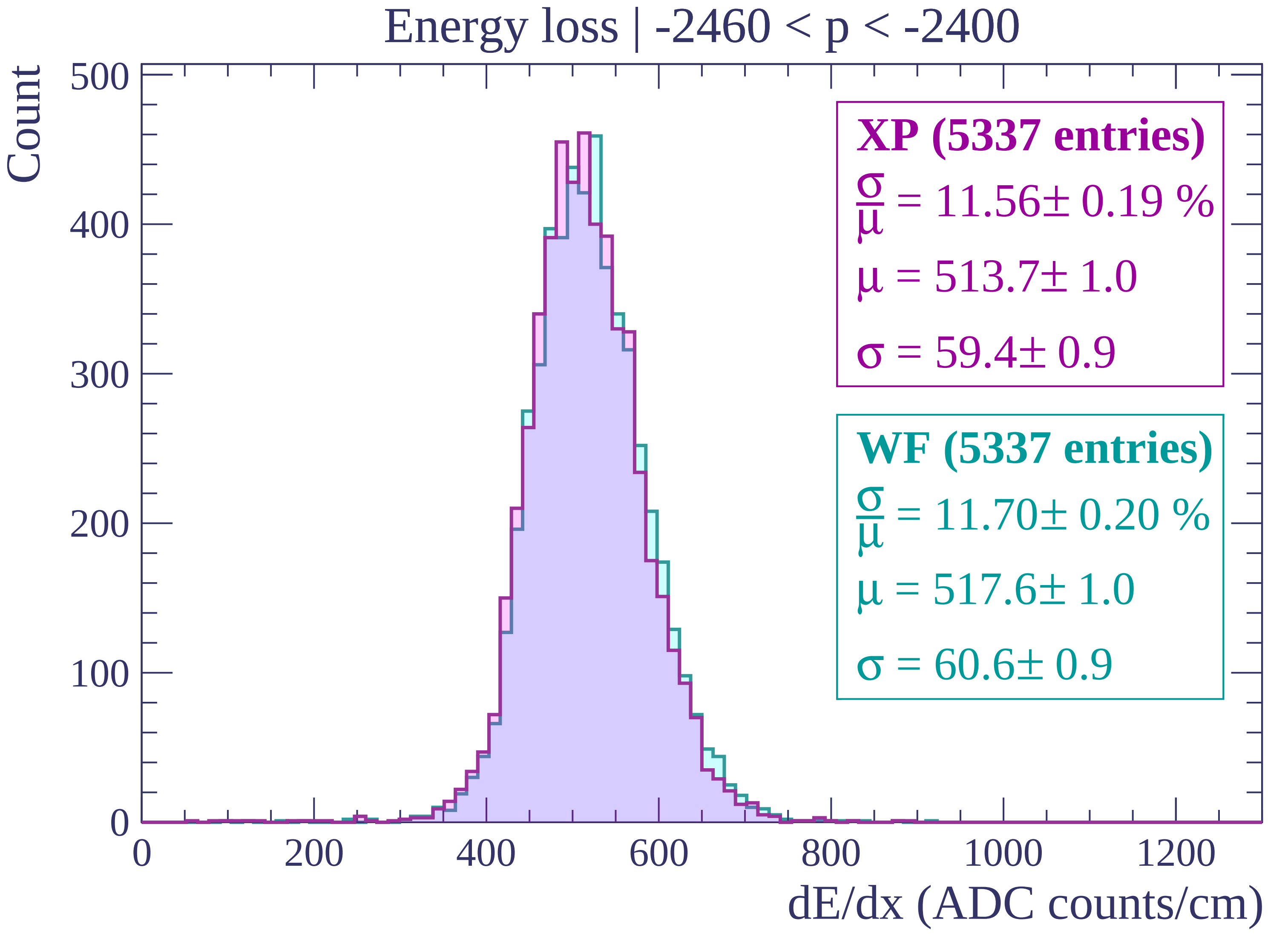
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4823 entries)
$\frac{\sigma}{\mu}$ = 11.62± 0.23 %
μ = 517.3± 1.0
σ = 60.1± 1.0
WF (4823 entries)
$\frac{\sigma}{\mu}$ = 11.76± 0.21 %
μ = 521.6± 1.0
σ = 61.4± 1.0



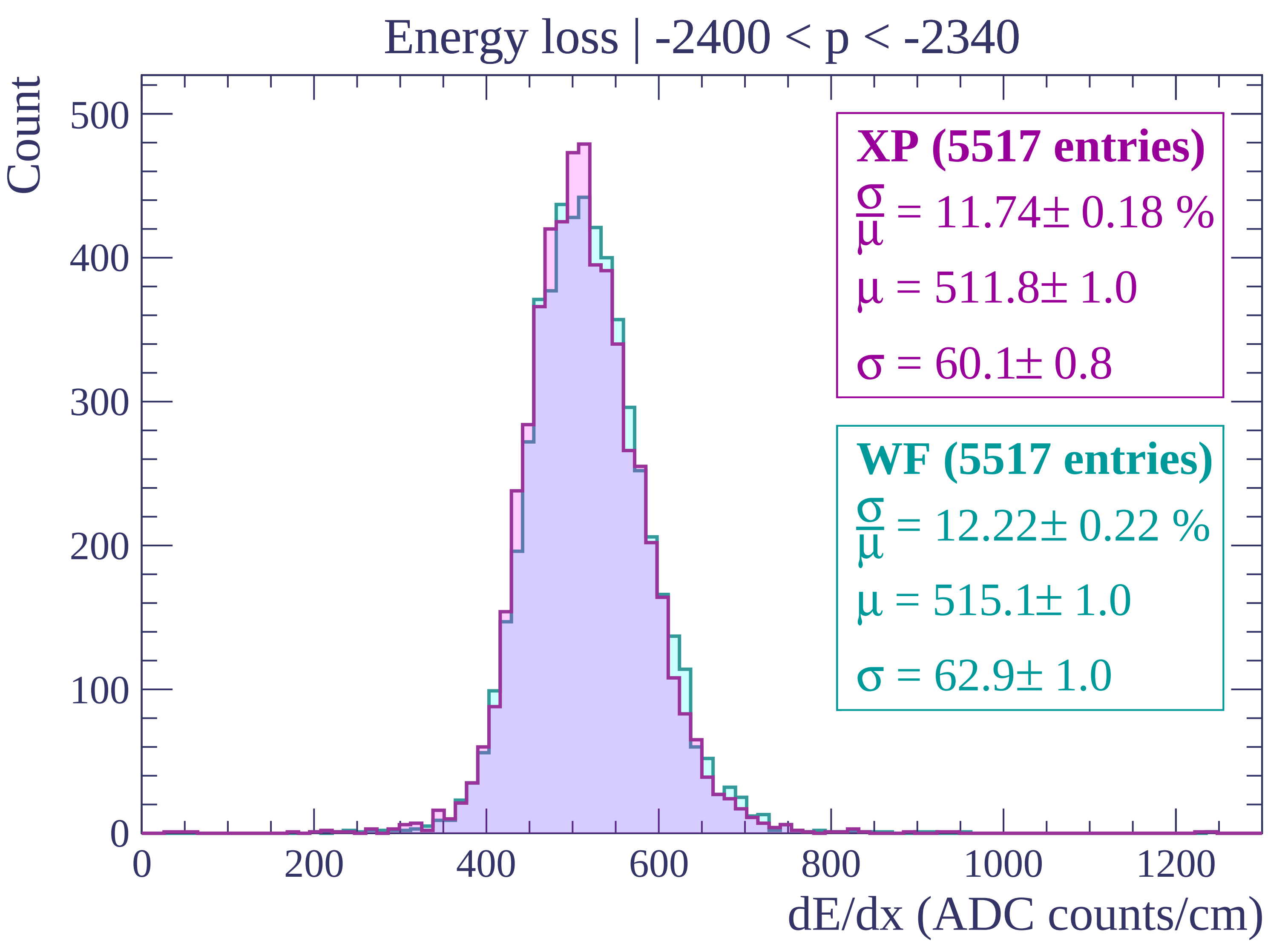
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4942 entries)
σ/μ = 11.20± 0.19 %
μ = 517.0± 1.0
σ = 57.9± 0.9
WF (4942 entries)
σ/μ = 11.52± 0.22 %
μ = 520.3± 1.0
σ = 60.0± 1.0


Energy loss | -2580 < p < -2520
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5059 entries)
σ/μ = 11.58± 0.20 %
μ = 516.1± 1.0
σ = 59.8± 0.9
WF (5059 entries)
σ/μ = 12.08± 0.21 %
μ = 520.4± 1.0
σ = 62.9± 1.0

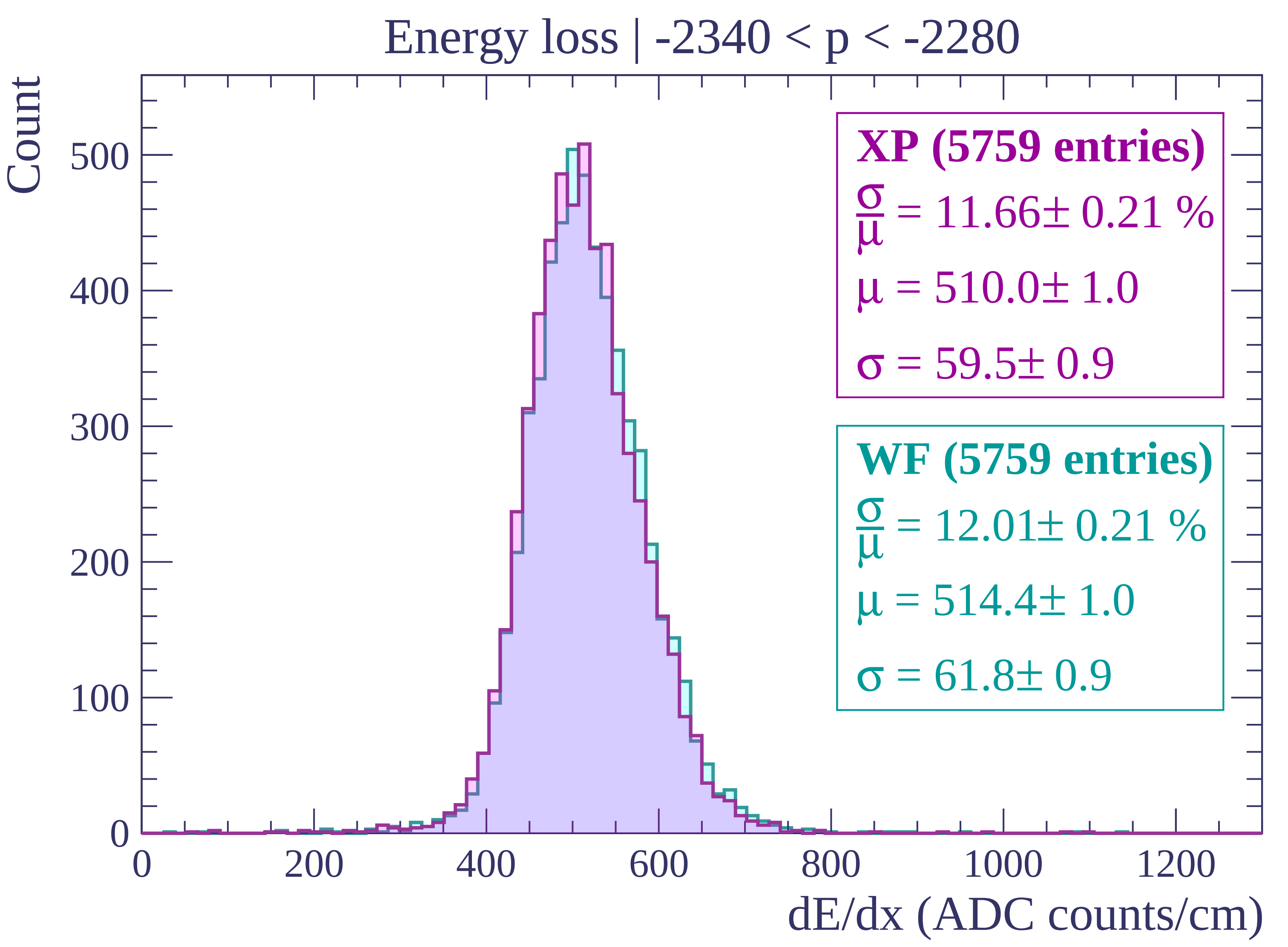

Energy loss | -2520 < p < -2460
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5238 entries)
σ/μ = 11.62± 0.20 %
μ = 515.5± 1.0
σ = 59.9± 0.9
WF (5238 entries)
σ/μ = 11.66± 0.20 %
μ = 517.8± 1.0
σ = 60.4± 0.9



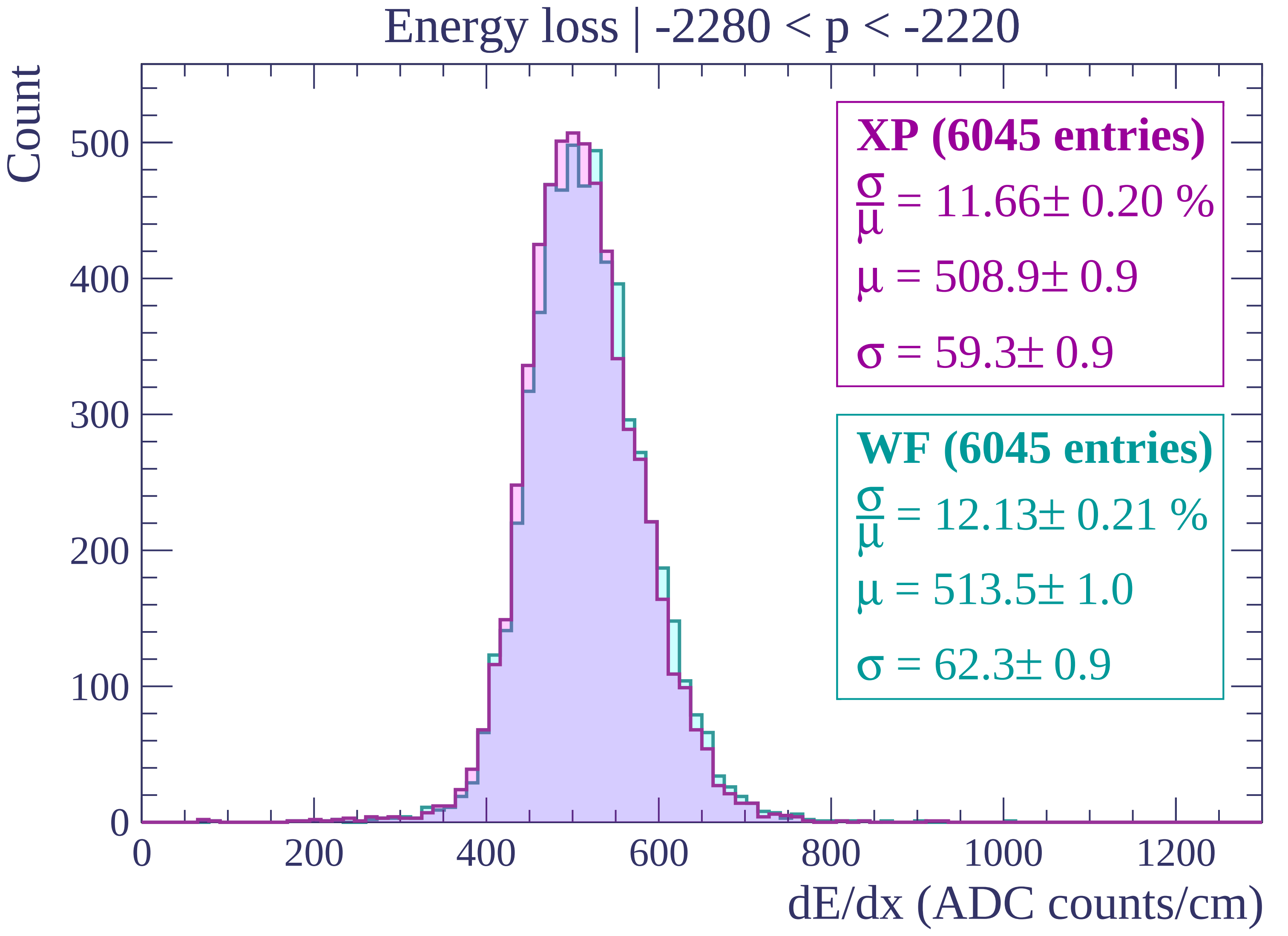
Energy loss | -2460 < p < -2400
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5337 entries)
σ/μ = 11.56± 0.19 %
μ = 513.7± 1.0
σ = 59.4± 0.9
WF (5337 entries)
σ/μ = 11.70± 0.20 %
μ = 517.6± 1.0
σ = 60.6± 0.9



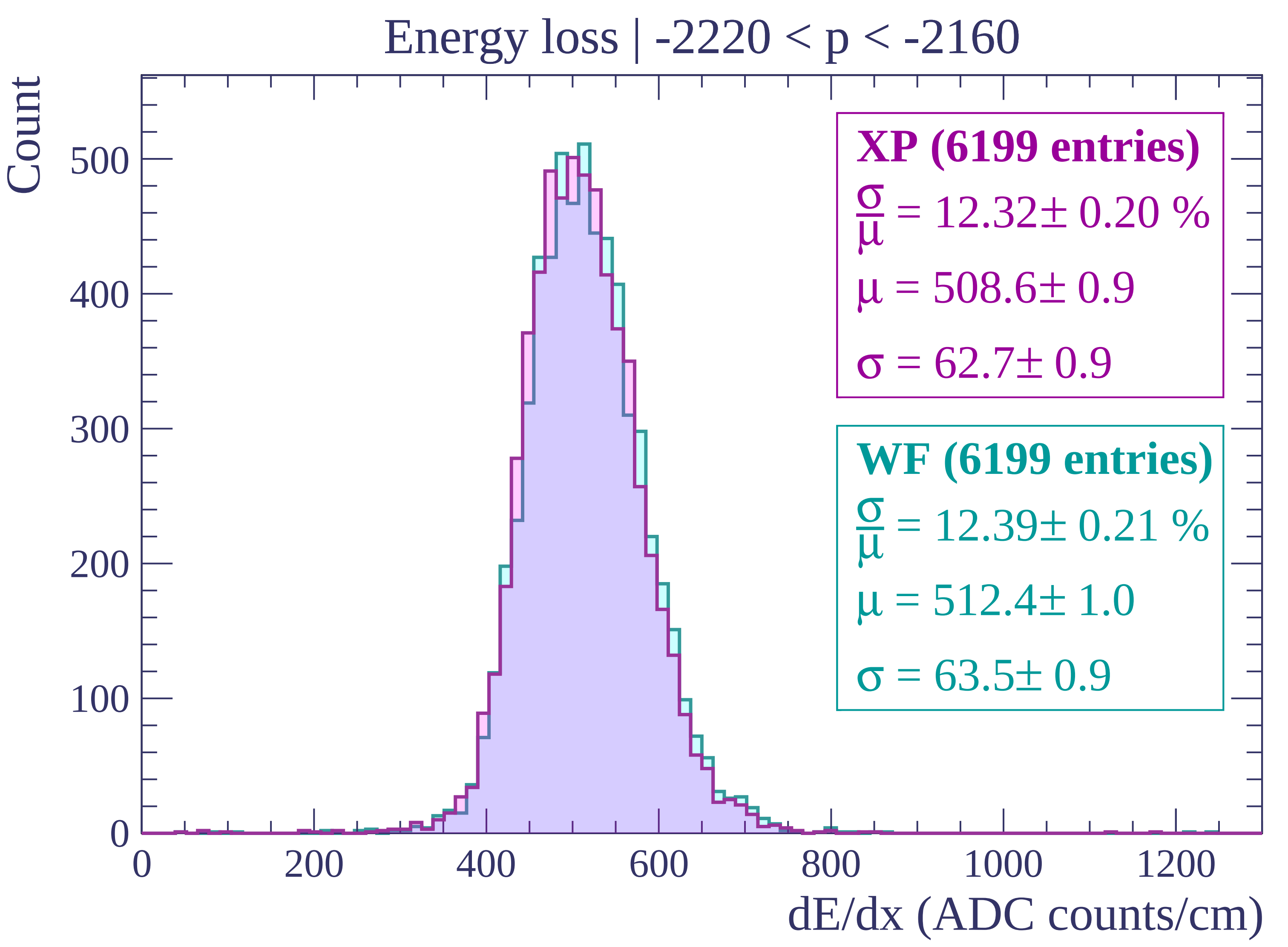
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5517 entries)
σ/μ = 11.74± 0.18 %
μ = 511.8± 1.0
σ = 60.1± 0.8
WF (5517 entries)
σ/μ = 12.22± 0.22 %
μ = 515.1± 1.0
σ = 62.9± 1.0



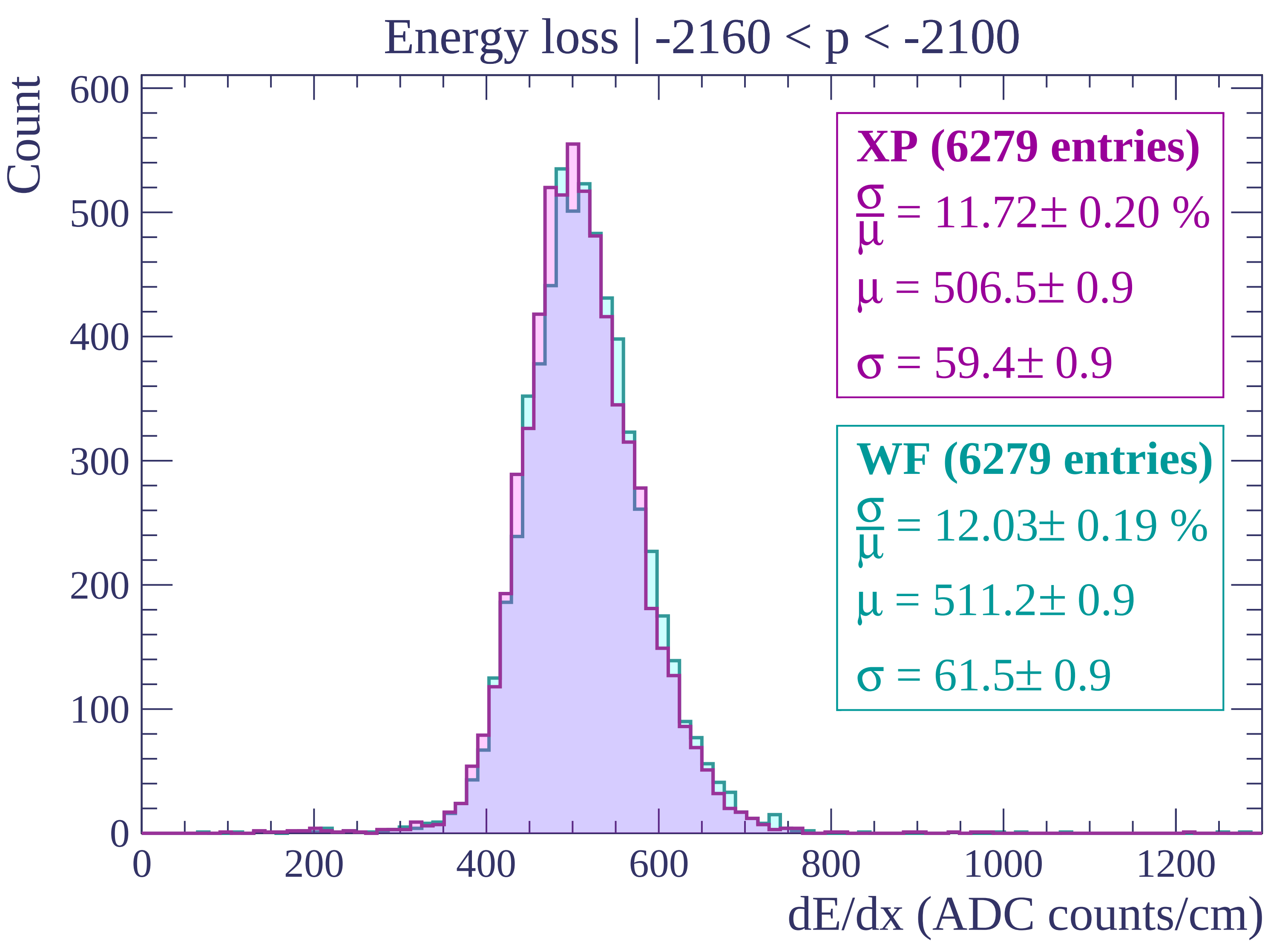
Energy loss | -2340 < p < -2280
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5759 entries)
σ/μ = 11.66± 0.21 %
μ = 510.0± 1.0
σ = 59.5± 0.9
WF (5759 entries)
σ/μ = 12.01± 0.21 %
μ = 514.4± 1.0
σ = 61.8± 0.9

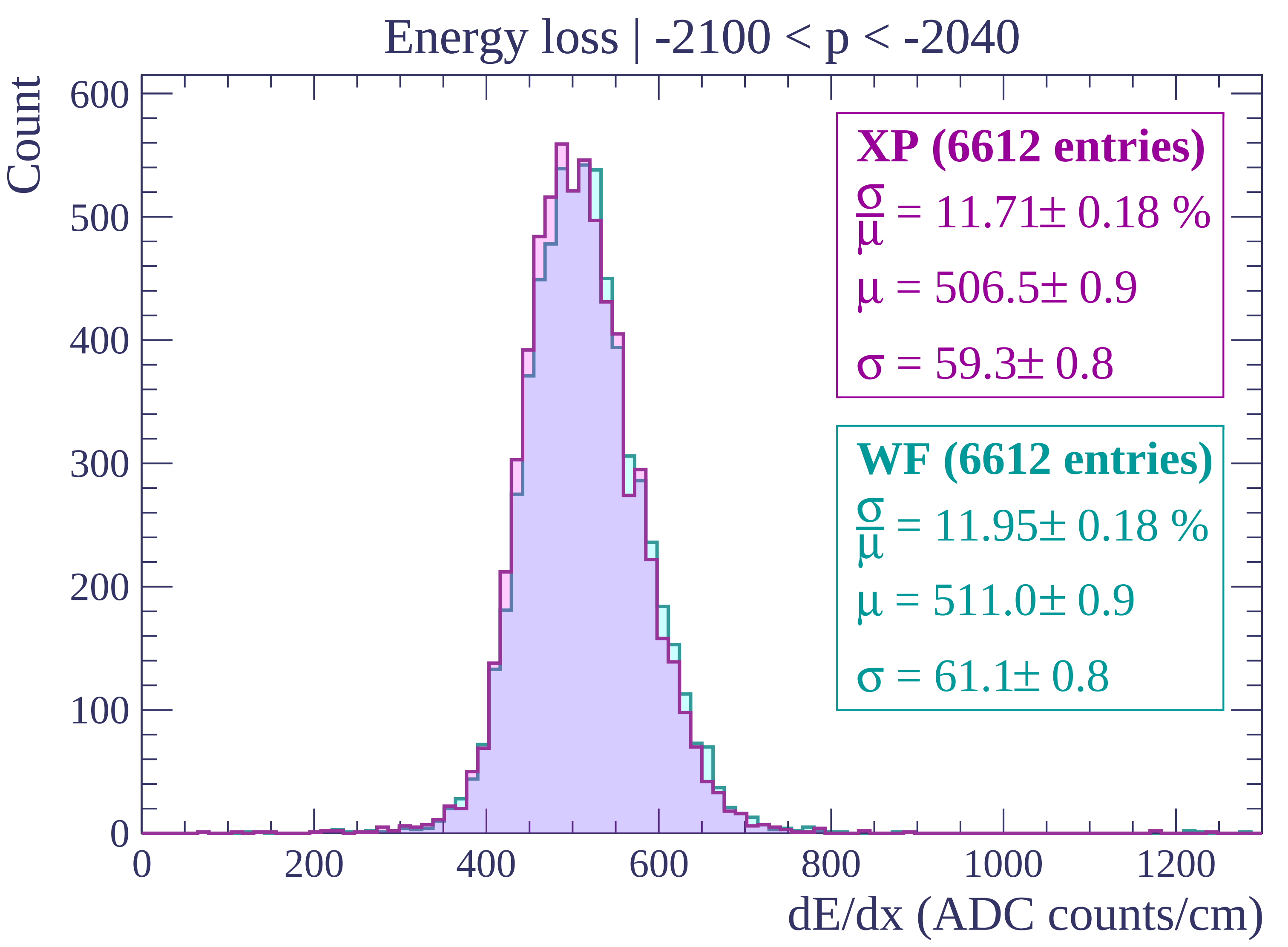


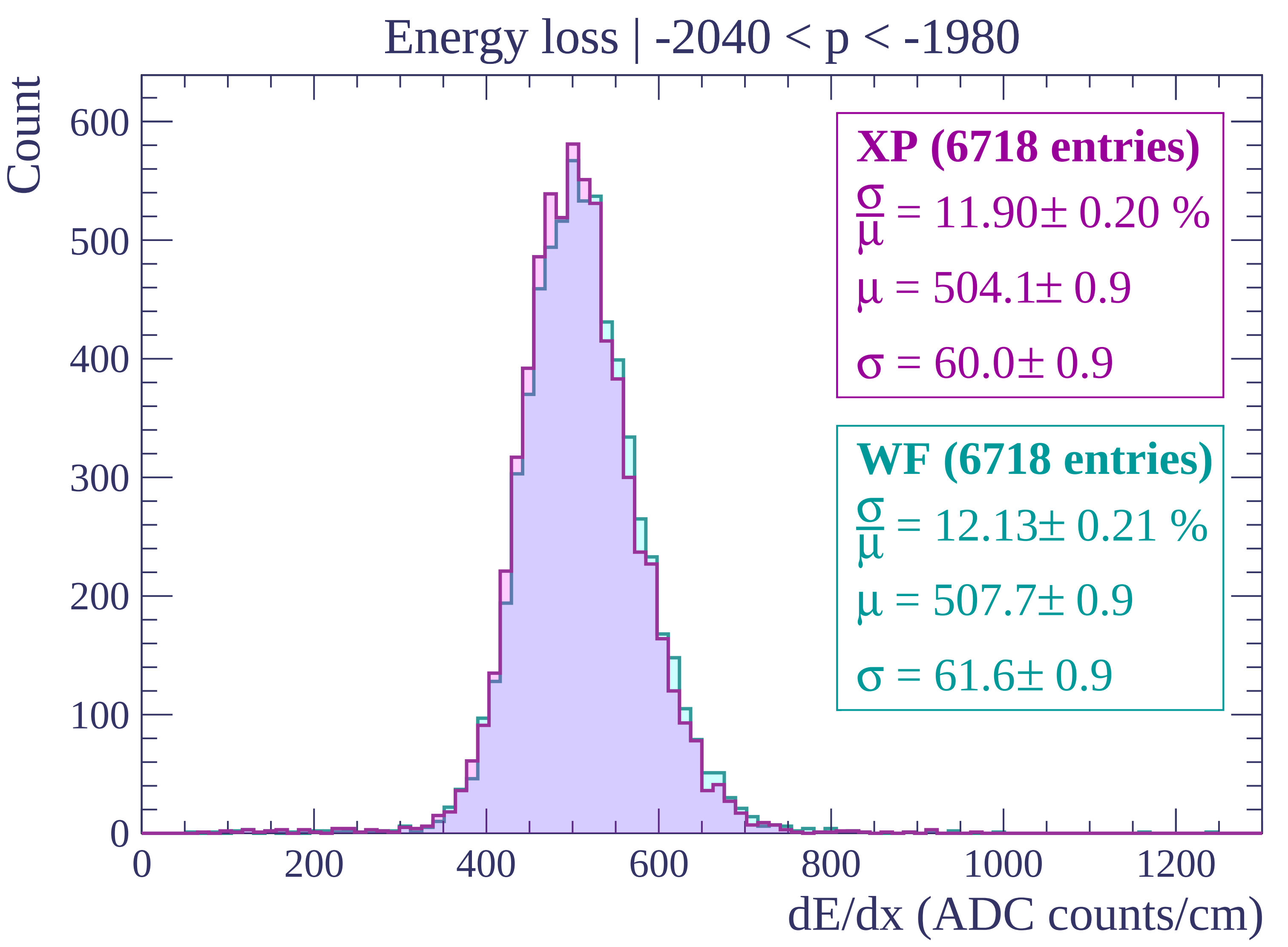
Energy loss | -2280 < p < -2220
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6045 entries)
σ/μ = 11.66± 0.20 %
μ = 508.9± 0.9
σ = 59.3± 0.9
WF (6045 entries)
σ/μ = 12.13± 0.21 %
μ = 513.5± 1.0
σ = 62.3± 0.9



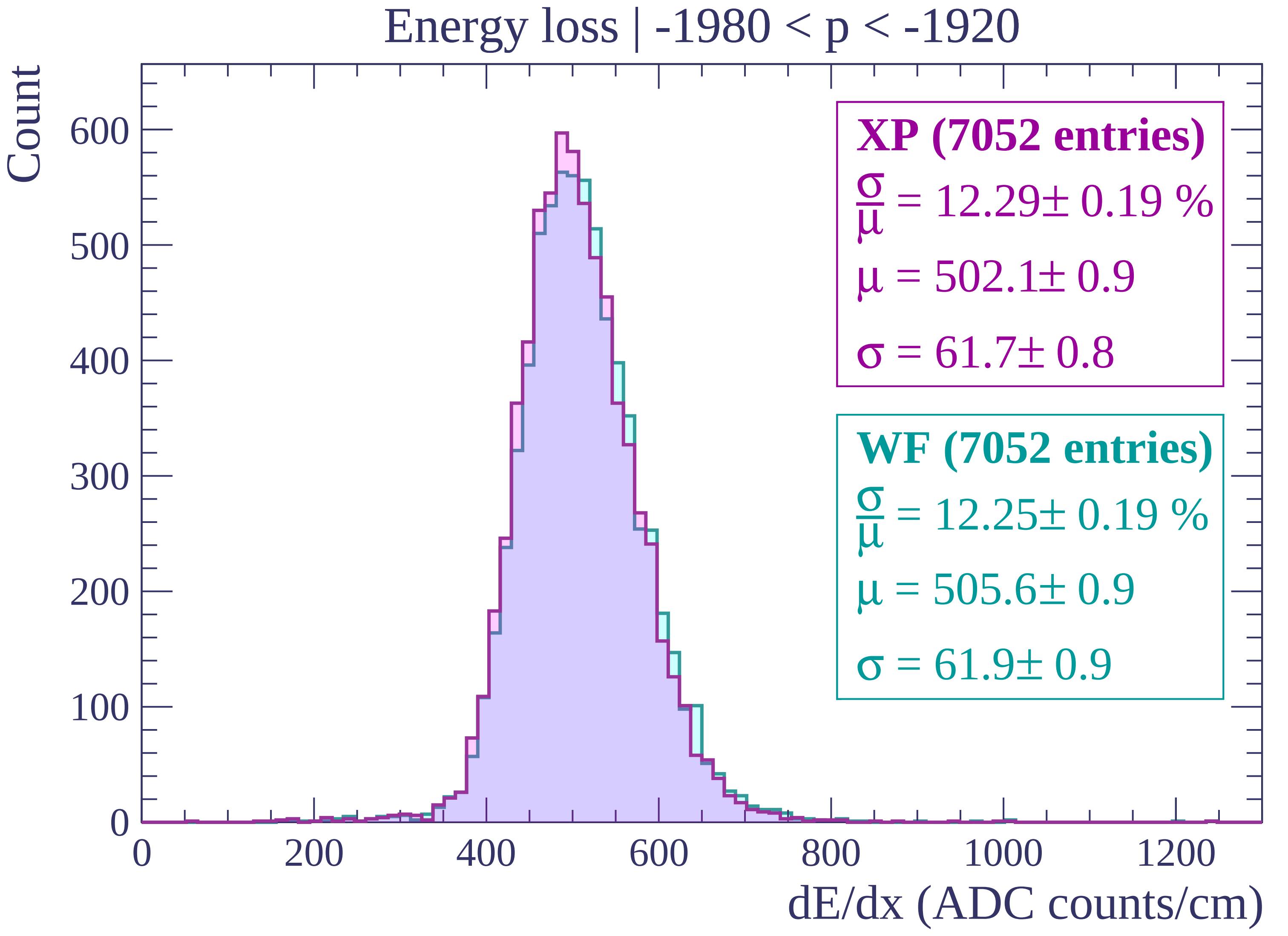
Energy loss | -2220 < p < -2160
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6199 entries)
$\frac{\sigma}{\mu}$ = 12.32± 0.20 %
μ = 508.6± 0.9
σ = 62.7± 0.9
WF (6199 entries)
$\frac{\sigma}{\mu}$ = 12.39± 0.21 %
μ = 512.4± 1.0
σ = 63.5± 0.9


Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (6279 entries)
$\frac{\sigma}{\mu}$ = 11.72± 0.20 %
μ = 506.5± 0.9
σ = 59.4± 0.9
WF (6279 entries)
$\frac{\sigma}{\mu}$ = 12.03± 0.19 %
μ = 511.2± 0.9
σ = 61.5± 0.9
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

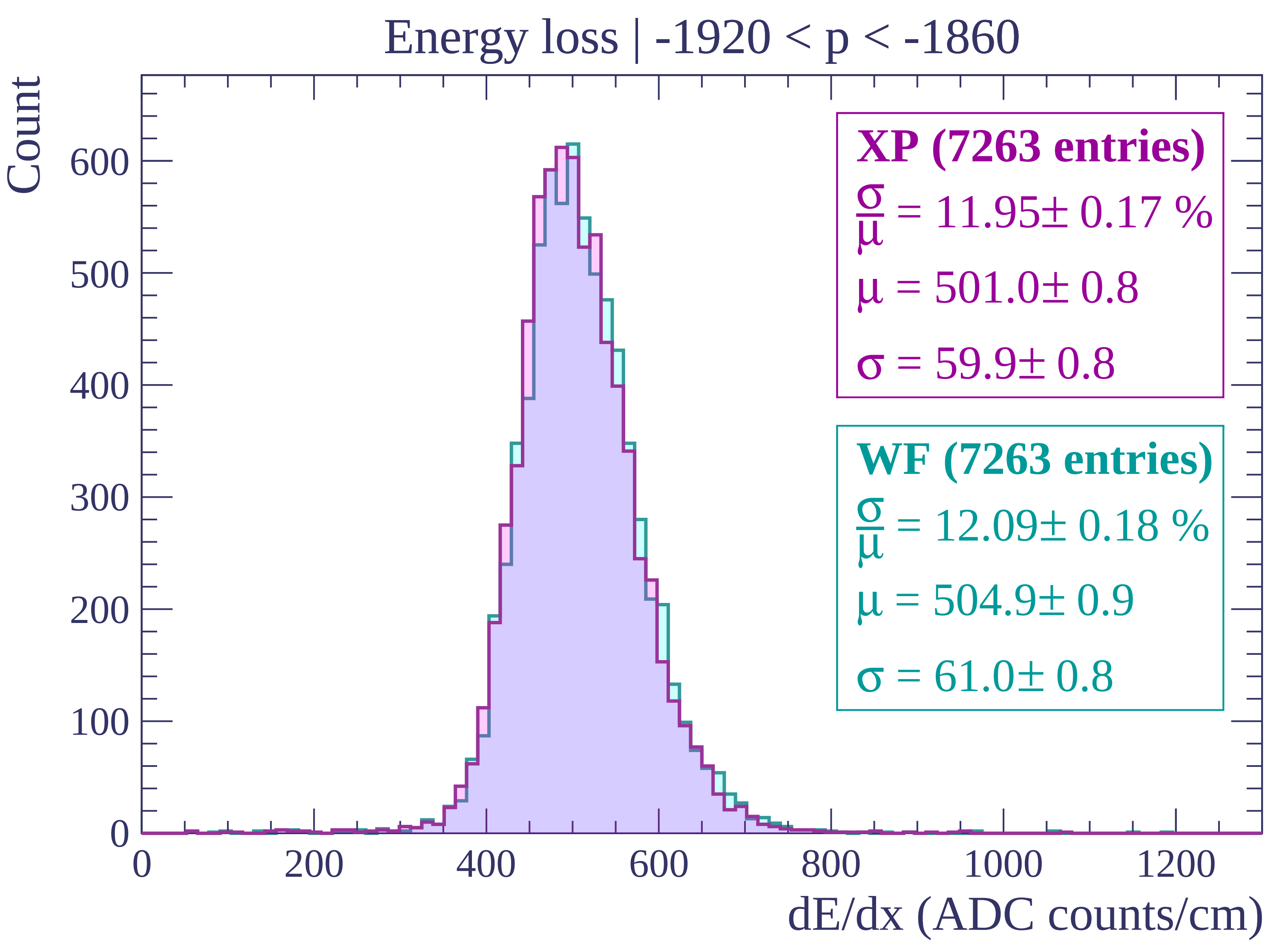

Energy loss | -2100 < p < -2040
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6612 entries)
σ/μ = 11.71± 0.18 %
μ = 506.5± 0.9
σ = 59.3± 0.8
WF (6612 entries)
σ/μ = 11.95± 0.18 %
μ = 511.0± 0.9
σ = 61.1± 0.8

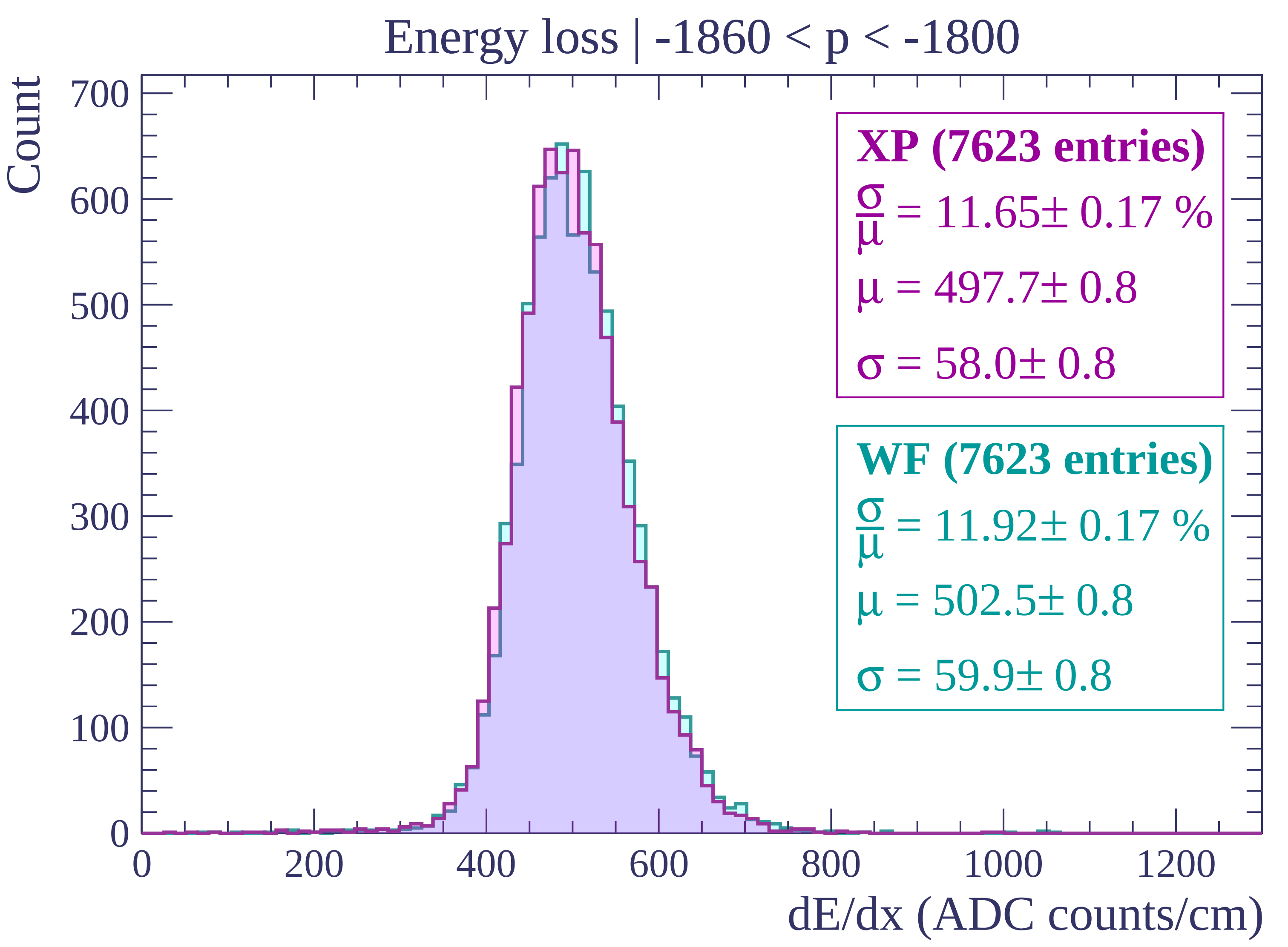


Energy loss | -2040 < p < -1980
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6718 entries)
σ/μ = 11.90± 0.20 %
μ = 504.1± 0.9
σ = 60.0± 0.9
WF (6718 entries)
σ/μ = 12.13± 0.21 %
μ = 507.7± 0.9
σ = 61.6± 0.9

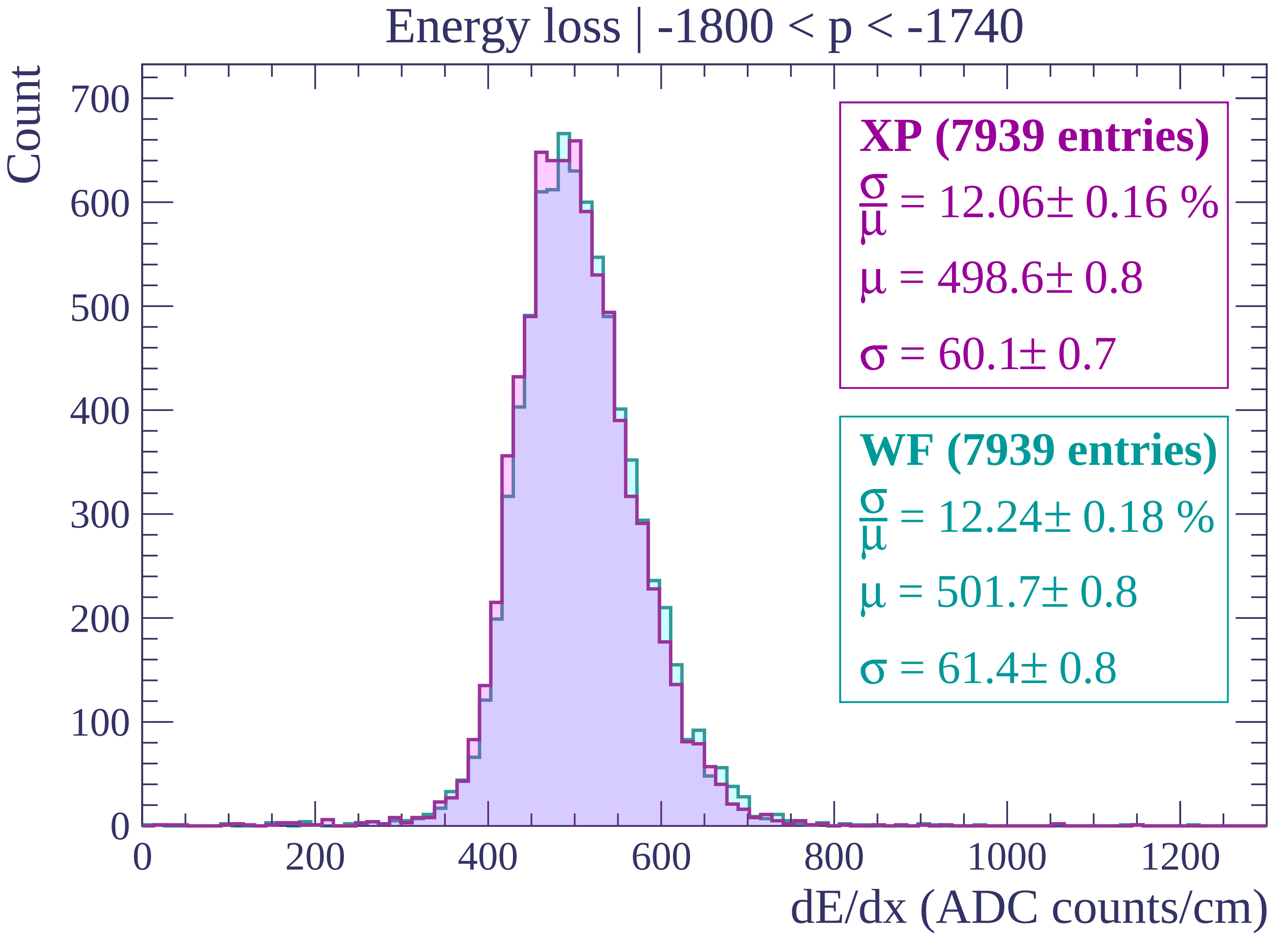


Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (7052 entries)
$\frac{\sigma}{\mu}$ = 12.29± 0.19 %
μ = 502.1± 0.9
σ = 61.7± 0.8
WF (7052 entries)
$\frac{\sigma}{\mu}$ = 12.25± 0.19 %
μ = 505.6± 0.9
σ = 61.9± 0.9

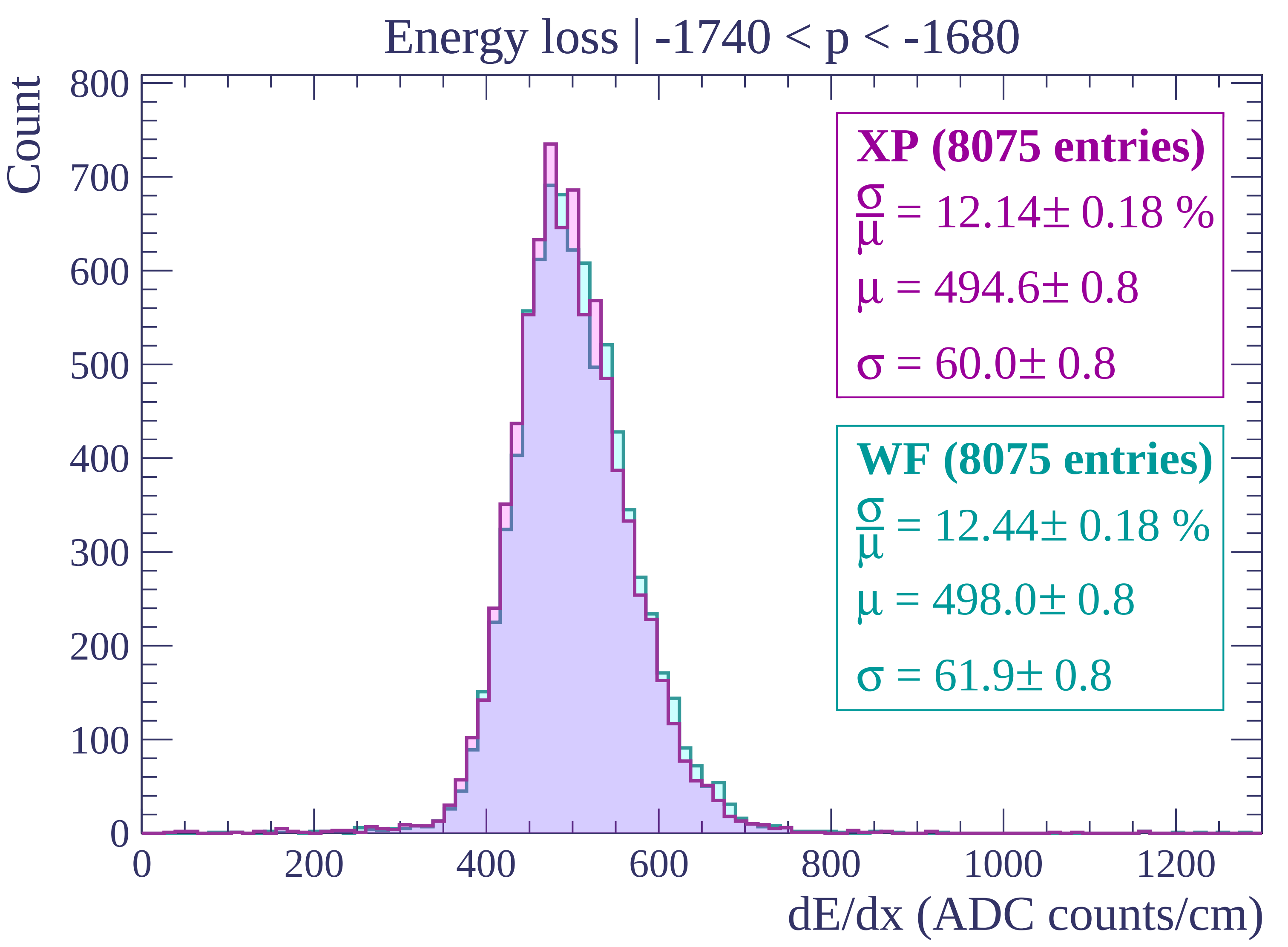

Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (7263 entries)
$\frac{\sigma}{\mu}$ = 11.95± 0.17 %
μ = 501.0± 0.8
σ = 59.9± 0.8
WF (7263 entries)
$\frac{\sigma}{\mu}$ = 12.09± 0.18 %
μ = 504.9± 0.9
σ = 61.0± 0.8



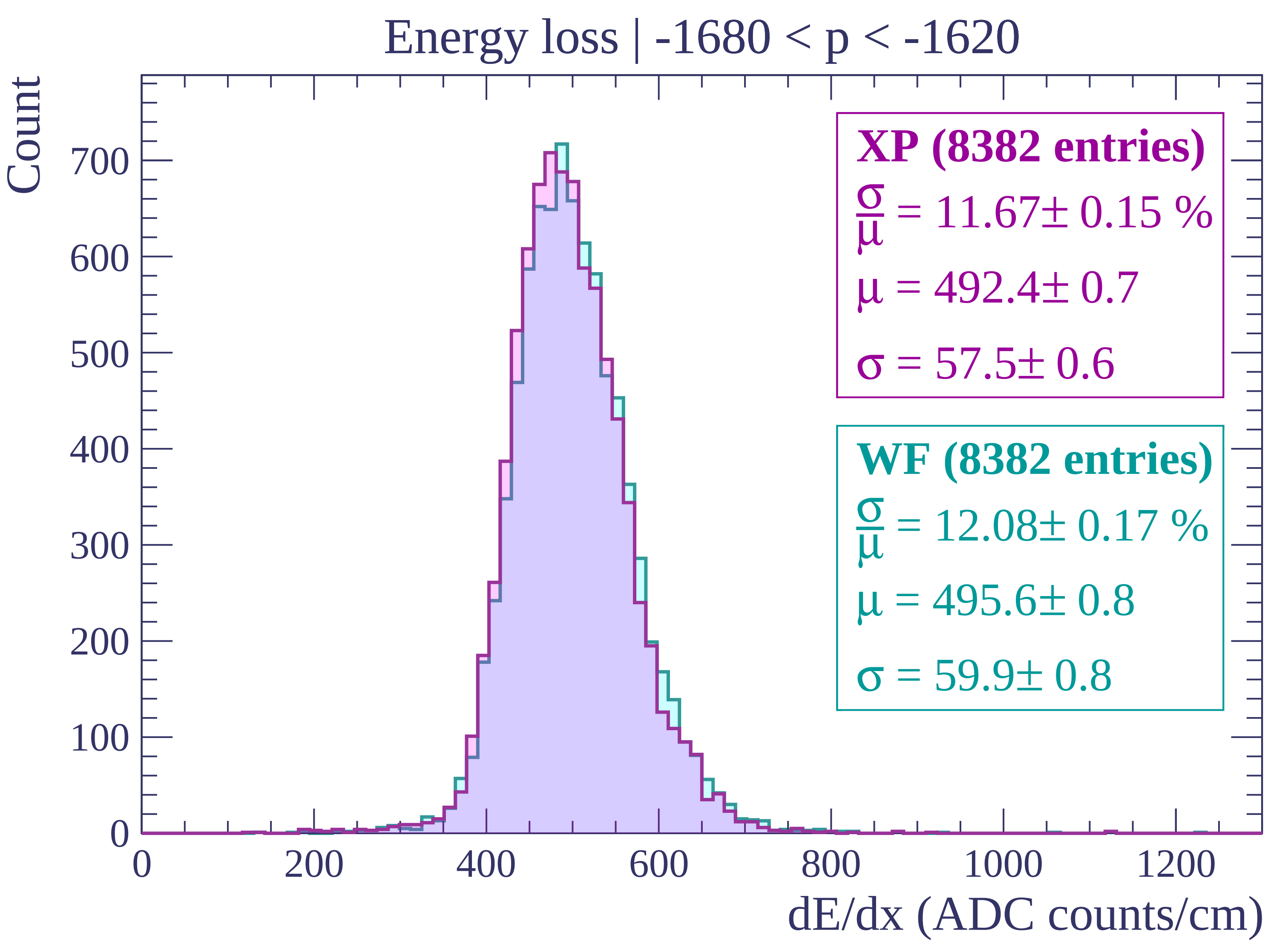
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (7623 entries)
$\frac{\sigma}{\mu}$ = 11.65± 0.17 %
μ = 497.7± 0.8
σ = 58.0± 0.8
WF (7623 entries)
$\frac{\sigma}{\mu}$ = 11.92± 0.17 %
μ = 502.5± 0.8
σ = 59.9± 0.8
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

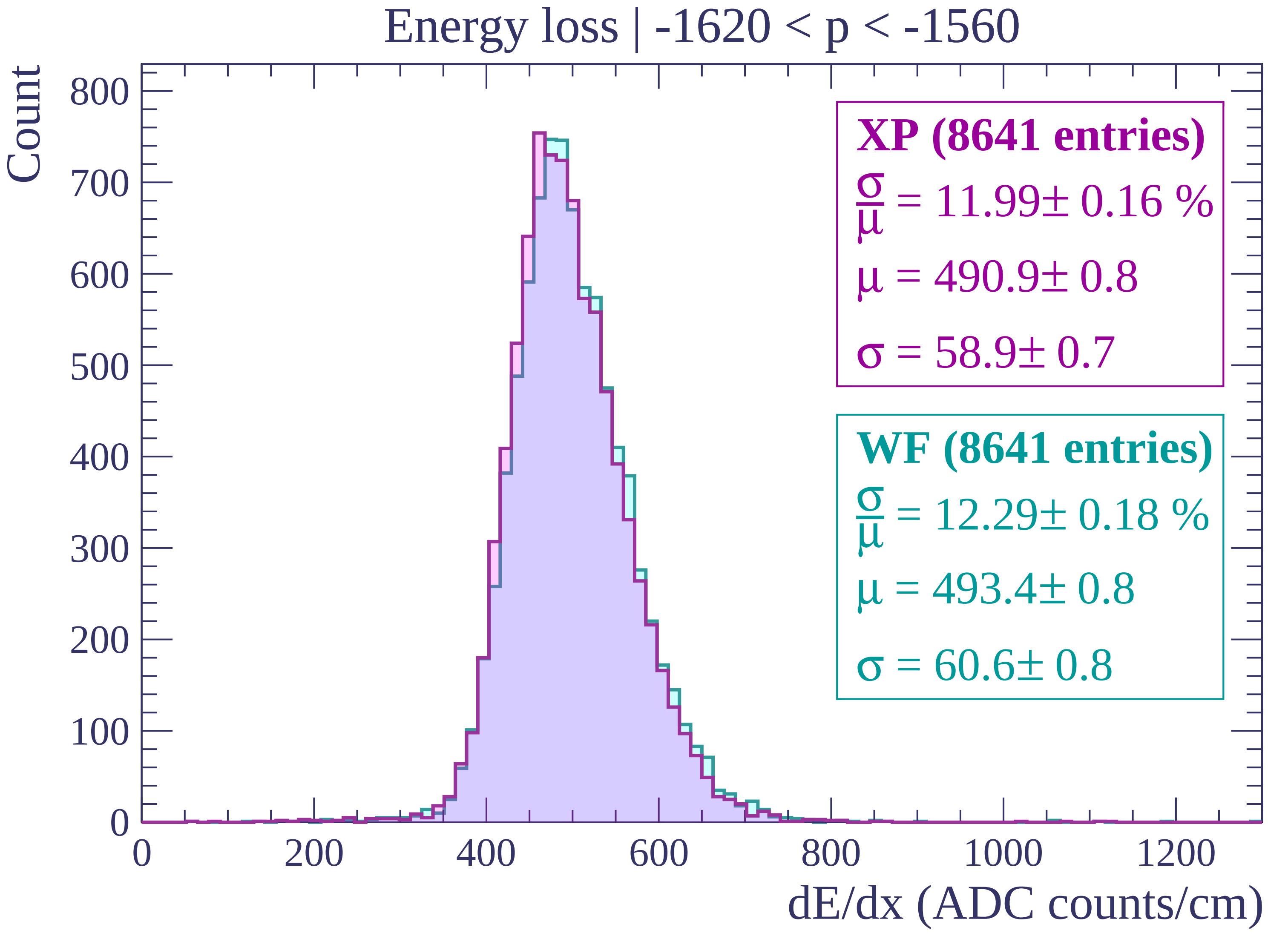


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (7939 entries)
$\frac{\sigma}{\mu} = 12.06 \pm 0.16$ %
$\mu = 498.6 \pm 0.8$
$\sigma = 60.1 \pm 0.7$
WF (7939 entries)
$\frac{\sigma}{\mu} = 12.24 \pm 0.18$ %
$\mu = 501.7 \pm 0.8$
$\sigma = 61.4 \pm 0.8$

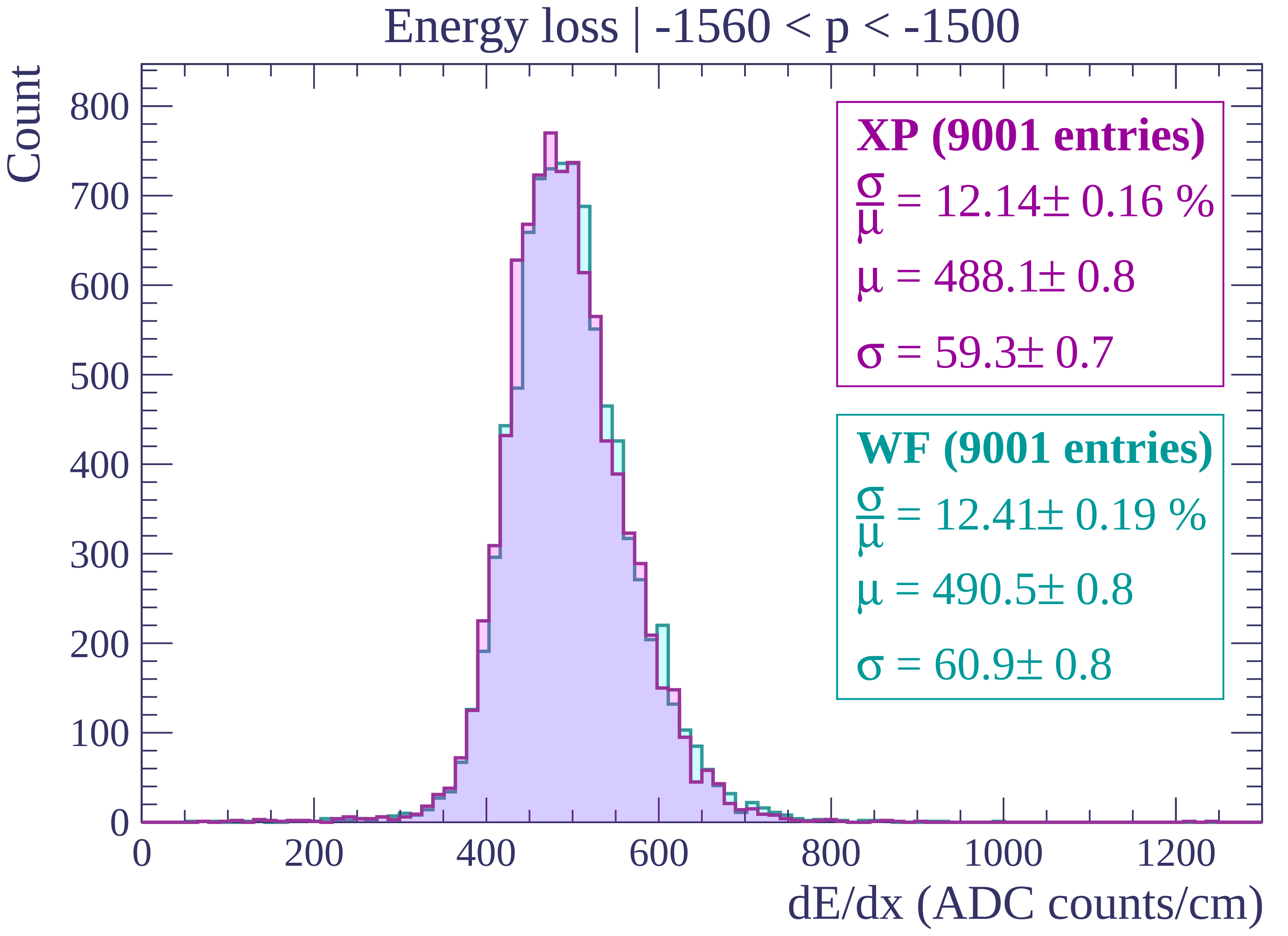


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
XP (8075 entries)
$\frac{\sigma}{\mu}$ = 12.14± 0.18 %
μ = 494.6± 0.8
σ = 60.0± 0.8
WF (8075 entries)
$\frac{\sigma}{\mu}$ = 12.44± 0.18 %
μ = 498.0± 0.8
σ = 61.9± 0.8
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

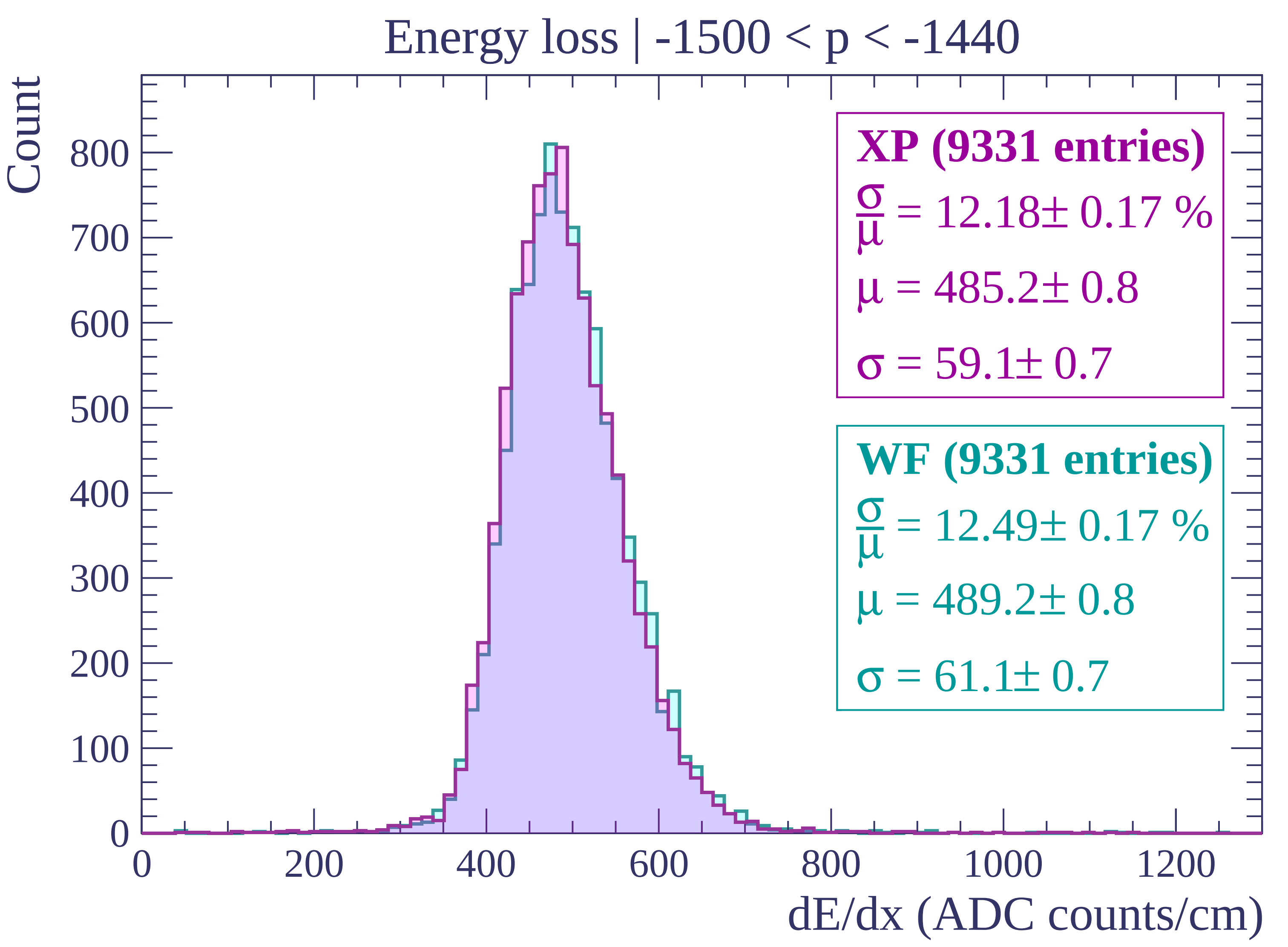

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
XP (8382 entries)
σ/μ = 11.67± 0.15 %
μ = 492.4± 0.7
σ = 57.5± 0.6
WF (8382 entries)
σ/μ = 12.08± 0.17 %
μ = 495.6± 0.8
σ = 59.9± 0.8



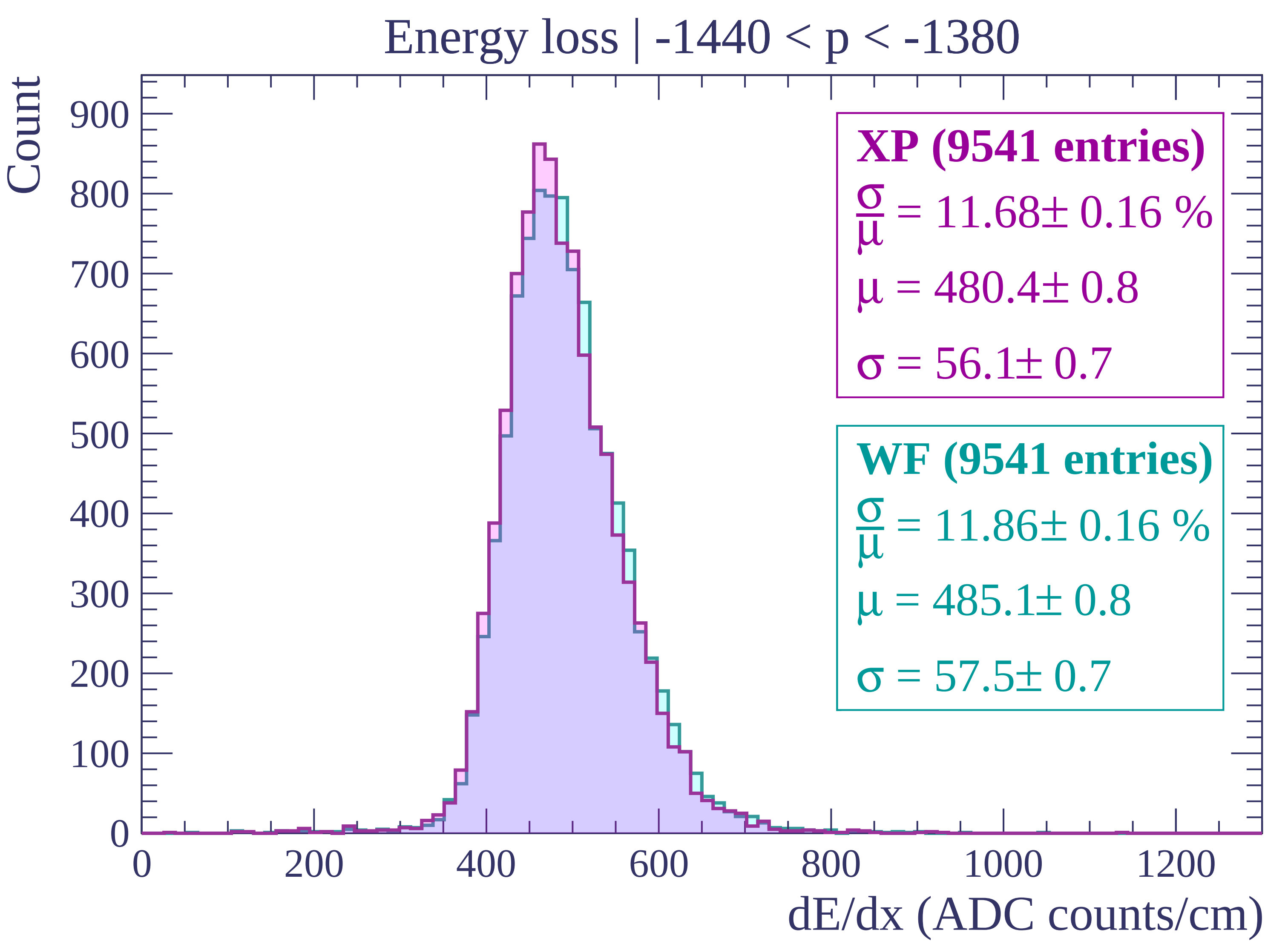
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
XP (8641 entries)
$\frac{\sigma}{\mu}$ = 11.99± 0.16 %
μ = 490.9± 0.8
σ = 58.9± 0.7
WF (8641 entries)
$\frac{\sigma}{\mu}$ = 12.29± 0.18 %
μ = 493.4± 0.8
σ = 60.6± 0.8
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

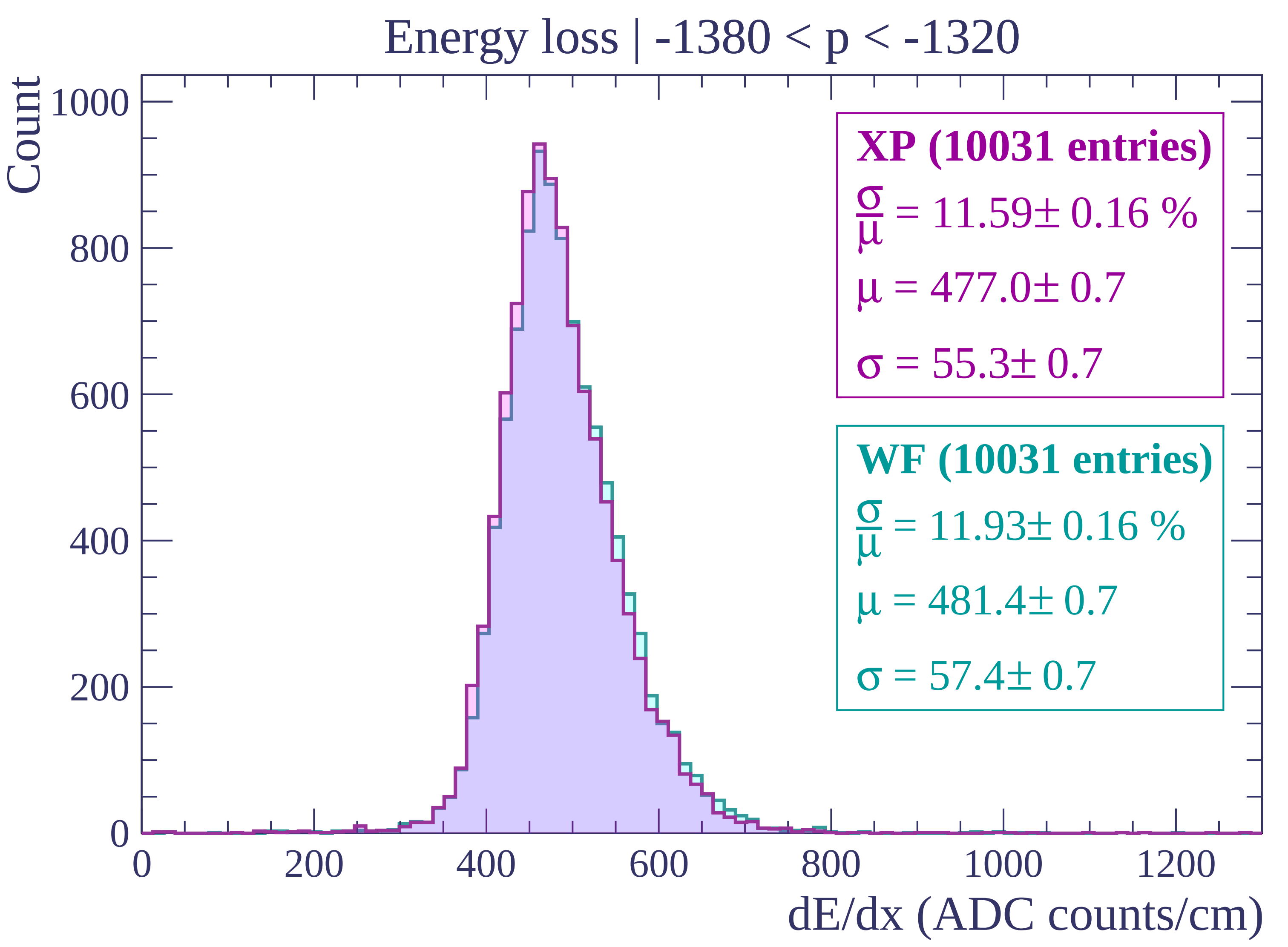


Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (9001 entries)
σ/μ = 12.14± 0.16 %
μ = 488.1± 0.8
σ = 59.3± 0.7
WF (9001 entries)
σ/μ = 12.41± 0.19 %
μ = 490.5± 0.8
σ = 60.9± 0.8

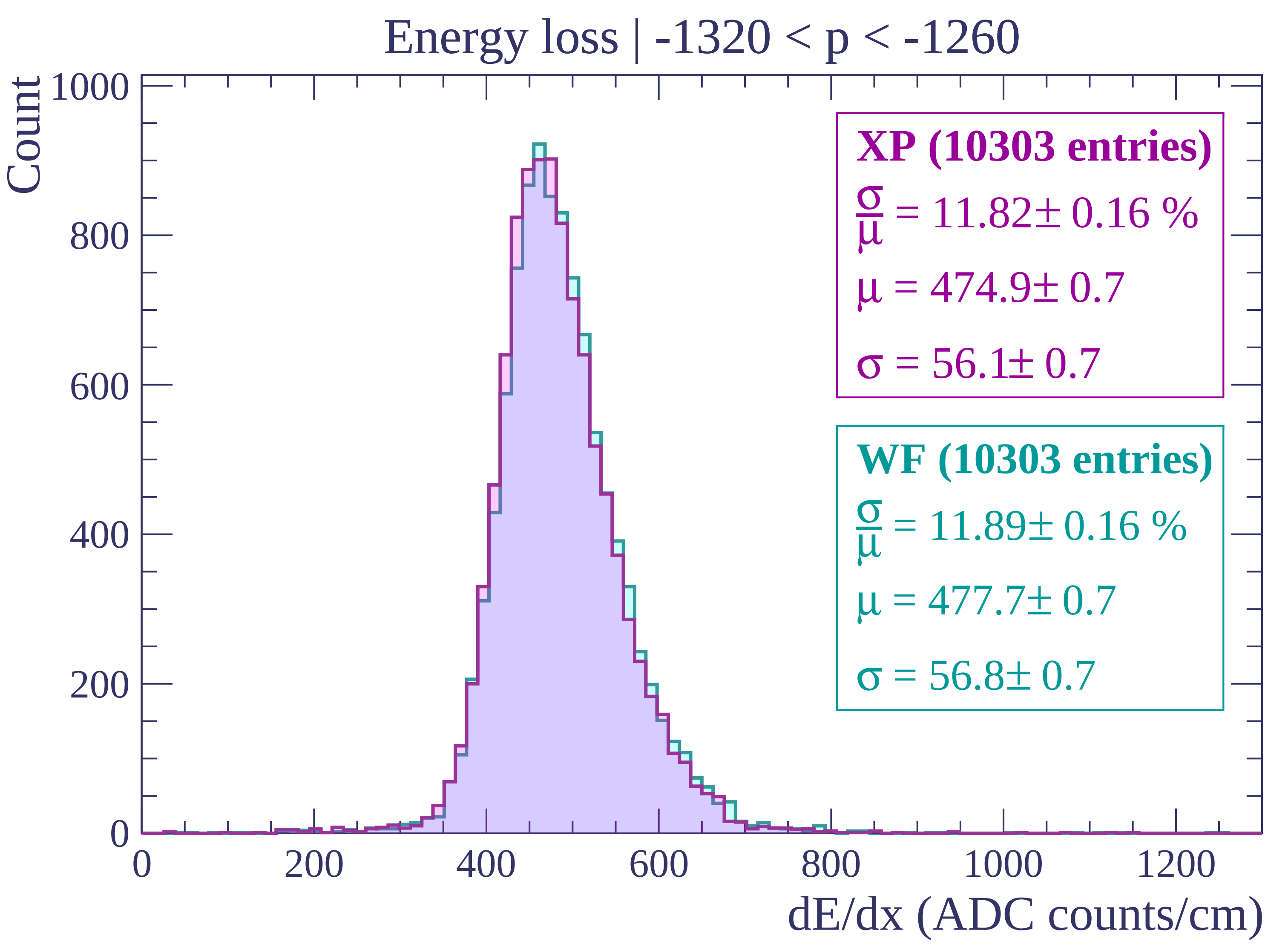


Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (9331 entries)
σ/μ = 12.18± 0.17 %
μ = 485.2± 0.8
σ = 59.1± 0.7
WF (9331 entries)
σ/μ = 12.49± 0.17 %
μ = 489.2± 0.8
σ = 61.1± 0.7

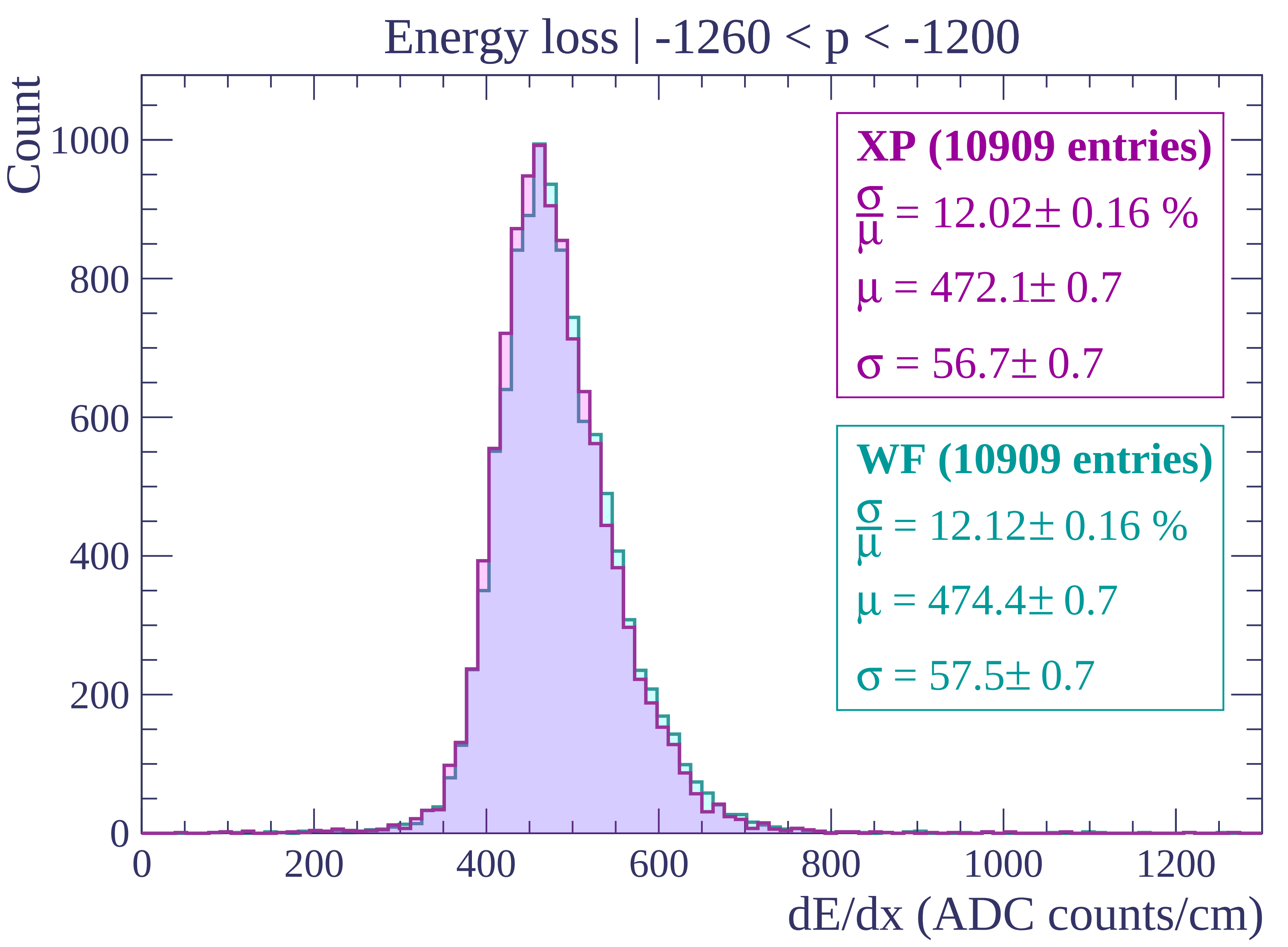

Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (9541 entries)
$\frac{\sigma}{\mu}$ = 11.68± 0.16 %
μ = 480.4± 0.8
σ = 56.1± 0.7
WF (9541 entries)
$\frac{\sigma}{\mu}$ = 11.86± 0.16 %
μ = 485.1± 0.8
σ = 57.5± 0.7



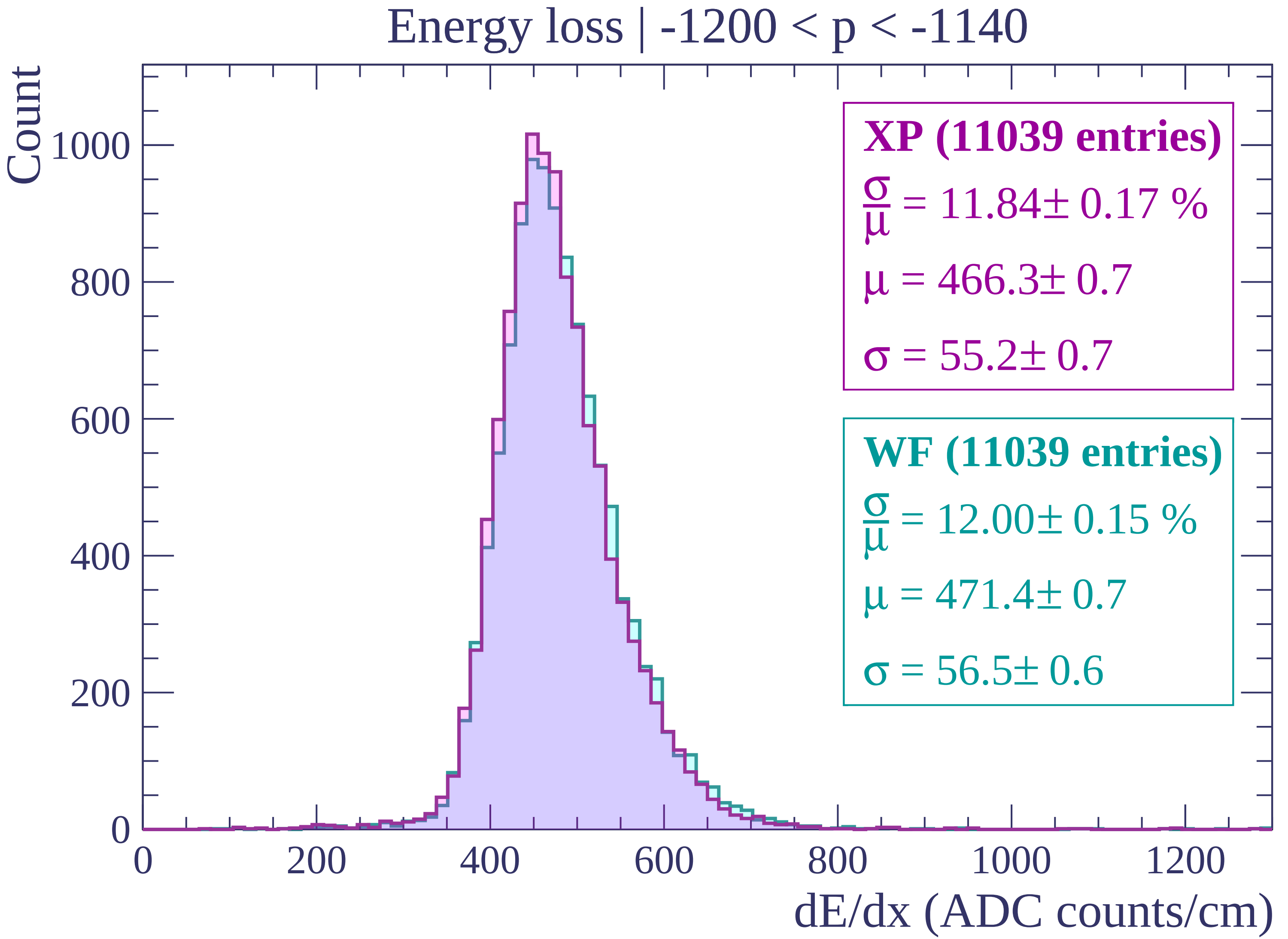
Energy loss | -1380 < p < -1320
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10031 entries)
$\frac{\sigma}{\mu}$ = 11.59± 0.16 %
μ = 477.0± 0.7
σ = 55.3± 0.7
WF (10031 entries)
$\frac{\sigma}{\mu}$ = 11.93± 0.16 %
μ = 481.4± 0.7
σ = 57.4± 0.7



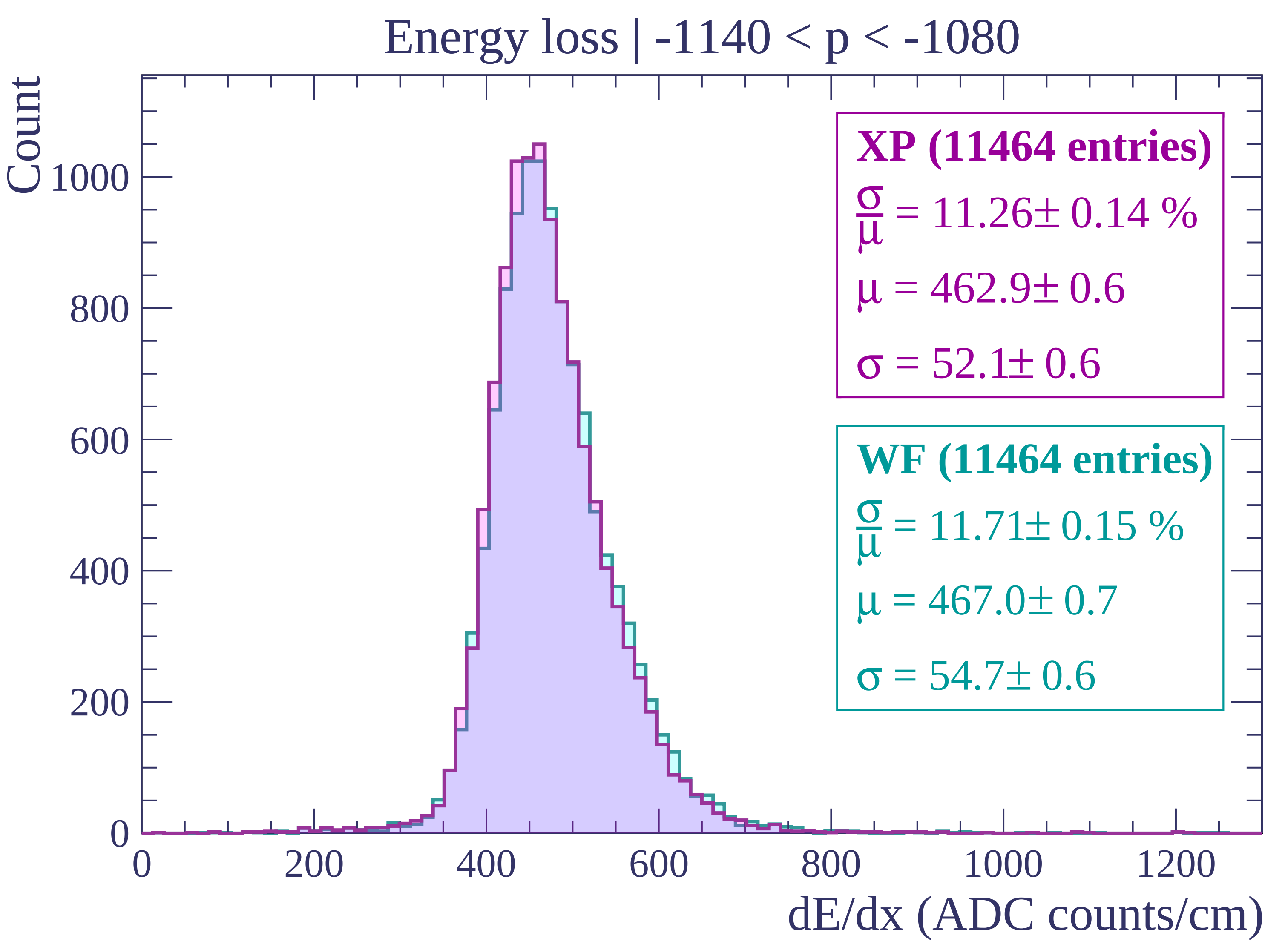
Energy loss | -1320 < p < -1260
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (10303 entries)
σ/μ = 11.82± 0.16 %
μ = 474.9± 0.7
σ = 56.1± 0.7
WF (10303 entries)
σ/μ = 11.89± 0.16 %
μ = 477.7± 0.7
σ = 56.8± 0.7

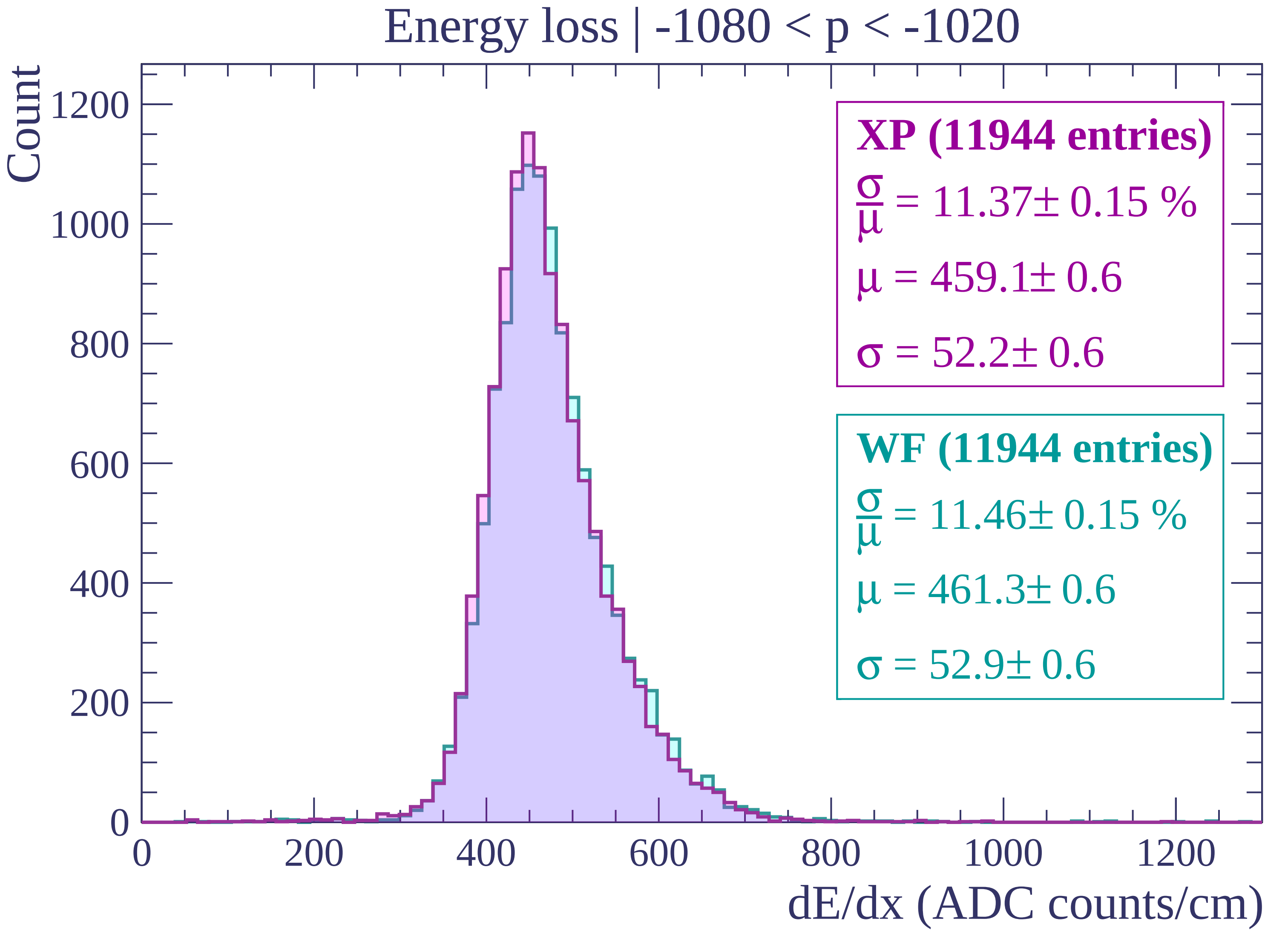


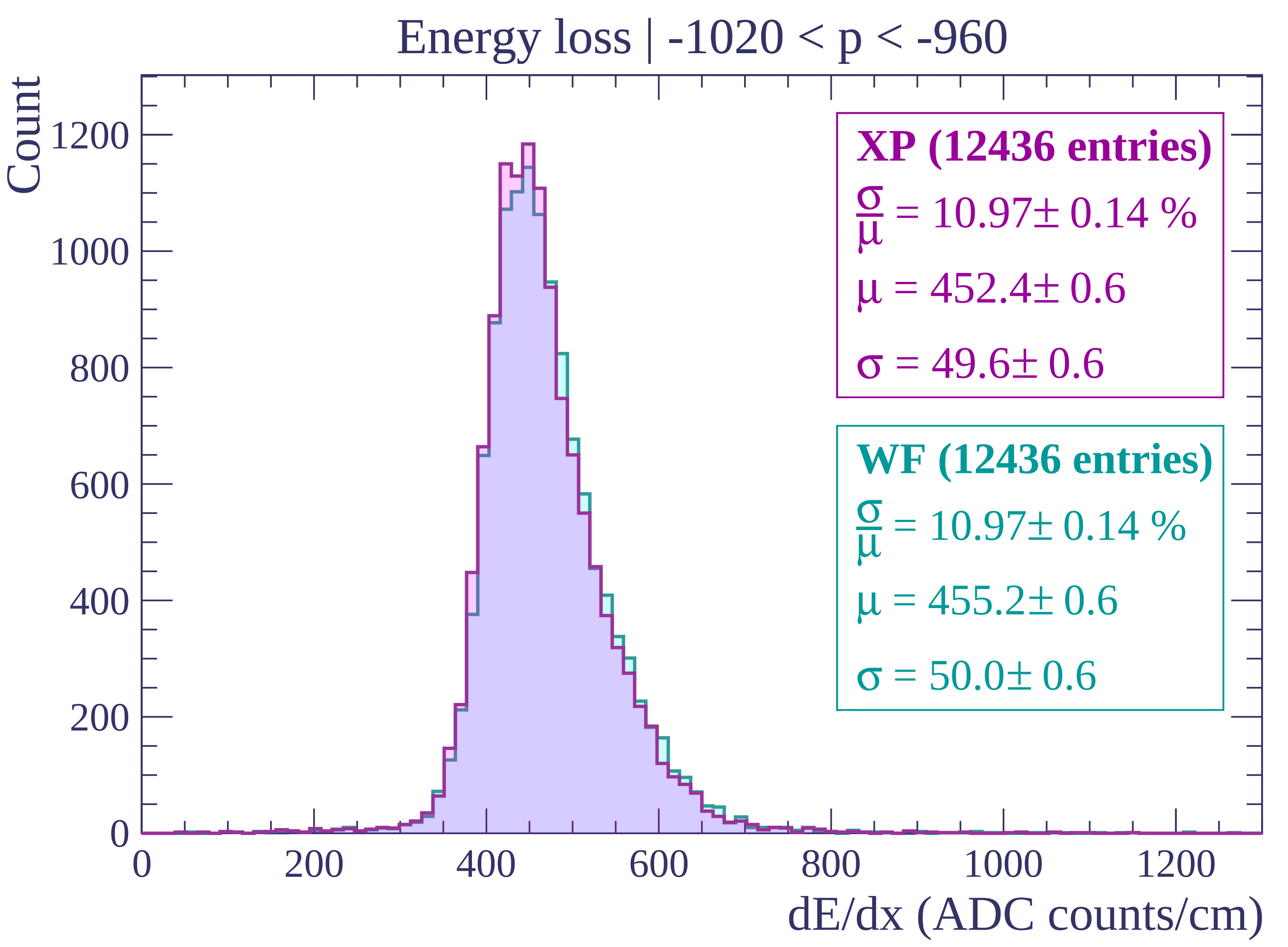
Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10909 entries)
σ/μ = 12.02± 0.16 %
μ = 472.1± 0.7
σ = 56.7± 0.7
WF (10909 entries)
σ/μ = 12.12± 0.16 %
μ = 474.4± 0.7
σ = 57.5± 0.7



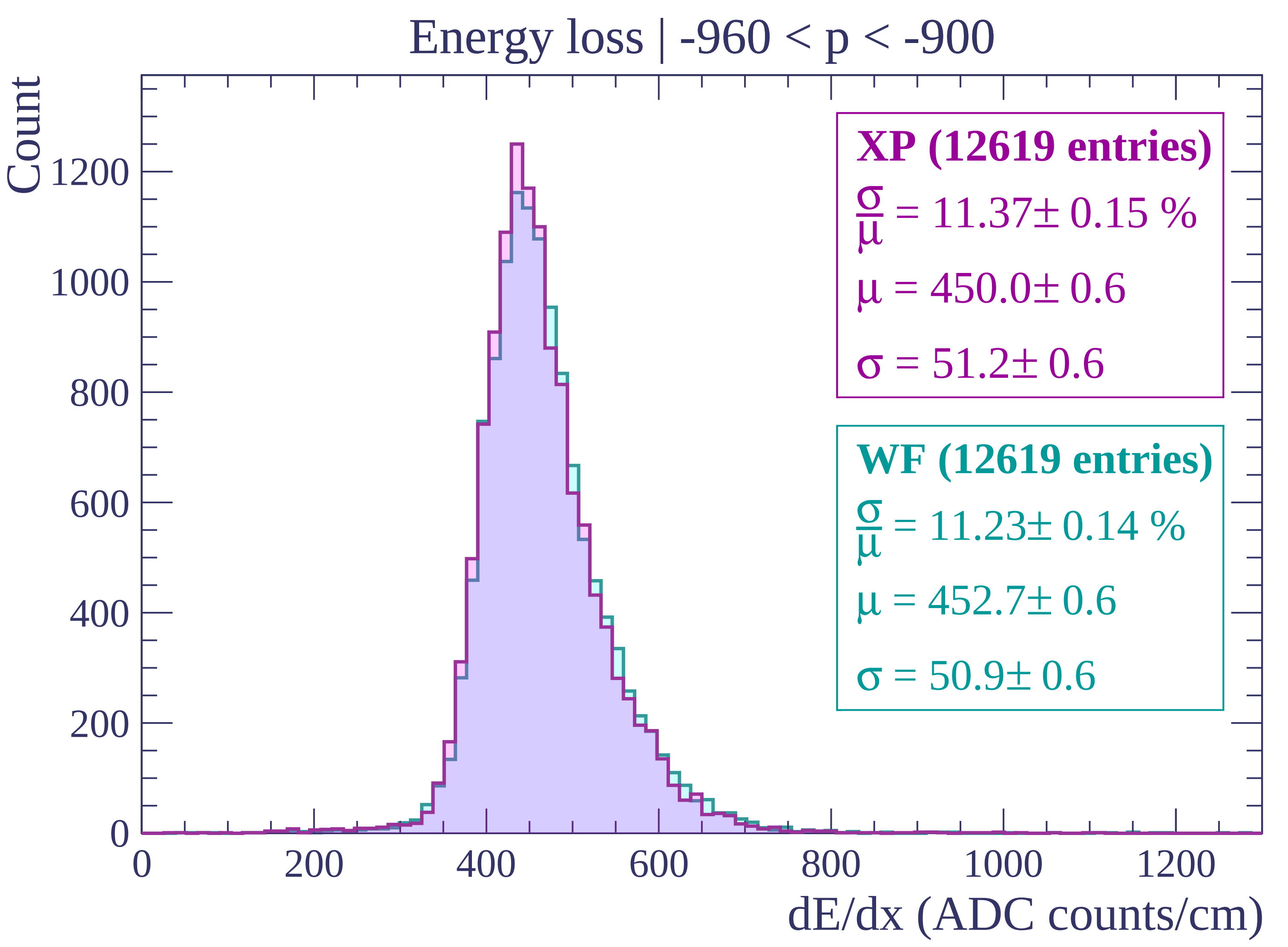
Energy loss | -1200 < p < -1140
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (11039 entries)
$\frac{\sigma}{\mu}$ = 11.84± 0.17 %
μ = 466.3± 0.7
σ = 55.2± 0.7
WF (11039 entries)
$\frac{\sigma}{\mu}$ = 12.00± 0.15 %
μ = 471.4± 0.7
σ = 56.5± 0.6


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11464 entries)
σ/μ = 11.26± 0.14 %
μ = 462.9± 0.6
σ = 52.1± 0.6
WF (11464 entries)
σ/μ = 11.71± 0.15 %
μ = 467.0± 0.7
σ = 54.7± 0.6

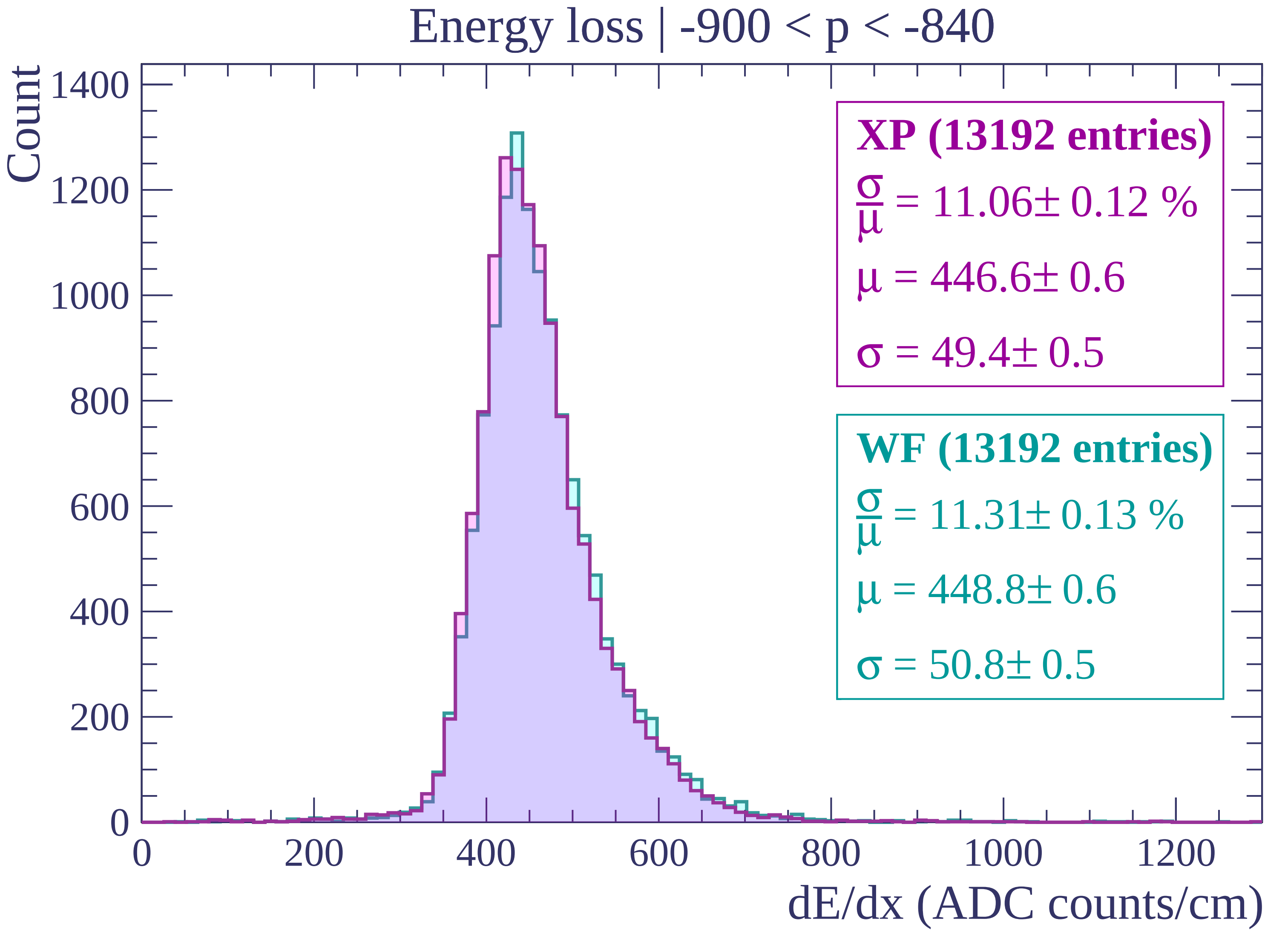

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11944 entries)
$\frac{\sigma}{\mu}$ = 11.37± 0.15 %
μ = 459.1± 0.6
σ = 52.2± 0.6
WF (11944 entries)
$\frac{\sigma}{\mu}$ = 11.46± 0.15 %
μ = 461.3± 0.6
σ = 52.9± 0.6

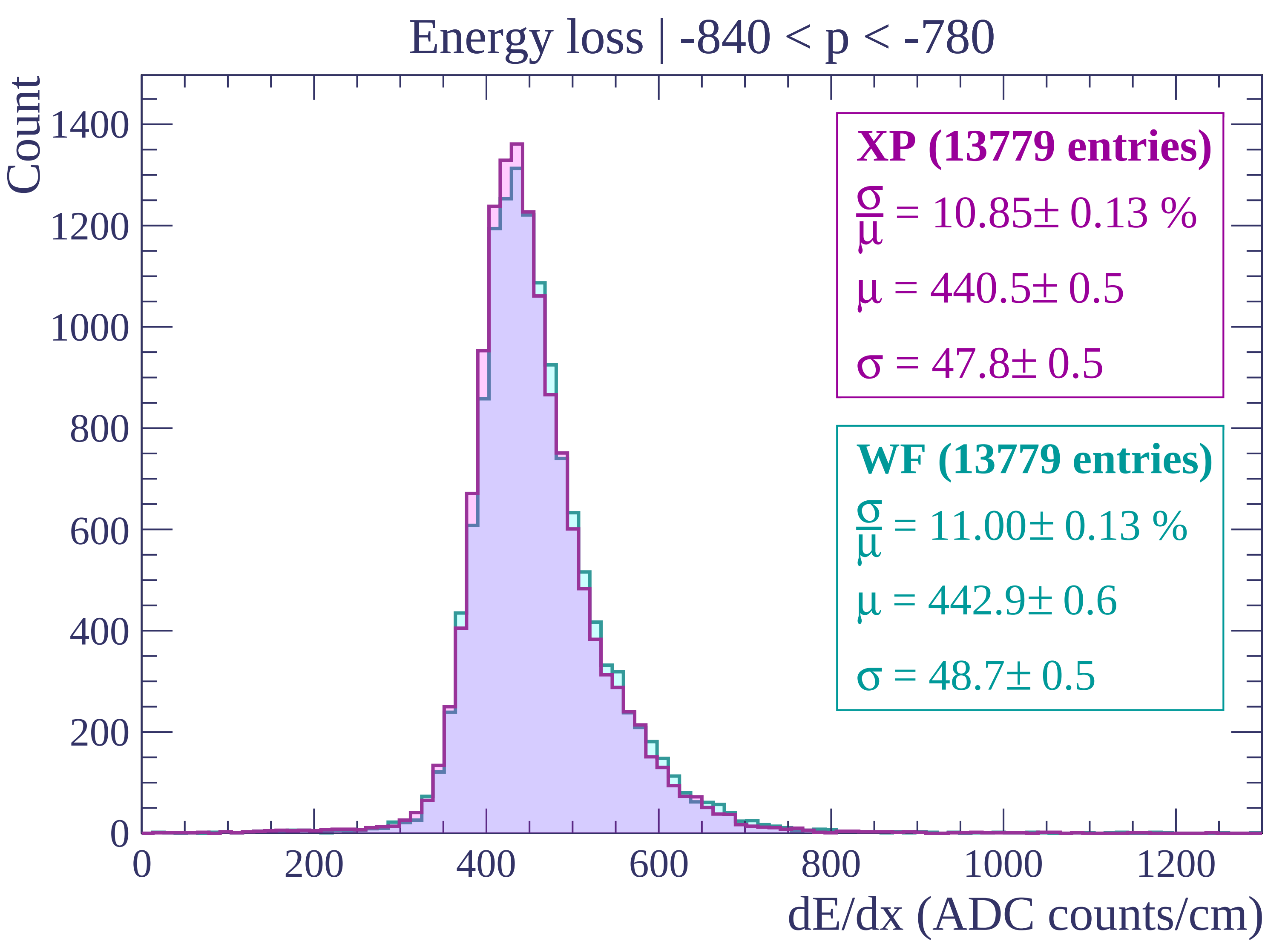


Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12436 entries)
σ/μ = 10.97± 0.14 %
μ = 452.4± 0.6
σ = 49.6± 0.6
WF (12436 entries)
σ/μ = 10.97± 0.14 %
μ = 455.2± 0.6
σ = 50.0± 0.6

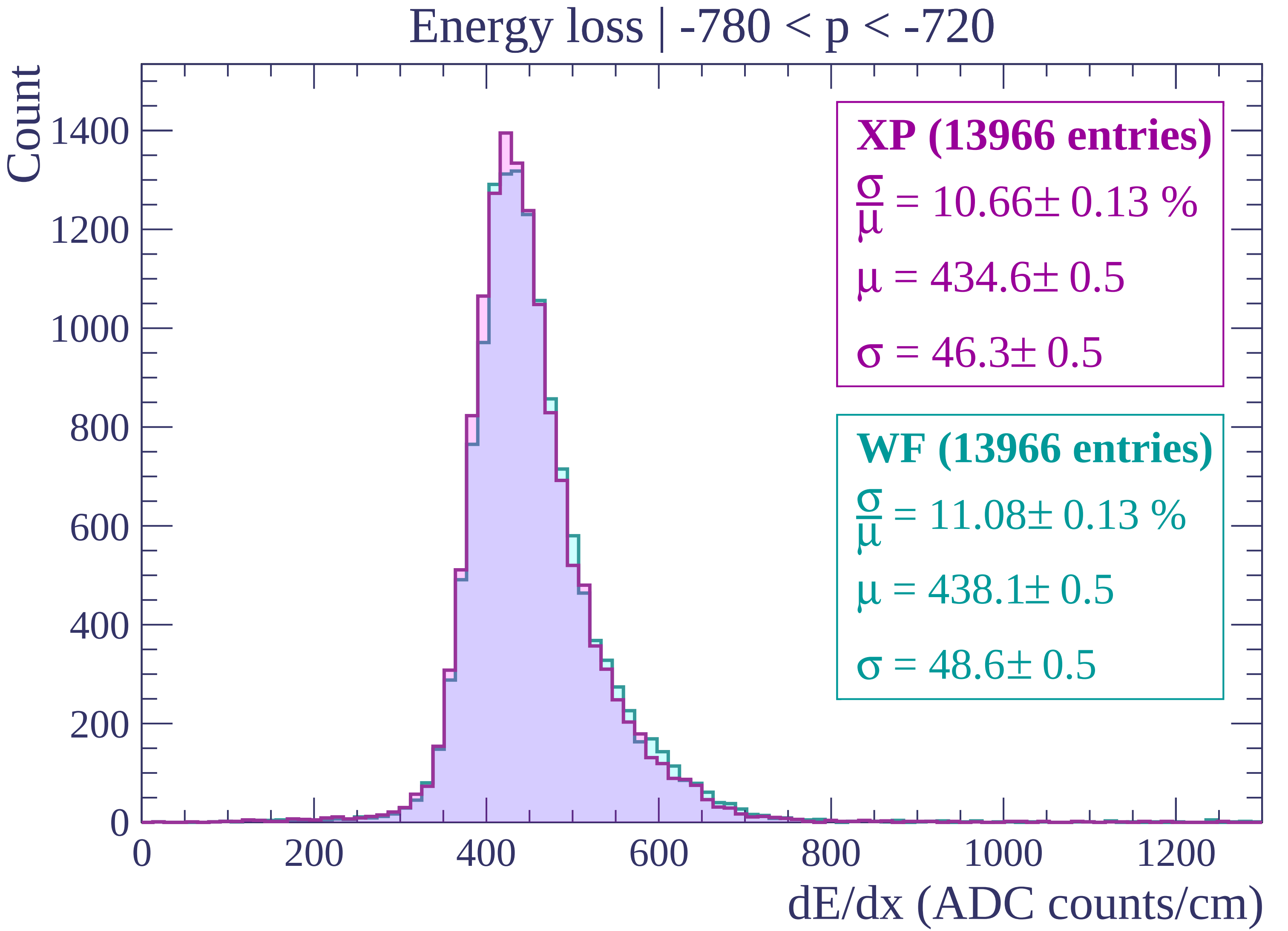


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (12619 entries)
σ/μ = 11.37± 0.15 %
μ = 450.0± 0.6
σ = 51.2± 0.6
WF (12619 entries)
σ/μ = 11.23± 0.14 %
μ = 452.7± 0.6
σ = 50.9± 0.6

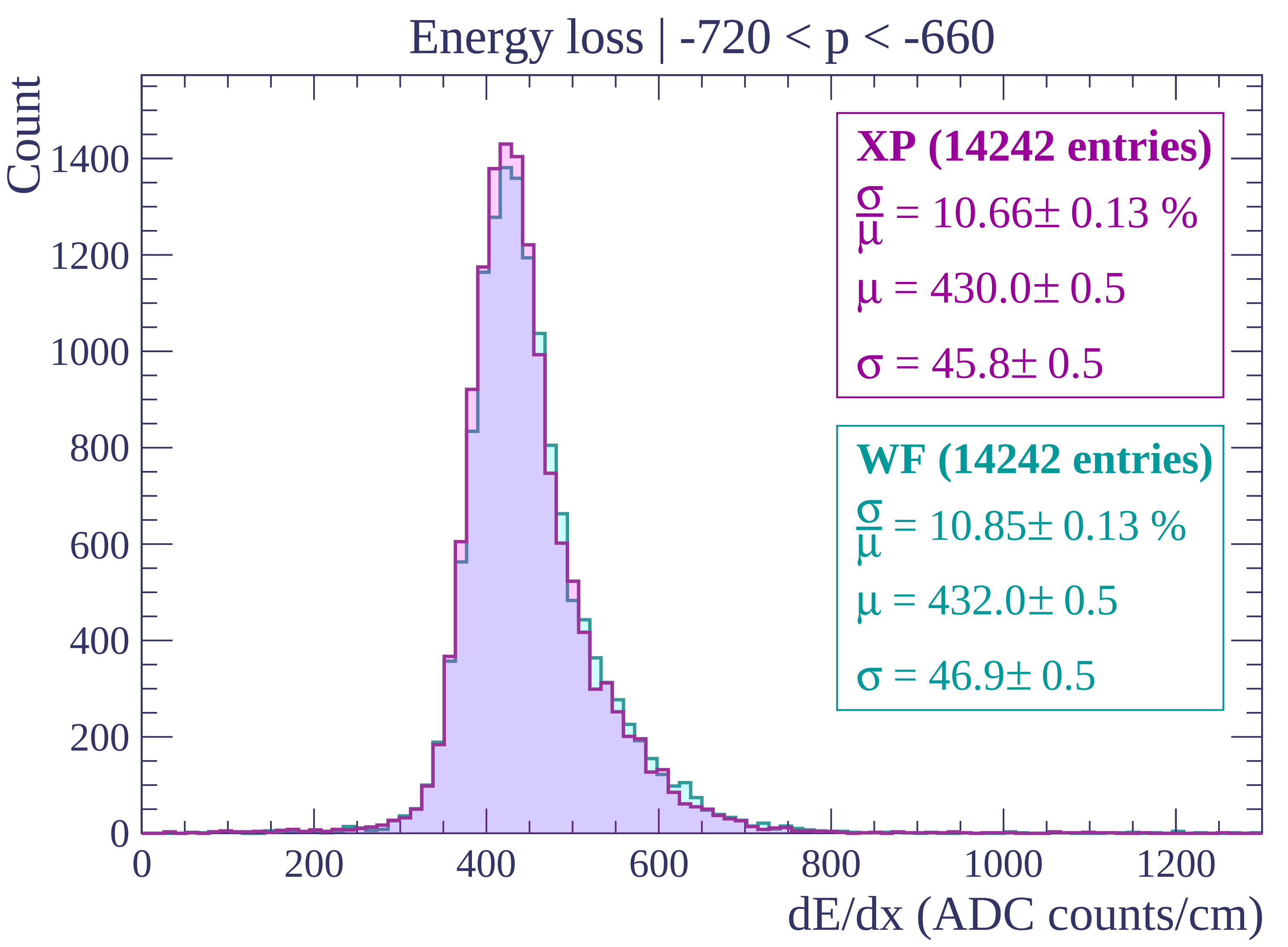

Energy loss | -900 < p < -840
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13192 entries)
σ/μ = 11.06± 0.12 %
μ = 446.6± 0.6
σ = 49.4± 0.5
WF (13192 entries)
σ/μ = 11.31± 0.13 %
μ = 448.8± 0.6
σ = 50.8± 0.5

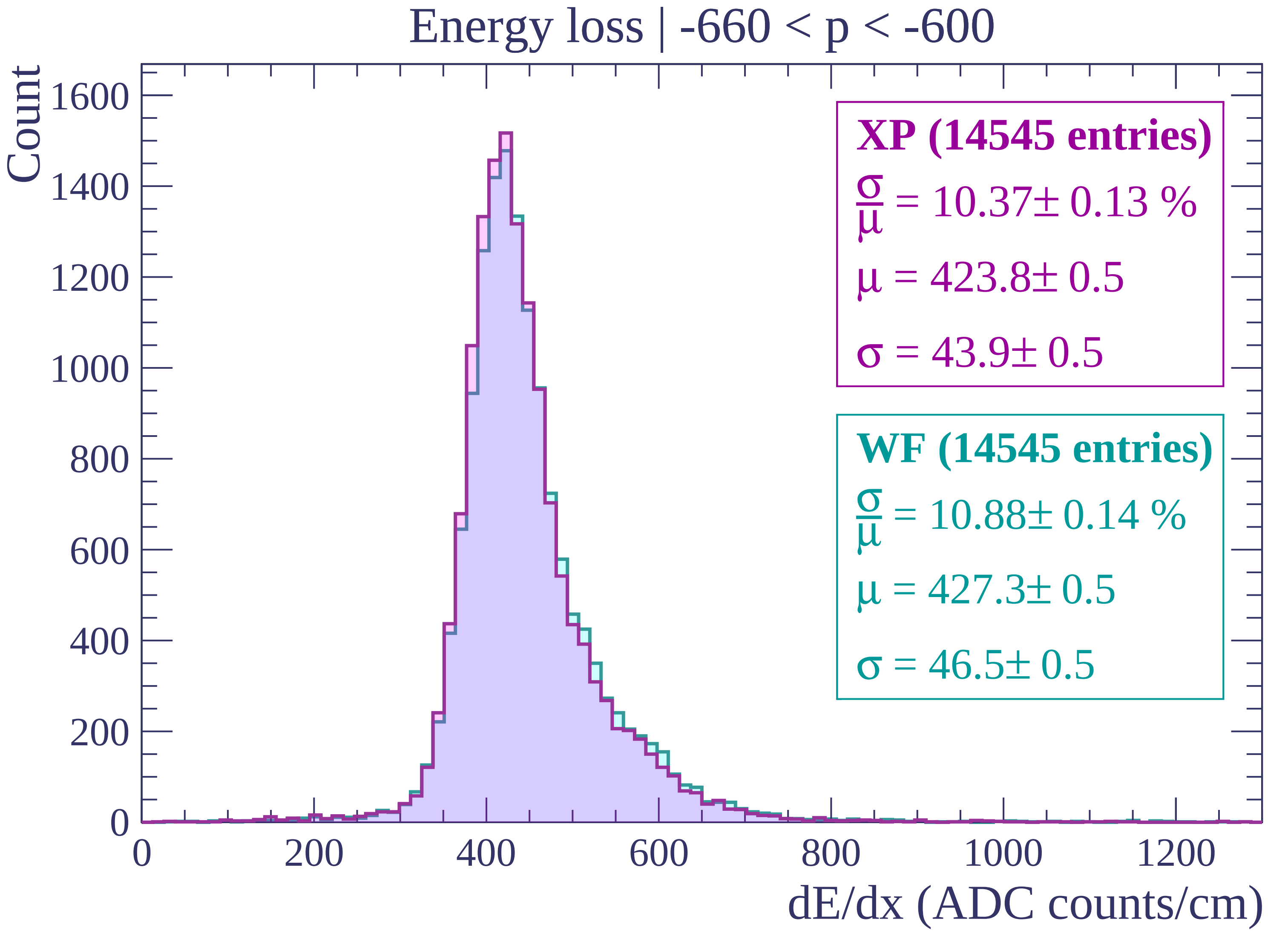


Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
XP (13779 entries)
σ/μ = 10.85± 0.13 %
μ = 440.5± 0.5
σ = 47.8± 0.5
WF (13779 entries)
σ/μ = 11.00± 0.13 %
μ = 442.9± 0.6
σ = 48.7± 0.5
0
200
400
600
800
1000
1200
1400

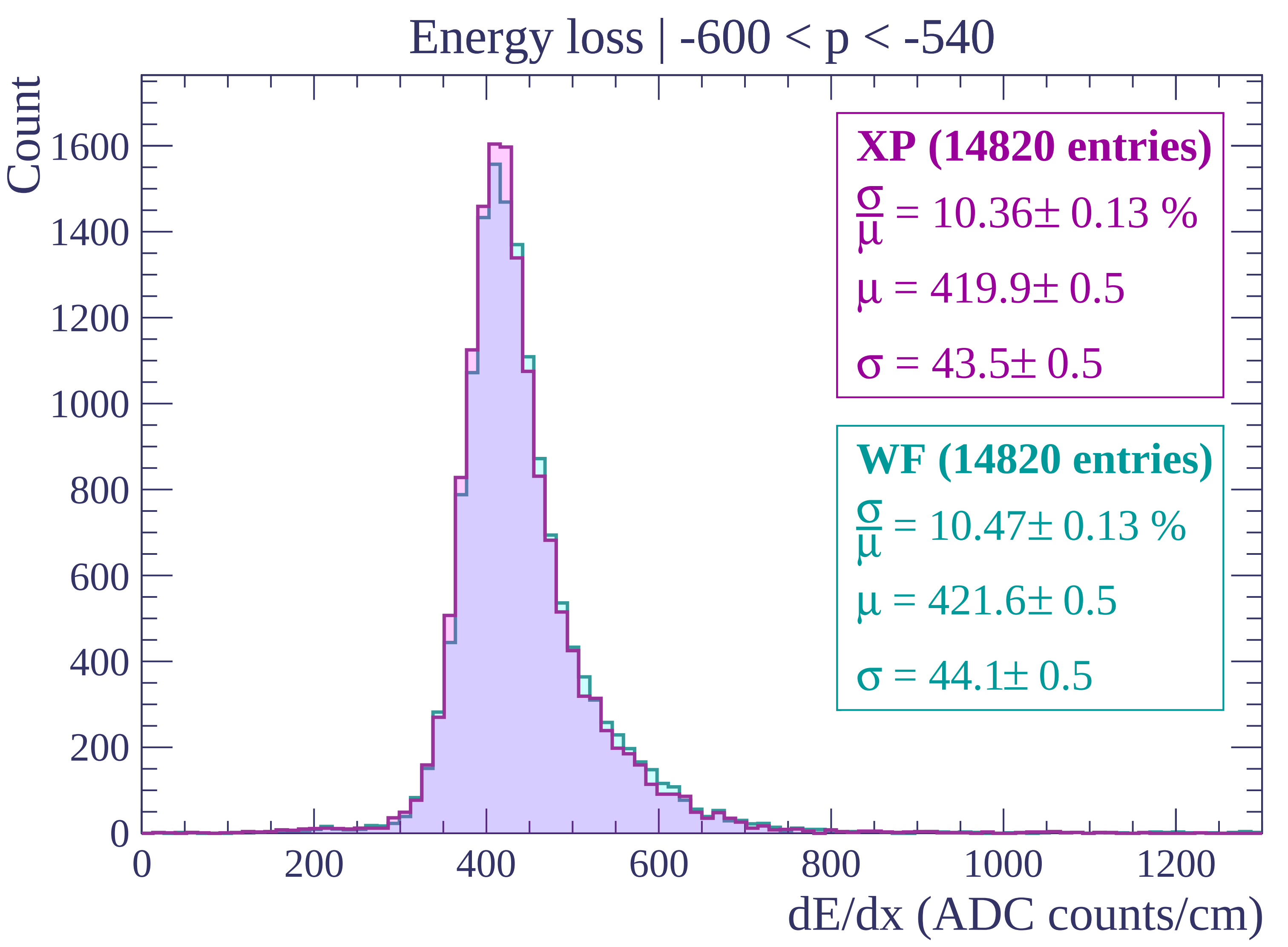


Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (13966 entries)
$\frac{\sigma}{\mu}$ = 10.66± 0.13 %
μ = 434.6± 0.5
σ = 46.3± 0.5
WF (13966 entries)
$\frac{\sigma}{\mu}$ = 11.08± 0.13 %
μ = 438.1± 0.5
σ = 48.6± 0.5

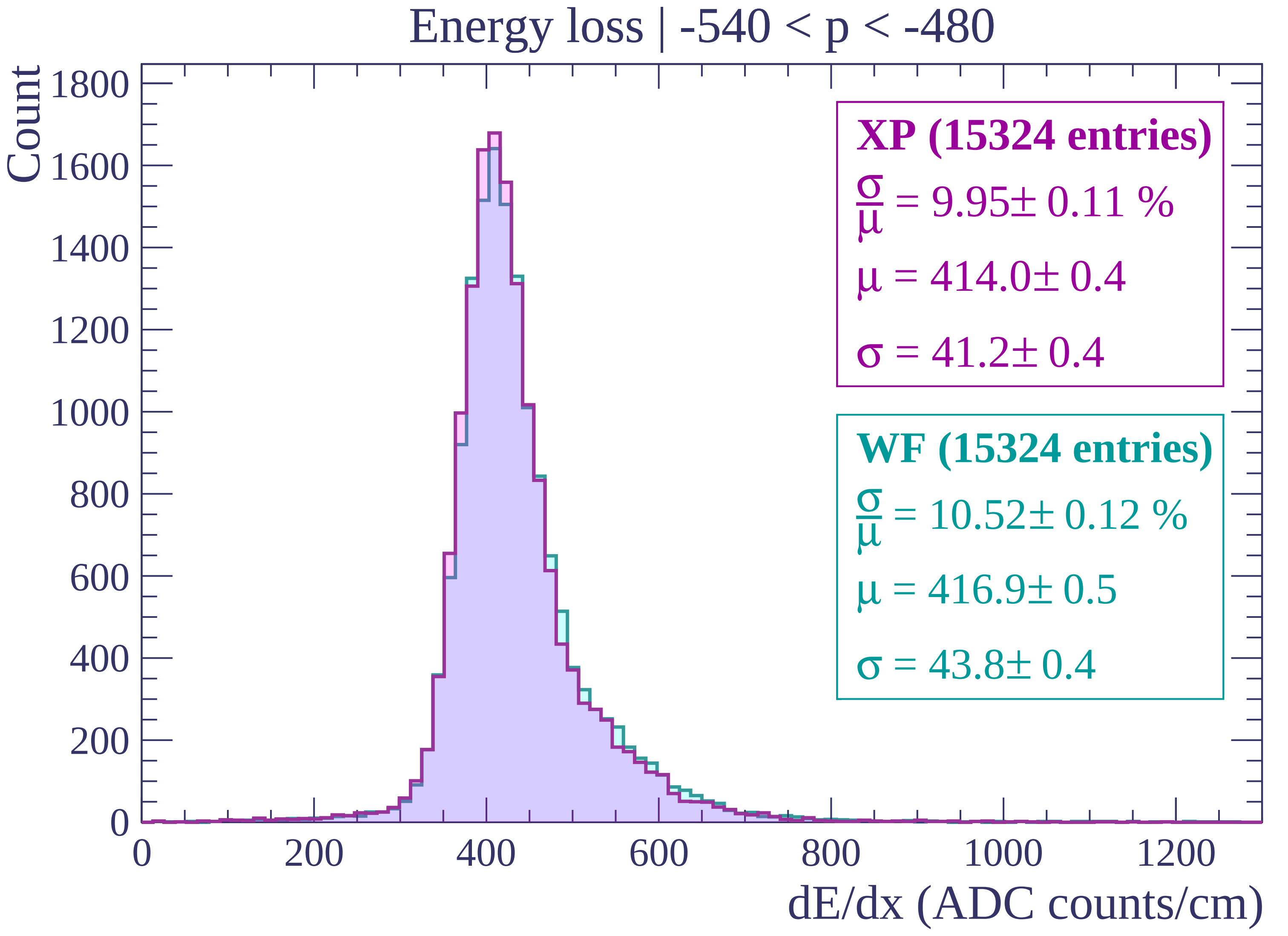

Energy loss | -720 < p < -660
Count
XP (14242 entries)
σ/μ = 10.66± 0.13 %
μ = 430.0± 0.5
σ = 45.8± 0.5
WF (14242 entries)
σ/μ = 10.85± 0.13 %
μ = 432.0± 0.5
σ = 46.9± 0.5
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



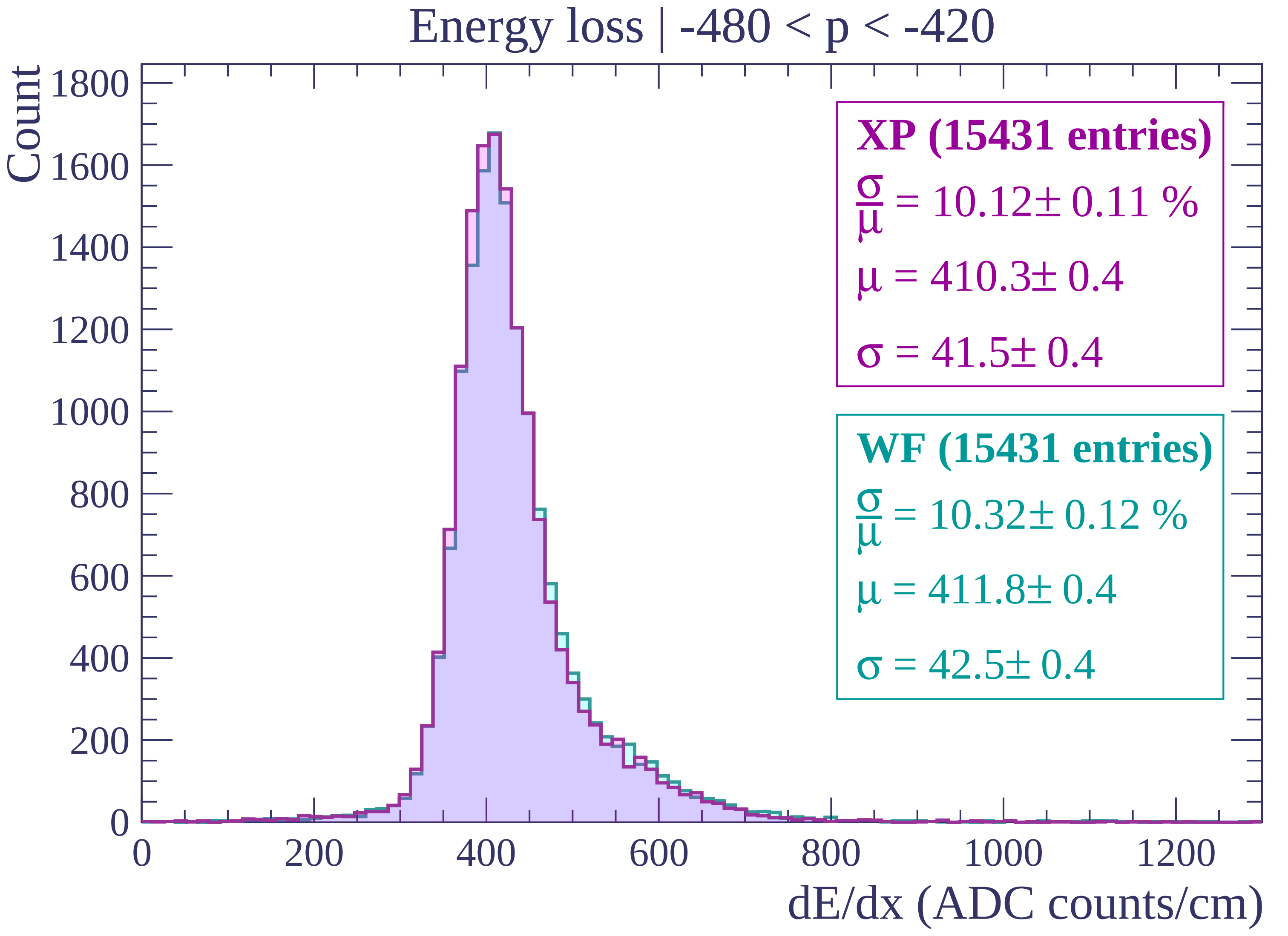
Energy loss | -660 < p < -600
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14545 entries)
σ/μ = 10.37± 0.13 %
μ = 423.8± 0.5
σ = 43.9± 0.5
WF (14545 entries)
σ/μ = 10.88± 0.14 %
μ = 427.3± 0.5
σ = 46.5± 0.5



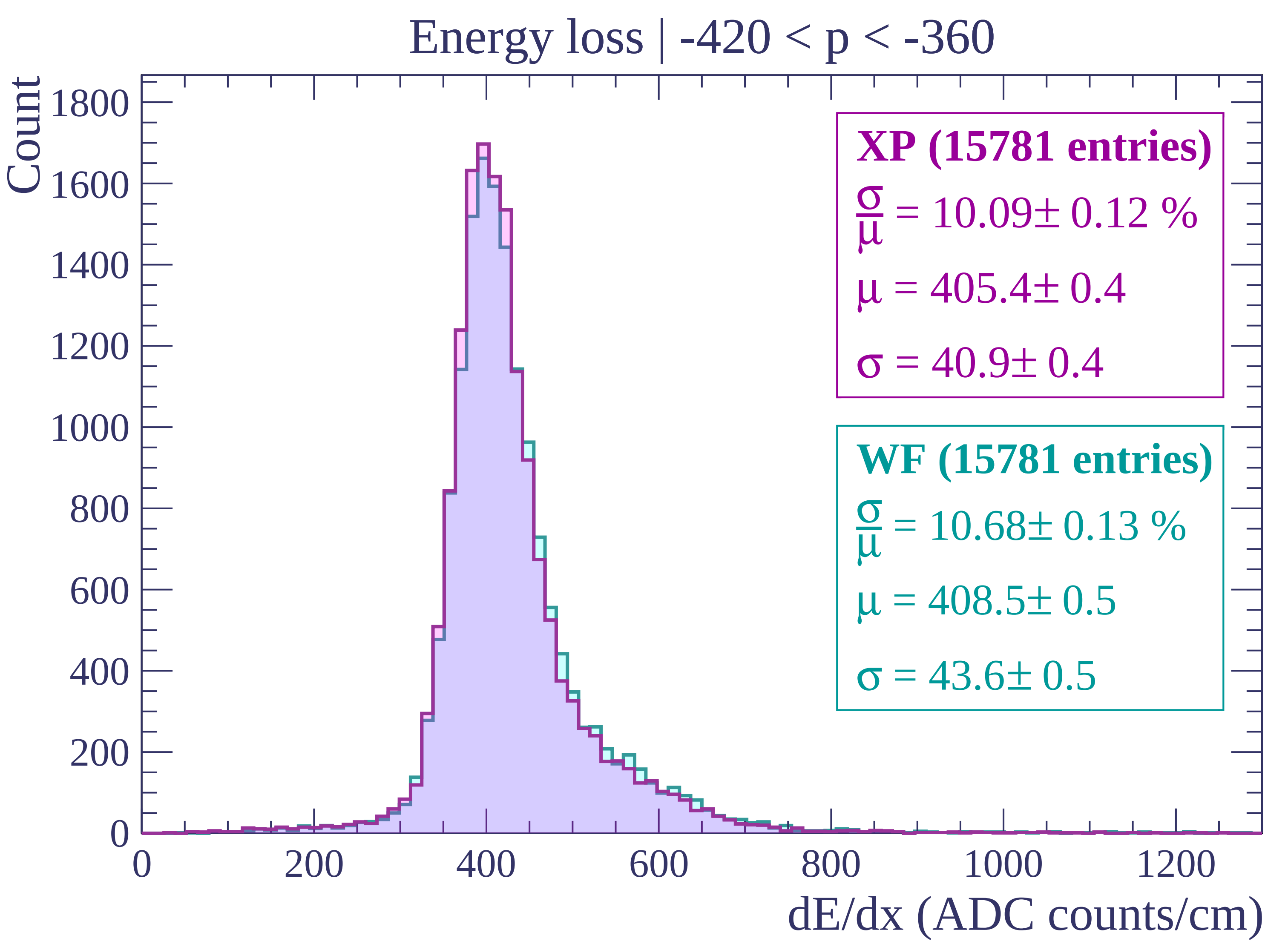
Energy loss | -600 < p < -540
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (14820 entries)
$\frac{\sigma}{\mu}$ = 10.36± 0.13 %
μ = 419.9± 0.5
σ = 43.5± 0.5
WF (14820 entries)
$\frac{\sigma}{\mu}$ = 10.47± 0.13 %
μ = 421.6± 0.5
σ = 44.1± 0.5



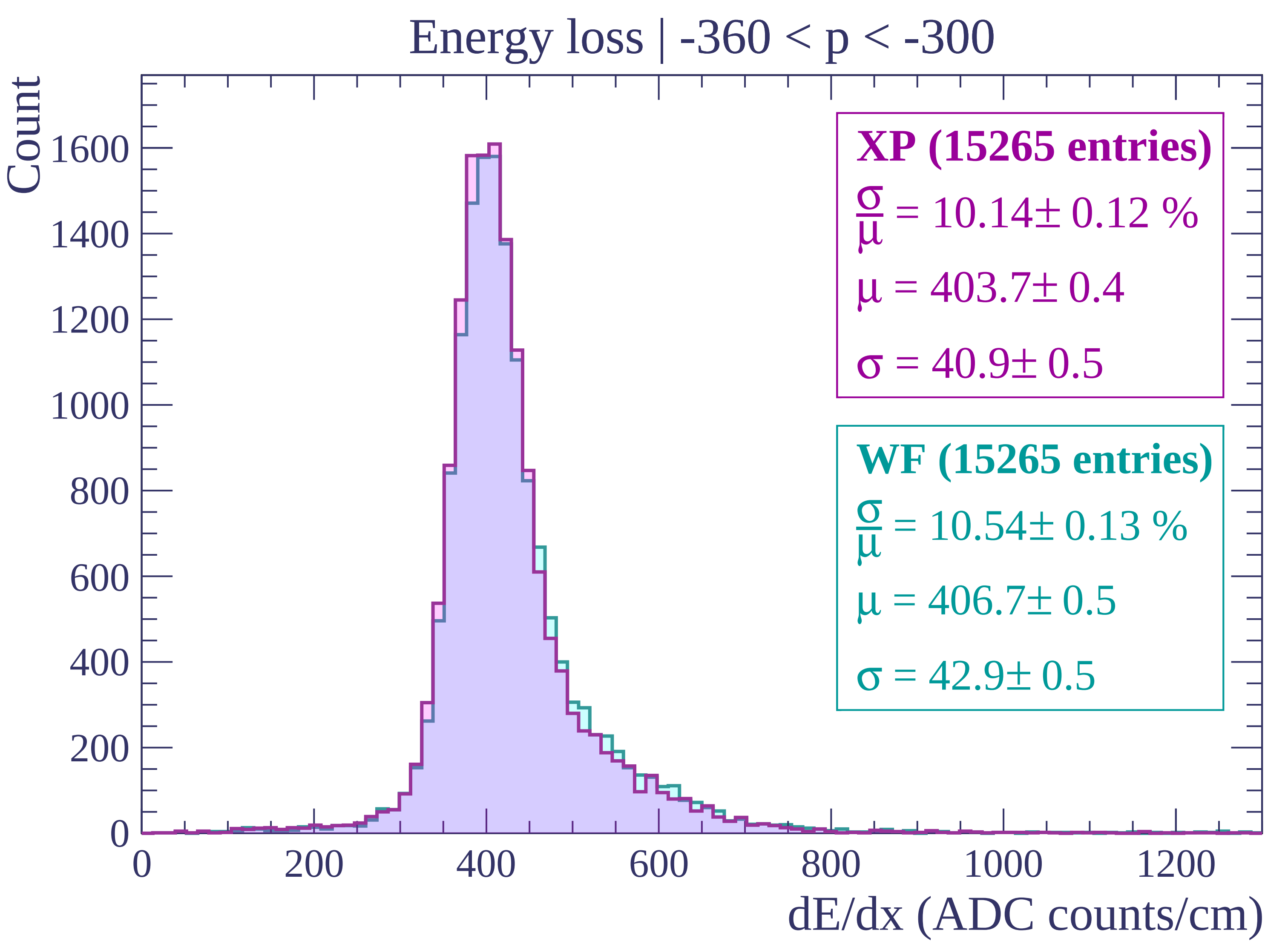
Energy loss | -540 < p < -480
Count
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (15324 entries)
σ/μ = 9.95± 0.11 %
μ = 414.0± 0.4
σ = 41.2± 0.4
WF (15324 entries)
σ/μ = 10.52± 0.12 %
μ = 416.9± 0.5
σ = 43.8± 0.4



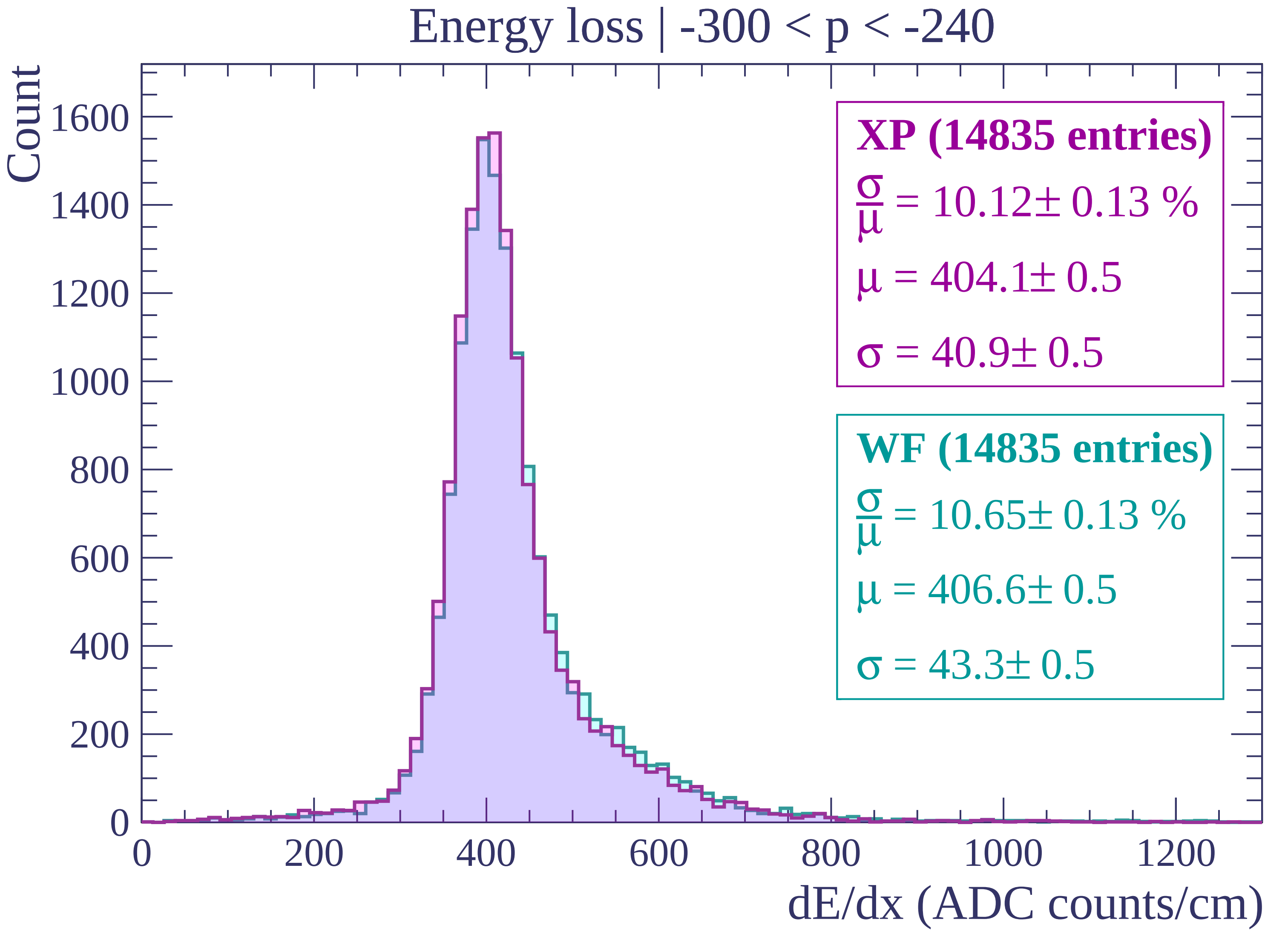
Energy loss | -480 < p < -420
Count
1800
1600
1400
1200
1000
800
600
400
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (15431 entries)
$\frac{\sigma}{\mu}$ = 10.12± 0.11 %
μ = 410.3± 0.4
σ = 41.5± 0.4
WF (15431 entries)
$\frac{\sigma}{\mu}$ = 10.32± 0.12 %
μ = 411.8± 0.4
σ = 42.5± 0.4

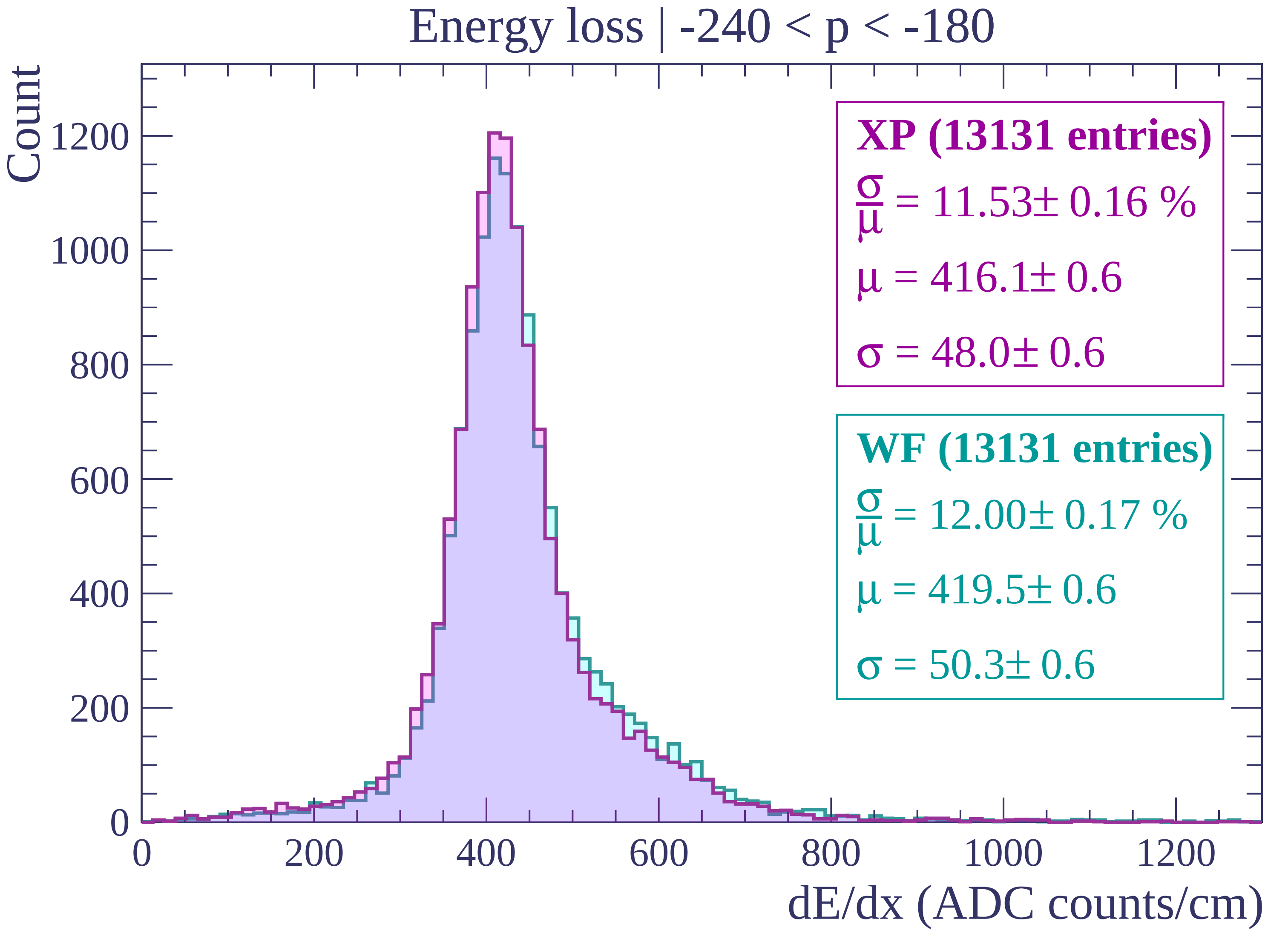


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP (15781 entries)
$\frac{\sigma}{\mu}$ = 10.09± 0.12 %
μ = 405.4± 0.4
σ = 40.9± 0.4
WF (15781 entries)
$\frac{\sigma}{\mu}$ = 10.68± 0.13 %
μ = 408.5± 0.5
σ = 43.6± 0.5

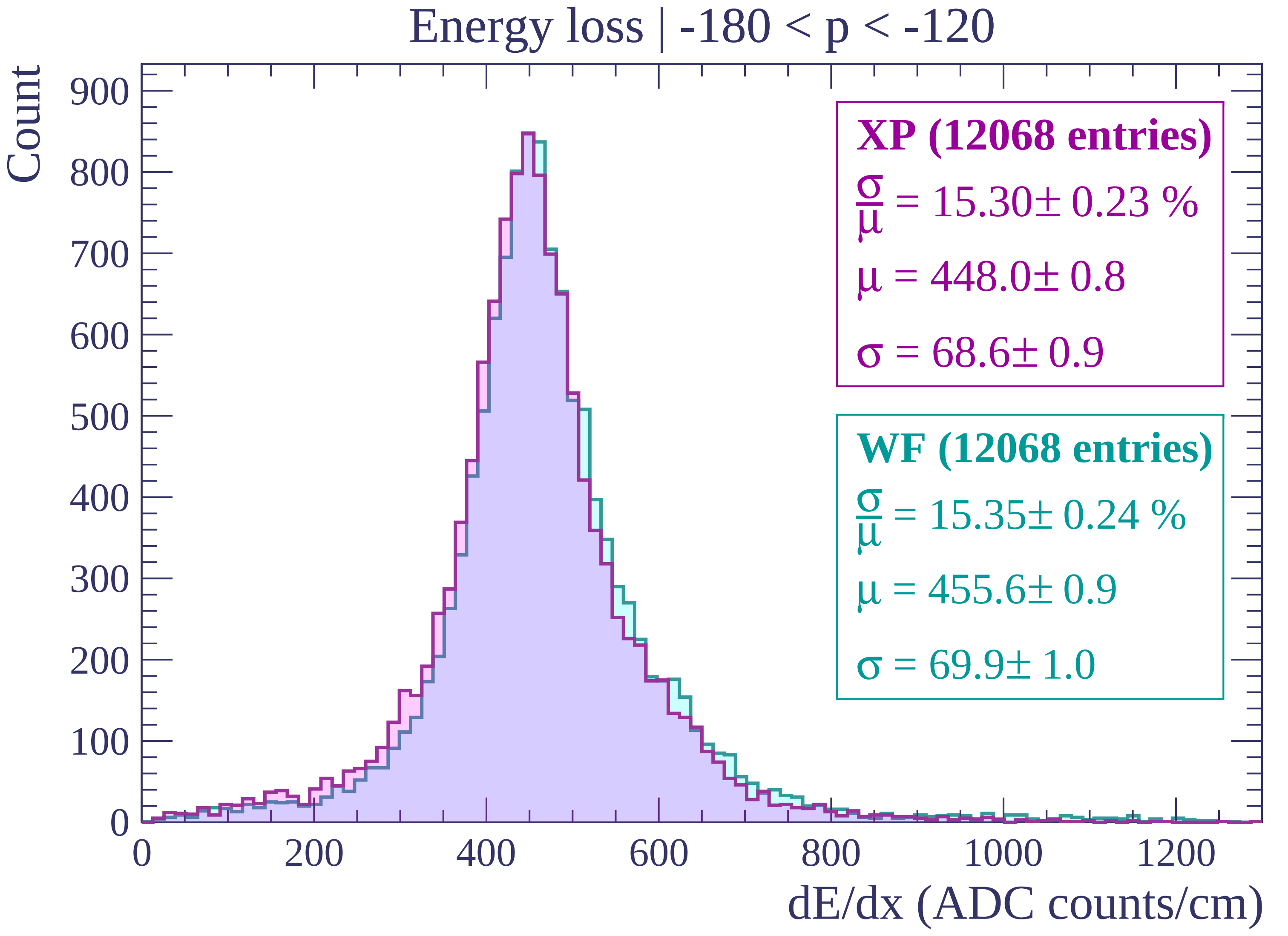


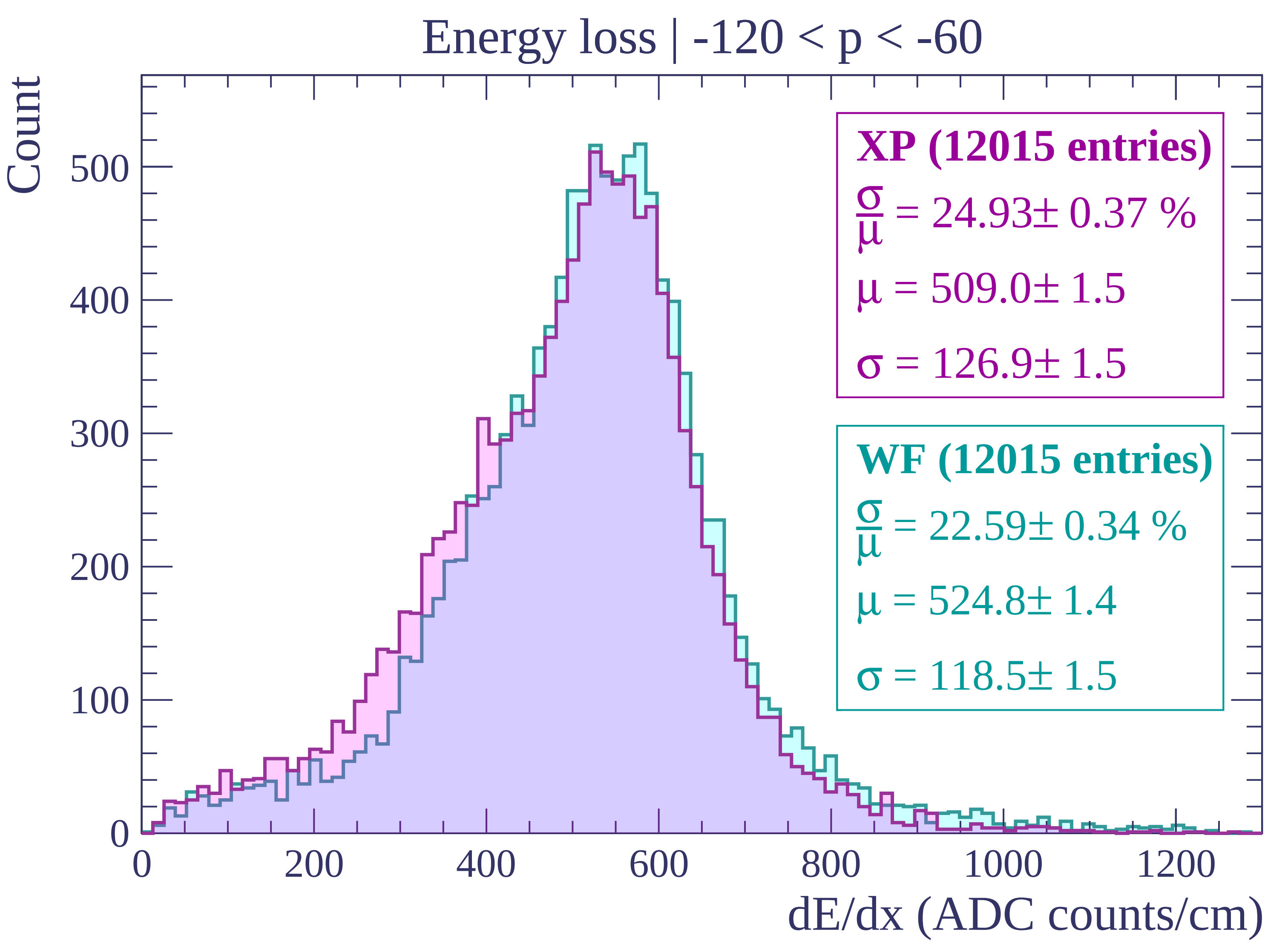
Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP (15265 entries)
$\frac{\sigma}{\mu}$ = 10.14± 0.12 %
μ = 403.7± 0.4
σ = 40.9± 0.5
WF (15265 entries)
$\frac{\sigma}{\mu}$ = 10.54± 0.13 %
μ = 406.7± 0.5
σ = 42.9± 0.5

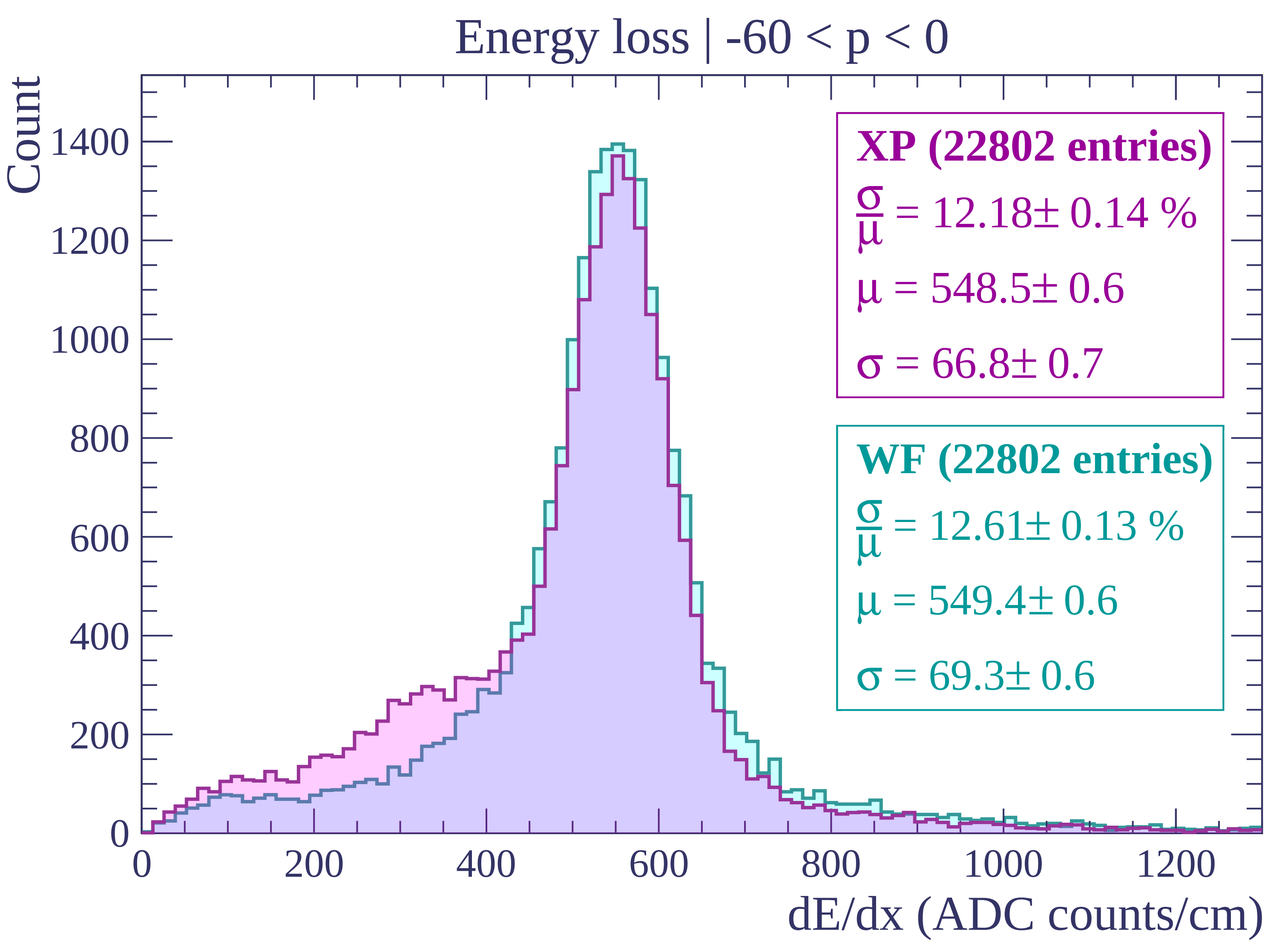

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
XP (14835 entries)
$\frac{\sigma}{\mu}$ = 10.12± 0.13 %
μ = 404.1± 0.5
σ = 40.9± 0.5
WF (14835 entries)
$\frac{\sigma}{\mu}$ = 10.65± 0.13 %
μ = 406.6± 0.5
σ = 43.3± 0.5



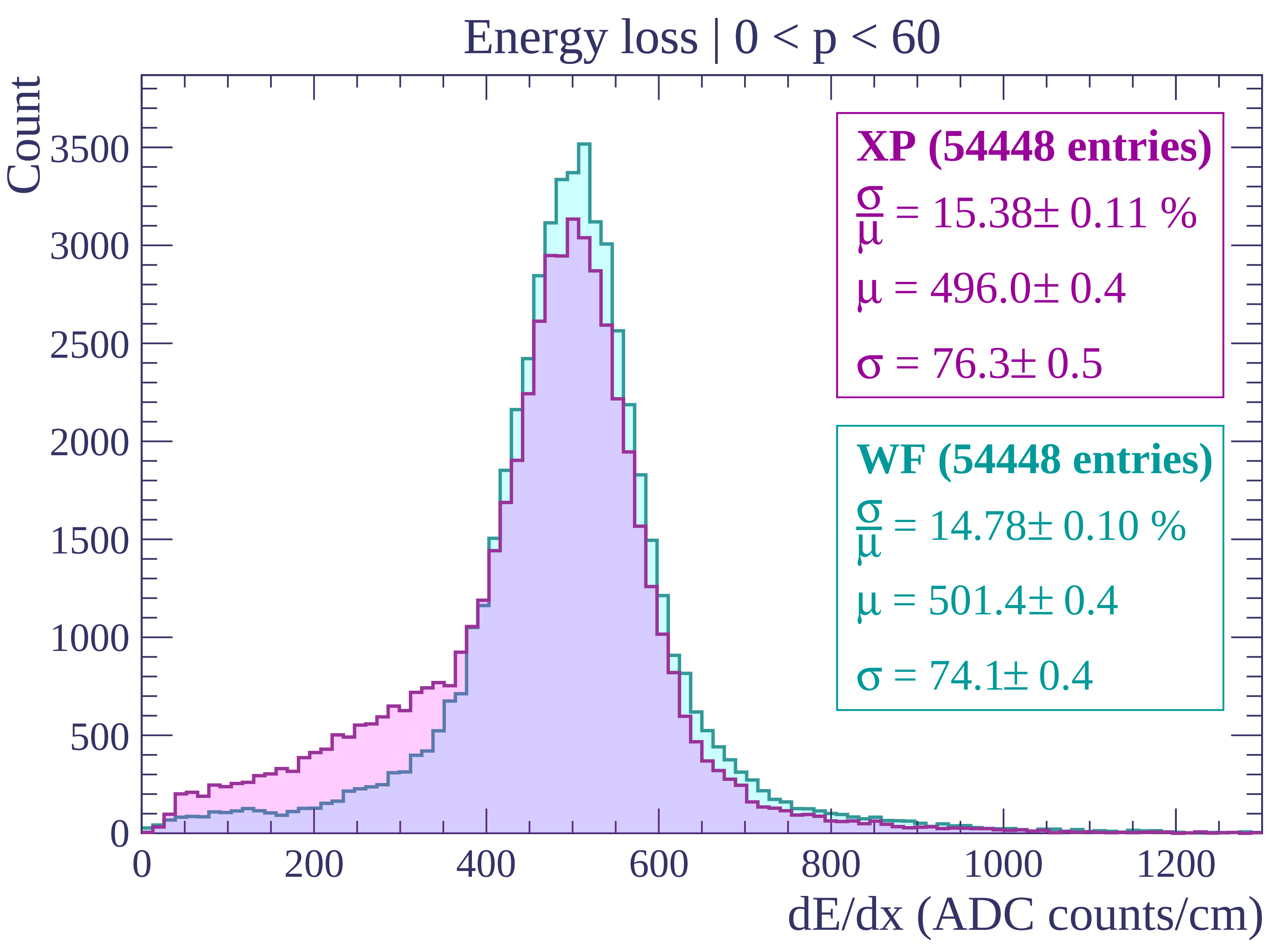
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (13131 entries)
σ/μ = 11.53± 0.16 %
μ = 416.1± 0.6
σ = 48.0± 0.6
WF (13131 entries)
σ/μ = 12.00± 0.17 %
μ = 419.5± 0.6
σ = 50.3± 0.6


Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
XP (12068 entries)
$\frac{\sigma}{\mu}$ = 15.30± 0.23 %
μ = 448.0± 0.8
σ = 68.6± 0.9
WF (12068 entries)
$\frac{\sigma}{\mu}$ = 15.35± 0.24 %
μ = 455.6± 0.9
σ = 69.9± 1.0
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

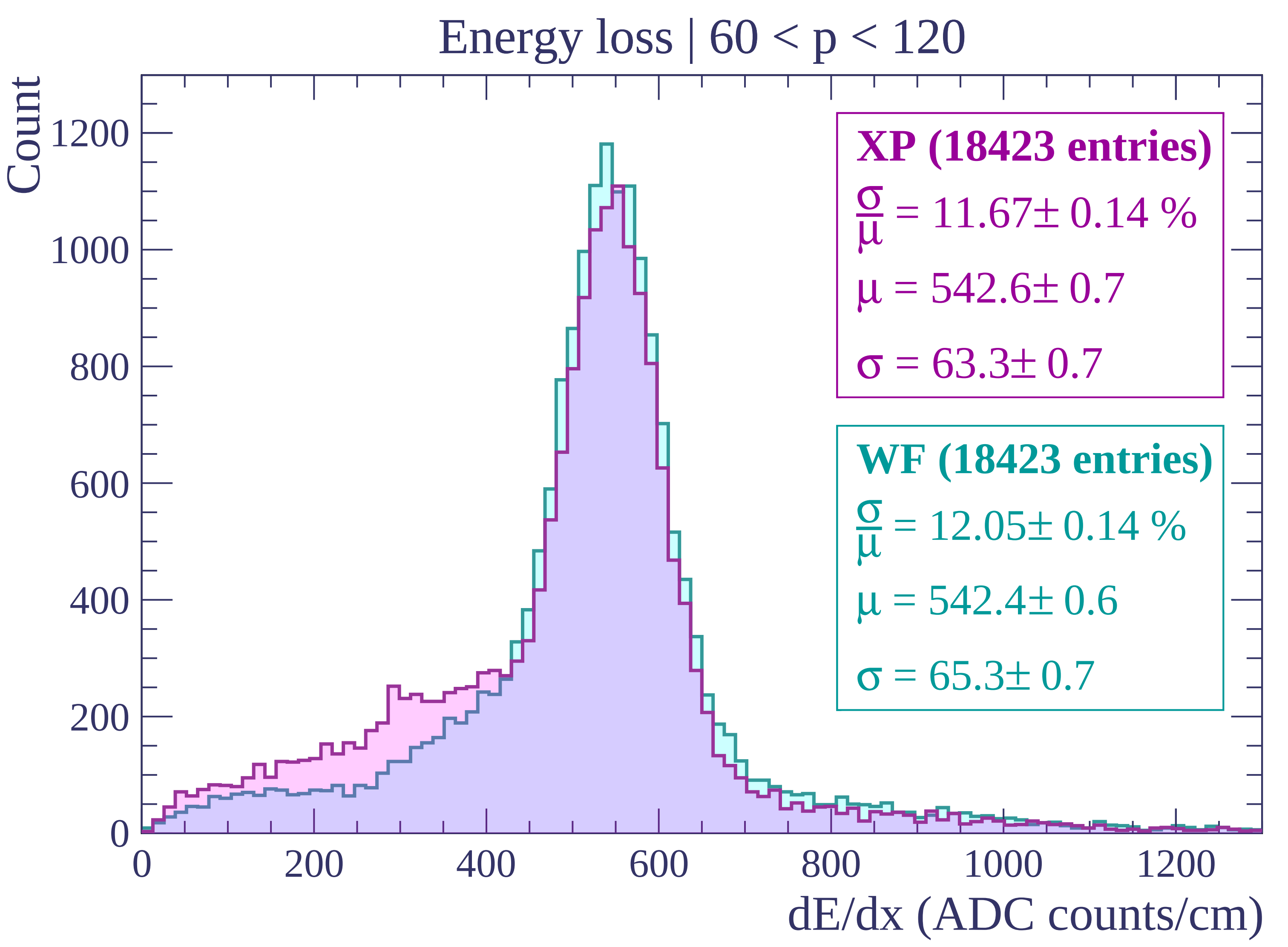

Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (12015 entries)
σ/μ = 24.93± 0.37 %
μ = 509.0± 1.5
σ = 126.9± 1.5
WF (12015 entries)
σ/μ = 22.59± 0.34 %
μ = 524.8± 1.4
σ = 118.5± 1.5



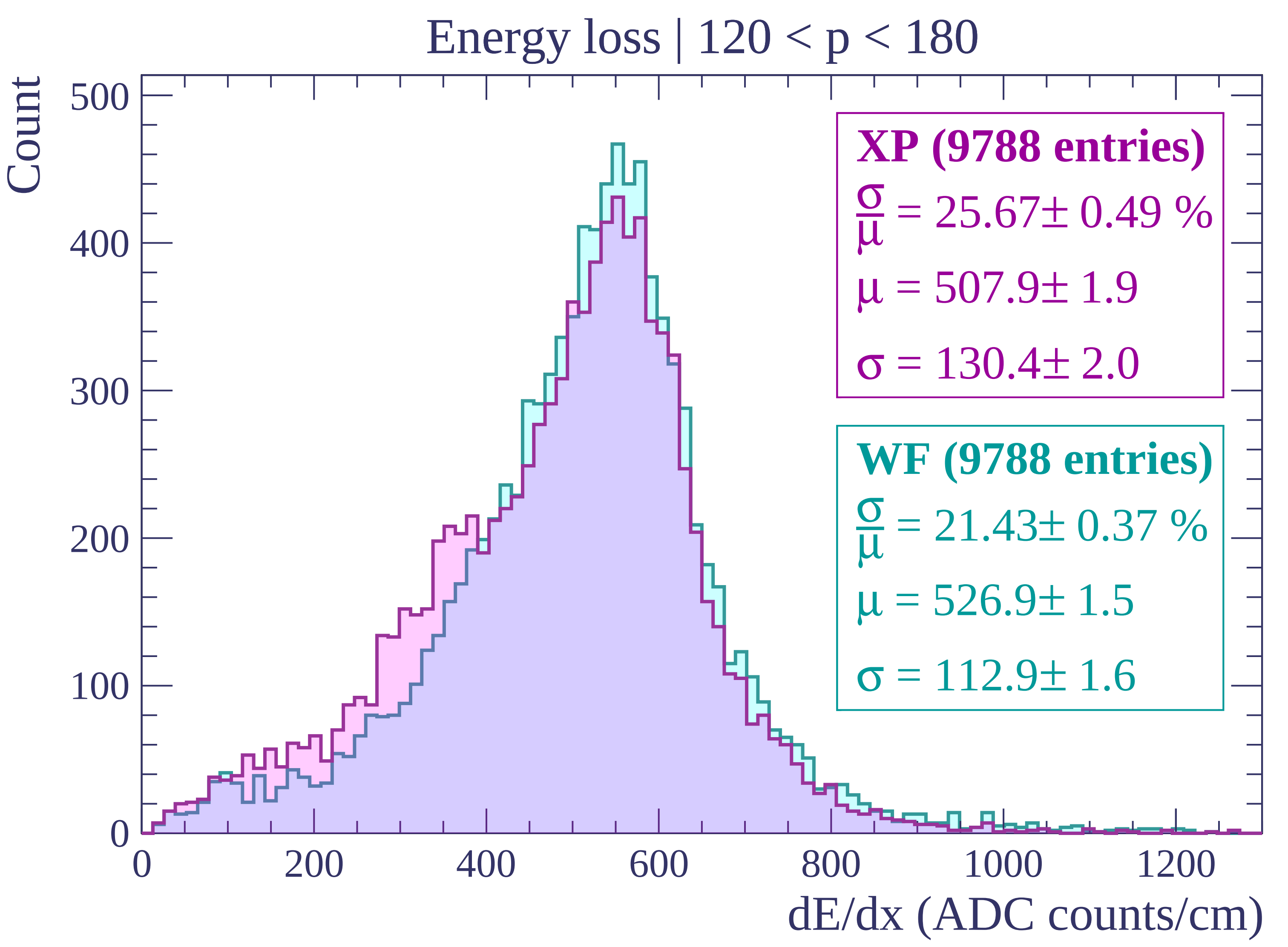
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP (22802 entries)
σ/μ = 12.18± 0.14 %
μ = 548.5± 0.6
σ = 66.8± 0.7
WF (22802 entries)
σ/μ = 12.61± 0.13 %
μ = 549.4± 0.6
σ = 69.3± 0.6

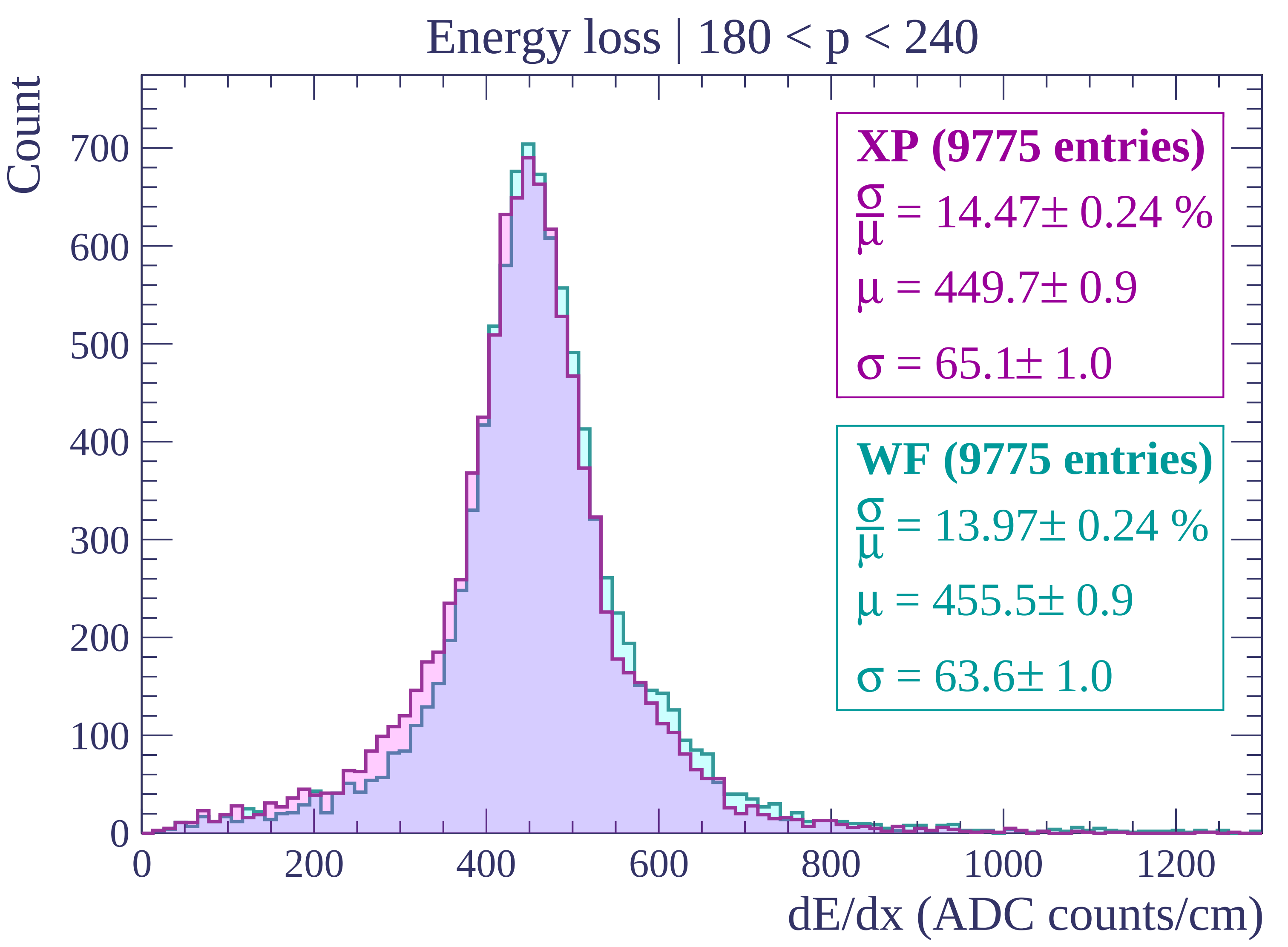


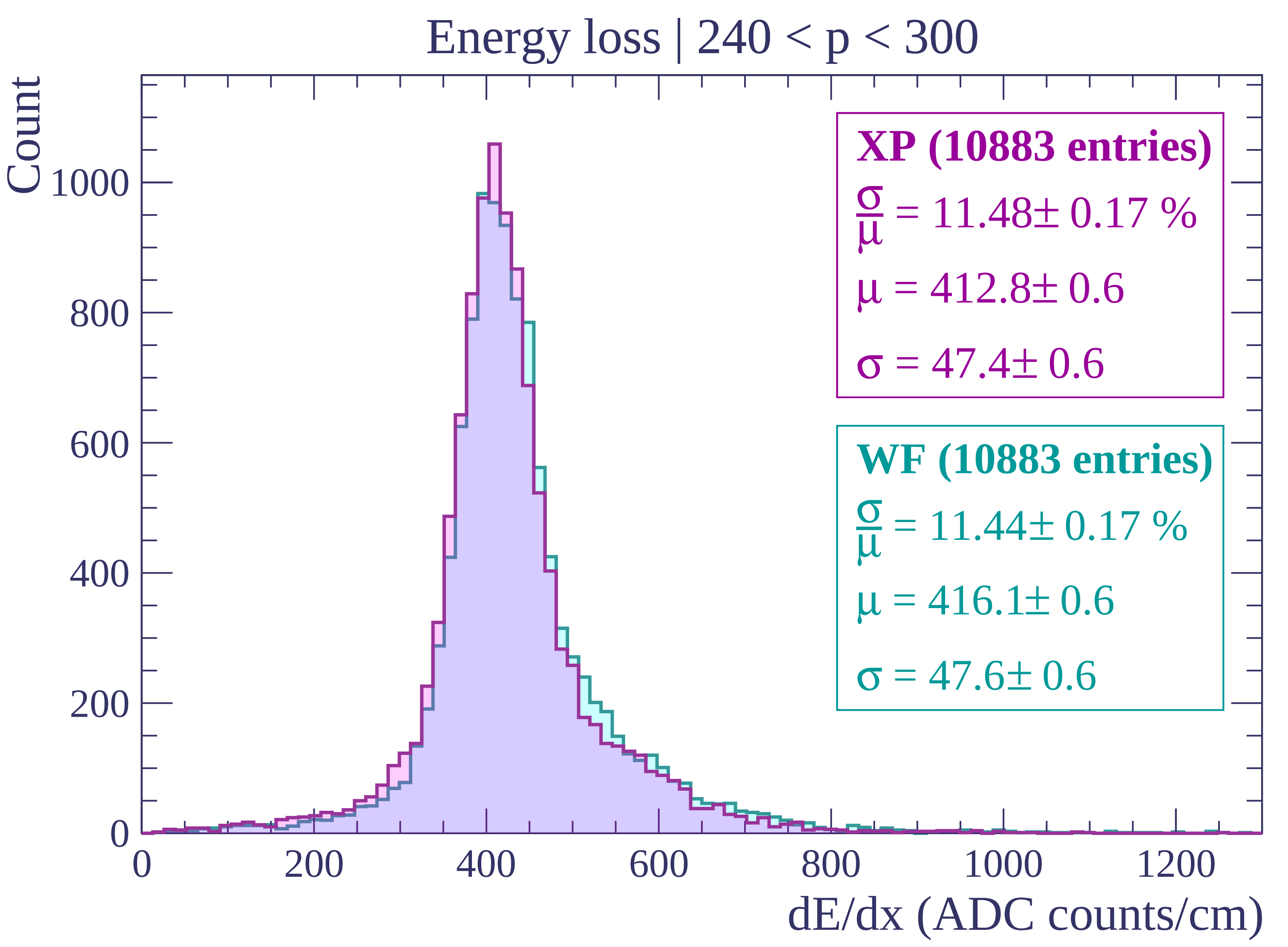
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP (54448 entries)
$\frac{\sigma}{\mu}$ = 15.38± 0.11 %
μ = 496.0± 0.4
σ = 76.3± 0.5
WF (54448 entries)
$\frac{\sigma}{\mu}$ = 14.78± 0.10 %
μ = 501.4± 0.4
σ = 74.1± 0.4



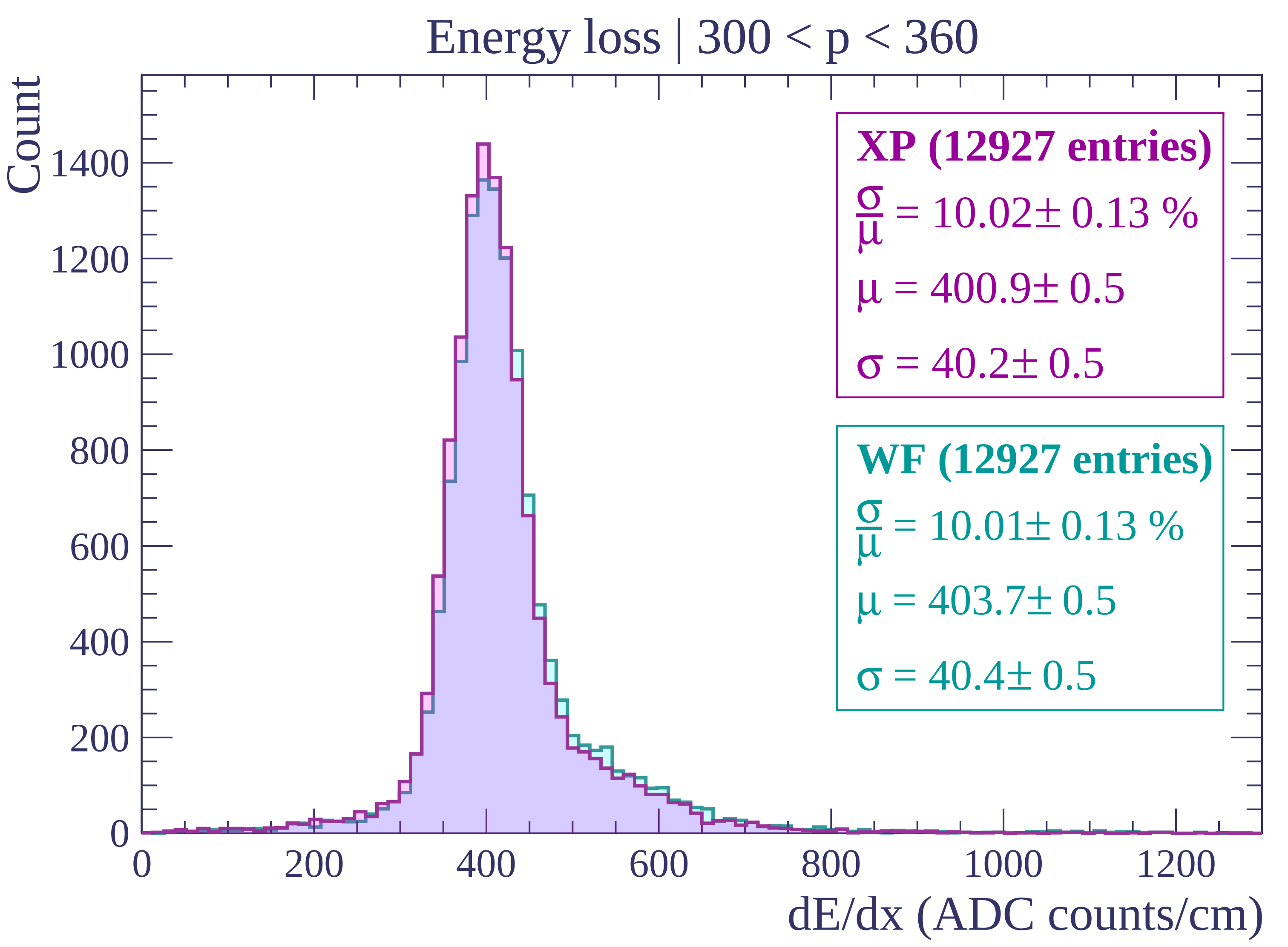
Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP (18423 entries)
$\frac{\sigma}{\mu}$ = 11.67± 0.14 %
μ = 542.6± 0.7
σ = 63.3± 0.7
WF (18423 entries)
$\frac{\sigma}{\mu}$ = 12.05± 0.14 %
μ = 542.4± 0.6
σ = 65.3± 0.7
0
200
400
600
800
1000
1200

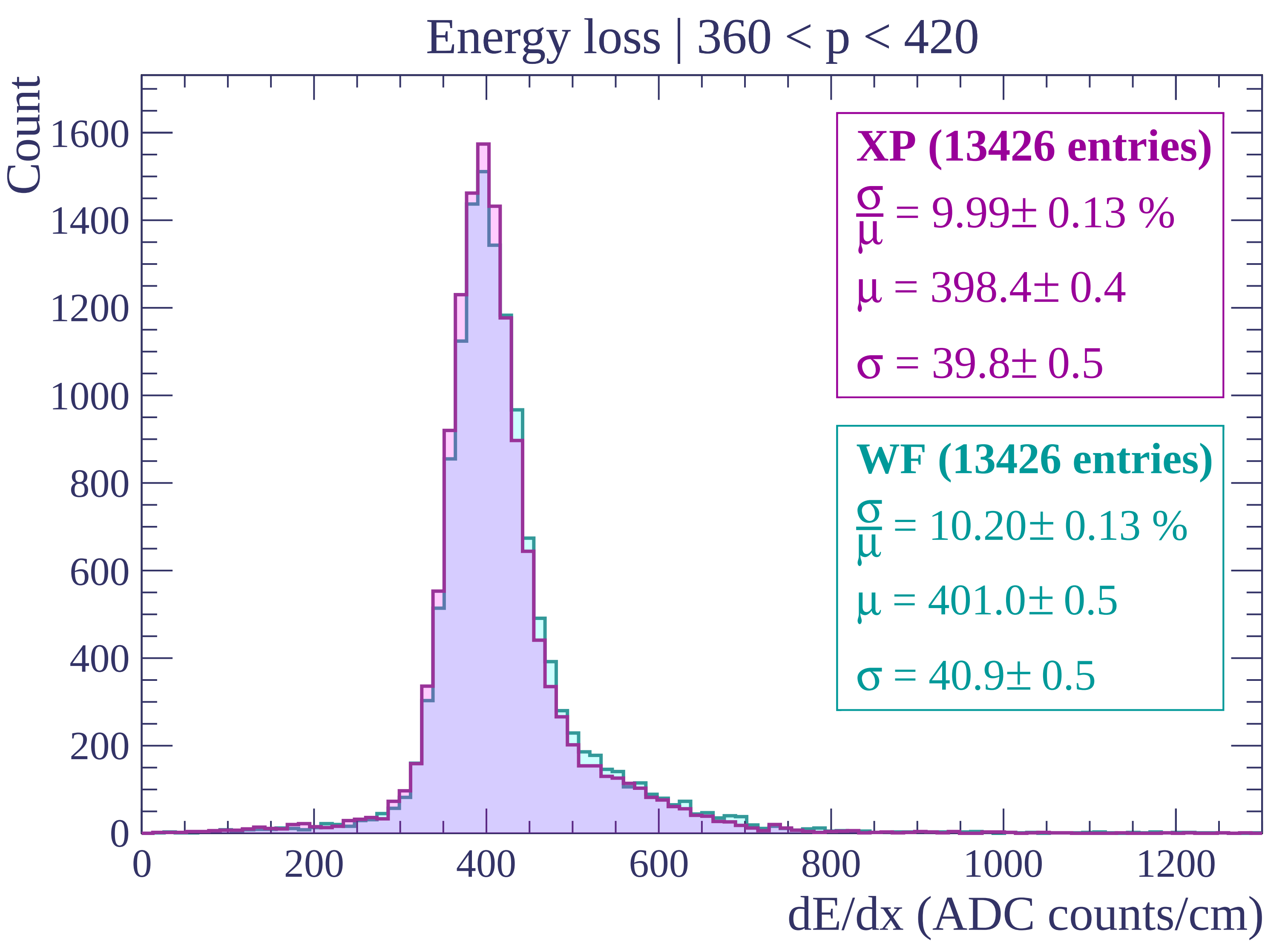

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (9788 entries)
σ/μ = 25.67± 0.49 %
μ = 507.9± 1.9
σ = 130.4± 2.0
WF (9788 entries)
σ/μ = 21.43± 0.37 %
μ = 526.9± 1.5
σ = 112.9± 1.6

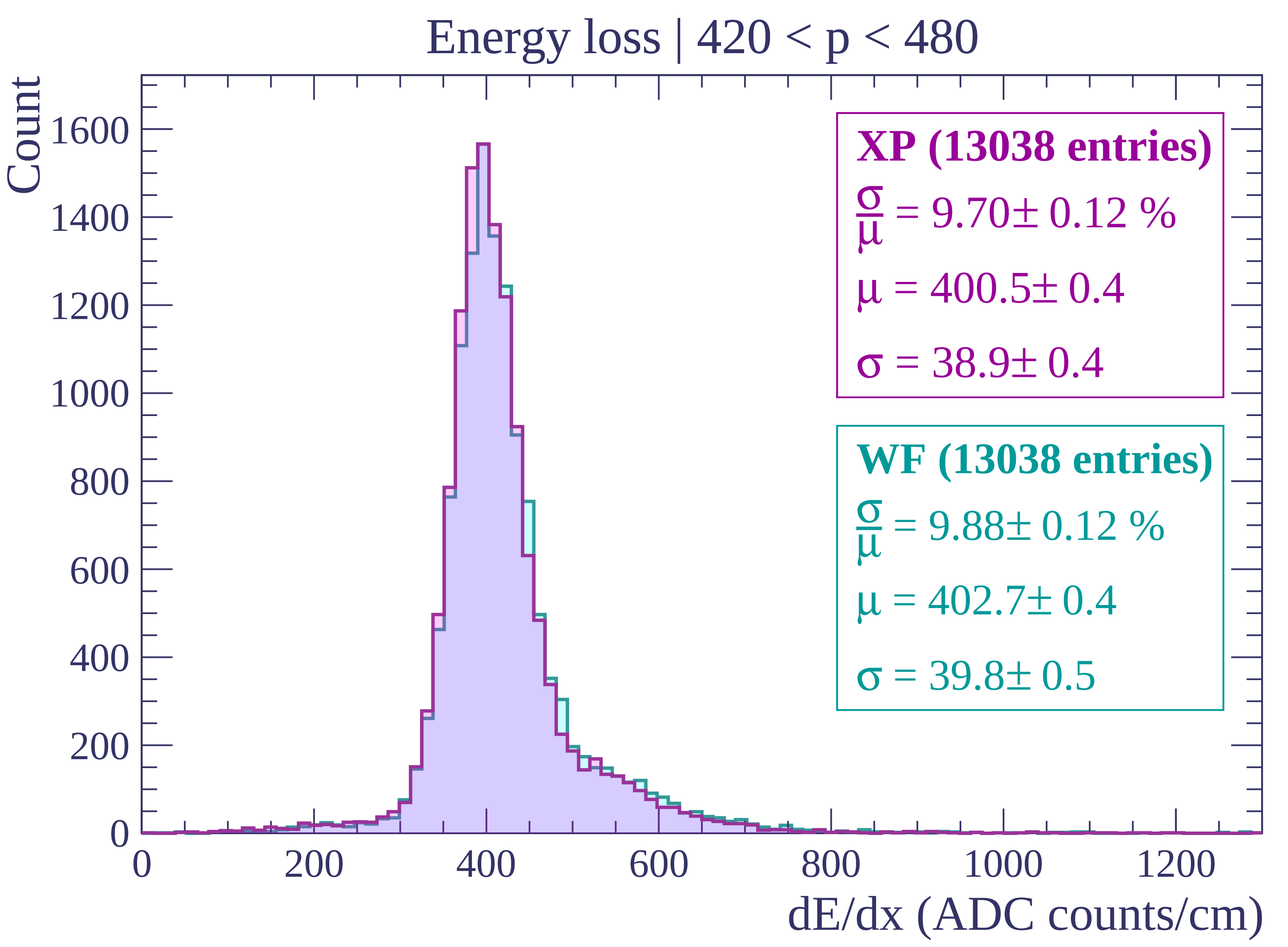

Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (9775 entries)
$\frac{\sigma}{\mu}$ = 14.47± 0.24 %
μ = 449.7± 0.9
σ = 65.1± 1.0
WF (9775 entries)
$\frac{\sigma}{\mu}$ = 13.97± 0.24 %
μ = 455.5± 0.9
σ = 63.6± 1.0

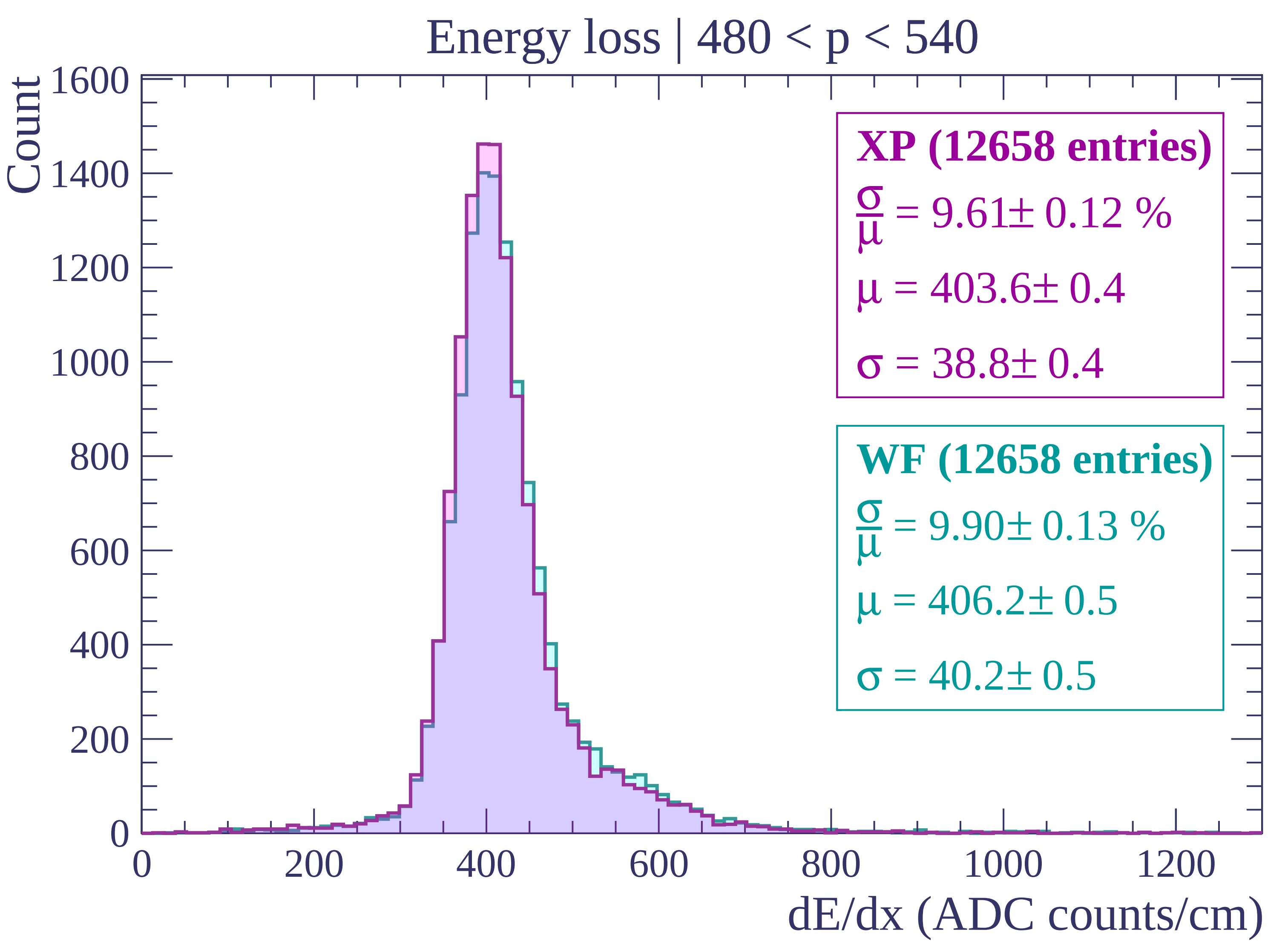


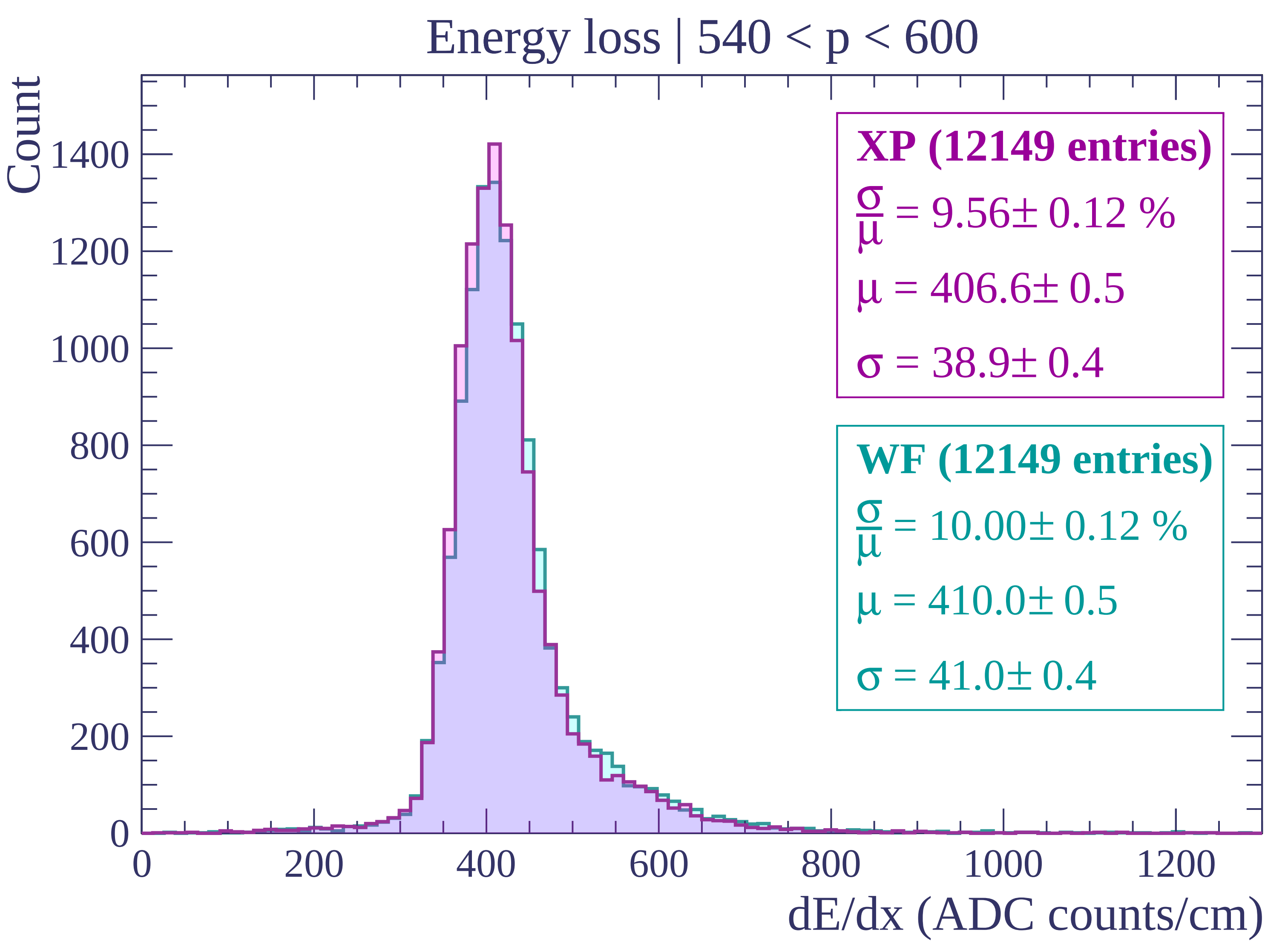
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10883 entries)
σ/μ = 11.48± 0.17 %
μ = 412.8± 0.6
σ = 47.4± 0.6
WF (10883 entries)
σ/μ = 11.44± 0.17 %
μ = 416.1± 0.6
σ = 47.6± 0.6

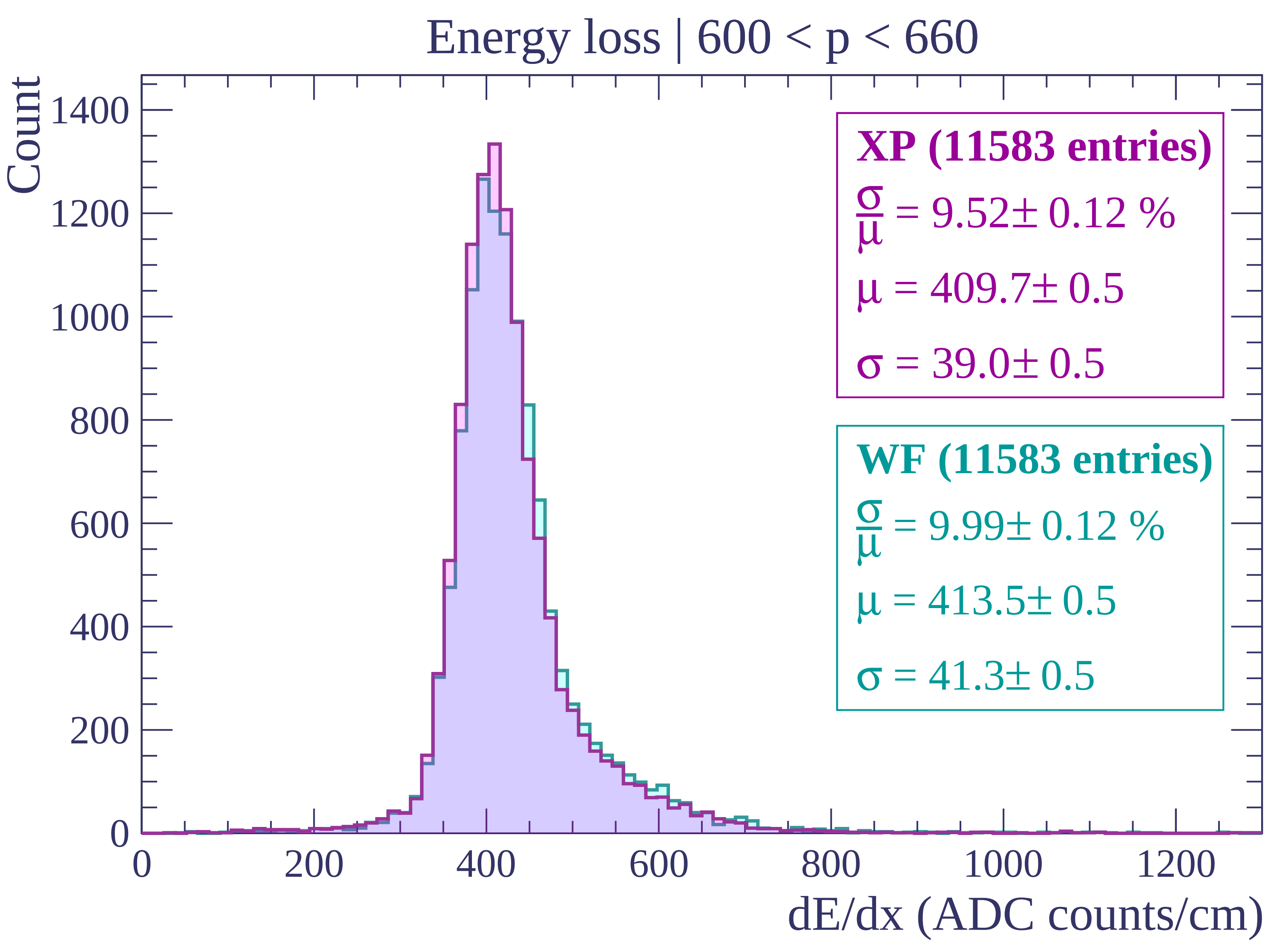

Energy loss | 300 < p < 360
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (12927 entries)
$\frac{\sigma}{\mu}$ = 10.02± 0.13 %
μ = 400.9± 0.5
σ = 40.2± 0.5
WF (12927 entries)
$\frac{\sigma}{\mu}$ = 10.01± 0.13 %
μ = 403.7± 0.5
σ = 40.4± 0.5


Energy loss | 360 < p < 420
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13426 entries)
σ/μ = 9.99± 0.13 %
μ = 398.4± 0.4
σ = 39.8± 0.5
WF (13426 entries)
σ/μ = 10.20± 0.13 %
μ = 401.0± 0.5
σ = 40.9± 0.5

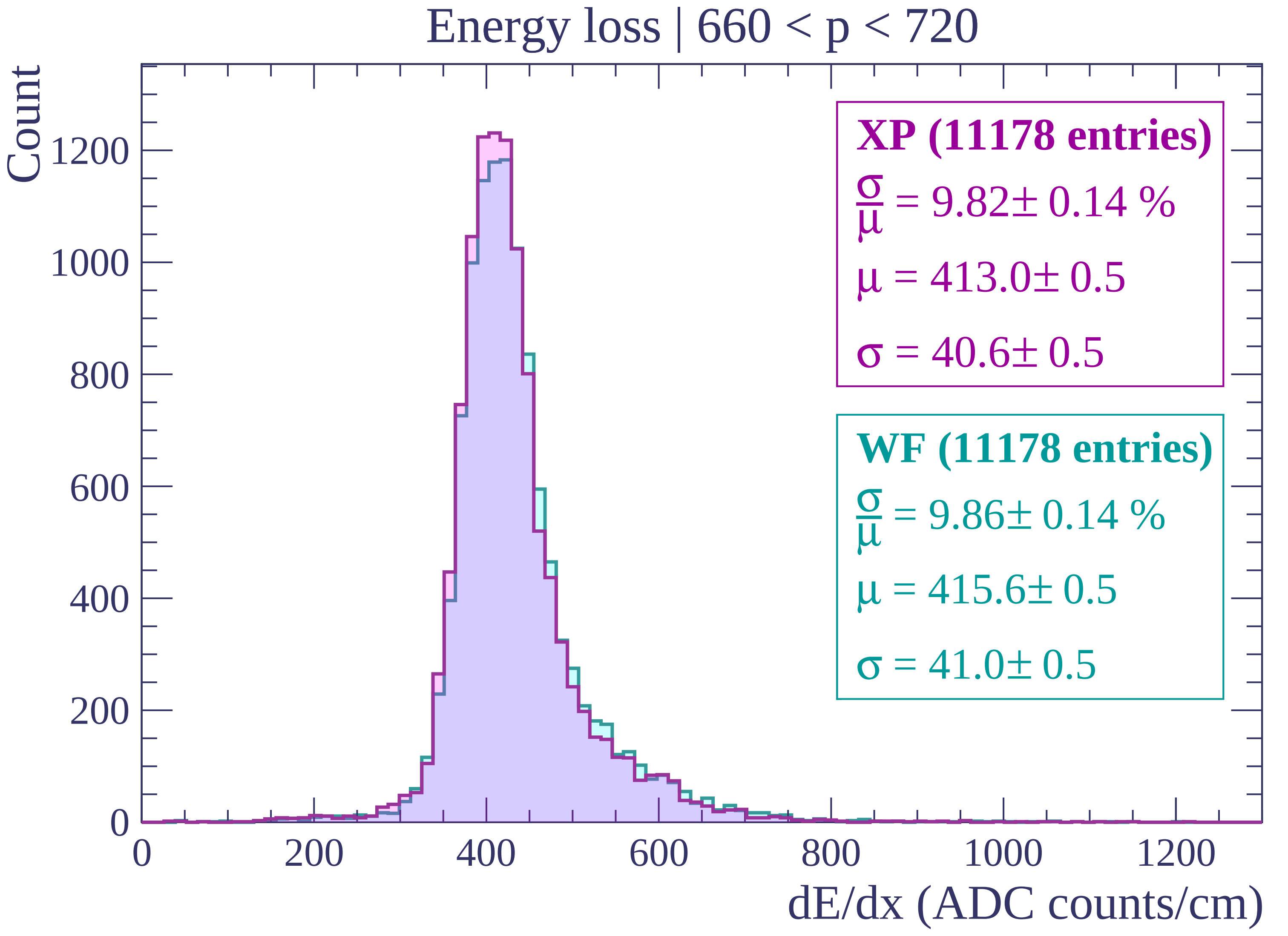

Energy loss | 420 < p < 480
Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (13038 entries)
σ/μ = 9.70± 0.12 %
μ = 400.5± 0.4
σ = 38.9± 0.4
WF (13038 entries)
σ/μ = 9.88± 0.12 %
μ = 402.7± 0.4
σ = 39.8± 0.5

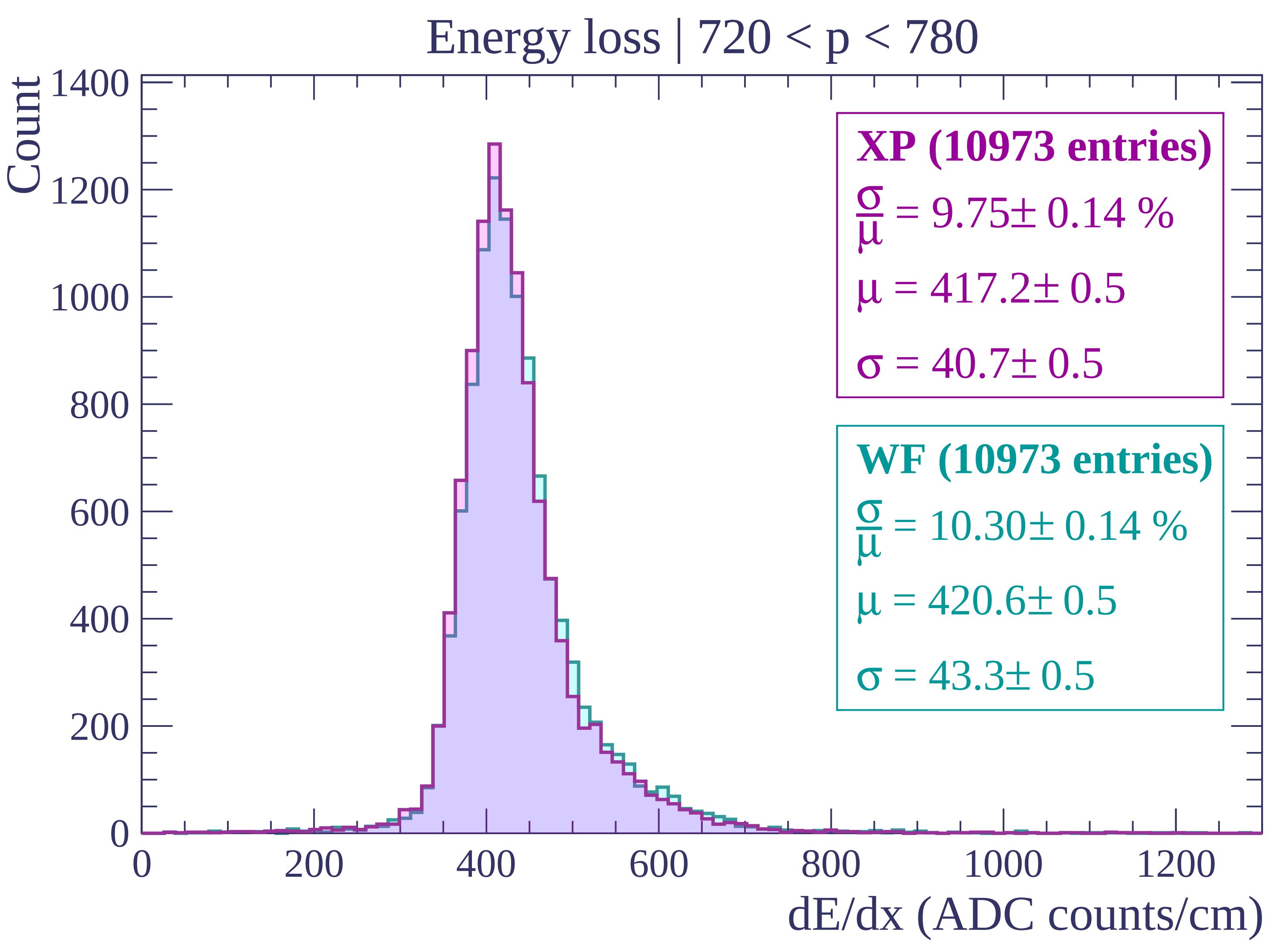

Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
XP (12658 entries)
σ/μ = 9.61± 0.12 %
μ = 403.6± 0.4
σ = 38.8± 0.4
WF (12658 entries)
σ/μ = 9.90± 0.13 %
μ = 406.2± 0.5
σ = 40.2± 0.5
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
0
200
400
600
800
1000
1200
XP (12149 entries)
$\frac{\sigma}{\mu}$ = 9.56± 0.12 %
μ = 406.6± 0.5
σ = 38.9± 0.4
WF (12149 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.12 %
μ = 410.0± 0.5
σ = 41.0± 0.4

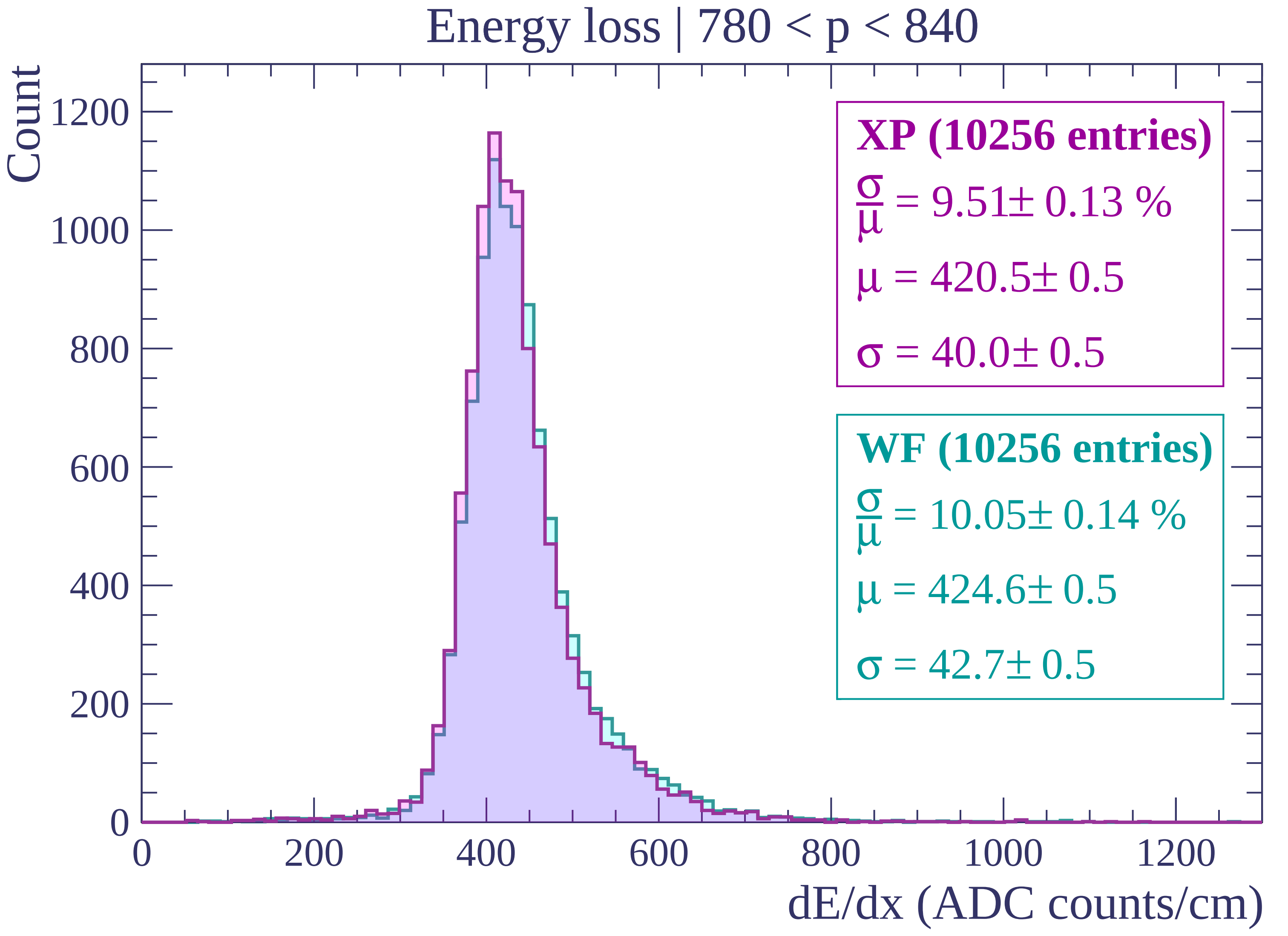


Energy loss | 600 < p < 660
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (11583 entries)
σ/μ = 9.52± 0.12 %
μ = 409.7± 0.5
σ = 39.0± 0.5
WF (11583 entries)
σ/μ = 9.99± 0.12 %
μ = 413.5± 0.5
σ = 41.3± 0.5

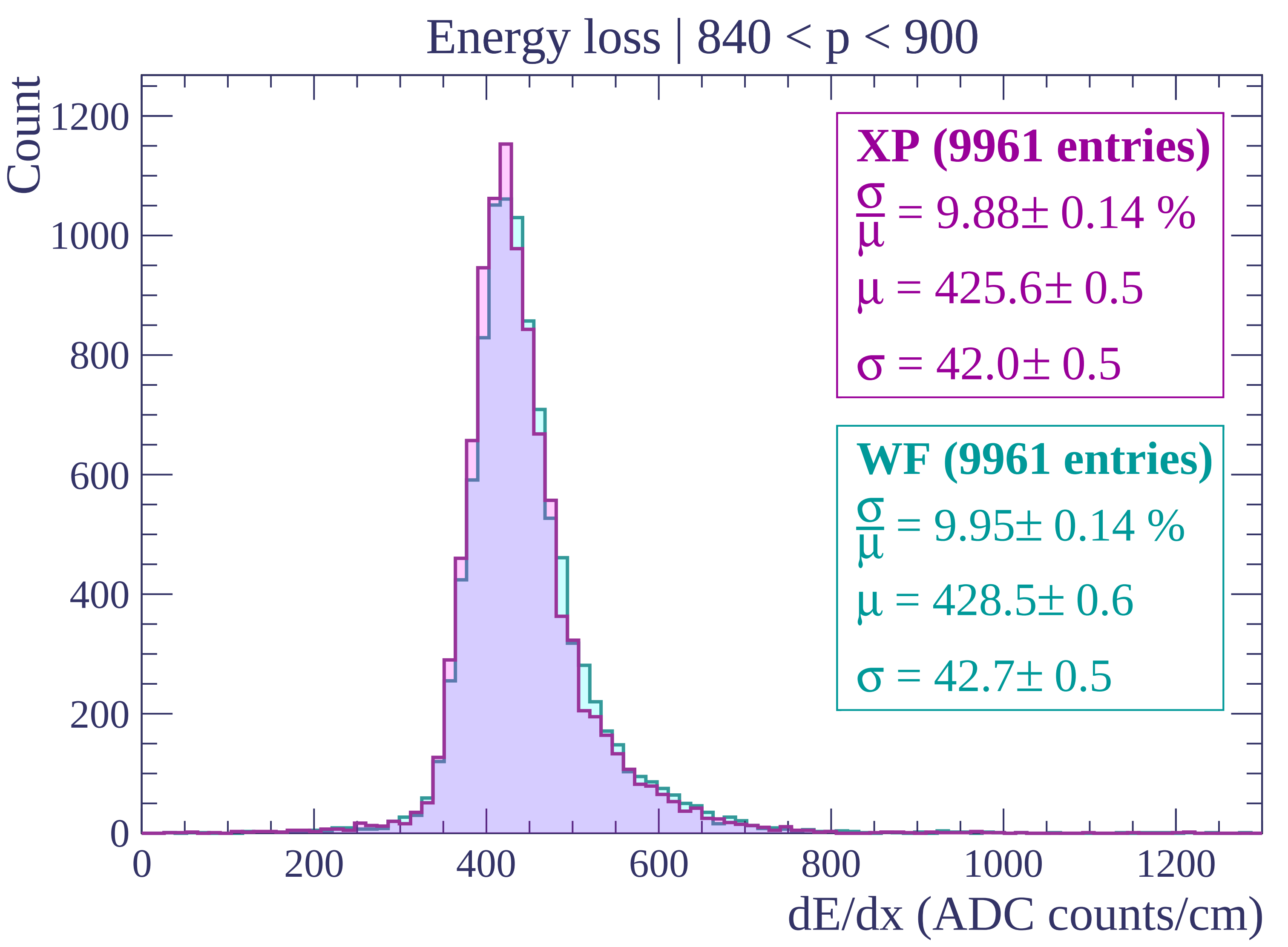

Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (11178 entries)
$\frac{\sigma}{\mu}$ = 9.82± 0.14 %
μ = 413.0± 0.5
σ = 40.6± 0.5
WF (11178 entries)
$\frac{\sigma}{\mu}$ = 9.86± 0.14 %
μ = 415.6± 0.5
σ = 41.0± 0.5

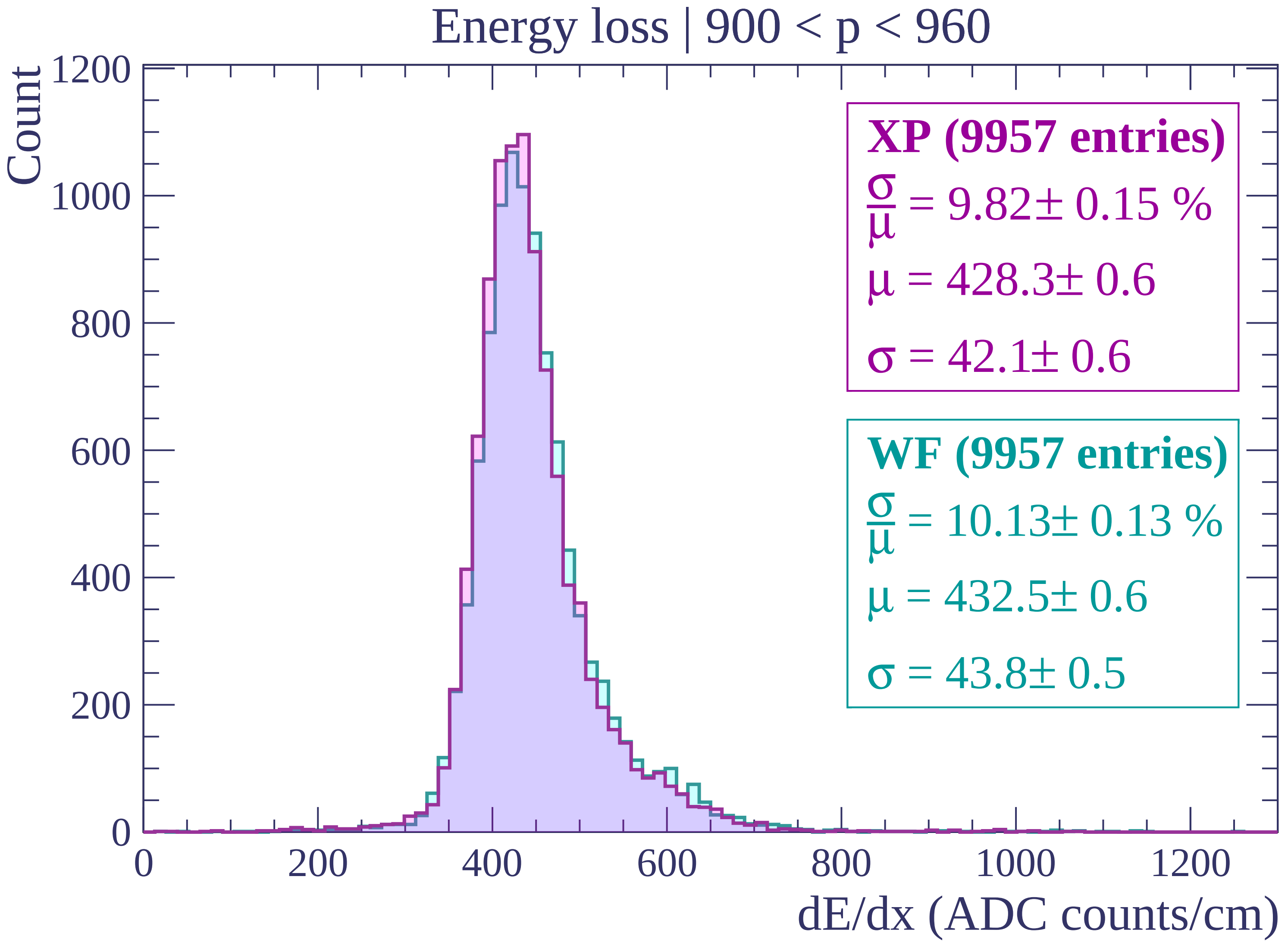


Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
XP (10973 entries)
$\frac{\sigma}{\mu}$ = 9.75± 0.14 %
μ = 417.2± 0.5
σ = 40.7± 0.5
WF (10973 entries)
$\frac{\sigma}{\mu}$ = 10.30± 0.14 %
μ = 420.6± 0.5
σ = 43.3± 0.5

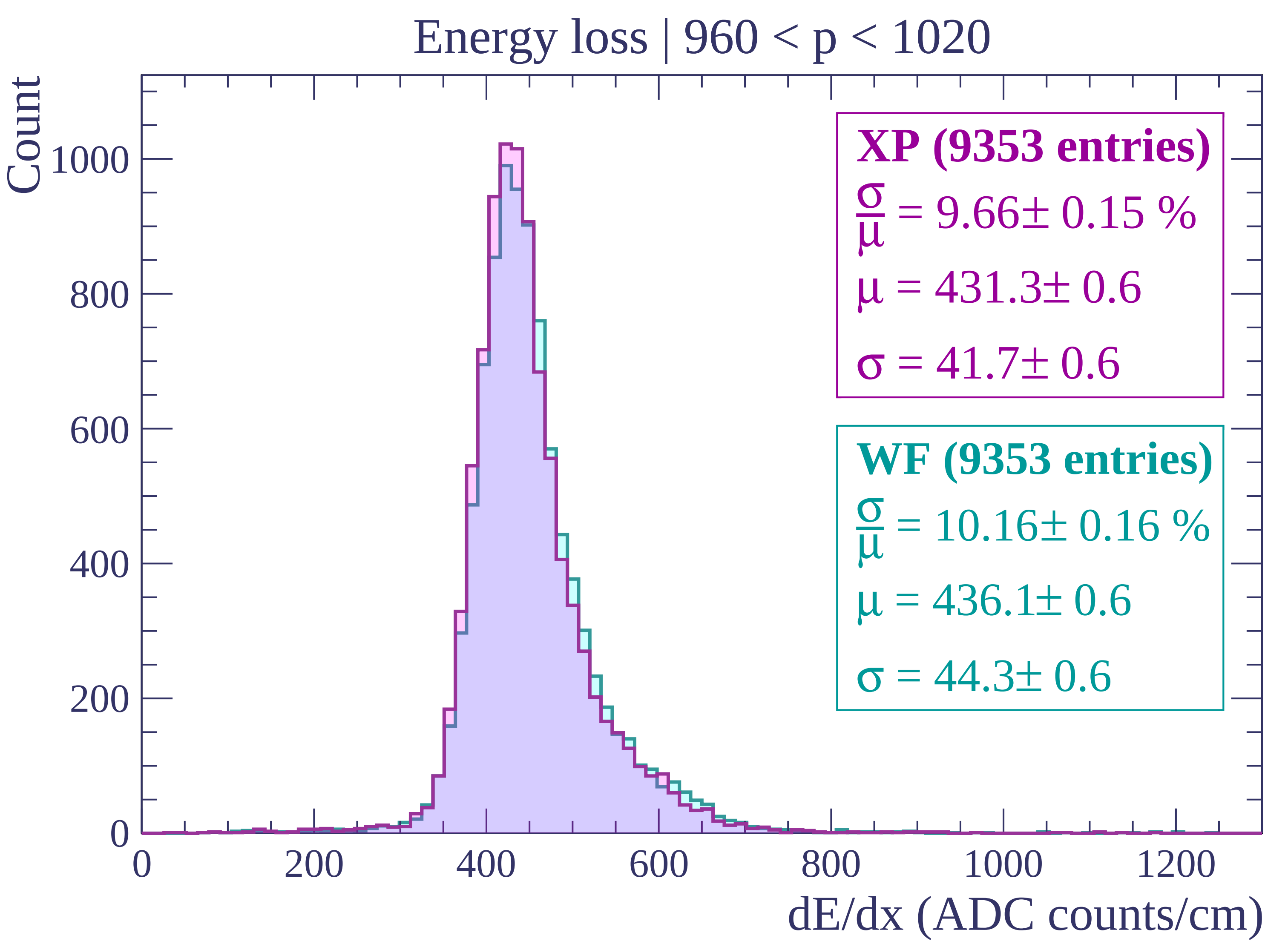


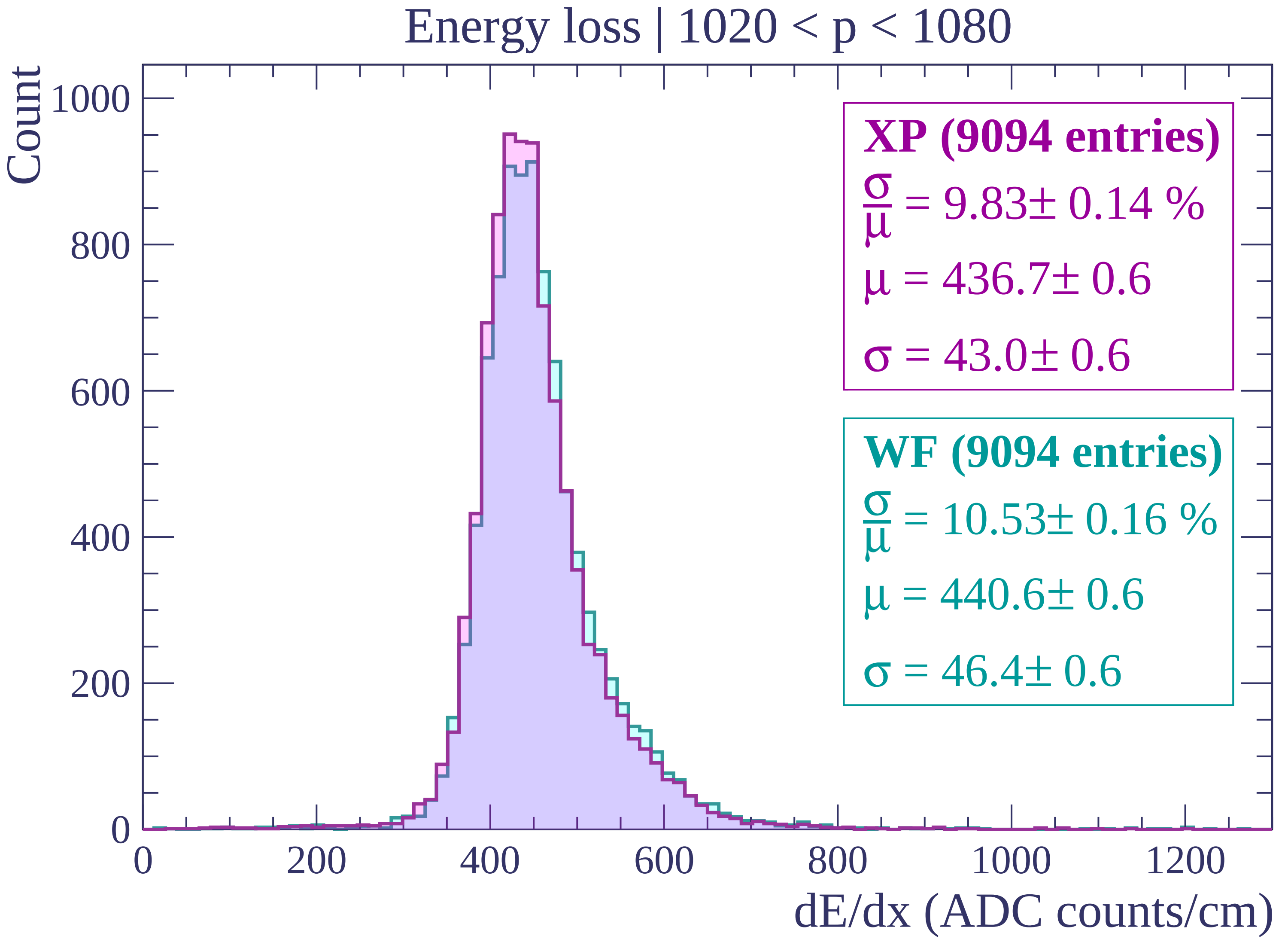
Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (10256 entries)
σ/μ = 9.51± 0.13 %
μ = 420.5± 0.5
σ = 40.0± 0.5
WF (10256 entries)
σ/μ = 10.05± 0.14 %
μ = 424.6± 0.5
σ = 42.7± 0.5

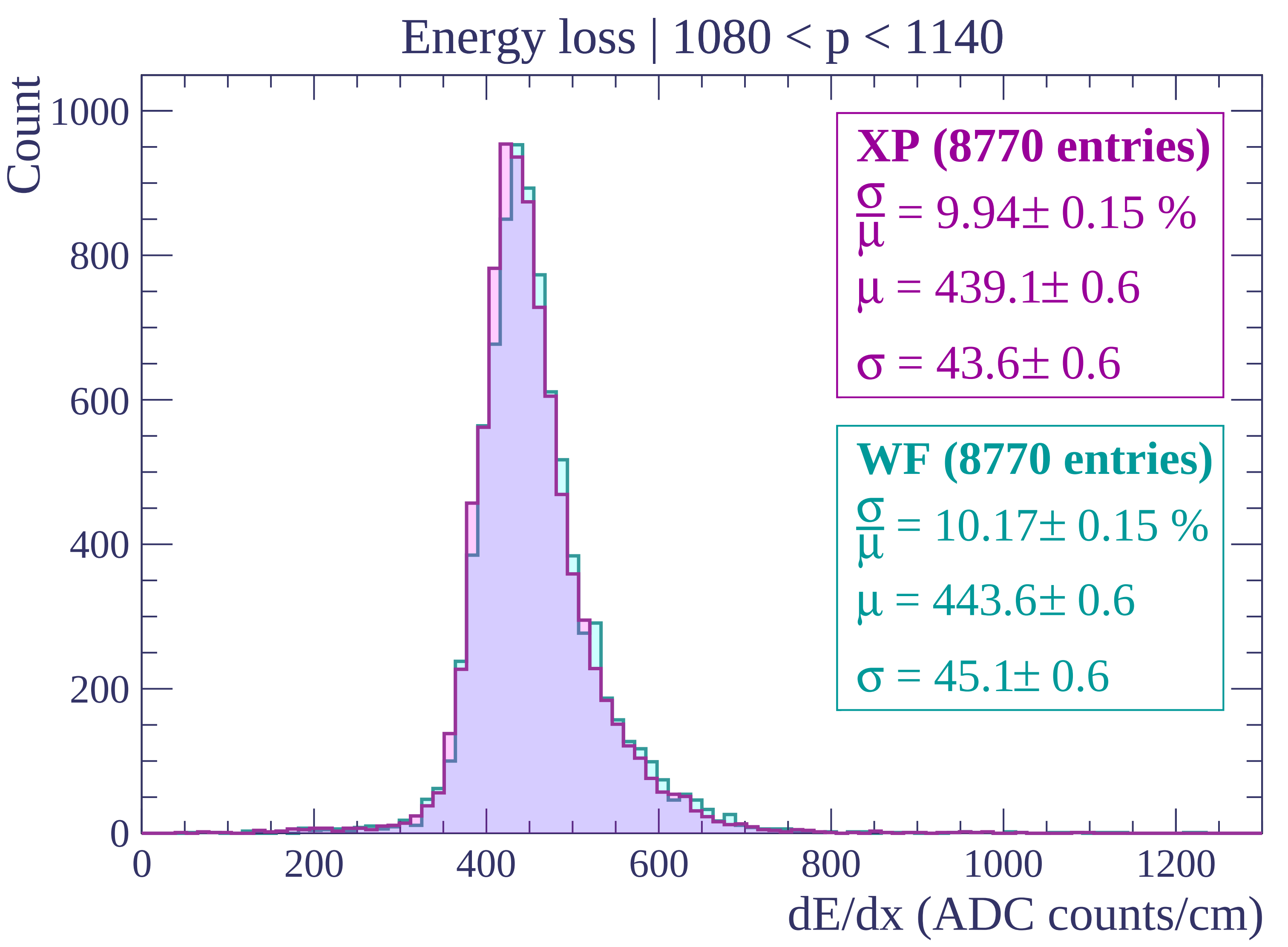

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
XP (9961 entries)
$\frac{\sigma}{\mu}$ = 9.88± 0.14 %
μ = 425.6± 0.5
σ = 42.0± 0.5
WF (9961 entries)
$\frac{\sigma}{\mu}$ = 9.95± 0.14 %
μ = 428.5± 0.6
σ = 42.7± 0.5
0
200
400
600
800
1000
1200



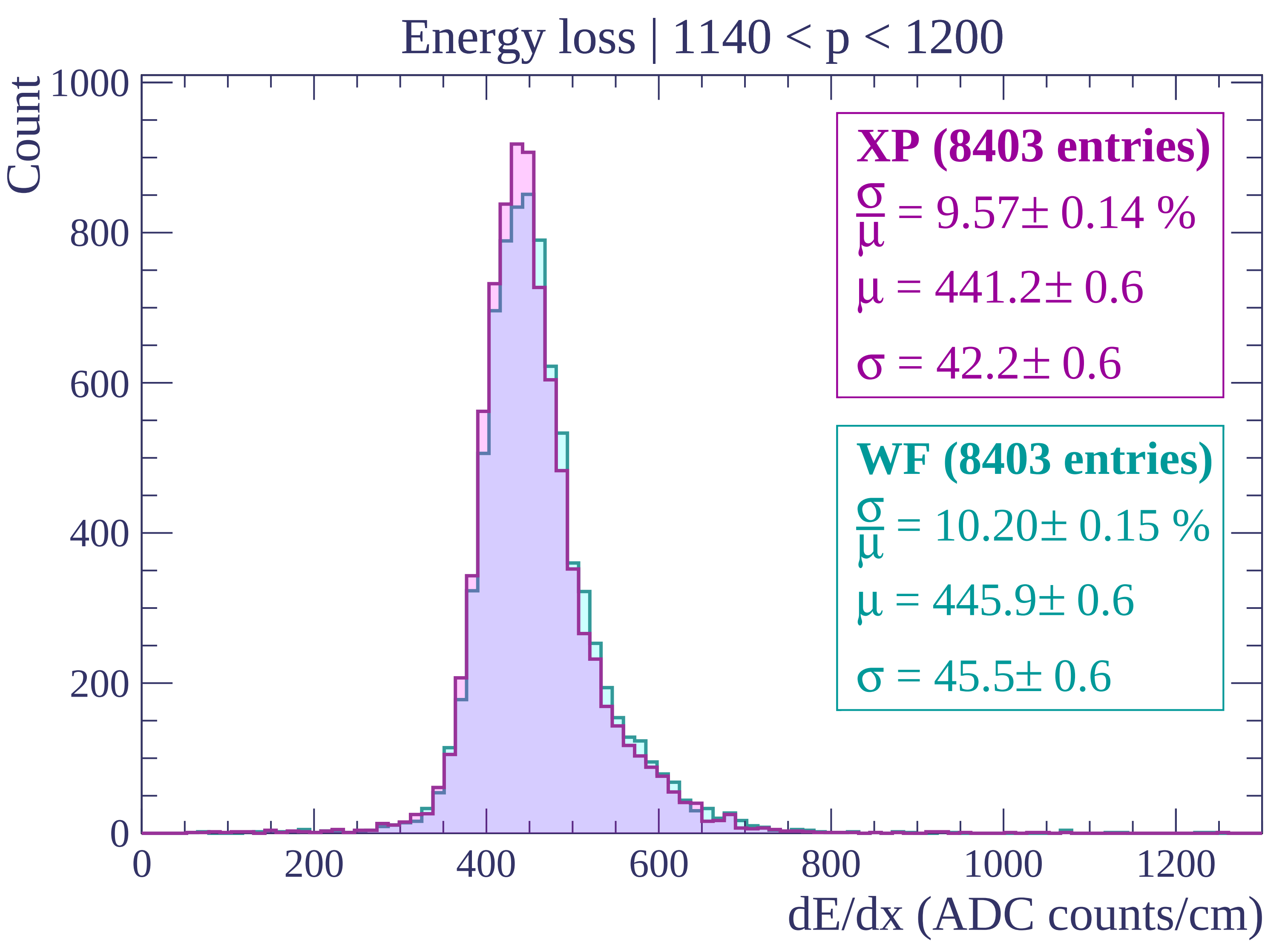
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (9957 entries)
σ/μ = 9.82± 0.15 %
μ = 428.3± 0.6
σ = 42.1± 0.6
WF (9957 entries)
σ/μ = 10.13± 0.13 %
μ = 432.5± 0.6
σ = 43.8± 0.5

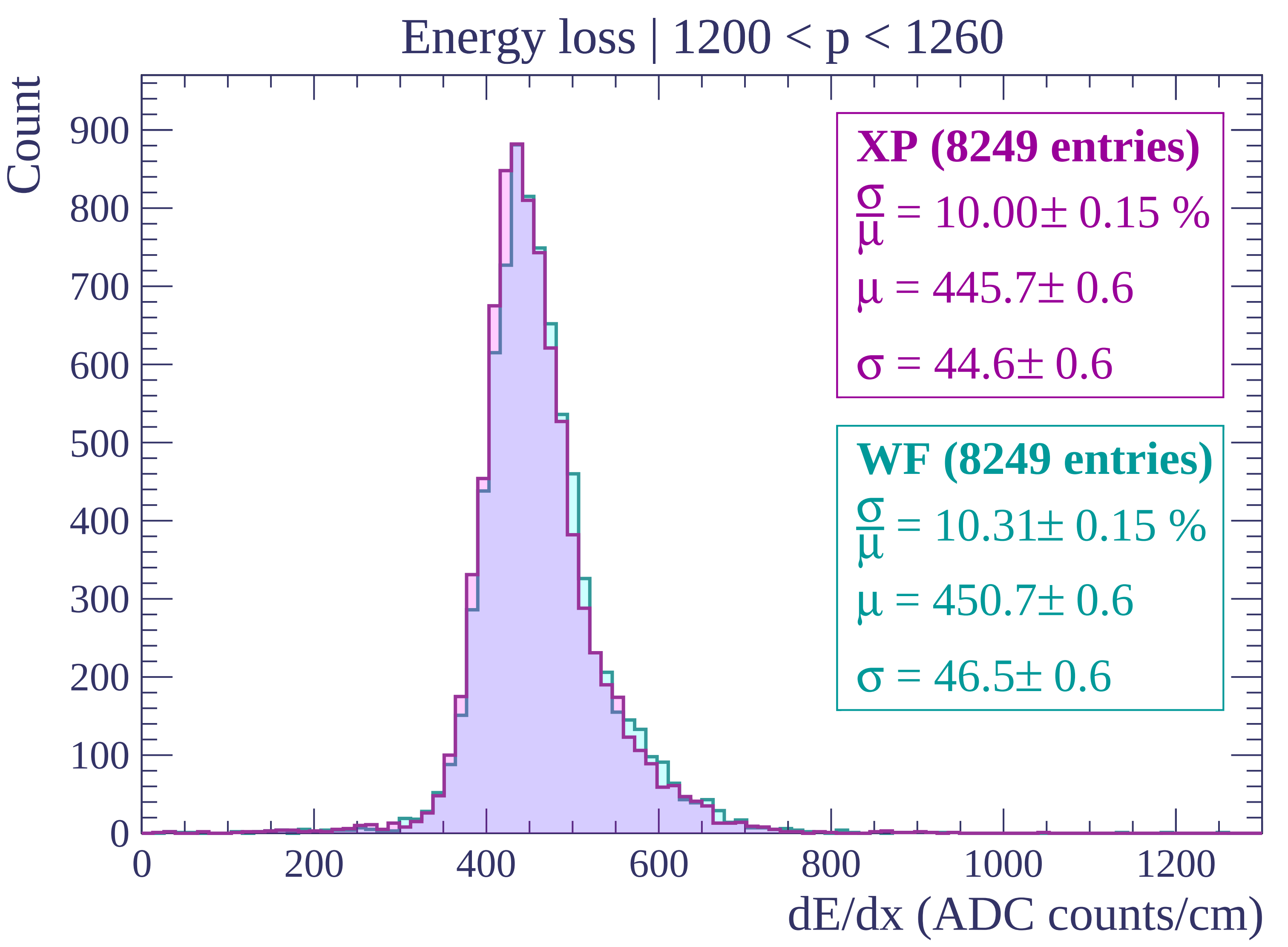

Energy loss | 960 < p < 1020
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9353 entries)
σ/μ = 9.66± 0.15 %
μ = 431.3± 0.6
σ = 41.7± 0.6
WF (9353 entries)
σ/μ = 10.16± 0.16 %
μ = 436.1± 0.6
σ = 44.3± 0.6

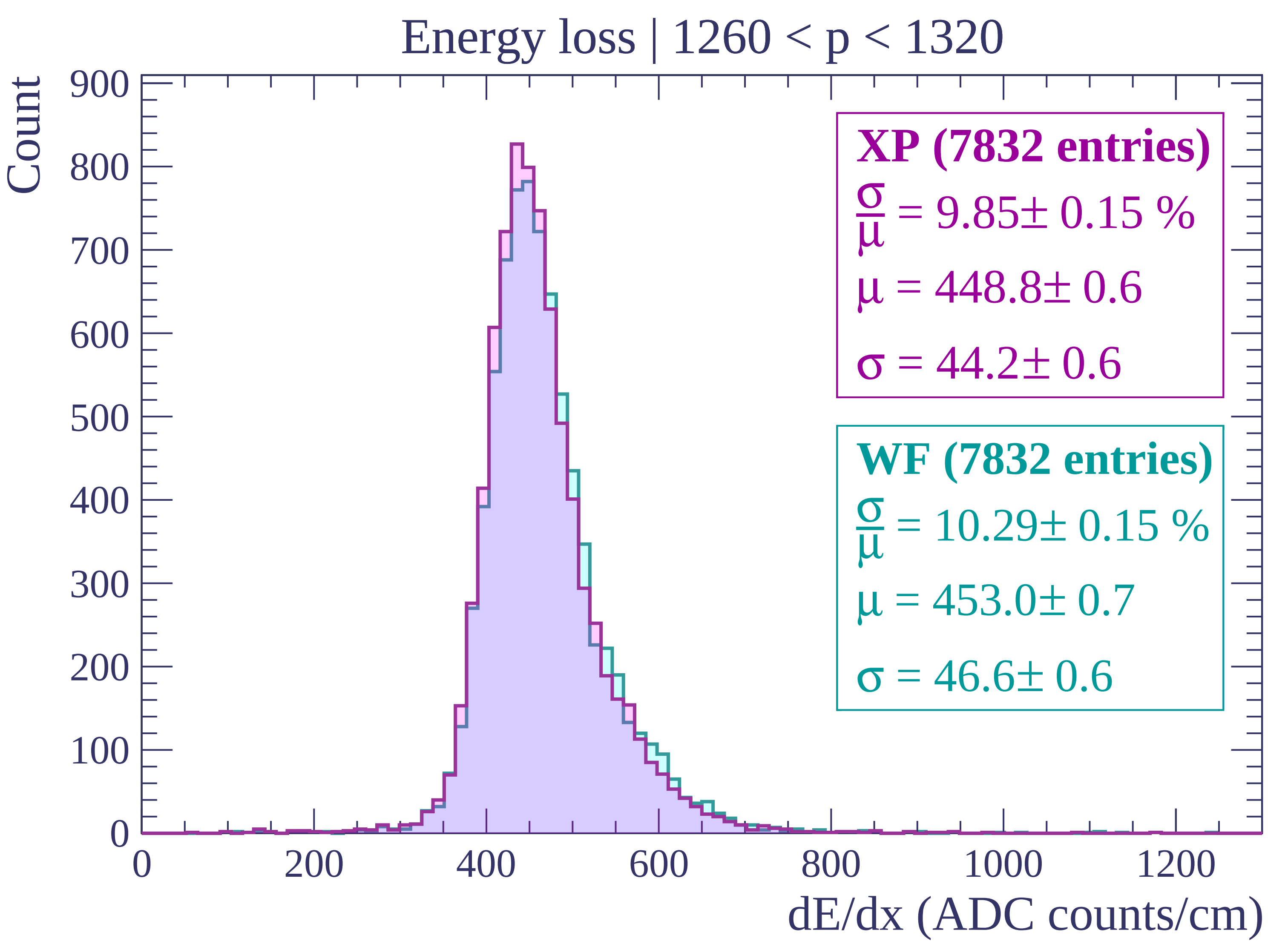

Energy loss | 1020 < p < 1080
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (9094 entries)
σ/μ = 9.83± 0.14 %
μ = 436.7± 0.6
σ = 43.0± 0.6
WF (9094 entries)
σ/μ = 10.53± 0.16 %
μ = 440.6± 0.6
σ = 46.4± 0.6



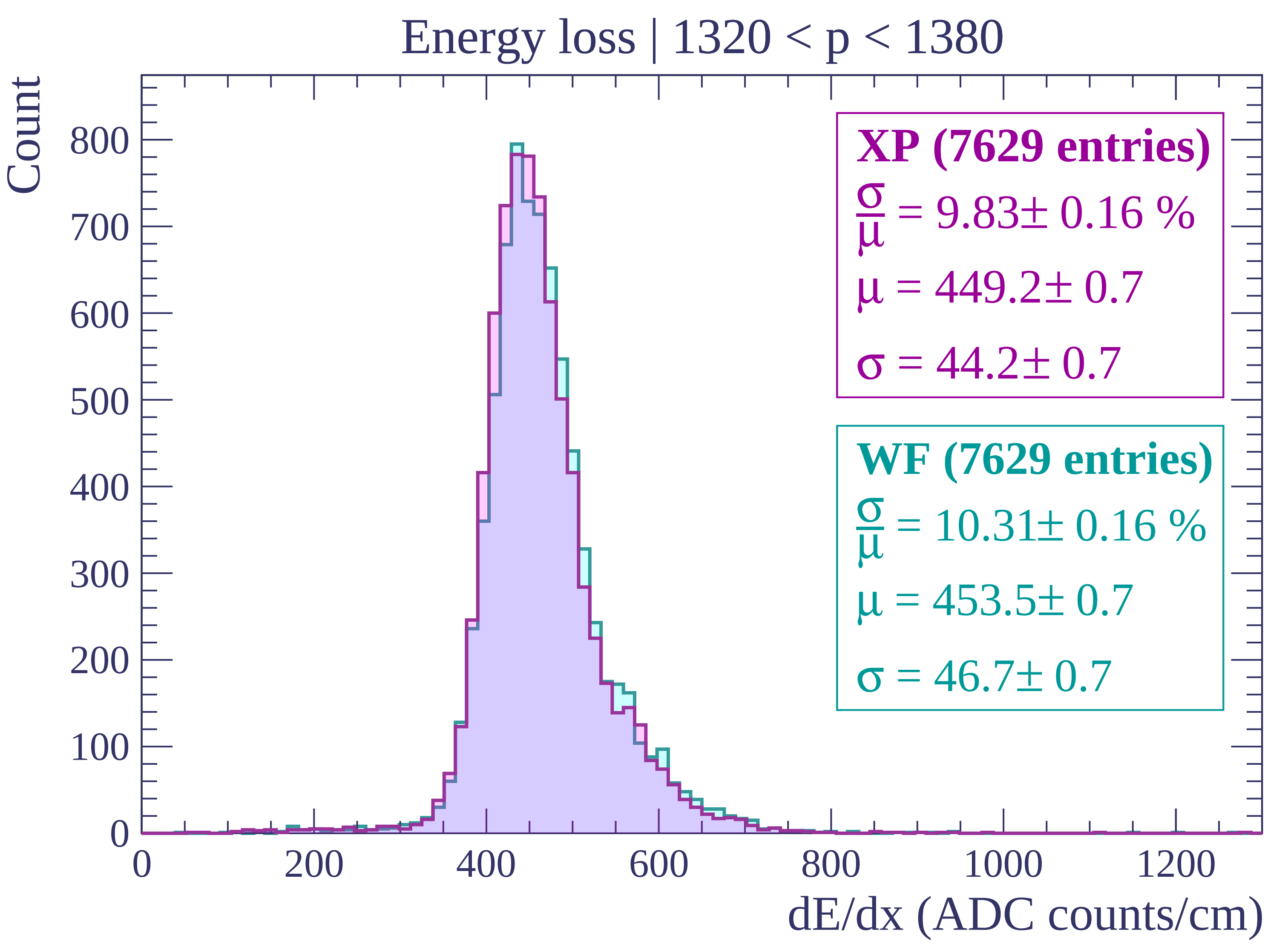
Energy loss | 1080 < p < 1140
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8770 entries)
$\frac{\sigma}{\mu}$ = 9.94± 0.15 %
μ = 439.1± 0.6
σ = 43.6± 0.6
WF (8770 entries)
$\frac{\sigma}{\mu}$ = 10.17± 0.15 %
μ = 443.6± 0.6
σ = 45.1± 0.6

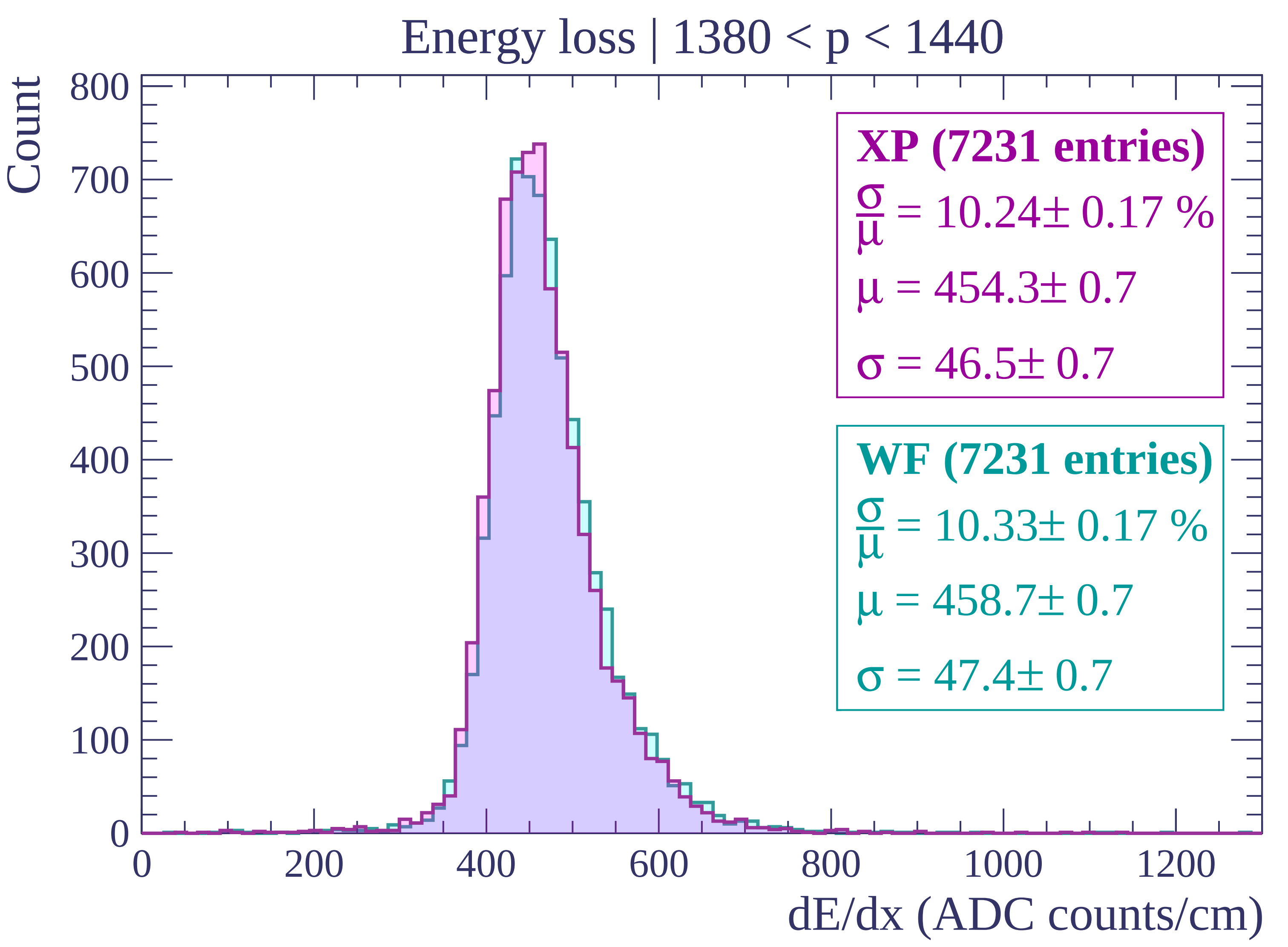

Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (8403 entries)
σ/μ = 9.57± 0.14 %
μ = 441.2± 0.6
σ = 42.2± 0.6
WF (8403 entries)
σ/μ = 10.20± 0.15 %
μ = 445.9± 0.6
σ = 45.5± 0.6
0
200
400
600
800
1000
1200


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (8249 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.15 %
μ = 445.7± 0.6
σ = 44.6± 0.6
WF (8249 entries)
$\frac{\sigma}{\mu}$ = 10.31± 0.15 %
μ = 450.7± 0.6
σ = 46.5± 0.6

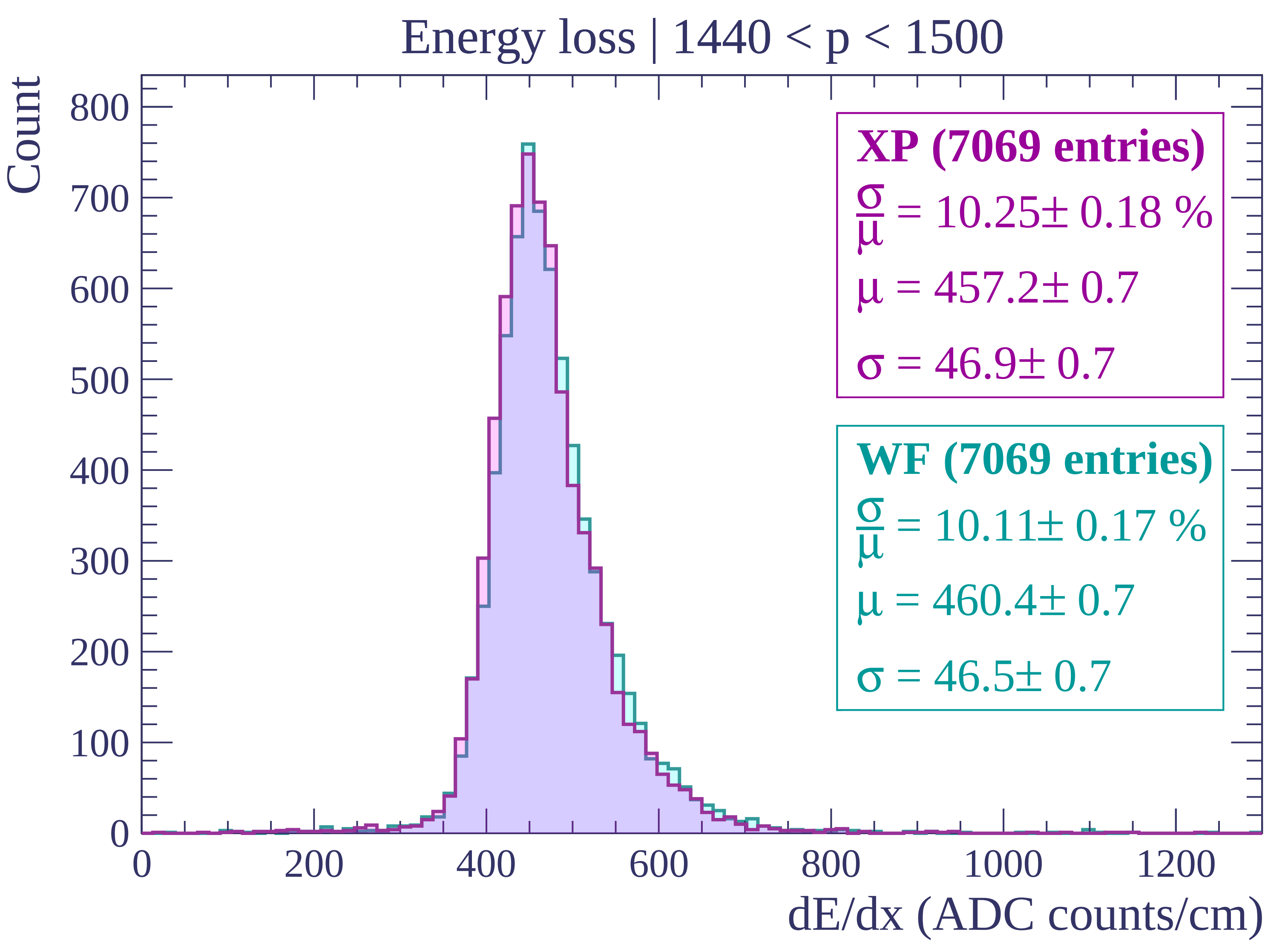


Energy loss | 1260 < p < 1320
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7832 entries)
σ/μ = 9.85± 0.15 %
μ = 448.8± 0.6
σ = 44.2± 0.6
WF (7832 entries)
σ/μ = 10.29± 0.15 %
μ = 453.0± 0.7
σ = 46.6± 0.6

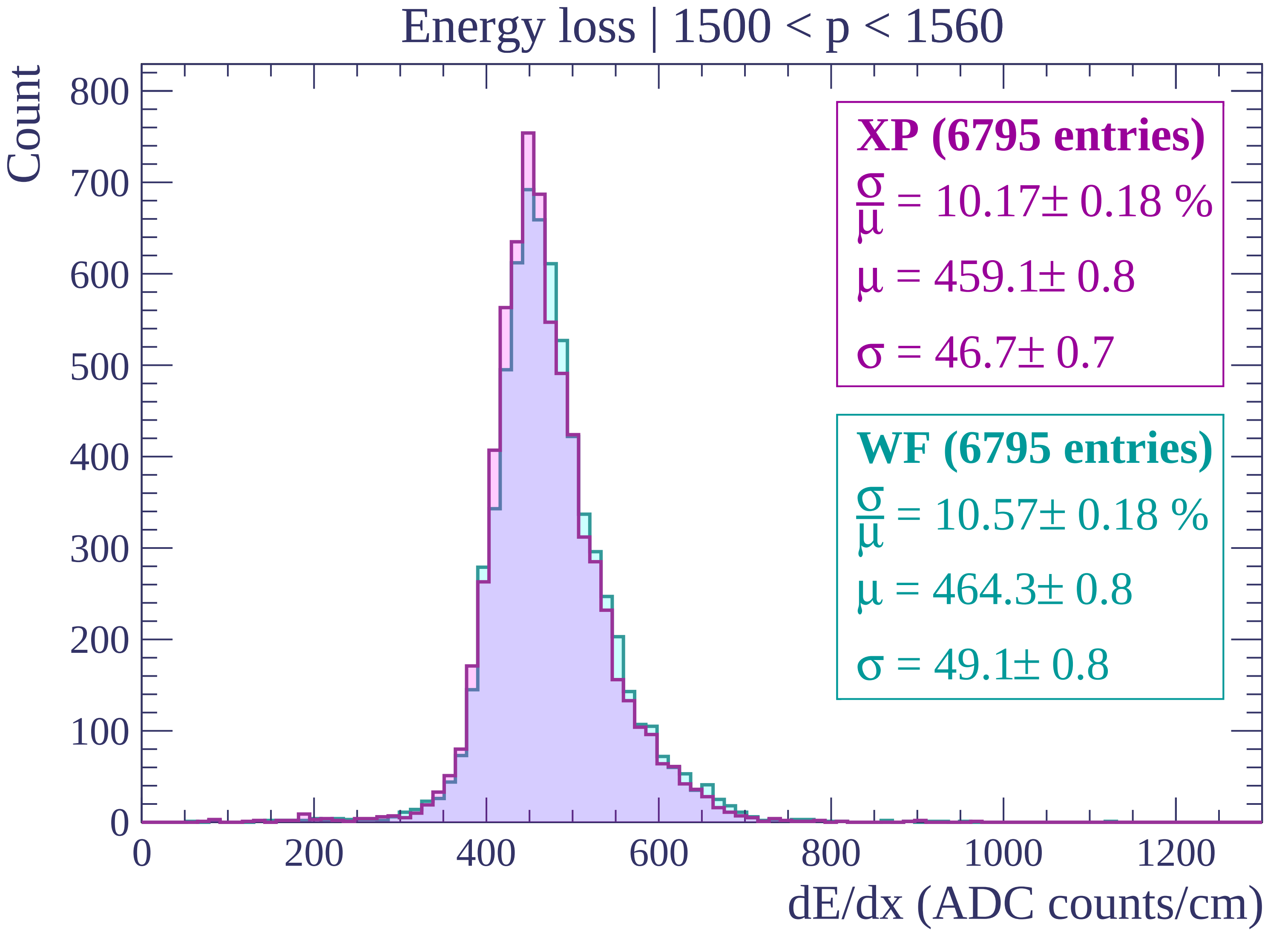

Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (7629 entries)
σ/μ = 9.83± 0.16 %
μ = 449.2± 0.7
σ = 44.2± 0.7
WF (7629 entries)
σ/μ = 10.31± 0.16 %
μ = 453.5± 0.7
σ = 46.7± 0.7

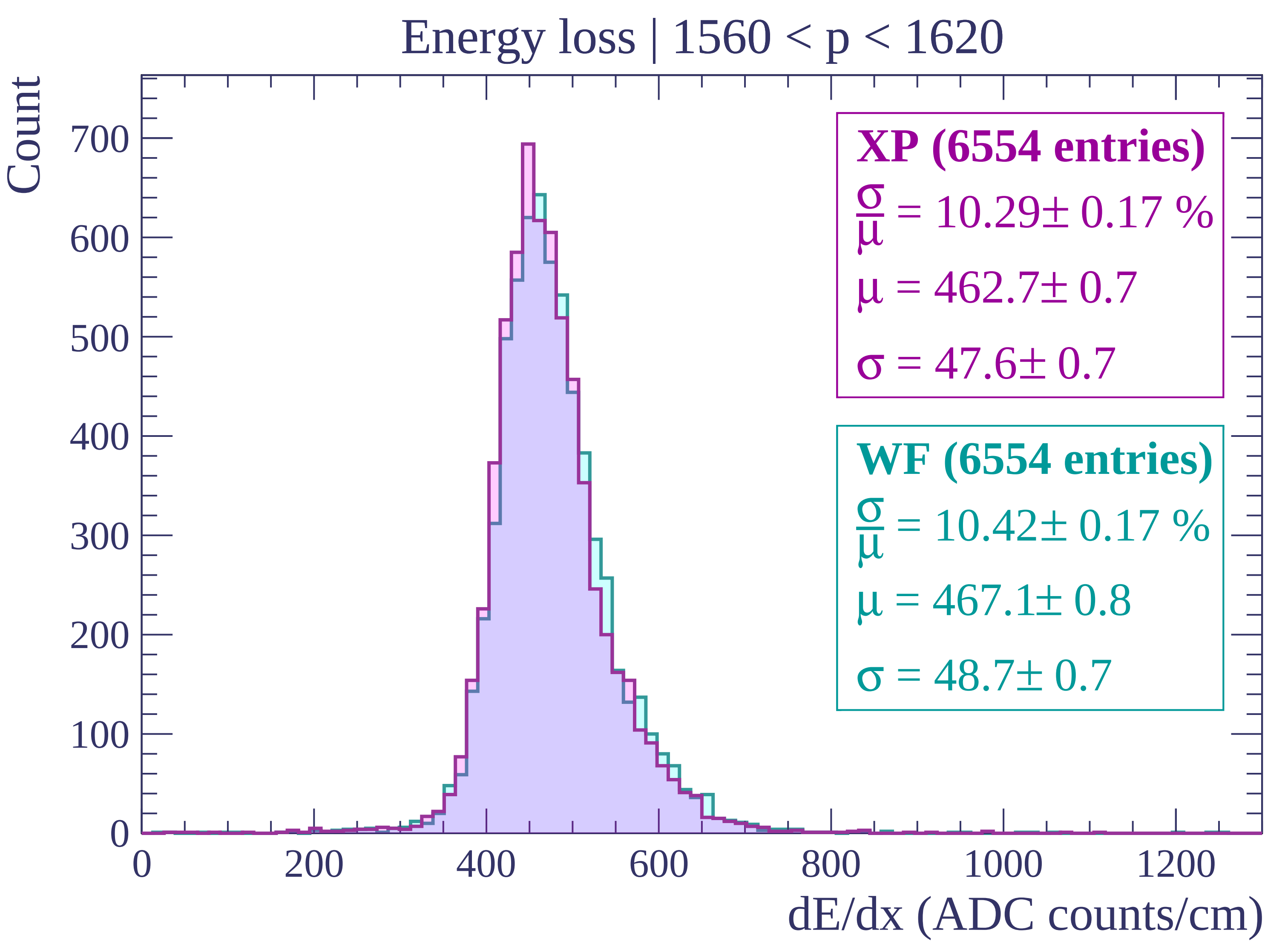


Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (7231 entries)
σ/μ = 10.24± 0.17 %
μ = 454.3± 0.7
σ = 46.5± 0.7
WF (7231 entries)
σ/μ = 10.33± 0.17 %
μ = 458.7± 0.7
σ = 47.4± 0.7

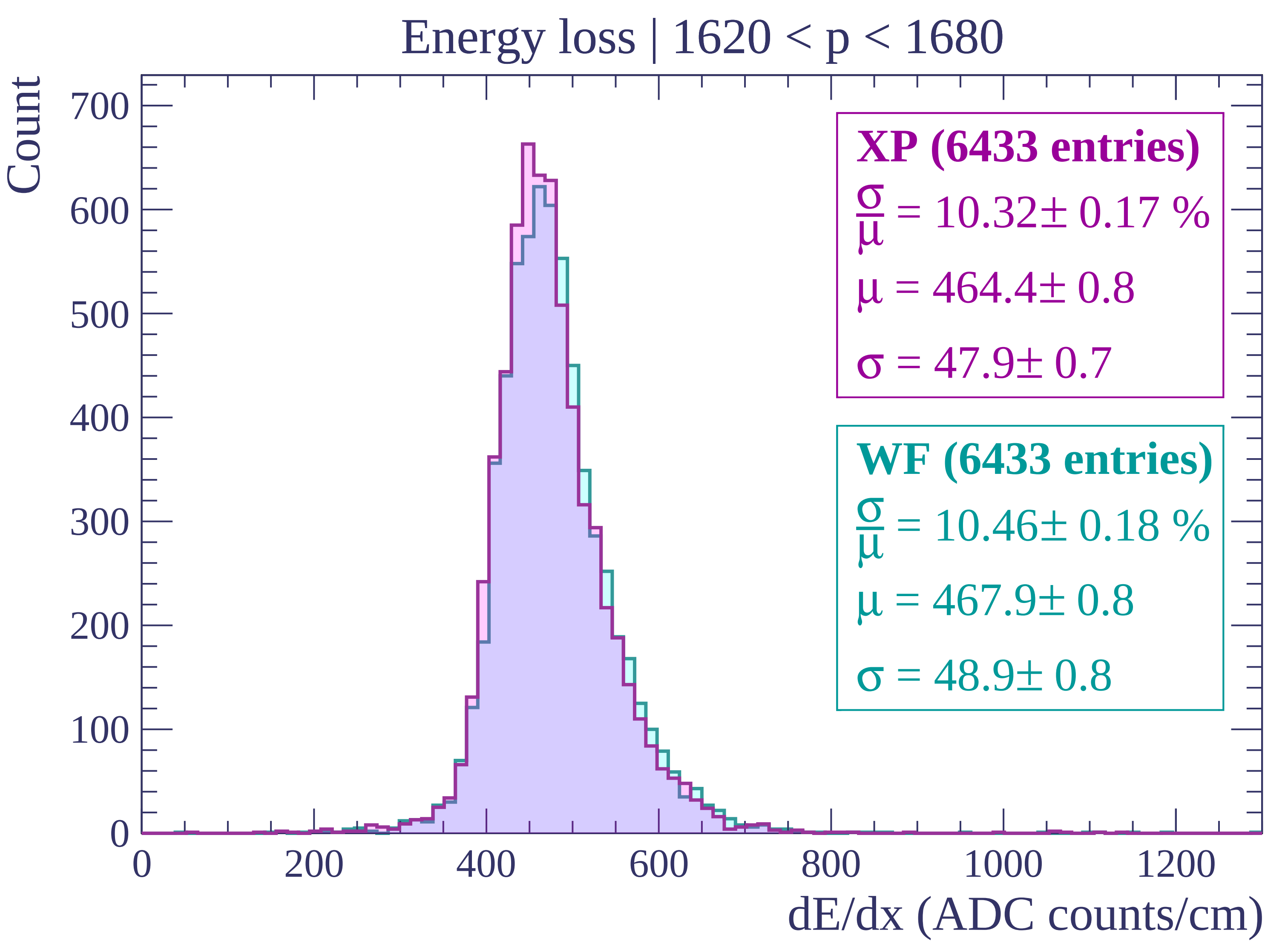


Energy loss | 1440 < p < 1500
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7069 entries)
σ/μ = 10.25± 0.18 %
μ = 457.2± 0.7
σ = 46.9± 0.7
WF (7069 entries)
σ/μ = 10.11± 0.17 %
μ = 460.4± 0.7
σ = 46.5± 0.7

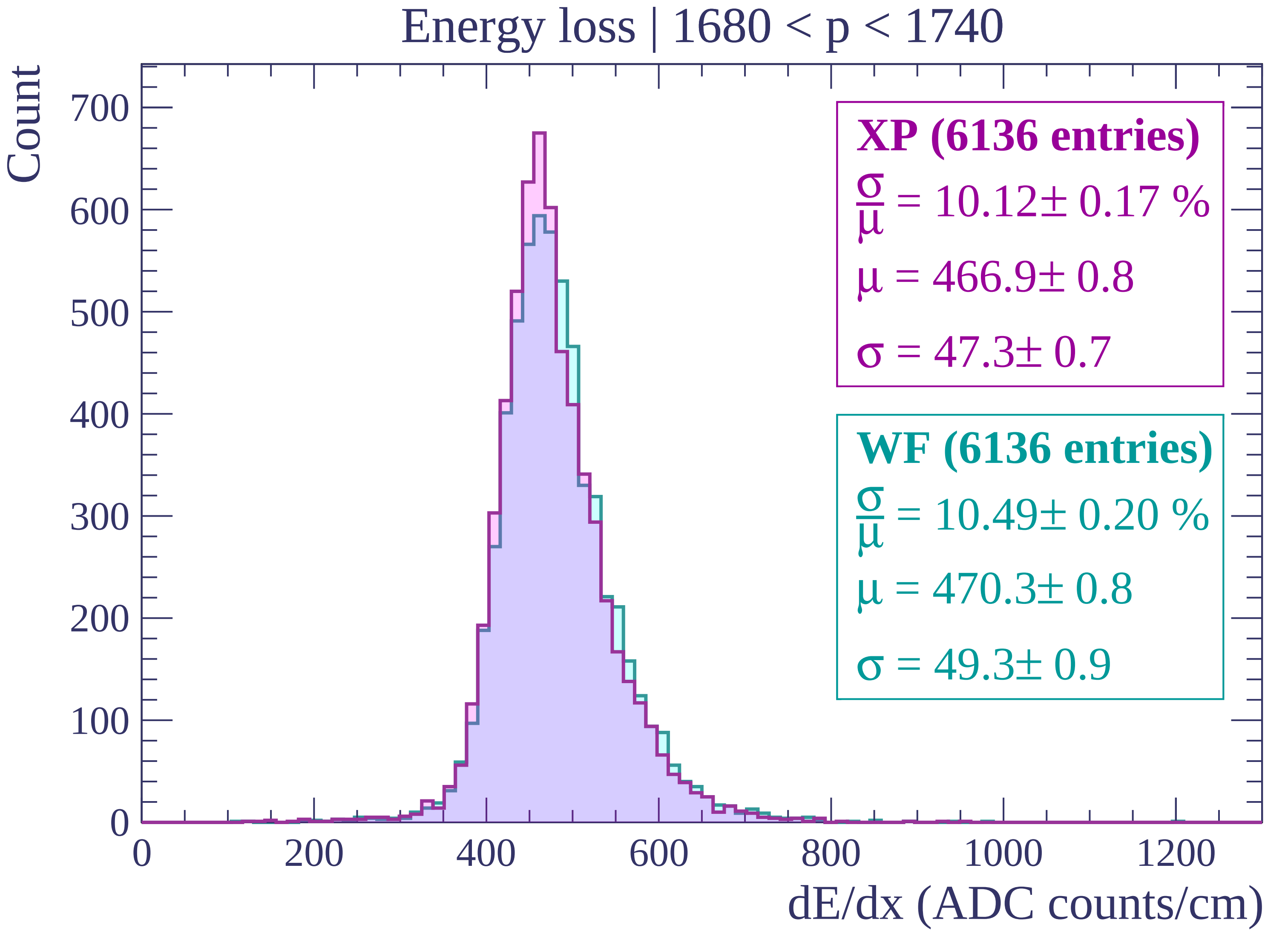

Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (6795 entries)
σ/μ = 10.17± 0.18 %
μ = 459.1± 0.8
σ = 46.7± 0.7
WF (6795 entries)
σ/μ = 10.57± 0.18 %
μ = 464.3± 0.8
σ = 49.1± 0.8

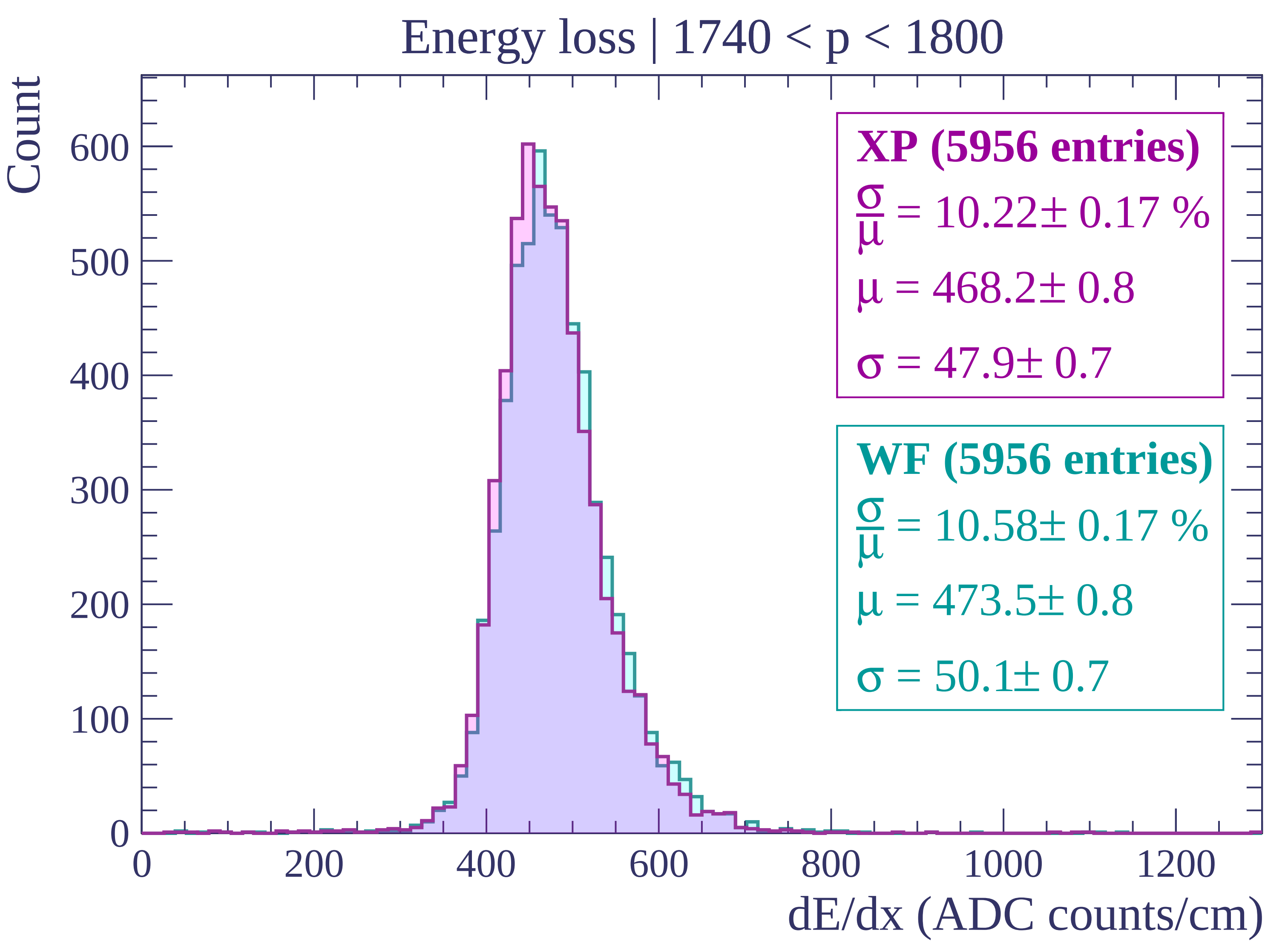


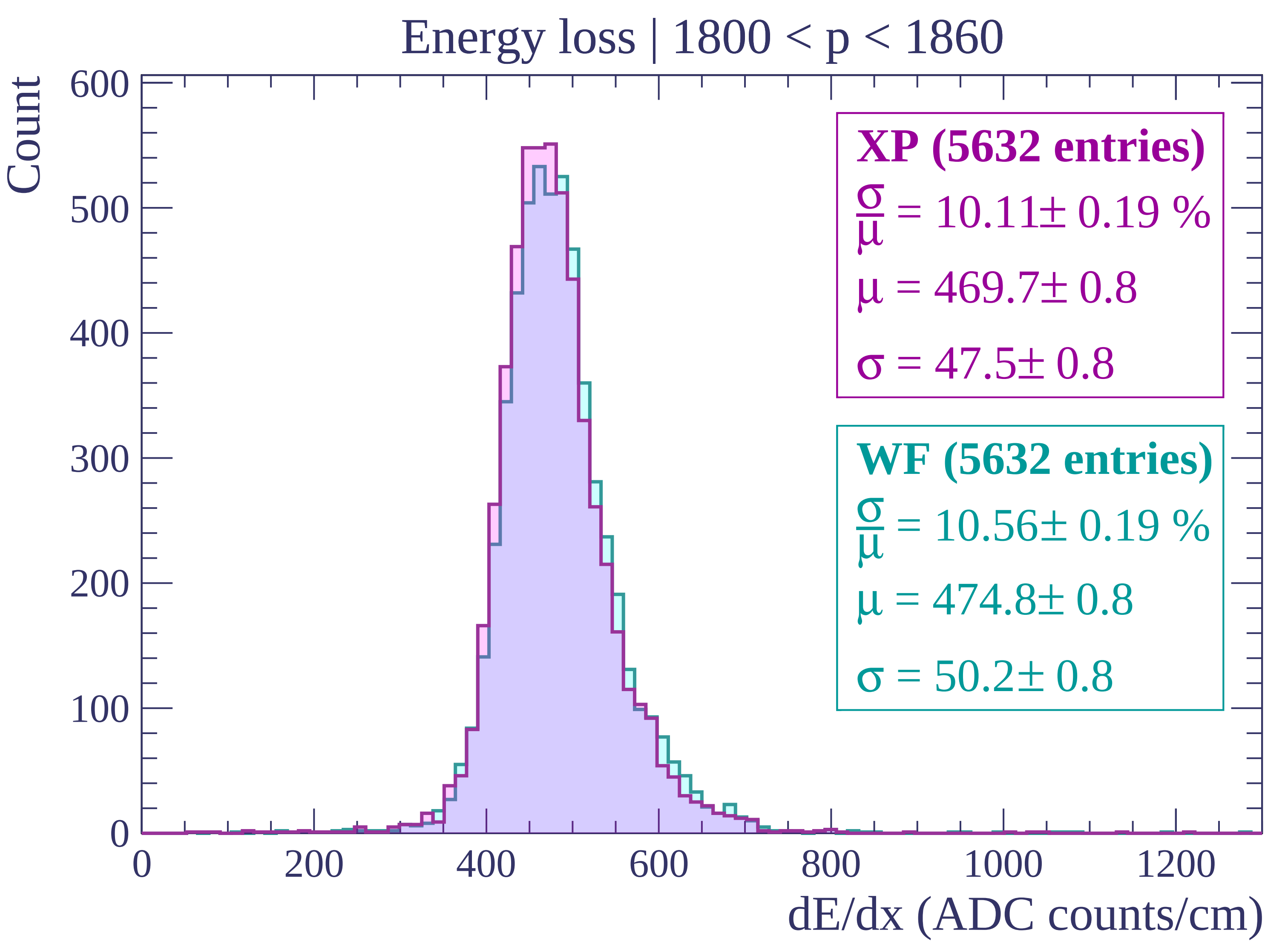
Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (6554 entries)
σ/μ = 10.29± 0.17 %
μ = 462.7± 0.7
σ = 47.6± 0.7
WF (6554 entries)
σ/μ = 10.42± 0.17 %
μ = 467.1± 0.8
σ = 48.7± 0.7



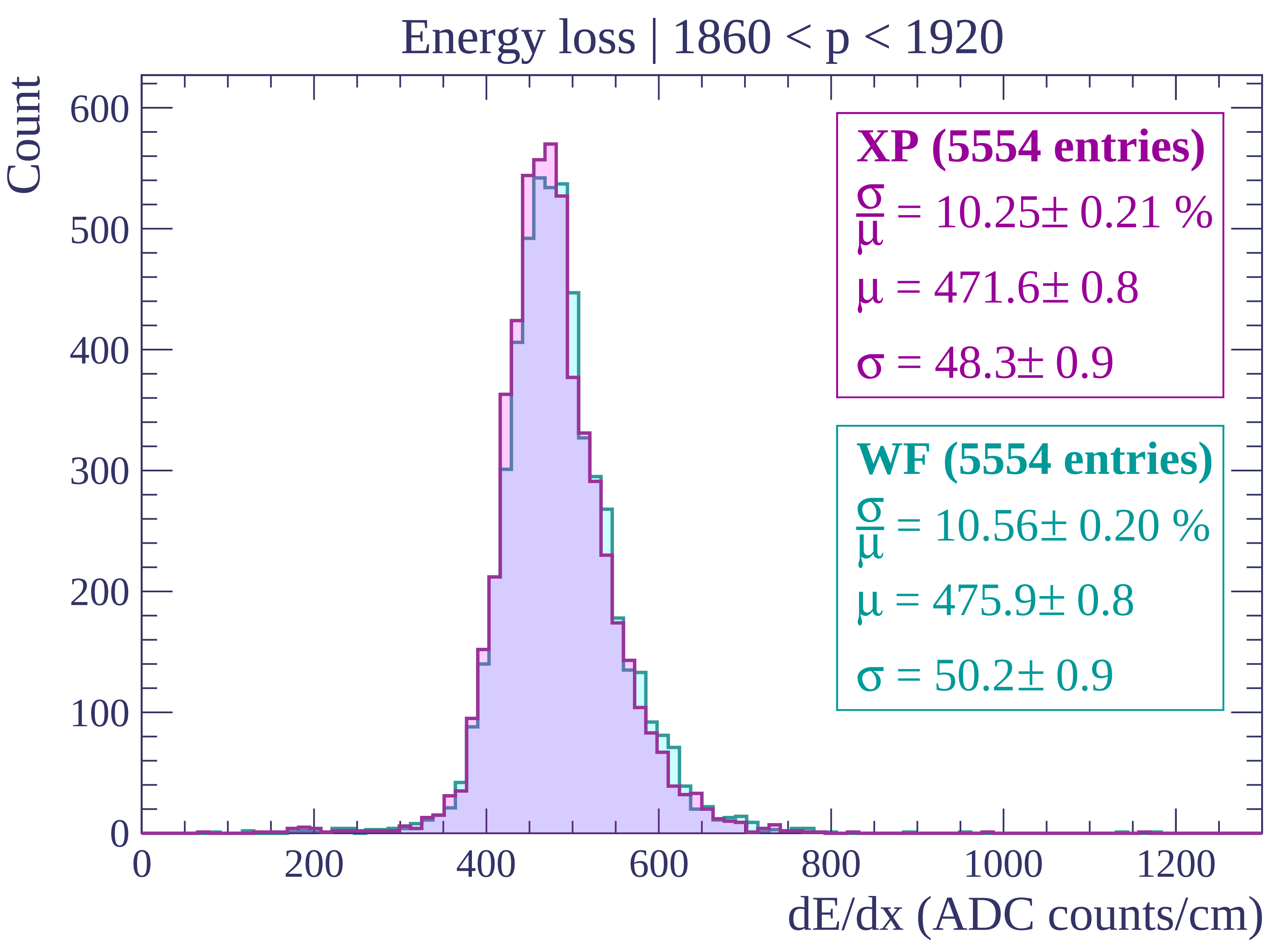
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (6433 entries)
σ/μ = 10.32± 0.17 %
μ = 464.4± 0.8
σ = 47.9± 0.7
WF (6433 entries)
σ/μ = 10.46± 0.18 %
μ = 467.9± 0.8
σ = 48.9± 0.8


Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (6136 entries)
σ/μ = 10.12± 0.17 %
μ = 466.9± 0.8
σ = 47.3± 0.7
WF (6136 entries)
σ/μ = 10.49± 0.20 %
μ = 470.3± 0.8
σ = 49.3± 0.9

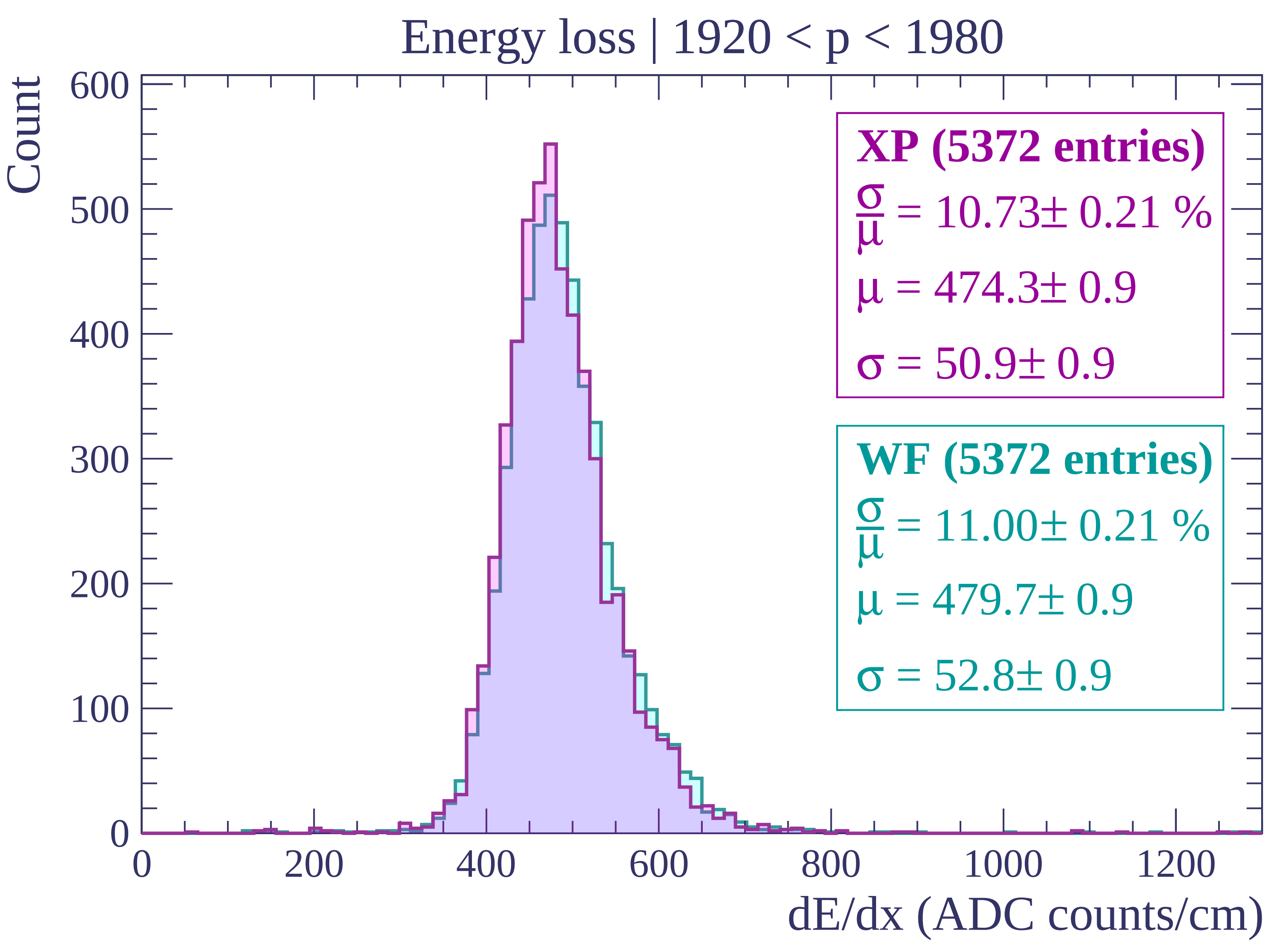

Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5956 entries)
σ/μ = 10.22± 0.17 %
μ = 468.2± 0.8
σ = 47.9± 0.7
WF (5956 entries)
σ/μ = 10.58± 0.17 %
μ = 473.5± 0.8
σ = 50.1± 0.7



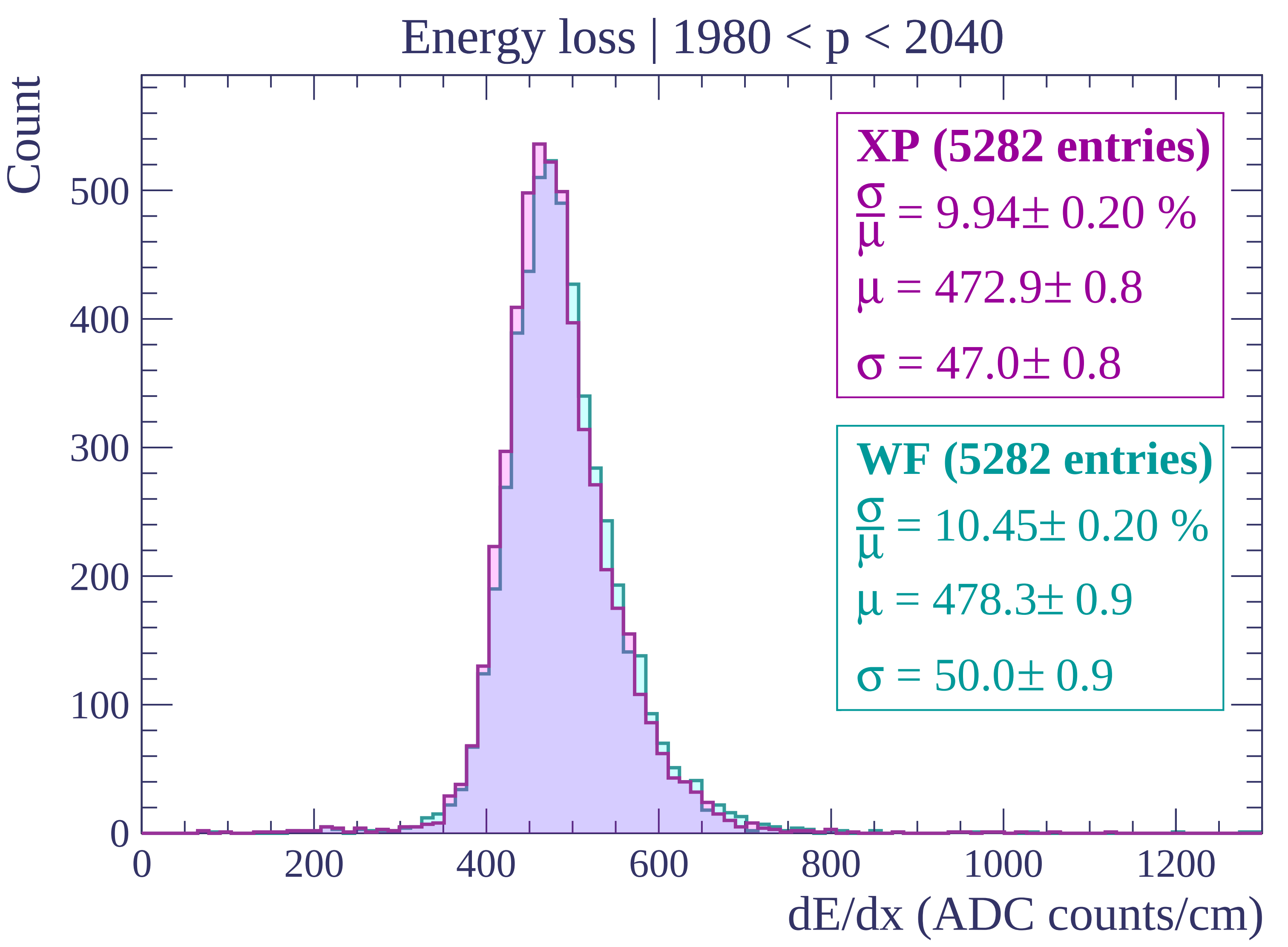
Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (5632 entries)
$\frac{\sigma}{\mu}$ = 10.11± 0.19 %
μ = 469.7± 0.8
σ = 47.5± 0.8
WF (5632 entries)
$\frac{\sigma}{\mu}$ = 10.56± 0.19 %
μ = 474.8± 0.8
σ = 50.2± 0.8

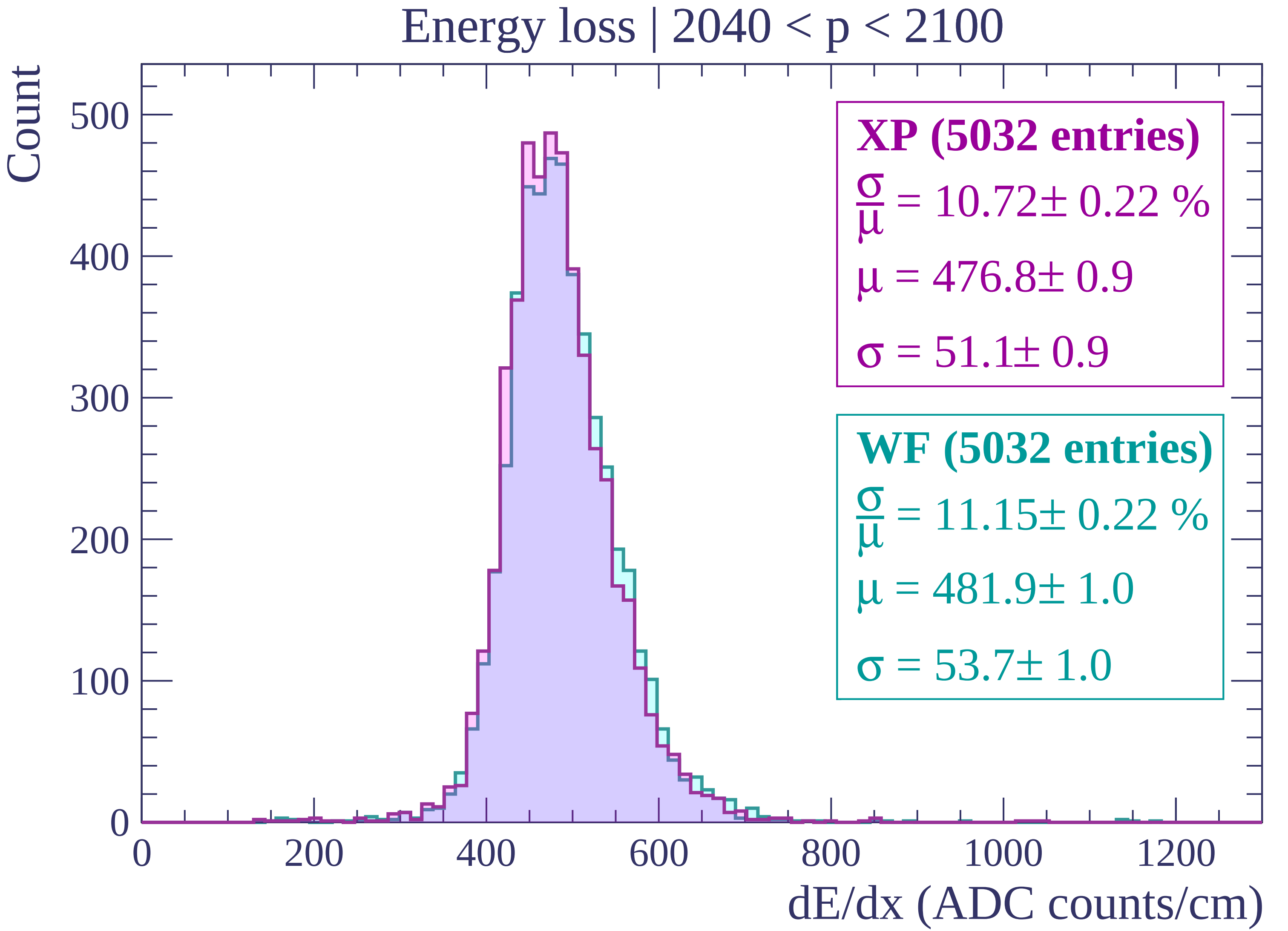


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5554 entries)
$\frac{\sigma}{\mu}$ = 10.25± 0.21 %
μ = 471.6± 0.8
σ = 48.3± 0.9
WF (5554 entries)
$\frac{\sigma}{\mu}$ = 10.56± 0.20 %
μ = 475.9± 0.8
σ = 50.2± 0.9

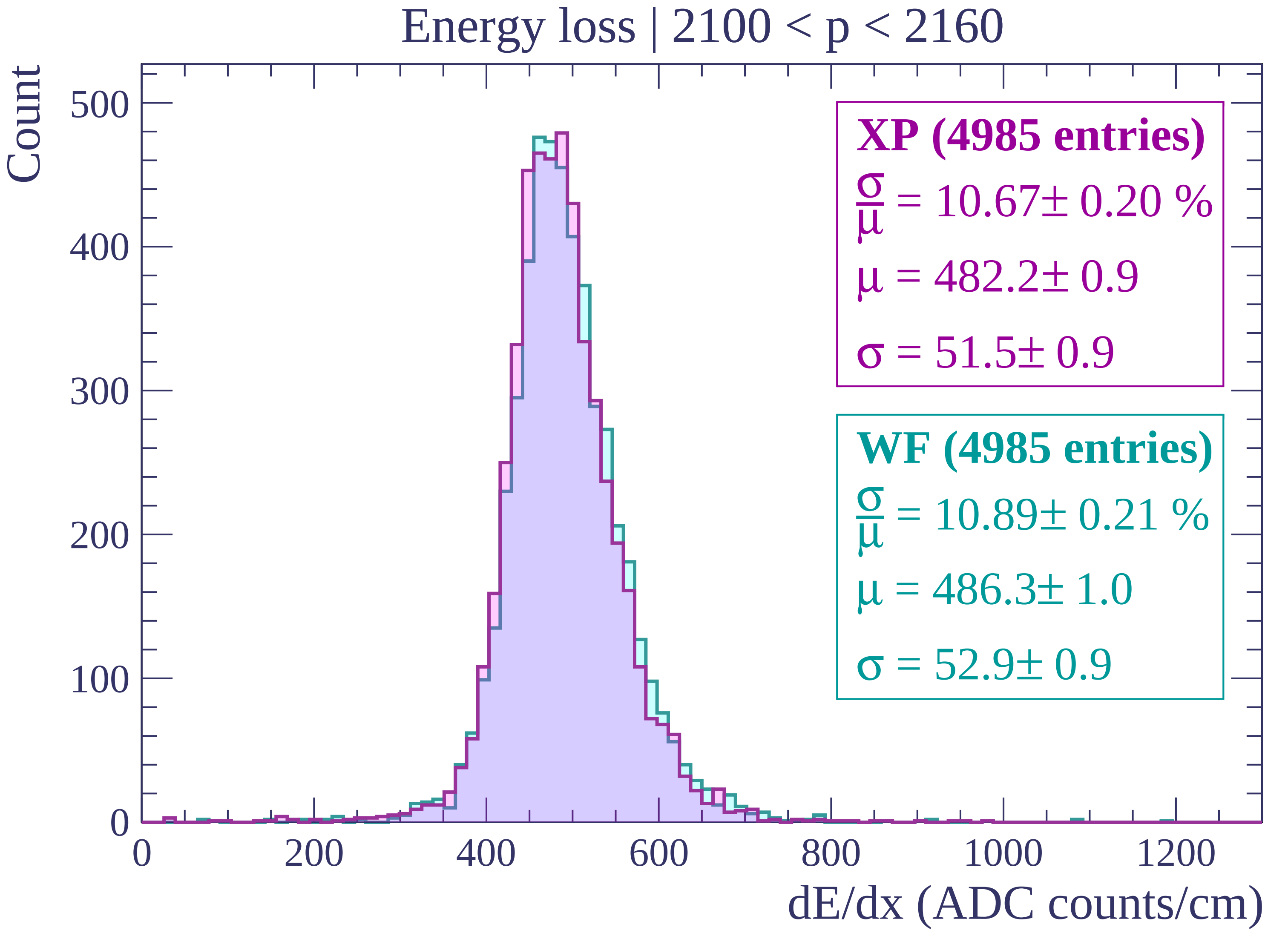


Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5372 entries)
$\frac{\sigma}{\mu}$ = 10.73± 0.21 %
μ = 474.3± 0.9
σ = 50.9± 0.9
WF (5372 entries)
$\frac{\sigma}{\mu}$ = 11.00± 0.21 %
μ = 479.7± 0.9
σ = 52.8± 0.9

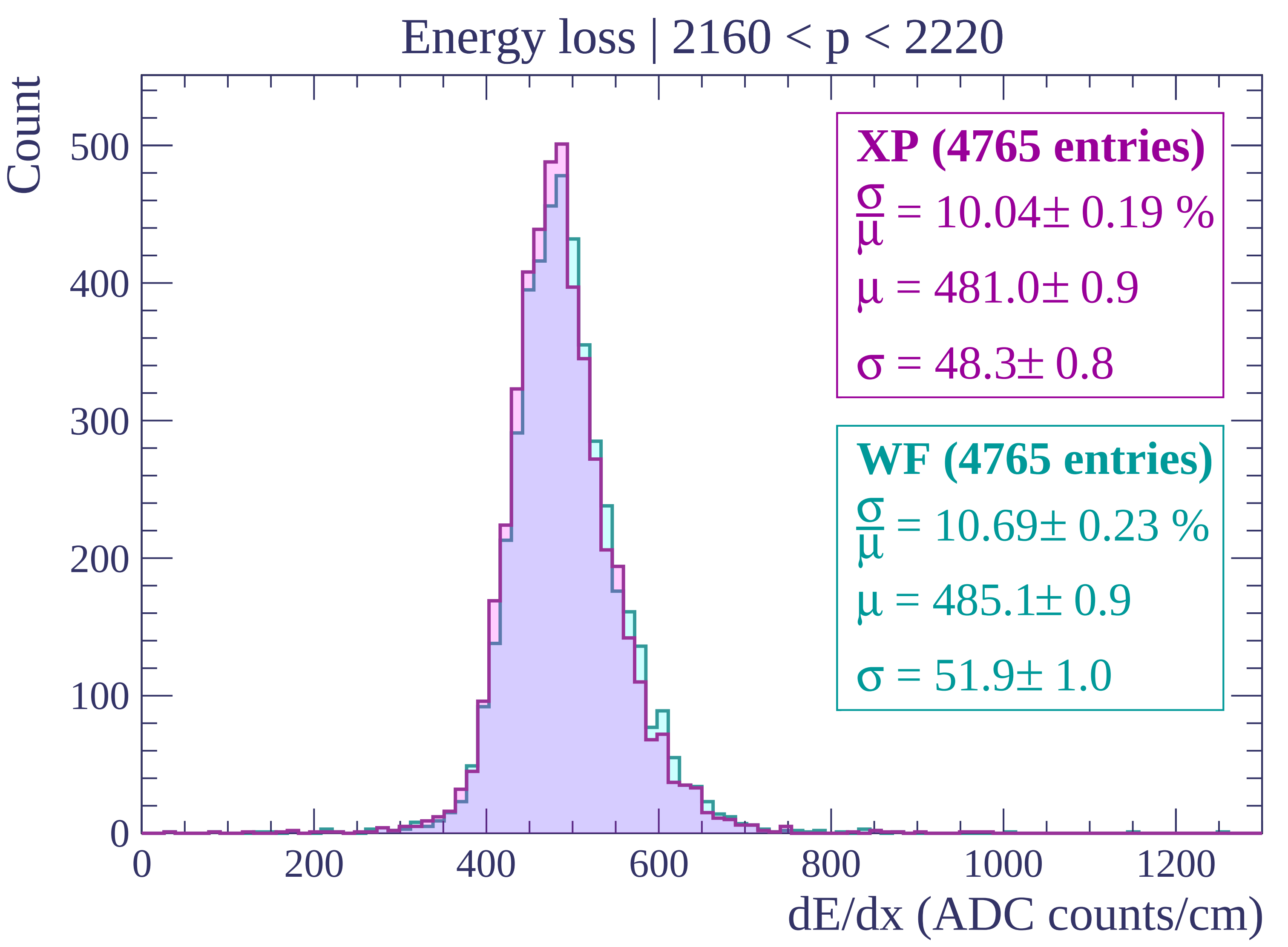

Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5282 entries)
σ/μ = 9.94± 0.20 %
μ = 472.9± 0.8
σ = 47.0± 0.8
WF (5282 entries)
σ/μ = 10.45± 0.20 %
μ = 478.3± 0.9
σ = 50.0± 0.9



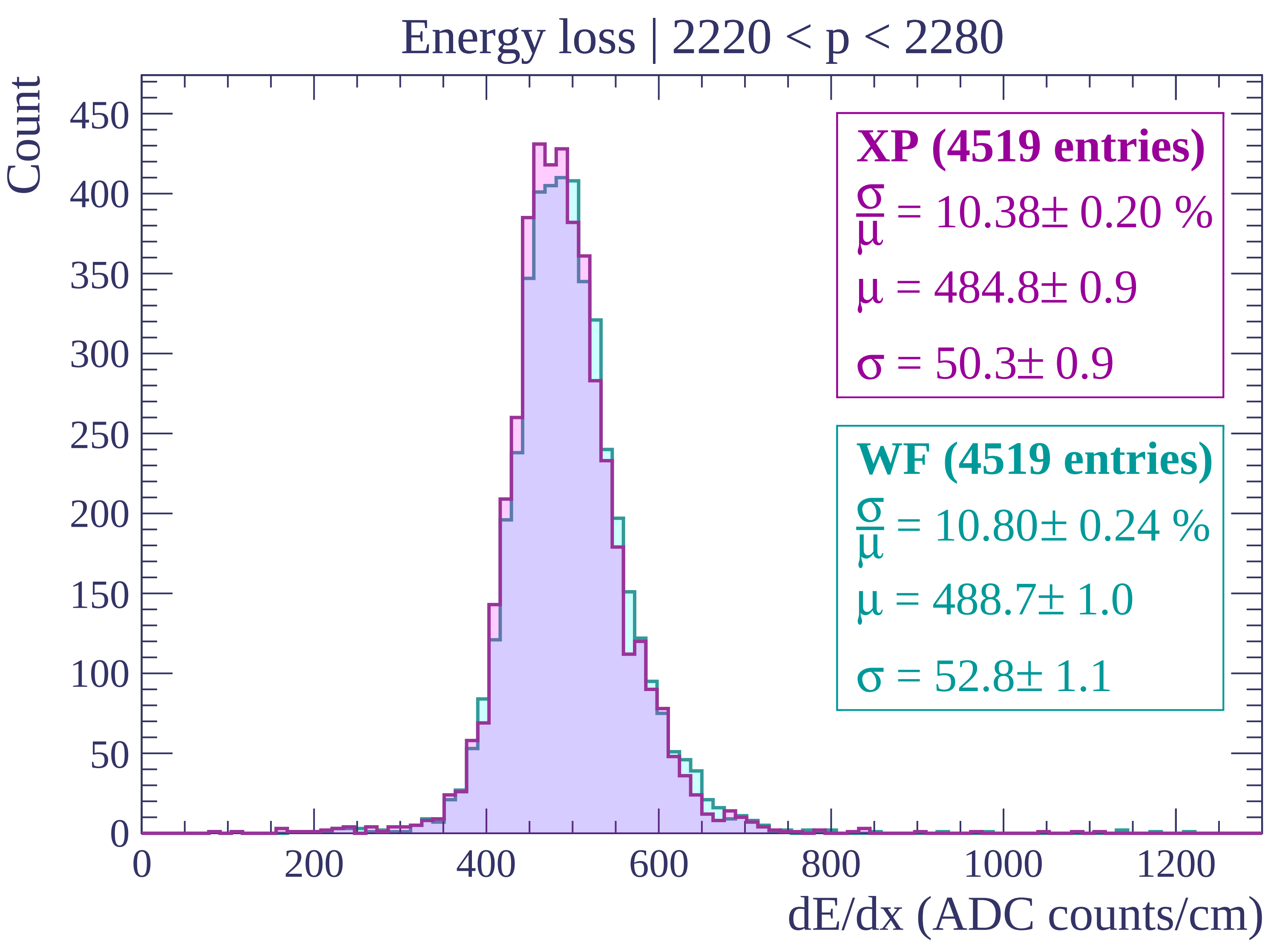
Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5032 entries)
σ/μ = 10.72± 0.22 %
μ = 476.8± 0.9
σ = 51.1± 0.9
WF (5032 entries)
σ/μ = 11.15± 0.22 %
μ = 481.9± 1.0
σ = 53.7± 1.0

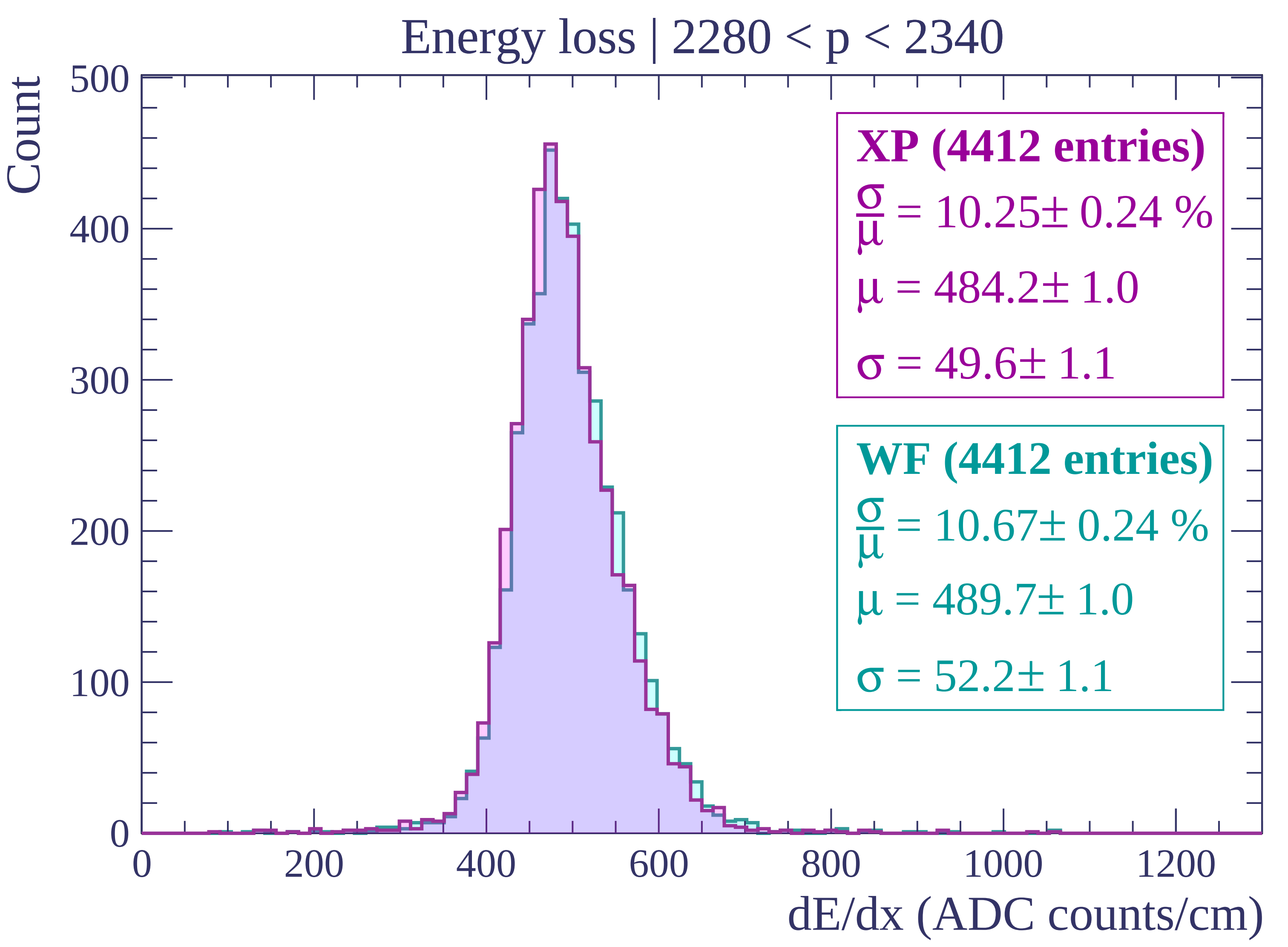


Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
XP (4985 entries)
$\frac{\sigma}{\mu}$ = 10.67± 0.20 %
μ = 482.2± 0.9
σ = 51.5± 0.9
WF (4985 entries)
$\frac{\sigma}{\mu}$ = 10.89± 0.21 %
μ = 486.3± 1.0
σ = 52.9± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200

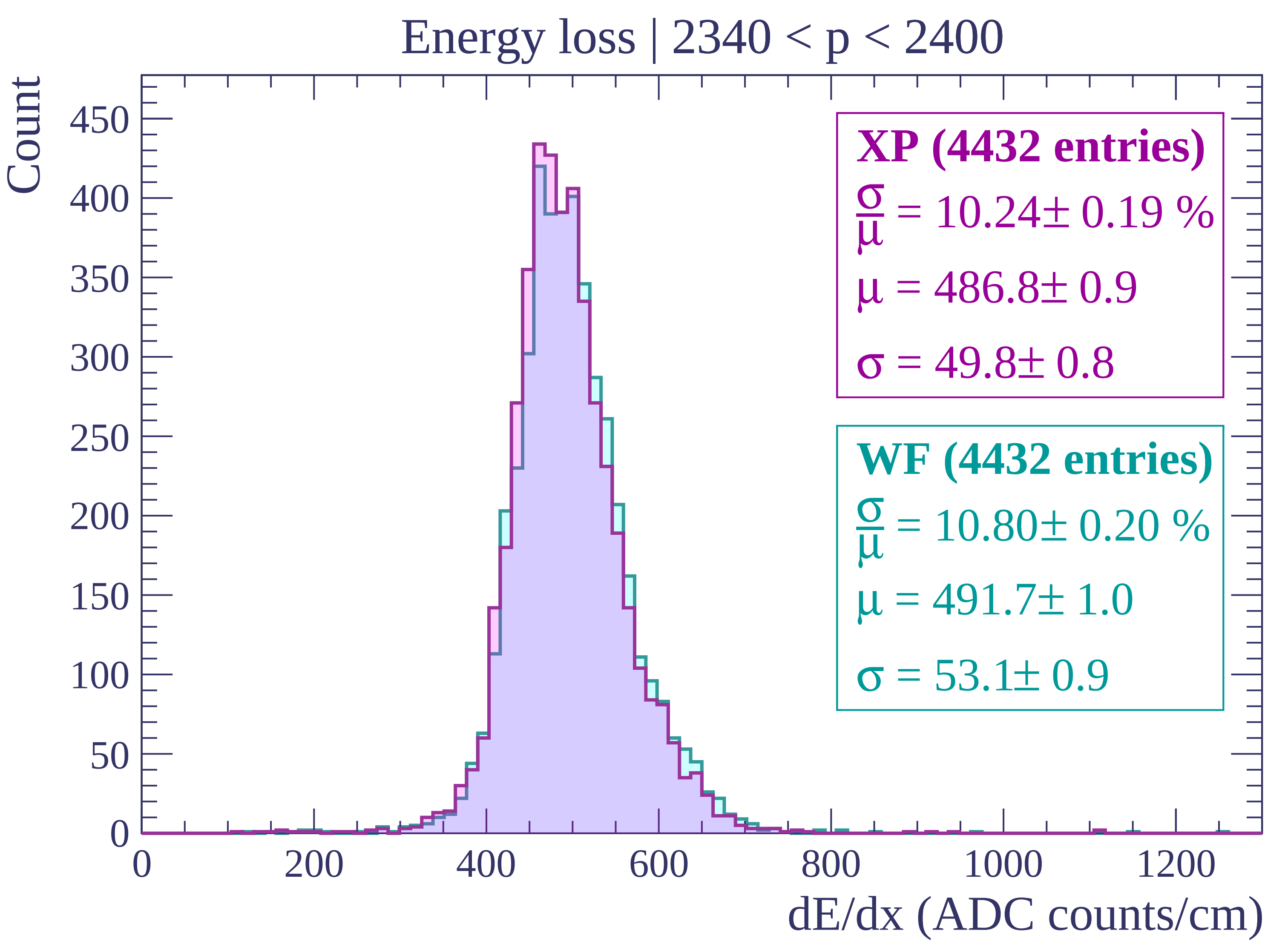


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4765 entries)
σ/μ = 10.04± 0.19 %
μ = 481.0± 0.9
σ = 48.3± 0.8
WF (4765 entries)
σ/μ = 10.69± 0.23 %
μ = 485.1± 0.9
σ = 51.9± 1.0

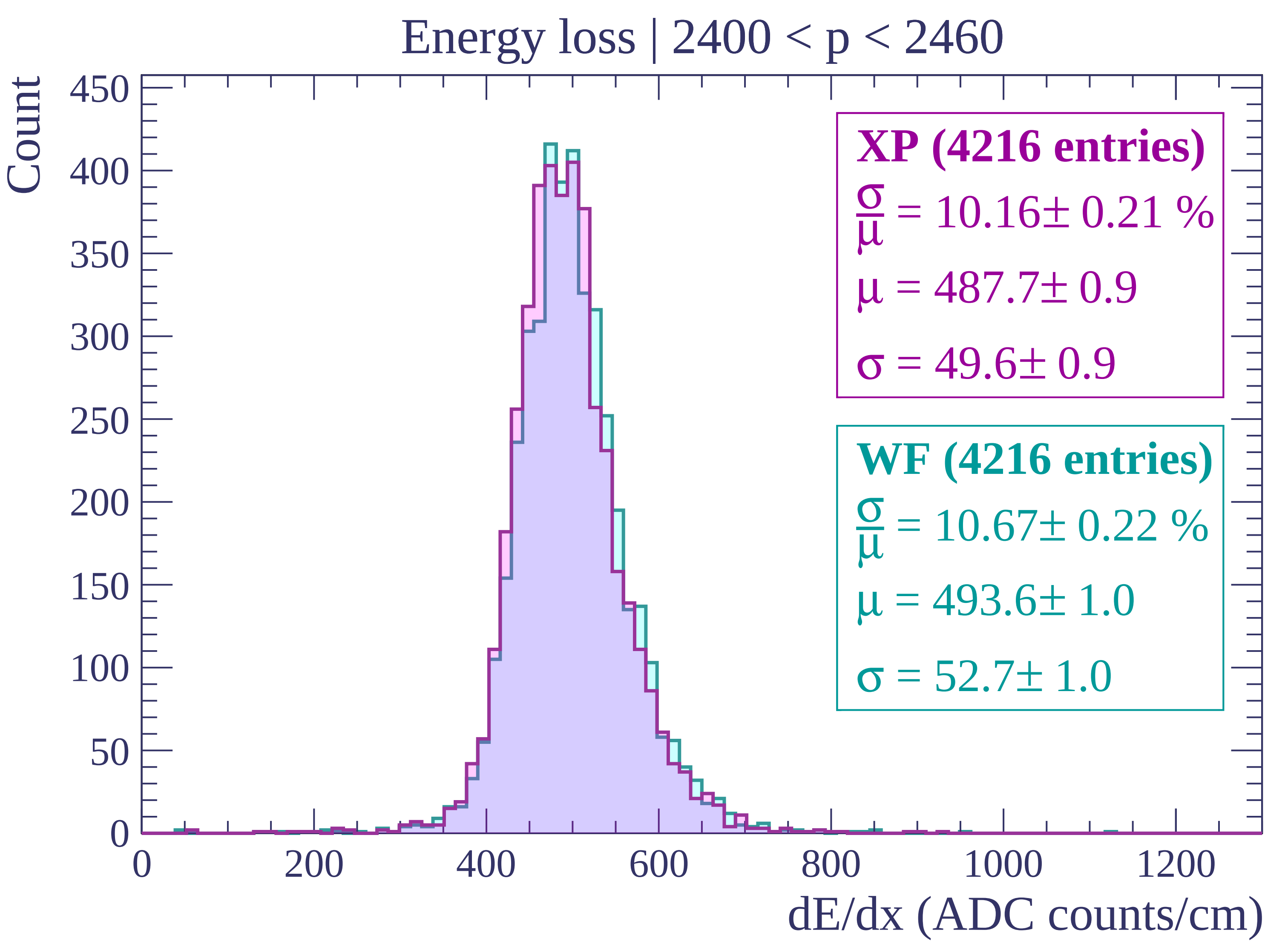

Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4519 entries)
σ/μ = 10.38± 0.20 %
μ = 484.8± 0.9
σ = 50.3± 0.9
WF (4519 entries)
σ/μ = 10.80± 0.24 %
μ = 488.7± 1.0
σ = 52.8± 1.1



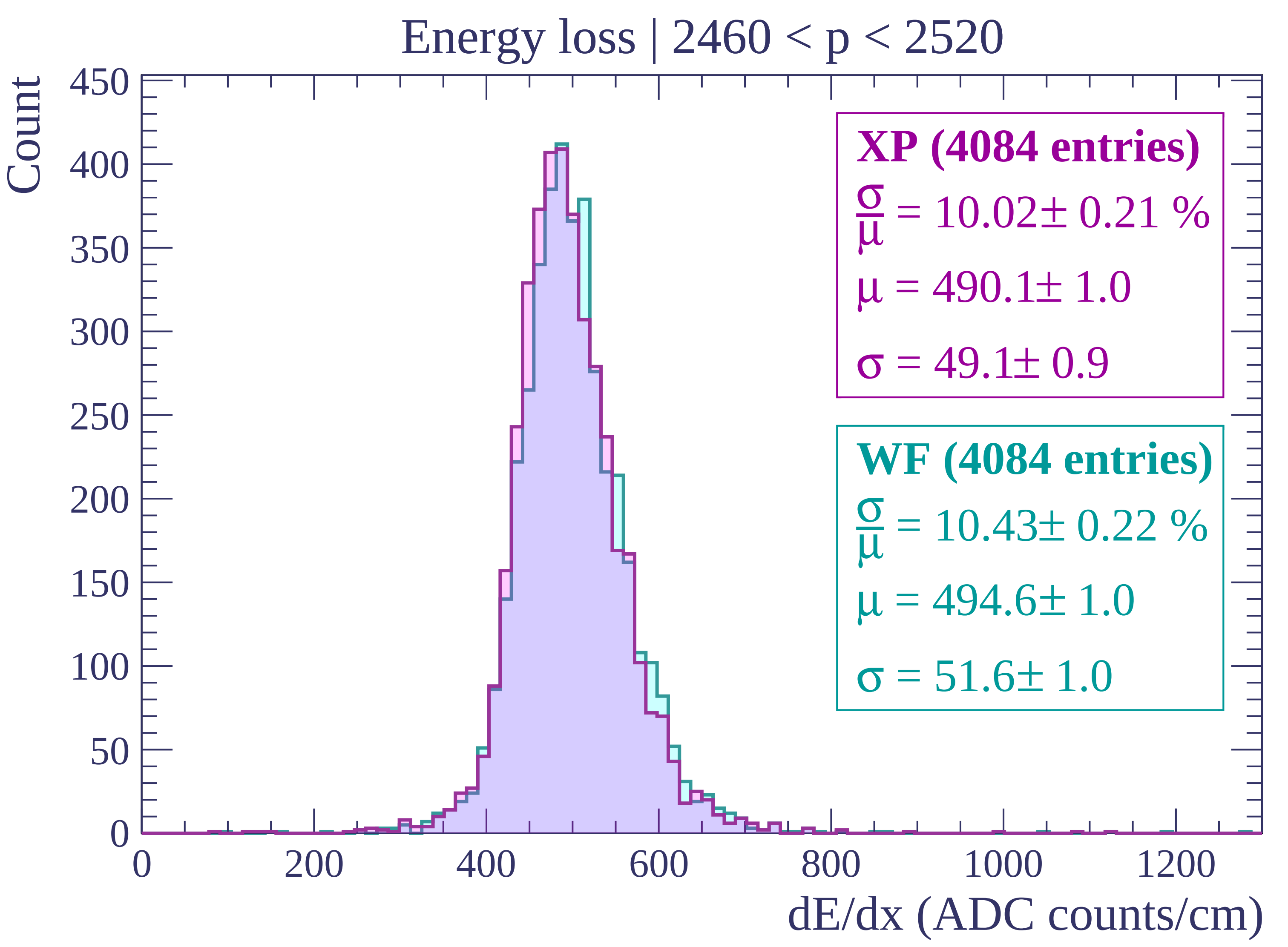
Energy loss | 2280 < p < 2340
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4412 entries)
$\frac{\sigma}{\mu}$ = 10.25± 0.24 %
μ = 484.2± 1.0
σ = 49.6± 1.1
WF (4412 entries)
$\frac{\sigma}{\mu}$ = 10.67± 0.24 %
μ = 489.7± 1.0
σ = 52.2± 1.1

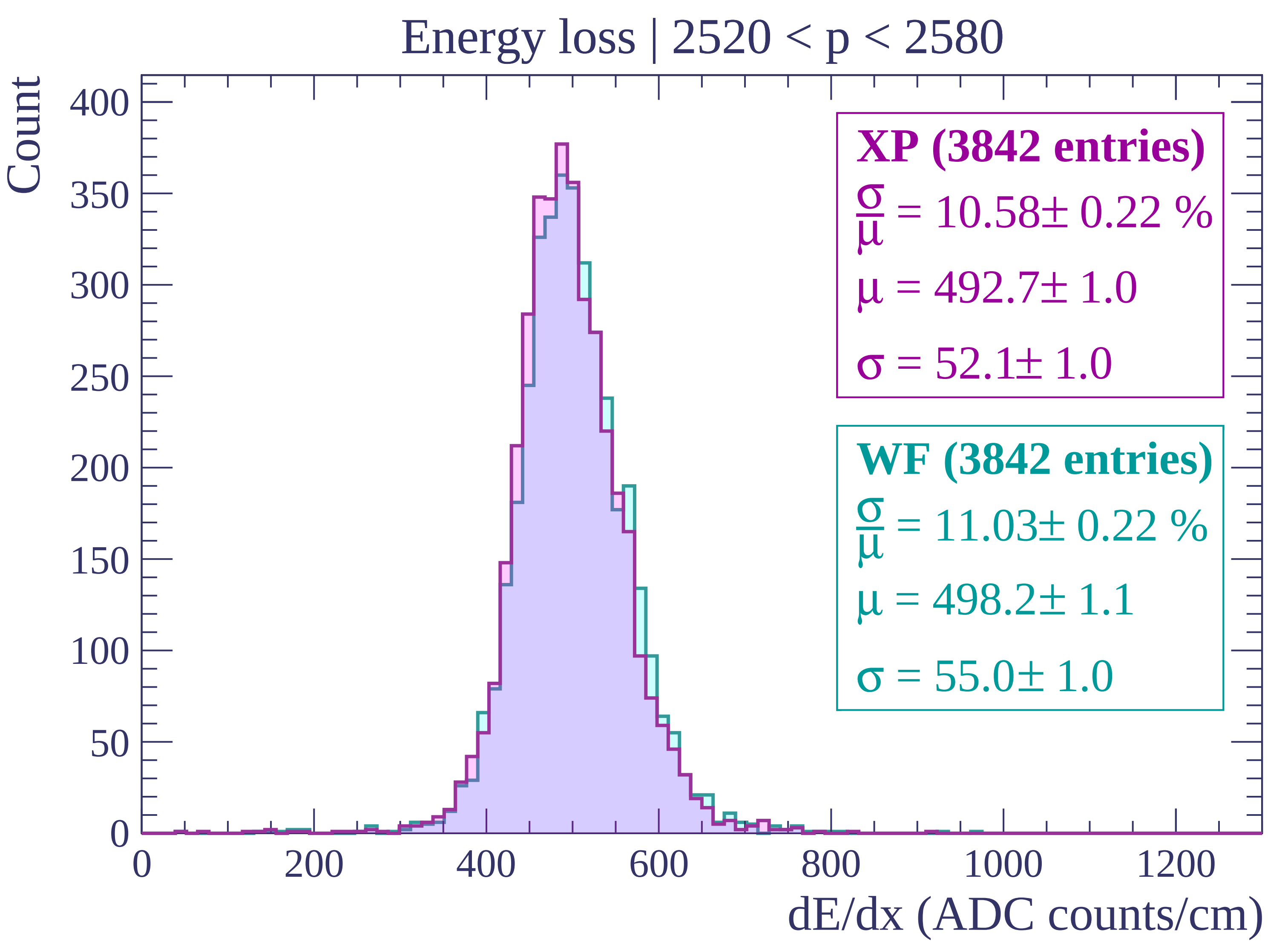


Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4432 entries)
$\frac{\sigma}{\mu}$ = 10.24± 0.19 %
μ = 486.8± 0.9
σ = 49.8± 0.8
WF (4432 entries)
$\frac{\sigma}{\mu}$ = 10.80± 0.20 %
μ = 491.7± 1.0
σ = 53.1± 0.9

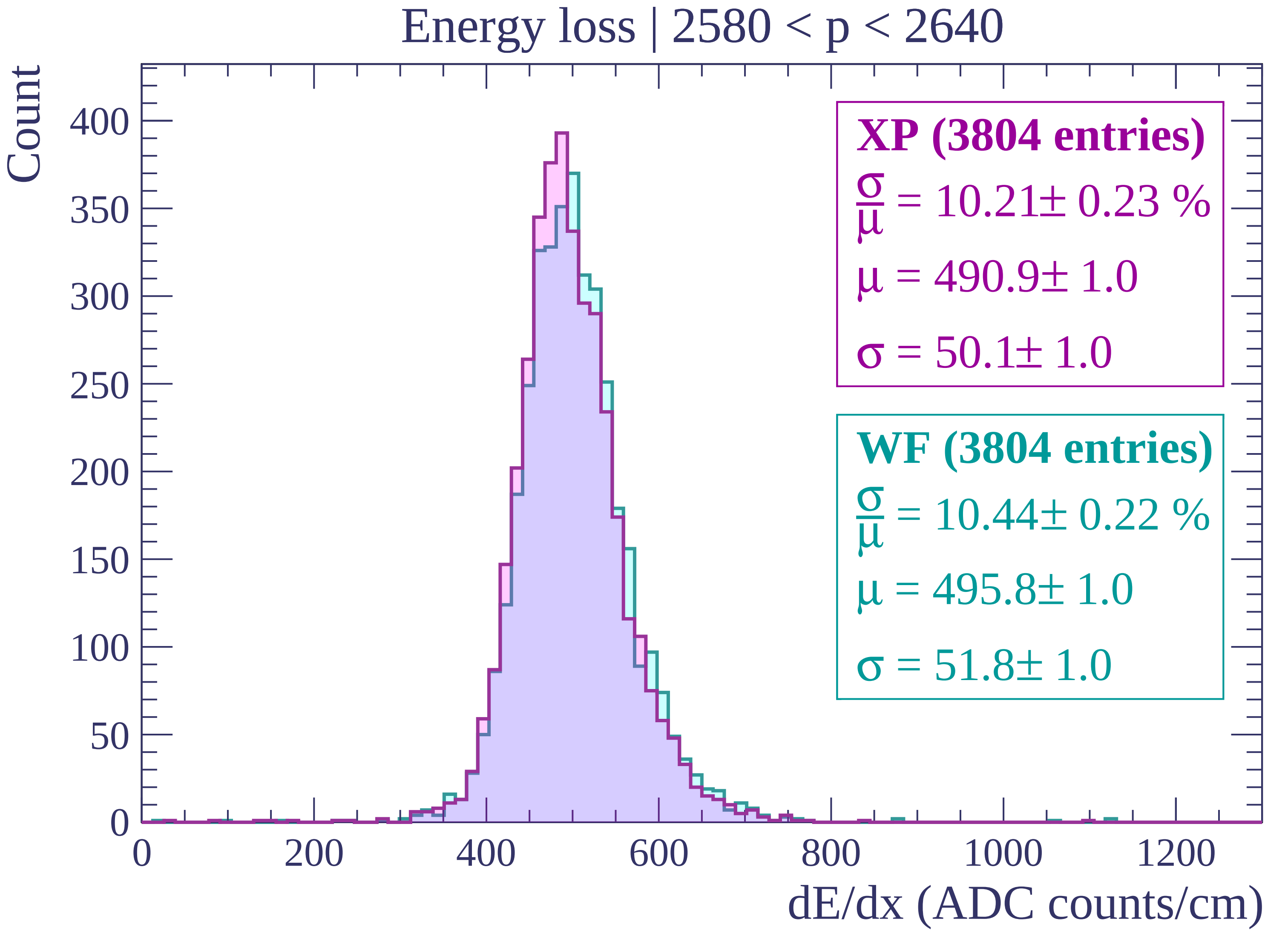


Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP (4216 entries)
σ/μ = 10.16± 0.21 %
μ = 487.7± 0.9
σ = 49.6± 0.9
WF (4216 entries)
σ/μ = 10.67± 0.22 %
μ = 493.6± 1.0
σ = 52.7± 1.0

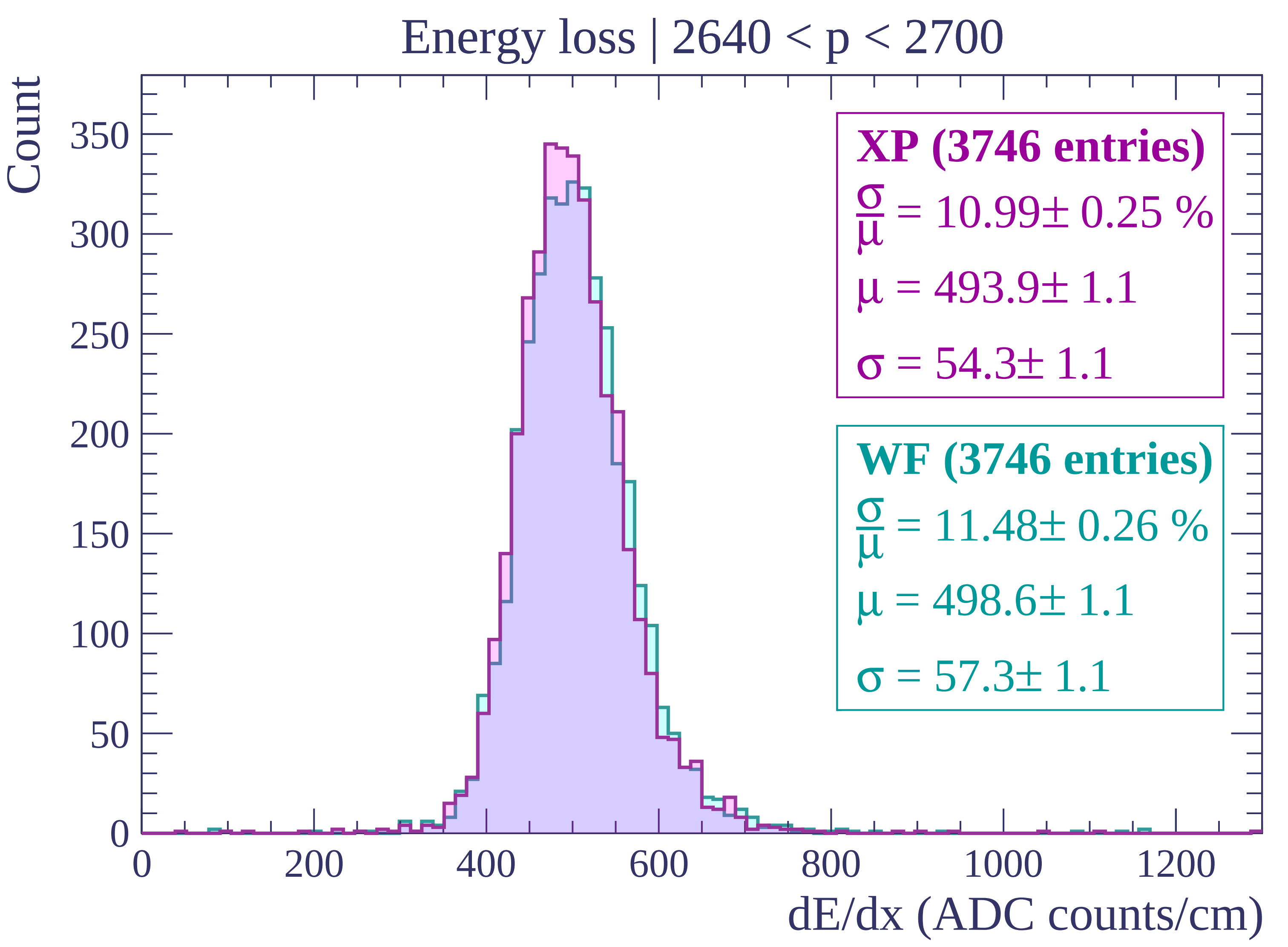

Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4084 entries)
$\frac{\sigma}{\mu}$ = 10.02± 0.21 %
μ = 490.1± 1.0
σ = 49.1± 0.9
WF (4084 entries)
$\frac{\sigma}{\mu}$ = 10.43± 0.22 %
μ = 494.6± 1.0
σ = 51.6± 1.0



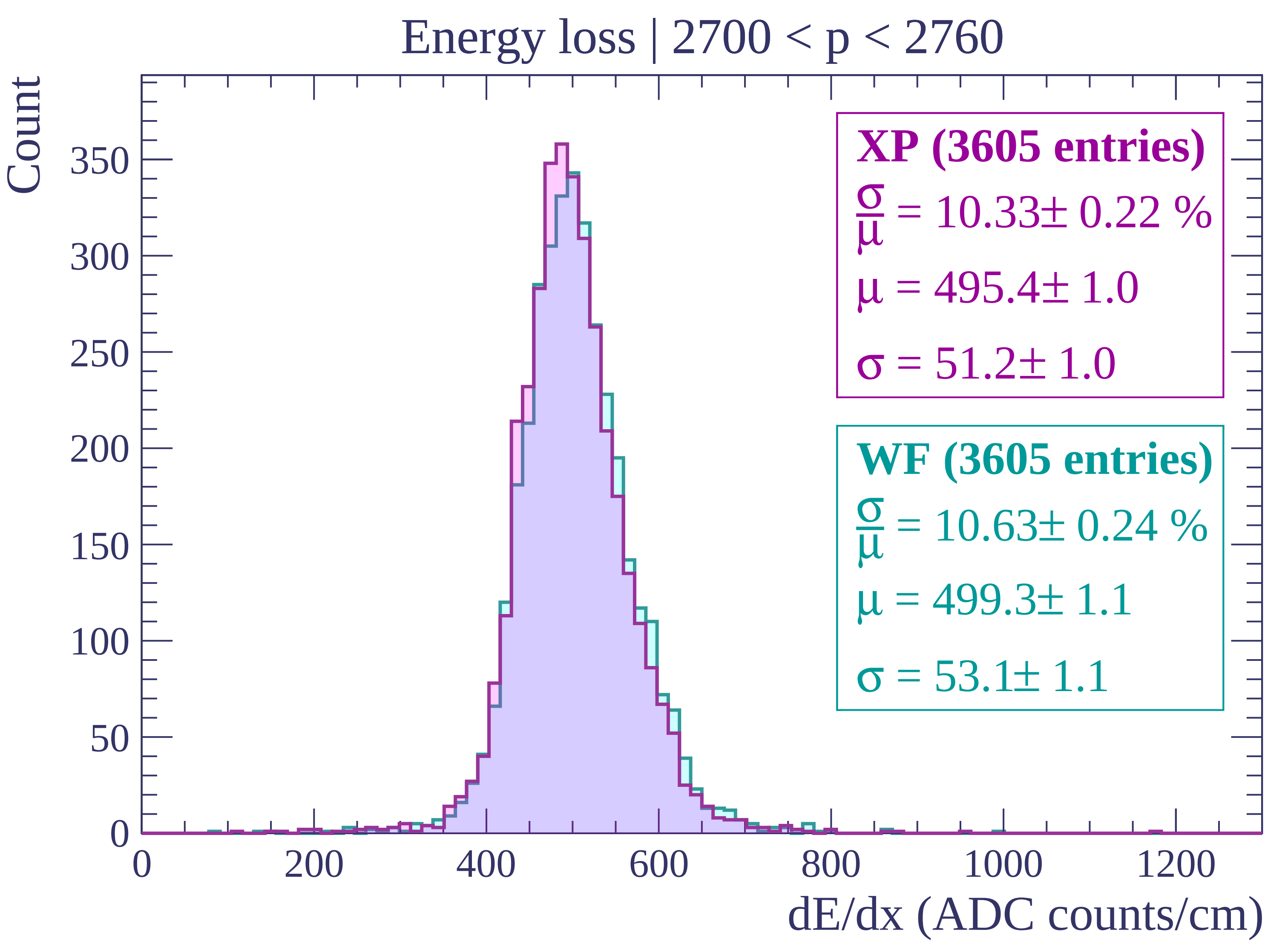
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
XP (3842 entries)
σ/μ = 10.58± 0.22 %
μ = 492.7± 1.0
σ = 52.1± 1.0
WF (3842 entries)
σ/μ = 11.03± 0.22 %
μ = 498.2± 1.1
σ = 55.0± 1.0

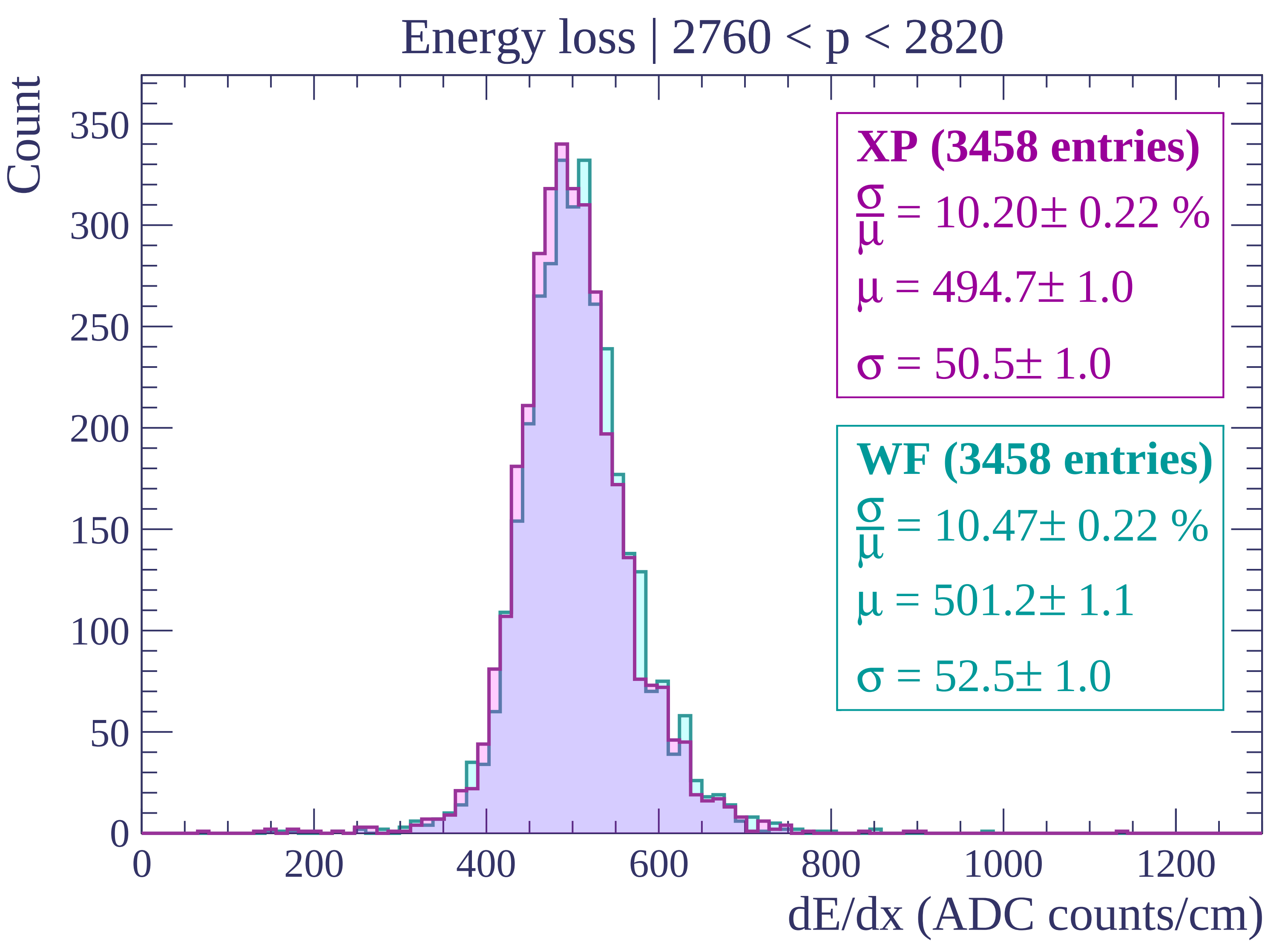


Energy loss | 2580 < p < 2640
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3804 entries)
σ/μ = 10.21± 0.23 %
μ = 490.9± 1.0
σ = 50.1± 1.0
WF (3804 entries)
σ/μ = 10.44± 0.22 %
μ = 495.8± 1.0
σ = 51.8± 1.0

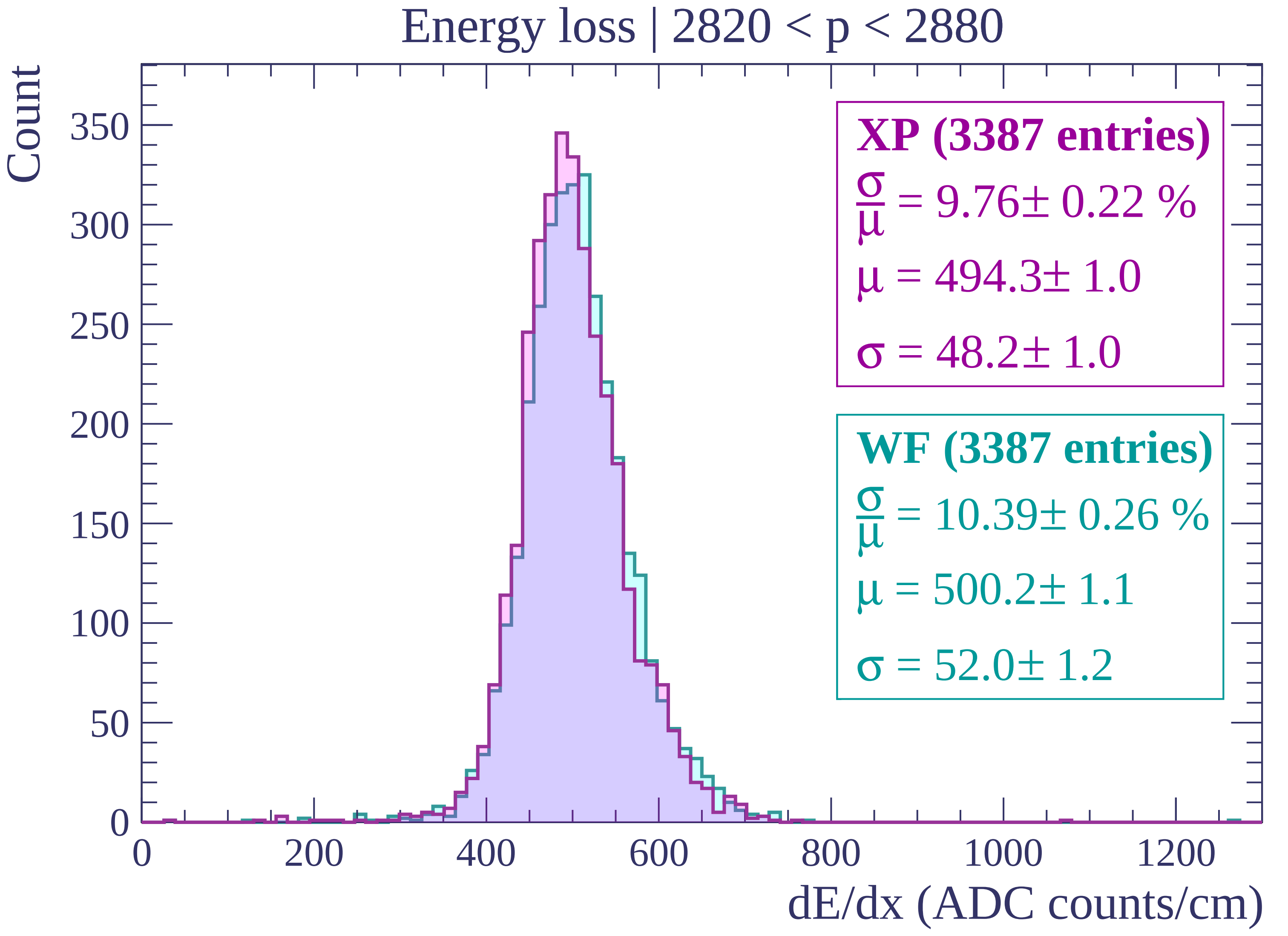


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
XP (3746 entries)
$\frac{\sigma}{\mu}$ = 10.99± 0.25 %
μ = 493.9± 1.1
σ = 54.3± 1.1
WF (3746 entries)
$\frac{\sigma}{\mu}$ = 11.48± 0.26 %
μ = 498.6± 1.1
σ = 57.3± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

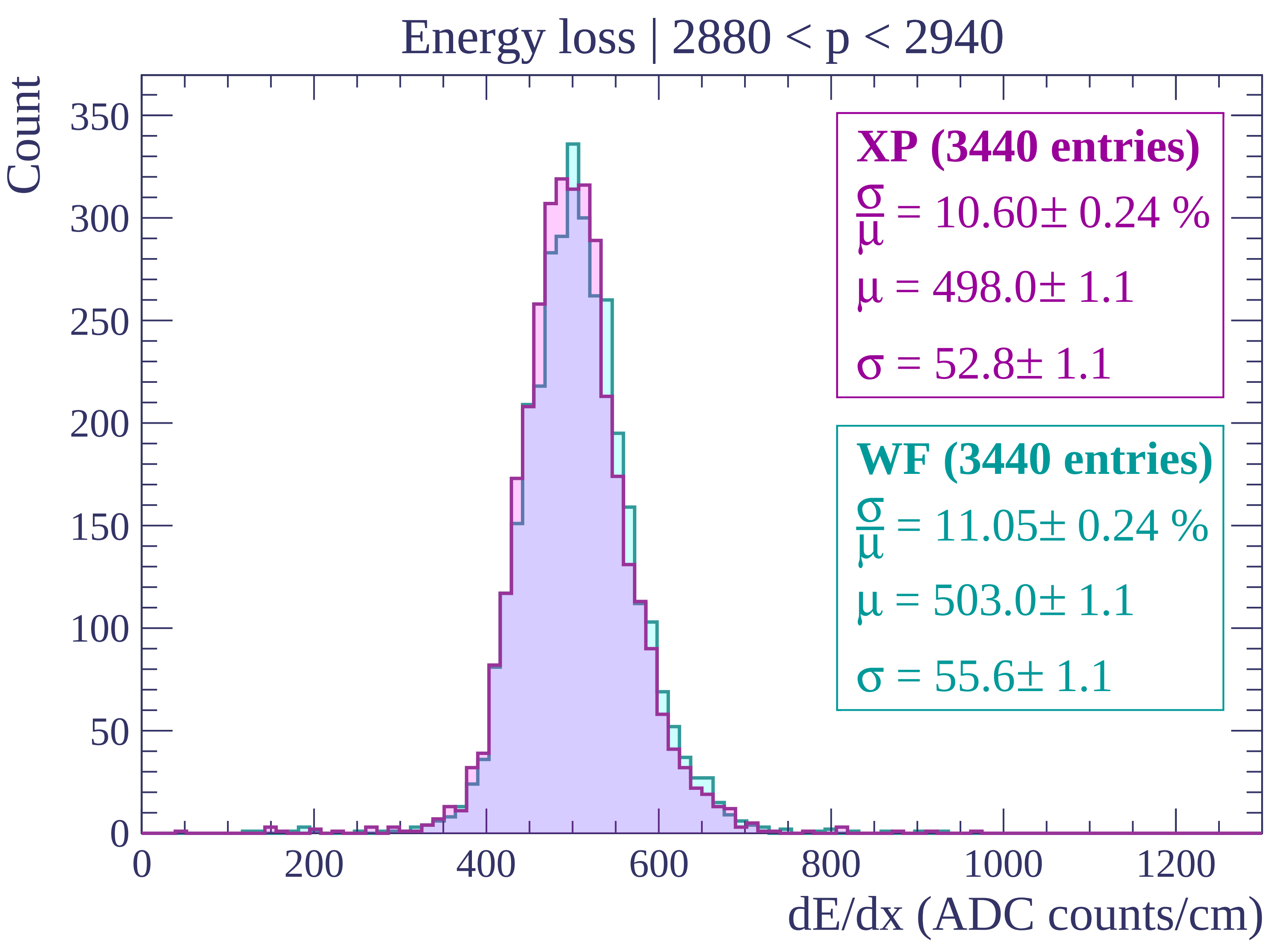

Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3605 entries)
σ/μ = 10.33± 0.22 %
μ = 495.4± 1.0
σ = 51.2± 1.0
WF (3605 entries)
σ/μ = 10.63± 0.24 %
μ = 499.3± 1.1
σ = 53.1± 1.1



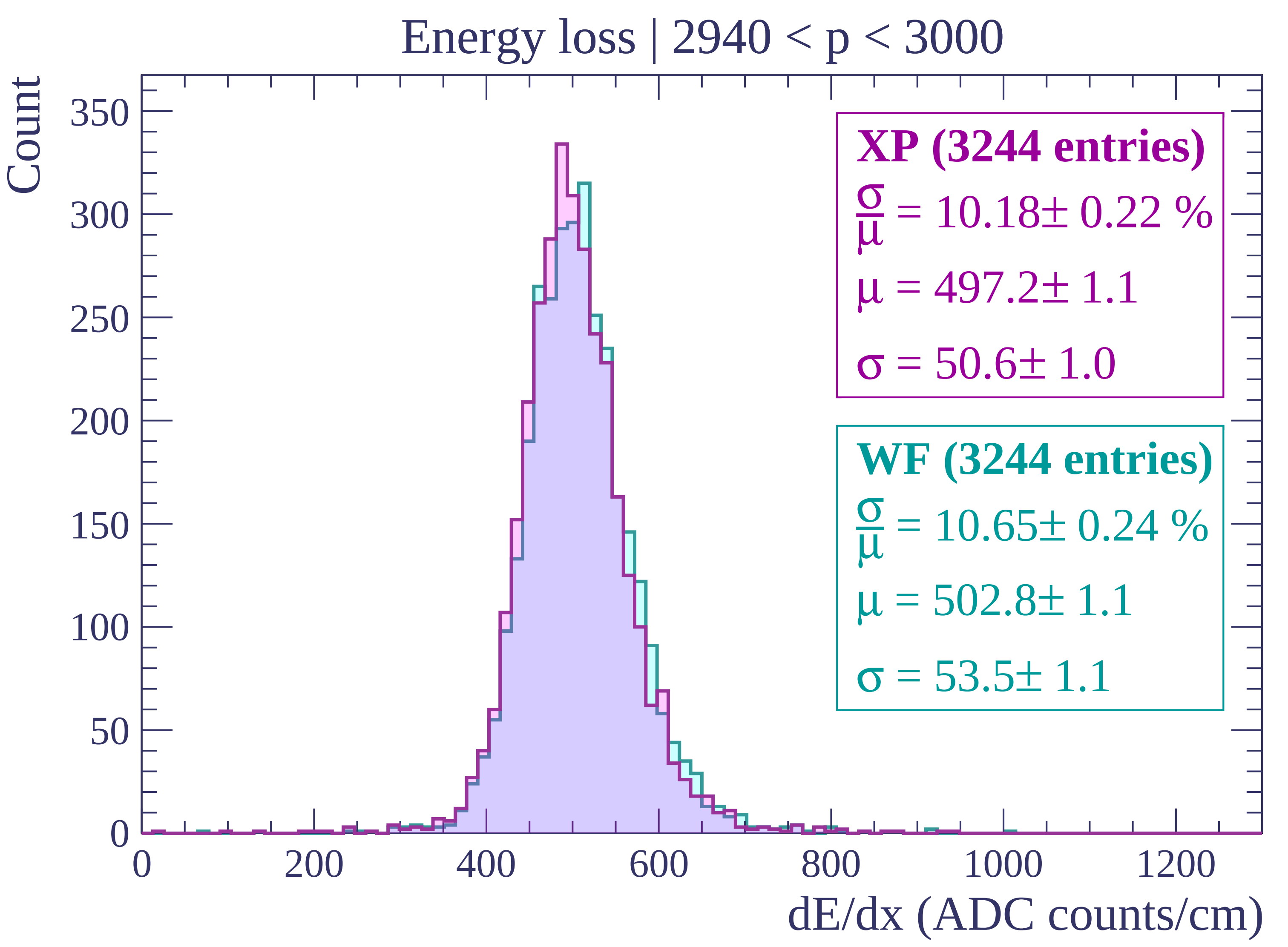
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3458 entries)
σ/μ = 10.20± 0.22 %
μ = 494.7± 1.0
σ = 50.5± 1.0
WF (3458 entries)
σ/μ = 10.47± 0.22 %
μ = 501.2± 1.1
σ = 52.5± 1.0



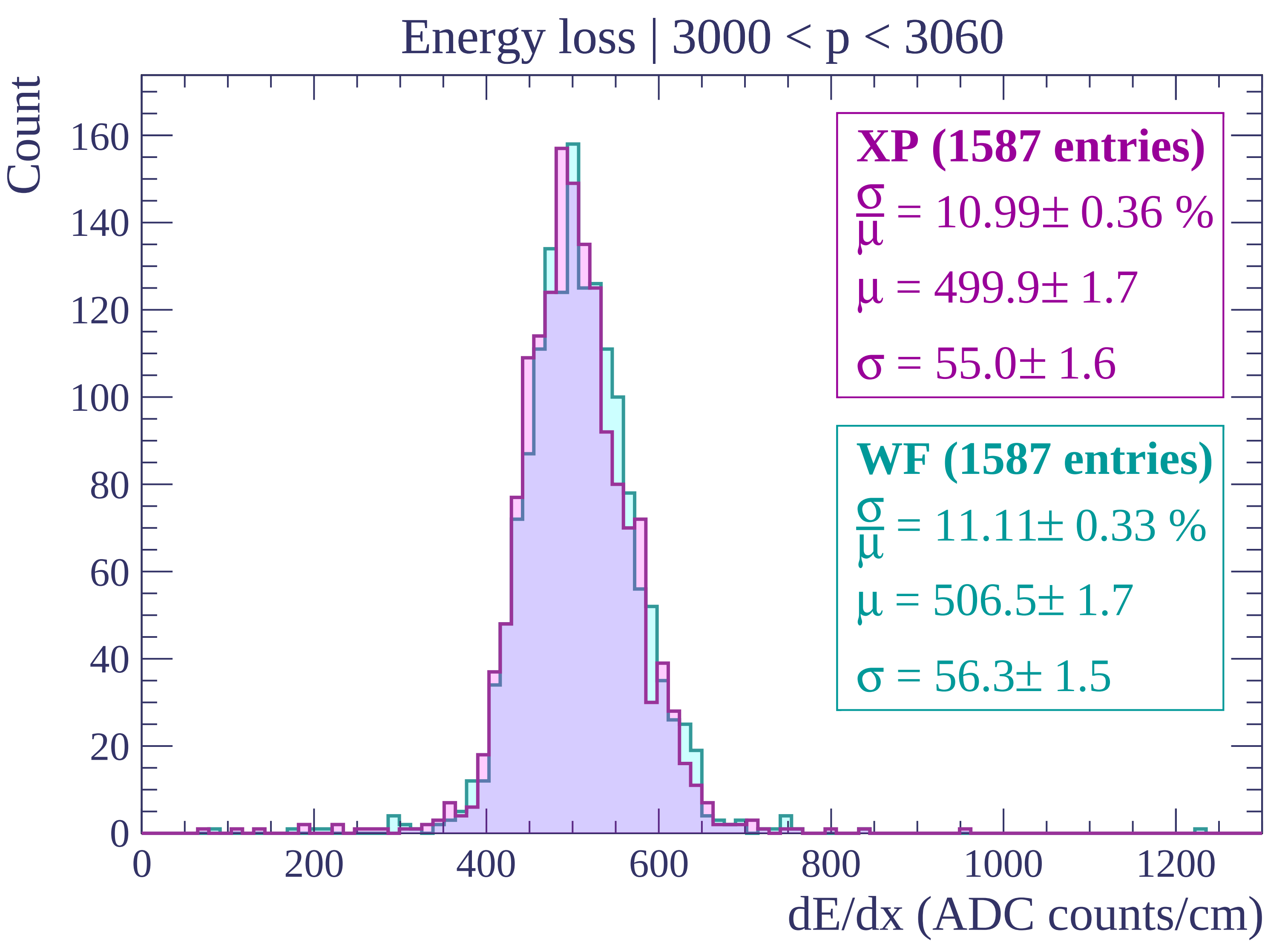
Energy loss | 2820 < p < 2880
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3387 entries)
σ/μ = 9.76± 0.22 %
μ = 494.3± 1.0
σ = 48.2± 1.0
WF (3387 entries)
σ/μ = 10.39± 0.26 %
μ = 500.2± 1.1
σ = 52.0± 1.2



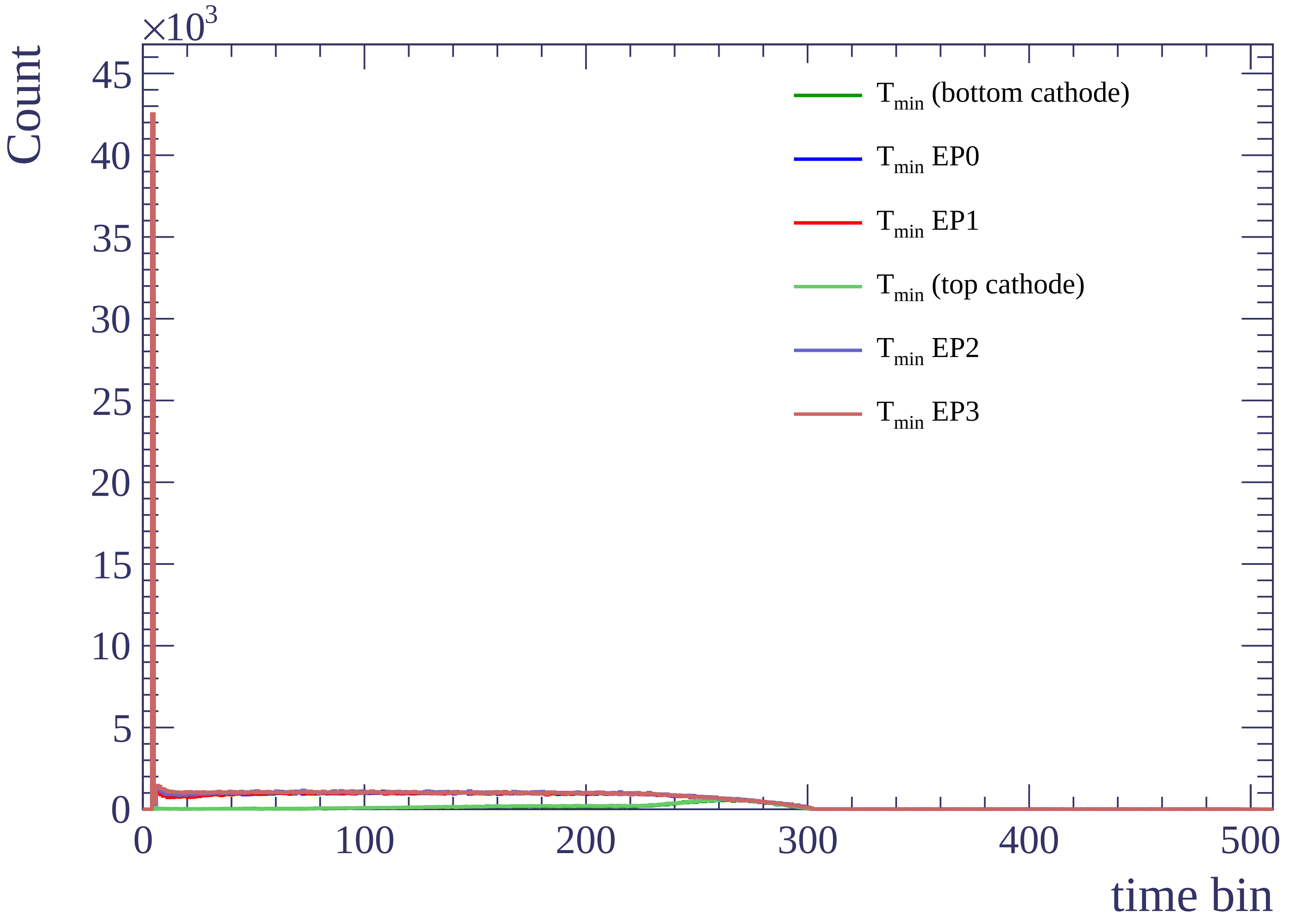
Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3440 entries)
σ/μ = 10.60± 0.24 %
μ = 498.0± 1.1
σ = 52.8± 1.1
WF (3440 entries)
σ/μ = 11.05± 0.24 %
μ = 503.0± 1.1
σ = 55.6± 1.1



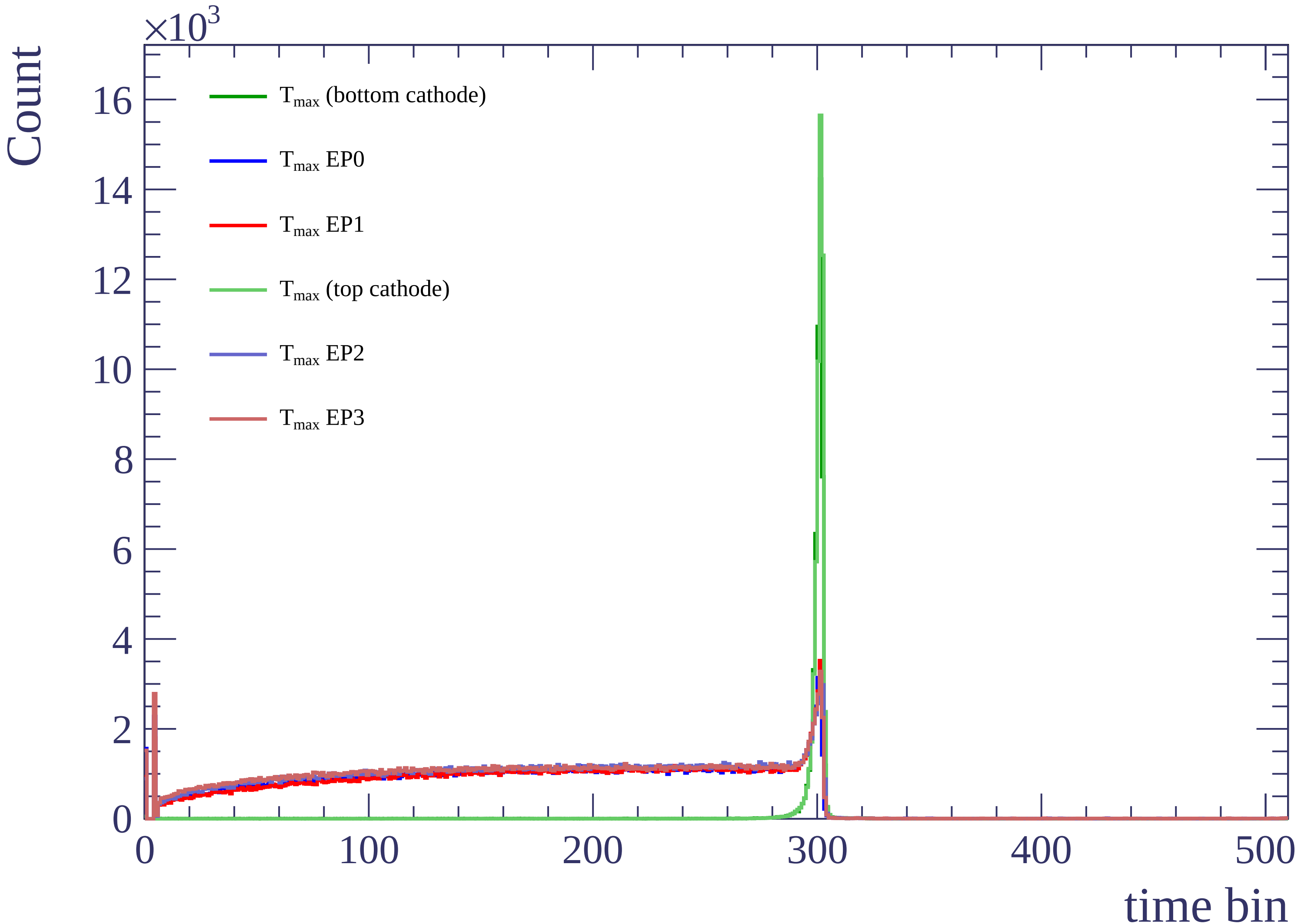
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP (3244 entries)
$\frac{\sigma}{\mu}$ = 10.18± 0.22 %
μ = 497.2± 1.1
σ = 50.6± 1.0
WF (3244 entries)
$\frac{\sigma}{\mu}$ = 10.65± 0.24 %
μ = 502.8± 1.1
σ = 53.5± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1587 entries)
σ/μ = 10.99± 0.36 %
μ = 499.9± 1.7
σ = 55.0± 1.6
WF (1587 entries)
σ/μ = 11.11± 0.33 %
μ = 506.5± 1.7
σ = 56.3± 1.5



Count
×10³
0
5
10
15
20
25
30
35
40
45
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



×10^3
Count
16
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3